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Preface

Linear Algebra is often the first place where a student encounters the true power of mathematical abstraction. In an introductory course, one learns to manipulate matrices, solve systems of linear equations, and perhaps calculate eigenvalues. But *Advanced* Linear Algebra is about something deeper: it is the study of structures.

The Philosophy of this Book

The central theme of these notes is the transition from concrete calculation to abstract reasoning. We begin not with numbers, but with axioms. By choosing to define mathematical objects—be they vectors, linear transformations, or inner products—entirely by how they behave rather than what they are, we unlock a universal language.

A theorem proved for an abstract vector space is a theorem proved simultaneously for geometric arrows, for polynomials, for continuous functions, and for quantum states. This “solve once, apply everywhere” philosophy is the heartbeat of modern mathematics.

How to Use These Notes

This book is designed to be a rigorous companion for a second course in linear algebra. Each chapter is structured to build a solid theoretical foundation:

- **Rigorous Definitions:** We start with precise mathematical statements to ensure no ambiguity in our logic.
- **Complete Proofs:** Every major result is accompanied by a full proof. Mathematics is not a spectator sport; I encourage you to attempt the proofs yourself before reading the solutions.
- **Illustrative Examples:** Abstract theory is balanced with concrete examples to build intuition.
- **Exercises:** Practice problems are provided at the end of sections to test your understanding.

Prerequisites

To get the most out of these notes, you should have a solid grasp of:

- **Introductory Linear Algebra:** You should be comfortable with matrix multiplication, determinants, and Gaussian elimination.

- **Calculus:** Familiarity with derivatives and integrals is assumed for some of our more advanced examples.
 - **Mathematical Maturity:** A basic understanding of logic and proof techniques (induction, contradiction, contrapositive) is essential. These are reviewed briefly in Chapter 0.
-

Welcome to the world of abstract linear algebra. I hope you find as much beauty in these structures as I do.

Chapter 0

Preliminaries

Most people meet linear algebra as a way of computing: row reduction, determinants, maybe eigenvalues. This book treats it as a subject you **prove**. Every statement comes with a reason, and those reasons are built from a small set of ideas: sets, functions, equivalence relations, fields, polynomials and matrices. This chapter collects that toolkit in one place.

You have probably seen some of this before. Some of it, such as negating a statement with several quantifiers, or checking that a rule on equivalence classes is well defined, is exactly where careful readers later stumble. So each section spends its time on the delicate points and moves quickly past the rest.

How to use this chapter. Skim the sections you know, but try the *A. Check your understanding* exercises at the end of each one. If you can do them without looking back, move on. If not, that section is worth a slower read.

What you need. Nothing from this book. School algebra is enough, plus some familiarity with numbers such as $\mathbb{Z}$, $\mathbb{Q}$ and $\mathbb{R}$. One convention to note now: in this book the natural numbers $\mathbb{N}$ start at 0.

0.0.1 Roadmap

- **Statements, implications and quantifiers.** How mathematical statements are built, and how to negate them correctly.
- **How proofs are built.** Direct proof, contrapositive, contradiction, cases, uniqueness, counterexamples and induction, and the conventions this book's proofs follow.
- **Sets.** Subsets, operations, indexed families, and the habit of proving two sets equal by double inclusion.
- **Functions.** Injective, surjective and bijective maps; composition; left, right and two-sided inverses.
- **Relations, equivalence classes and quotients.** Sorting a set into classes, and defining functions on classes without ambiguity.
- **Complex numbers.** Arithmetic, conjugates, the modulus, polar form and roots of unity.
- **Fields.** The number systems in which linear algebra works, from $\mathbb{Q}$, $\mathbb{R}$ and $\mathbb{C}$ to the finite fields $\mathbb{F}_p$.
- **Polynomials.** Degree, evaluation, and why a polynomial is not the same thing as a function.

- **Matrices.** Addition, the three ways to read a matrix product, transpose, trace and inverses.
- **Groups and permutations.** One structure shared by invertible matrices, the non-zero elements of a field, and rearrangements of a set.

0.0.2 Named moves

These proof techniques appear throughout the book. Each is introduced in this chapter.

- **Contrapositive.** To prove “if P then Q ”, prove “if not Q then not P ”.
- **Assume two.** To prove something is unique, suppose two objects have the property and show they are equal.
- **Double inclusion.** To prove $A = B$, prove $A \subseteq B$ and $B \subseteq A$.
- **Suppose $f(a) = f(b)$.** The standard first line when proving that a function is injective.
- **Check well-definedness.** A rule given through a representative must not depend on which representative is chosen.
- **Induction on n .** Find a smaller instance of the problem inside the larger one.

0.1 Statements, Implications and Quantifiers

Linear algebra is a subject of proofs. Every theorem in this book has the shape “if these hypotheses hold, then this conclusion holds”, and most definitions hide a “for every” or a “there exists”. Before we can prove anything, we need to read such sentences exactly: when they are true, what they claim, and what their opposite says. This section builds that reading skill. It is the grammar that the rest of the book is written in.

0.1.1 Statements and connectives

A **statement** is a sentence that is either true or false, but not both. “7 is prime” is a statement, and it is true. “ $2 + 2 = 5$ ” is a statement, and it is false. “Is 7 prime?” is not a statement, and neither is “ $x > 3$ ” on its own: its truth depends on x , which nobody has chosen yet. Sentences with a free variable are handled by quantifiers later in this section.

We build longer statements from shorter ones with a few fixed words. Everyday language uses these words loosely, so mathematics pins each one down by saying exactly when the combined statement is true.

A connective builds a new statement whose truth depends only on the truth of its parts.

Definition 0.1.1 — Negation, conjunction, disjunction

Let P and Q be statements.

- The **negation** $\neg P$ (“not P ”) is true when P is false, and false when P is true.
- The **conjunction** $P \wedge Q$ (“ P and Q ”) is true when **both** P and Q are true, and false otherwise.
- The **disjunction** $P \vee Q$ (“ P or Q ”) is true when **at least one** of P , Q is true,

and false only when both are false.

A **truth table** lists every combination of truth values of the parts, one per row, and records the value of the compound statement. With T for true and F for false:

P	Q	$\neg P$	$P \wedge Q$	$P \vee Q$
T	T	F	T	T
T	F	F	F	T
F	T	T	F	T
F	F	T	F	F

Two statement forms are **logically equivalent**, written $\equiv$, if they have the same truth value in every row of the truth table. For example, $\neg(\neg P) \equiv P$: negating twice returns the original value in both rows.

A few examples, checked against the definition:

- “7 is prime **and** 7 is even” is false, because its second part is false.
- “7 is prime **or** 7 is even” is true, because its first part is true.
- “ $2 < 3$ **or** $3 < 4$ ” is true. Both parts are true, and that is allowed.

Warning. In mathematics “or” is **inclusive**: $P \vee Q$ is true when both parts are true. The menu phrase “soup or salad”, which forbids having both, is a different connective. So “6 is even or 6 is divisible by 3” is true, even though both halves hold.

0.1.2 Implication

Almost every theorem is an implication: “if P , then Q ”. It is worth being completely precise about when such a sentence counts as true, because the answer surprises many readers in one row of the table.

Think of an implication as a **promise**. A parent says: “If you finish your homework, then you may watch a film.” When has the parent broken the promise? Only when the homework is finished and the film is still refused. If the homework is not finished, the promise says nothing about what happens, so whatever happens, the promise was not broken.

An implication is false only when the hypothesis holds and the conclusion fails.

Definition 0.1.2 — Implication, converse, contrapositive

Let P and Q be statements.

- The **implication** $P \Rightarrow Q$ (“if P then Q ”) is false when P is true and Q is false, and it is true in **every** other case. We call P the **hypothesis** and Q the **conclusion**.
- The **converse** of $P \Rightarrow Q$ is the implication $Q \Rightarrow P$.
- The **contrapositive** of $P \Rightarrow Q$ is the implication $\neg Q \Rightarrow \neg P$.

In words: the first clause says an implication can only fail in one way, by a true hypothesis leading to a false conclusion. The converse swaps the roles of hypothesis and conclusion. The

contrapositive swaps them **and** negates both.

P	Q	$P \Rightarrow Q$
T	T	T
T	F	F
F	T	T
F	F	T

The implication $P \Rightarrow Q$ is read in many ways, and all of them mean the same thing:

- “ P implies Q ”, “ Q whenever P ”, “ Q if P ”;
- “ P only if Q ”;
- “ P is **sufficient** for Q ”, “ Q is **necessary** for P ”.

The phrase “only if” trips people up. “ P only if Q ” says that P cannot happen without Q , which is exactly $P \Rightarrow Q$.

The last two rows of the table, where the hypothesis is false, are called **vacuous** cases. An implication with a false hypothesis is **vacuously true**.

Example. (A vacuously true implication)

Let n be an integer. Is the statement “if n is odd and n is even, then $n = 7$ ” true?

Solution. Yes. No integer is both odd and even, so the hypothesis “ n is odd and n is even” is false for every integer n . By [Definition 0.1.2](#), an implication with a false hypothesis is true, whatever the conclusion says. So the statement is true for every integer n , including $n = 7$ and $n = 0$. It is true, but it tells us nothing.

Why this definition. The two vacuous rows may look like a convention picked for convenience. They are forced once we want implications to work inside “for every” statements. Consider the true fact

for every integer n , if $n > 2$ then $n^2 > 4$.

For this to be true, the implication must be true for **every** integer n . Try $n = -3$: the hypothesis $-3 > 2$ is false, and the conclusion $9 > 4$ is true. Try $n = 1$: the hypothesis $1 > 2$ is false, and the conclusion $1 > 4$ is false. If either row were declared false, this perfectly good theorem would become false. So both vacuous rows must be T. In the same way, a theorem with hypotheses says nothing about objects that fail them, and that is exactly how we want to read it.

Remark. The implication $P \Rightarrow Q$ does **not** claim that P causes Q , or that the two are related at all. “If $2 + 2 = 4$, then 7 is prime” is a true implication, because both parts are true. In practice we only ever prove implications where the hypothesis is used, but the truth value does not depend on that.

The contrapositive is the most useful rewriting of an implication, because it is always interchangeable with the original.

Theorem 0.1.1 — Contrapositive equivalence

For all statements P and Q , the implication $P \Rightarrow Q$ is logically equivalent to its contrapositive $\neg Q \Rightarrow \neg P$.

Proof. Let P and Q be statements. We compute both sides in each of the four rows, using Definition 0.1.1 for the negations and Definition 0.1.2 for the implications.

P	Q	$\neg Q$	$\neg P$	$P \Rightarrow Q$	$\neg Q \Rightarrow \neg P$
T	T	F	F	T	T
T	F	T	F	F	F
F	T	F	T	T	T
F	F	T	T	T	T

In the second row, $\neg Q$ is true and $\neg P$ is false, so $\neg Q \Rightarrow \neg P$ is false. In every other row the hypothesis $\neg Q$ is false or the conclusion $\neg P$ is true, so the contrapositive is true. The last two columns agree in every row. This proves that the two statements are logically equivalent. ■

So to prove “if P then Q ” we may instead prove “if Q fails then P fails”. The next section turns this into a proof technique.

The converse is a different matter. It is a separate statement, and it can be false while the original is true.

Warning. The converse is not the contrapositive. “If $x = 2$, then $x^2 = 4$ ” is true for every real number x . Its converse “if $x^2 = 4$, then $x = 2$ ” is false: $x = -2$ satisfies the hypothesis and fails the conclusion. Proving $P \Rightarrow Q$ never proves $Q \Rightarrow P$.

Later in the book the converse question comes up constantly. In Chapter 2 we will see that if a square matrix A is invertible, then $A\mathbf{x} = \text{hasonlythesolution}\mathbf{x} = \text{Theconverse}$, “if $A\mathbf{x} = \text{hasonlythesolution}\mathbf{x} =$, then A is invertible”, also turns out to be true for square matrices. But it is a separate theorem, it needs its own proof, and that proof takes real work.

Quick check. Consider “if an integer n is divisible by 6, then n is even”. Write its converse and its contrapositive, and decide which of the three statements are true for every integer n .

Solution. The converse is “if n is even, then n is divisible by 6”. The contrapositive is “if n is not even, then n is not divisible by 6”, that is, “if n is odd, then $6 \nmid n$ ”.

The original is true: if $n = 6k$ then $n = 2(3k)$ is even. The contrapositive is true, by Theorem 0.1.1. The converse is false: $n = 2$ is even but not divisible by 6.

When an implication and its converse are both true, we say the two statements are equivalent.

The **biconditional** $P \Leftrightarrow Q$ (“ P if and only if Q ”, often written “ P iff Q ”) is the statement $(P \Rightarrow Q) \wedge (Q \Rightarrow P)$. It is true exactly when P and Q have the same truth value:

P	Q	$P \Rightarrow Q$	$Q \Rightarrow P$	$P \Leftrightarrow Q$
T	T	T	T	T
T	F	F	T	F
F	T	T	F	F
F	F	T	T	T

The phrase splits in two: “ P if Q ” is $Q \Rightarrow P$, and “ P only if Q ” is $P \Rightarrow Q$. Other phrasings are “ P is necessary and sufficient for Q ” and “ P and Q are equivalent”. Proving an “iff” therefore always means proving two implications.

0.1.3 Quantifiers

The sentence “ $x^2 \geq 0$ ” is not yet a statement, because x is not specified. A sentence $P(x)$ that becomes a statement once a value of x is substituted is called a **predicate** (or open sentence) in the variable x . There are two standard ways to turn a predicate into a statement: claim it for every value, or claim it for at least one.

“For every” makes a claim about each element; “there exists” asks for a single witness.

Definition 0.1.3 — Quantifiers

Let $P(x)$ be a predicate whose variable x ranges over a fixed collection S .

- The **universal** statement $\forall x \in S : P(x)$ (“for every x in S , $P(x)$ ”) is true when $P(x)$ is true for **every** element x of S , and false otherwise.
- The **existential** statement $\exists x \in S : P(x)$ (“there exists x in S such that $P(x)$ ”) is true when $P(x)$ is true for **at least one** element x of S , and false otherwise.

The symbols $\forall$ and $\exists$ are the **universal** and **existential quantifiers**.

In words: a universal statement is a promise about every element, so a single element where P fails makes it false. An existential statement needs one element that works. It does not say which one, and it does not say there is only one.

The collection S is part of the statement. Changing S can change the truth value, so always ask: *ranging over what?*

Example. (Quantified statements)

Decide whether each statement is true.

- $\forall x \in \mathbb{R} : x^2 \geq 0$.
- $\exists x \in \mathbb{R} : x^2 = -1$.

- (c) $\exists n \in \mathbb{Z} : n^2 = 2$.
 (d) $\forall n \in \mathbb{Z} : n^2 \neq 2$.

Solution.

- (a) True. Every real number x is either ≥ 0 or < 0 , and in both cases $x^2 = x \cdot x \geq 0$.
 (b) False. By (a), $x^2 \geq 0 > -1$ for every real x , so no real x is a witness. Over the complex numbers (introduced later in this chapter), $x = i$ is a witness, so the same predicate becomes true when the range changes from $\mathbb{R}$ to $\mathbb{C}$.
 (c) False. If $|n| \leq 1$ then $n^2 \leq 1 < 2$, and if $|n| \geq 2$ then $n^2 \geq 4 > 2$. So no integer squares to 2.
 (d) True. This is the statement shown in (c), said the other way round: every integer fails to square to 2.

Parts (c) and (d) already show the pattern of the next subsection: “there is no witness” is the same as “everything fails”.

A degenerate case. What if the range contains no elements at all? Take “every integer n with $n^2 < 0$ satisfies $n = 7$ ”. There are no such integers, so there is nothing that could fail, and the statement is true. It is the quantifier version of vacuous truth: it reads $\forall n \in \mathbb{Z} : (n^2 < 0 \Rightarrow n = 7)$, and every instance has a false hypothesis. By contrast, “there exists an integer n with $n^2 < 0$ ” is false, because there is no witness. This case matters later: facts about “every element of an empty list” will be vacuously true, and that is what makes certain edge cases of definitions work.

Remark. We write $\exists! x \in S : P(x)$ for “there exists a **unique** x in S with $P(x)$ ”. It means two things: some x satisfies P , and any two elements satisfying P are equal. The next section shows how each half is proved.

0.1.4 The order of quantifiers

Most definitions in this book use several quantifiers in a row. Their order is part of the meaning.

Example. (Swapping two quantifiers)

Compare the two statements

$$(i) \quad \forall x \in \mathbb{R} \exists y \in \mathbb{R} : y > x, \quad (ii) \quad \exists y \in \mathbb{R} \forall x \in \mathbb{R} : y > x.$$

Decide which is true.

Solution. Statement (i) says: whichever real x you are given, you can find a real y larger than it. This is true. Given x , take $y = x + 1$, and then $y > x$.

Statement (ii) says: there is one real number y that is larger than every real x . This is false. Whatever y is proposed, the choice $x = y + 1$ gives $y > x$ false. So no y is a witness.

The difference is **who chooses first**. In (i), x is chosen first and y may depend on it: $y = x + 1$ changes with x . In (ii), y must be fixed before x is known, so it has to work for all x at once. Reading quantifiers as a game helps: $\forall$ is a move by an opponent, $\exists$ is your reply, and you may only use information already on the table.

Warning. Never swap a $\forall$ and an $\exists$ without a proof. The implication “ $\exists y \forall x \Rightarrow \forall x \exists y$ ” is always valid, since the single y works for each x . The reverse direction fails, as [Example \(Swapping two quantifiers\)](#) shows. Two quantifiers of the **same** kind, such as $\forall x \forall y$, may be swapped freely.

Quick check. Over the positive real numbers, statement (A) is “ $\forall \varepsilon > 0 \exists \delta > 0 : \delta < \varepsilon$ ”. Statement (B) is obtained by swapping the quantifiers: “ $\exists \delta > 0 \forall \varepsilon > 0 : \delta < \varepsilon$ ”. Are (A) and (B) both true?

Solution. No. (A) is true: given $\varepsilon > 0$, take $\delta = \varepsilon/2$, which is positive and smaller than ε . (B) is false: whatever $\delta > 0$ is proposed, the choice $\varepsilon = \delta$ makes $\delta < \varepsilon$ false. In (A) the number δ may depend on ε ; in (B) it may not.

0.1.5 Negation

Much of mathematics consists of showing that something **fails**: a function is not injective, a list is not independent, a claimed theorem is false. To do this we need the negation of the statement written in a usable form. Putting “it is not the case that” in front is correct but useless; we want the negation with the “not” pushed all the way inside, next to a simple condition we can check.

First, the connectives.

Theorem 0.1.2 — De Morgan’s laws for statements

For all statements P and Q :

- (a) $\neg(P \wedge Q) \equiv \neg P \vee \neg Q$;
- (b) $\neg(P \vee Q) \equiv \neg P \wedge \neg Q$;
- (c) $\neg(P \Rightarrow Q) \equiv P \wedge \neg Q$.

Proof. Let P and Q be statements. We compare truth tables, using [Definition 0.1.1](#) and [Definition 0.1.2](#).

P	Q	$\neg(P \wedge Q)$	$\neg P \vee \neg Q$	$\neg(P \vee Q)$	$\neg P \wedge \neg Q$	$\neg(P \Rightarrow Q)$	$P \wedge \neg Q$
T	T	F	F	F	F	F	F
T	F	T	T	F	F	T	T
F	T	T	T	F	F	F	F
F	F	T	T	T	T	F	F

For (a), $P \wedge Q$ is true only in the first row, so its negation is false only there. The statement $\neg P \vee \neg Q$ is false only when $\neg P$ and $\neg Q$ are both false, which is again only the first row. For (b), $\neg(P \vee Q)$ is true only in the last row, where both parts are false, and so is $\neg P \wedge \neg Q$. For (c), $P \Rightarrow Q$ is false only in the second row, so $\neg(P \Rightarrow Q)$ is true only there, and $P \wedge \neg Q$ is also true only there. In each case the two columns agree in all four rows, as claimed. ■

Law (c) deserves a sentence. The negation of an implication is **not** another implication. To refute “if P then Q ”, you must exhibit a situation where P holds **and** Q fails.

Now the quantifiers. A universal statement fails exactly when some element fails it, and an existential statement fails exactly when every element fails it. We record this as the rule we will use every time.

To negate, flip each quantifier and negate what is inside.

Definition 0.1.4 — Negation of quantifiers

Let $P(x)$ be a predicate with x ranging over S . Then

$$\neg(\forall x \in S : P(x)) \equiv \exists x \in S : \neg P(x), \quad \neg(\exists x \in S : P(x)) \equiv \forall x \in S : \neg P(x).$$

For a statement with several quantifiers, apply these rules from the outside in: each $\forall$ becomes $\exists$, each $\exists$ becomes $\forall$, the ranges stay the same, and the innermost condition is negated using [Theorem 0.1.2](#).

In words: “not every x works” means “some x fails”, and “no x works” means “every x fails”. Both rules are just [Definition 0.1.3](#) read carefully. The universal statement is false precisely when it is not the case that every element satisfies P , which is precisely when at least one element satisfies $\neg P$. The **ranges do not change**: the negation of “every real number squared is non-negative” talks about real numbers, not about some other collection.

Example. (Negating simple quantified statements)

Negate each statement, and say which of the statement and its negation is true.

- (a) $\forall x \in \mathbb{R} : x^2 \geq 0$.
- (b) $\exists n \in \mathbb{Z} : n$ is even and $n > 10$.
- (c) $\forall x \in \mathbb{R} \exists y \in \mathbb{R} : y > x$.

Solution.

- (a) The negation is $\exists x \in \mathbb{R} : x^2 < 0$. The original is true, by [Example \(Quantified statements\)](#), so the negation is false.
- (b) By [Definition 0.1.4](#) and law (a) of [Theorem 0.1.2](#), the negation is $\forall n \in \mathbb{Z} : n$ is odd or $n \leq 10$. The original is true, with witness $n = 12$, so the negation

is false: $n = 12$ is neither odd nor ≤ 10 .

- (c) Flip both quantifiers and negate the inside: $\exists x \in \mathbb{R} \forall y \in \mathbb{R} : y \leq x$. This says that some real number is at least as large as every real number. The original is true by [Example \(Swapping two quantifiers\)](#), so the negation is false.

The next two examples negate definitions that will appear over and over. We use the word **function** informally here, as a rule f that assigns to each a in a collection A one value $f(a)$ in a collection B ; functions are treated properly later in this chapter.

Example. (What “not injective” means)

A function f from A to B is called **injective** if

$$\forall a_1 \in A \forall a_2 \in A : (f(a_1) = f(a_2) \Rightarrow a_1 = a_2).$$

Write down, with no “not” in front, what it means for f to be **not** injective. Then show that $f(x) = x^2$, from $\mathbb{R}$ to $\mathbb{R}$, is not injective.

Solution. By [Definition 0.1.4](#), both $\forall$ become $\exists$, and the inside implication is negated. By law (c) of [Theorem 0.1.2](#), $\neg(f(a_1) = f(a_2) \Rightarrow a_1 = a_2) \equiv f(a_1) = f(a_2) \wedge a_1 \neq a_2$. So f is not injective if and only if

$$\exists a_1 \in A \exists a_2 \in A : f(a_1) = f(a_2) \text{ and } a_1 \neq a_2.$$

In words, two **different** inputs have the same output. For $f(x) = x^2$, take $a_1 = 1$ and $a_2 = -1$. Then $f(1) = 1 = f(-1)$ and $1 \neq -1$, so f is not injective.

Example. (What “not surjective” means)

A function f from A to B is called **surjective** if

$$\forall b \in B \exists a \in A : f(a) = b.$$

Negate this, and show that $f(x) = x^2$, from $\mathbb{R}$ to $\mathbb{R}$, is not surjective.

Solution. Flipping the two quantifiers and negating the equation gives: f is not surjective if and only if

$$\exists b \in B \forall a \in A : f(a) \neq b.$$

In words, **one** value in B is never reached. For $f(x) = x^2$, take $b = -1$. For every real a we have $f(a) = a^2 \geq 0 > -1$ by [Example \(Quantified statements\)](#), so $f(a) \neq -1$. Hence f is not surjective.

The most important negation in the early chapters is that of a statement ending in “all zero”. We meet it here without any vector spaces, using pairs of real numbers added and scaled entry

by entry.

Example. (Negating “only the zero combination”)

Consider the pairs $(1, 2)$, $(2, 4)$, $(0, 1)$, and the statement

$$\forall a, b, c \in \mathbb{R} : \left(a(1, 2) + b(2, 4) + c(0, 1) = (0, 0) \Rightarrow a = 0 \wedge b = 0 \wedge c = 0 \right).$$

Negate it, and decide which of the statement and its negation is true.

Solution. Flip the quantifier, and use law (c) of [Theorem 0.1.2](#) for the implication and law (a) for the conjunction. The negation is

$$\exists a, b, c \in \mathbb{R} : a(1, 2) + b(2, 4) + c(0, 1) = (0, 0) \text{ and } (a \neq 0 \vee b \neq 0 \vee c \neq 0).$$

The last condition says that a, b, c are **not all zero**: at least one of them is non-zero.

The negation is true. Take $a = 2$, $b = -1$, $c = 0$. Then

$$2(1, 2) - (2, 4) + 0(0, 1) = (2 - 2 + 0, 4 - 4 + 0) = (0, 0),$$

and $a = 2 \neq 0$. So the original statement is false.

In Chapter 1 the original statement is what “the list is linearly independent” means, and its negation is “linearly dependent”. The small words in the negation carry the weight.

Warning. **“Not all zero” is not “all non-zero”.** The negation of “ $a = b = c = 0$ ” is “at least one of a, b, c is non-zero”. In [Example \(Negating “only the zero combination”\)](#) the witness is $(2, -1, 0)$, which has a zero entry. In fact **no** witness has all three entries non-zero: comparing entries, $a + 2b = 0$ and $2a + 4b + c = 0$, so $c = -2(a + 2b) = 0$. Negating with “all non-zero” would wrongly conclude that the original statement is true.

Quick check. Negate “every student passed some exam”, with no “not” in front of the whole sentence.

Solution. Write it as “ $\forall$ students $s \exists$ exam $e : s$ passed e ”. Flipping both quantifiers and negating the inside gives “ $\exists$ student $s \forall$ exams $e : s$ did not pass e ”. In words: **some student passed no exam**. The tempting answer “no student passed any exam” is the negation of “some student passed some exam”, a different statement.

Remark. Each form of statement suggests how to prove or refute it. To prove $\forall x \in S : P(x)$, take an arbitrary $x \in S$ and show $P(x)$. To prove $\exists x \in S : P(x)$, produce a witness. To refute a $\forall$, produce one counterexample. To refute an $\exists$, show every candidate fails. The next section develops these moves into proof techniques.

0.1.6 Exercises

0.1.6.1 A. Check your understanding

Exercise A1.

- (a) Decide whether each statement is true: (i) “ $3 < 2$ or 3 is odd”; (ii) “if $3 < 2$, then $3 = 5$ ”; (iii) “if 3 is odd, then $3 < 2$ ”.
- (b) Write the converse and the contrapositive of “if an integer n is divisible by 4, then n is even”. Which of the three statements are true for every integer n ?
- (c) Negate $\forall x \in \mathbb{R} \exists n \in \mathbb{Z} : n > x$, with no $\neg$ in front. Which of the statement and its negation is true?
- (d) True or false: $\neg(P \wedge Q) \equiv \neg P \wedge \neg Q$ for all statements P, Q . Give a reason.

Solution.

- (a) (i) True: the second part is true, and a disjunction needs only one true part. (ii) True: the hypothesis $3 < 2$ is false, so the implication is vacuously true. (iii) False: the hypothesis is true and the conclusion is false, which is the one case where an implication fails.
- (b) The converse is “if n is even, then n is divisible by 4”. The contrapositive is “if n is odd, then n is not divisible by 4”. The original is true: $n = 4k$ gives $n = 2(2k)$. The contrapositive is true by [Theorem 0.1.1](#). The converse is false: $n = 2$ is even and not divisible by 4.
- (c) By [Definition 0.1.4](#), the negation is $\exists x \in \mathbb{R} \forall n \in \mathbb{Z} : n \leq x$. The original is true: given x , some integer exceeds x (for instance the integer part of $|x|$ plus 1). So the negation is false.
- (d) False. Take P true and Q false. Then $P \wedge Q$ is false, so $\neg(P \wedge Q)$ is true, while $\neg P \wedge \neg Q$ is false because $\neg P$ is false. The correct law is $\neg(P \wedge Q) \equiv \neg P \vee \neg Q$, by [Theorem 0.1.2](#).

0.1.6.2 B. Practice

Exercise B1: Negating definitions.

Write the negation of each definition as a quantified statement with no $\neg$ in front, and then say it in words.

- (a) A function f from $\mathbb{R}$ to $\mathbb{R}$ is **bounded** if $\exists M \in \mathbb{R} \forall x \in \mathbb{R} : |f(x)| \leq M$.
- (b) A function f from A to B is injective if $\forall a_1, a_2 \in A : (a_1 \neq a_2 \Rightarrow f(a_1) \neq f(a_2))$. Compare your answer with [Example \(What “not injective” means\)](#).
- (c) “Every pair of reals is a combination of $(1, 0)$ and $(1, 1)$ ”: $\forall p, q \in \mathbb{R} \exists a, b \in \mathbb{R} : (p, q) = a(1, 0) + b(1, 1)$, where pairs are added and scaled entry by entry.

Solution.

- (a) By [Definition 0.1.4](#), f is not bounded if and only if $\forall M \in \mathbb{R} \exists x \in \mathbb{R} : |f(x)| > M$. In words: whatever bound M is proposed, some value of f exceeds it in absolute value.

- (b) Flip both quantifiers and negate the implication using law (c) of [Theorem 0.1.2](#): $\exists a_1, a_2 \in A : a_1 \neq a_2 \wedge f(a_1) = f(a_2)$. In words: two different inputs share an output. This is the same condition as in [Example \(What “not injective” means\)](#). That is expected: the given implication is the contrapositive of $f(a_1) = f(a_2) \Rightarrow a_1 = a_2$, so by [Theorem 0.1.1](#) the two definitions agree, and so do their negations.
- (c) Flip all the quantifiers and negate the equation: $\exists p, q \in \mathbb{R} \forall a, b \in \mathbb{R} : (p, q) \neq a(1, 0) + b(1, 1)$. In words: some pair is not a combination of $(1, 0)$ and $(1, 1)$, whatever coefficients are used. (The original is in fact true, since $(p - q)(1, 0) + q(1, 1) = (p, q)$, so this negation is false.)

Exercise B2: Integers versus reals.

For each statement, decide whether it is true when x and y range over $\mathbb{Z}$, and whether it is true when they range over $\mathbb{R}$. Justify your answer.

- (a) $\forall x \exists y : x < y < x + 1$.
 (b) $\forall x (x \neq 0 \Rightarrow \exists y : xy = 1)$.
 (c) $\exists x \forall y : xy = y$.
 (d) $\forall x \exists y : y^2 = x$.

Solution.

- (a) Over $\mathbb{Z}$: false. Take $x = 0$. No integer y satisfies $0 < y < 1$. Over $\mathbb{R}$: true. Given x , take $y = x + \frac{1}{2}$; then $x < y < x + 1$.
- (b) Over $\mathbb{Z}$: false. Take $x = 2$, which is non-zero. If $2y = 1$ for an integer y , then 1 would be even, which is false. Over $\mathbb{R}$: true. Given $x \neq 0$, take $y = 1/x$, which exists because $x \neq 0$, and then $xy = 1$.
- (c) True over both. Take $x = 1$, which lies in $\mathbb{Z}$ and in $\mathbb{R}$. Then $xy = y$ for every y . Here one x works for all y , so the order $\exists x \forall y$ causes no difficulty.
- (d) False over both. Take $x = -1$. For every y , integer or real, $y^2 \geq 0 > -1$ by [Example \(Quantified statements\)](#), so $y^2 \neq -1$.

0.1.6.3 C. Going deeper

Exercise C1: Exporting a hypothesis.

Prove that $P \Rightarrow (Q \Rightarrow R)$ is logically equivalent to $(P \wedge Q) \Rightarrow R$ for all statements P, Q, R . Explain in one sentence why this lets a theorem with two hypotheses be stated either way.

Solution. Let P, Q, R be statements. By [Definition 0.1.2](#), an implication is false exactly when its hypothesis is true and its conclusion is false.

The statement $(P \wedge Q) \Rightarrow R$ is false exactly when $P \wedge Q$ is true and R is false, that is, when P is true, Q is true and R is false.

The statement $P \Rightarrow (Q \Rightarrow R)$ is false exactly when P is true and $Q \Rightarrow R$ is false. By the same definition, $Q \Rightarrow R$ is false exactly when Q is true and R is false. So $P \Rightarrow (Q \Rightarrow R)$ is false exactly when P is true, Q is true and R is false.

Both statements are false in exactly the same row of the eight-row truth table, and true in the other seven. Hence they are logically equivalent.

Consequently “if P , then (if Q , then R)” and “if P and Q , then R ” say the same thing, so a theorem may list its hypotheses one after another or all together.

Exercise C2: When can a converse fail too?.

- (a) Determine whether there exist statements P and Q such that both $P \Rightarrow Q$ and its converse $Q \Rightarrow P$ are false. Justify your answer.
- (b) Give predicates $P(n)$ and $Q(n)$ about integers such that both $\forall n \in \mathbb{Z} : (P(n) \Rightarrow Q(n))$ and $\forall n \in \mathbb{Z} : (Q(n) \Rightarrow P(n))$ are false.
- (c) Explain why (a) and (b) do not contradict each other.

Solution.

- (a) No such statements exist. Suppose $P \Rightarrow Q$ is false. By [Definition 0.1.2](#), P is true and Q is false. Then $Q \Rightarrow P$ has a false hypothesis, so it is true. Hence an implication and its converse can never both be false.
- (b) Let $P(n)$ be “ n is even” and $Q(n)$ be “ n is divisible by 3”. The first universal statement is false: $n = 2$ is even and not divisible by 3. The second is false: $n = 3$ is divisible by 3 and not even.
- (c) In (a), P and Q are single statements, each with one fixed truth value. In (b), each universal statement is refuted by its own counterexample, and the two counterexamples are **different** integers. At $n = 2$ the converse $Q(2) \Rightarrow P(2)$ is vacuously true, and at $n = 3$ the original $P(3) \Rightarrow Q(3)$ is vacuously true, in line with (a). A “for every” statement can fail at one place while its converse fails somewhere else.

0.2 How Proofs Are Built

The previous section taught us to read statements: what an implication claims, what the quantifiers demand, and how to negate. Now we write proofs of them. A proof is a chain of statements, each one justified by a definition, a hypothesis or an earlier result, that ends in the statement we want. This section collects the handful of shapes that almost every proof in this book takes, shows each on a small fact about integers, and ends with the conventions that the book’s proofs follow.

Throughout, a theorem will usually come with two blocks. The **Idea** says how the argument was found, often by working backwards from the goal. The **Proof** then presents the argument forwards, one justified step at a time. Read the Idea first, then check the Proof line by line.

0.2.1 Direct proof

Most statements have the form “for every x , if $P(x)$ then $Q(x)$ ”. The direct way to prove this follows the shape of the statement:

1. **Let** x be arbitrary. By [Definition 0.1.3](#), a universal statement is a claim about each element, so we must not pick a convenient one.
2. **Suppose** $P(x)$. By [Definition 0.1.2](#), the implication can only fail when $P(x)$ is true, so that is the only case we need to handle.
3. **Unwind** $P(x)$ and $Q(x)$ into their definitions. Then push forward from what we have towards what we want, often working backwards from the goal on scrap paper first.

Before we can prove anything about integers, we need definitions to unwind.

Definition 0.2.1 — Even and odd

An integer n is **even** if there **exists** an integer k with $n = 2k$. An integer n is **odd** if there **exists** an integer k with $n = 2k + 1$.

So $0 = 2 \cdot 0$ is even, $-6 = 2 \cdot (-3)$ is even, and $7 = 2 \cdot 3 + 1$ is odd. Each definition is an existential statement: to show a number is even, we must produce the integer k . We use the school fact that **every integer is even or odd, and not both**; an exercise at the end of this section proves a more general fact about remainders.

Proposition 0.2.1 — Sum of even integers

For all integers m and n , if m and n are even, then $m + n$ is even.

Idea. Givens: $m = 2a$ and $n = 2b$ for some integers a, b . Goal: $m + n = 2k$ for some integer k . So we must *find* k . Adding the givens gives $m + n = 2a + 2b = 2(a + b)$, and $k = a + b$ is the witness.

Proof. Let m and n be even integers. By [Definition 0.2.1](#), there exist integers a and b with $m = 2a$ and $n = 2b$. Then

$$m + n = 2a + 2b = 2(a + b).$$

Since $a + b$ is an integer, [Definition 0.2.1](#) shows that $m + n$ is even. This proves the claim. ■

The pattern is worth naming. The hypothesis “ m is even” is an **existence** statement, so using it hands us an integer a to work with. The goal “ $m + n$ is even” is also an existence statement, so proving it means **producing** an integer, here $a + b$. The two letters a and b must be different: m and n are arbitrary, so their halves have nothing to do with each other.

Warning. A proof of a “for every” statement cannot start with a specific example. Writing “ $4 + 6 = 10$, which is even” checks one case and proves nothing about all even integers. Equally, writing $m = 2k$ and $n = 2k$ with the **same** k silently assumes $m = n$, so it only proves that $2m$ is even.

0.2.2 Proof by contrapositive

Sometimes the direct route stalls. Consider the statement “if n^2 is even, then n is even”. Going directly, we would suppose $n^2 = 2k$ and try to say something about n . But $n = \pm\sqrt{2k}$ is not visibly of the form $2 \cdot (\text{integer})$, and we are stuck.

By [Theorem 0.1.1](#), “if P then Q ” is equivalent to “if not Q then not P ”. Here the contrapositive is “if n is not even, then n^2 is not even”, that is, “if n is odd, then n^2 is odd”. Now the hypothesis hands us a formula for n , and squaring a formula is easy.

Proposition 0.2.2 — Even squares

For every integer n , if n^2 is even, then n is even.

Idea. The hypothesis n^2 even says little about n , but “ n odd” gives $n = 2k + 1$ explicitly. So prove the contrapositive: from $n = 2k + 1$, expand n^2 and look for the form $2(\dots) + 1$.

Proof. Let n be an integer. By [Theorem 0.1.1](#), it suffices to prove that if n is not even, then n^2 is not even. Suppose n is not even. Since every integer is even or odd, n is odd, so by [Definition 0.2.1](#) there is an integer k with $n = 2k + 1$. Then

$$n^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1.$$

Since $2k^2 + 2k$ is an integer, n^2 is odd by [Definition 0.2.1](#). Since no integer is both even and odd, n^2 is not even. This proves the contrapositive, and hence the statement. ■

A contrapositive proof is a direct proof of a different, equivalent implication. Use it when the negated conclusion gives more to work with than the hypothesis does.

0.2.3 Proof by contradiction

To prove a statement S by **contradiction**, suppose that S is false and deduce something impossible: a statement together with its negation. Why does this work? We have shown that the implication $\neg S \Rightarrow (R \wedge \neg R)$ is true, for some statement R . Its conclusion $R \wedge \neg R$ is false in every row, so by [Definition 0.1.2](#) the implication can only be true if its hypothesis $\neg S$ is false. Hence S is true.

Contradiction is the natural tool for statements that say something **cannot** happen, or that there is **no** object of some kind. Such statements give nothing to start from directly, while their negation hands us an object to study.

Statements about fractions and primes need the language of divisibility.

Definition 0.2.2 — Divisibility and primes

Let a and b be integers. We say a **divides** b , written $a \mid b$, if there **exists** an integer k with $b = ak$. An integer p is **prime** if $p \geq 2$ and the only **positive** divisors of p are 1

and p .

For example, $3 \mid 12$ since $12 = 3 \cdot 4$, and $3 \nmid 7$. The first primes are 2, 3, 5, 7, 11. The integer 1 is **not** prime, because the definition requires $p \geq 2$; this convention is what makes factorizations into primes unique. In this language, n is even exactly when $2 \mid n$.

A real number is **rational** if it equals a/b for some integers a and b with $b \neq 0$, and **irrational** otherwise. We use the school fact that every rational number can be written **in lowest terms**: as a/b with $b \geq 1$, where no integer $d \geq 2$ divides both a and b . (We justify this fact at the end of this section.)

Theorem 0.2.3 — Irrationality of $\sqrt{2}$

There is no rational number whose square is 2. That is, $\sqrt{2}$ is irrational.

Idea. “There is no rational x with $x^2 = 2$ ” gives nothing to hold, so suppose there is one, and write it in lowest terms a/b . Clearing denominators gives $a^2 = 2b^2$, an integer equation. Then Proposition 0.2.2 applies twice: first a is even, then (after substituting $a = 2c$) b is even. That contradicts lowest terms.

Proof. Suppose, for a contradiction, that some rational number x satisfies $x^2 = 2$. Write $x = a/b$ in lowest terms, so $b \geq 1$ and no integer $d \geq 2$ divides both a and b . Then $a^2/b^2 = 2$, and multiplying by b^2 , which is non-zero, gives

$$a^2 = 2b^2.$$

Therefore a^2 is even by Definition 0.2.1, and hence a is even by Proposition 0.2.2. Write $a = 2c$ with c an integer. Substituting gives $4c^2 = 2b^2$, so $b^2 = 2c^2$ is even. By Proposition 0.2.2 again, b is even. Thus 2 divides both a and b , which contradicts the choice of a/b in lowest terms. Hence no rational number has square 2. ■

The contradiction came from the lowest-terms condition, which is the hypothesis we chose to impose. That is typical: in a contradiction proof, look for the assumption you have not yet used, because the contradiction usually comes from there.

Remark. If a contradiction proof of “ $P \Rightarrow Q$ ” assumes P and $\neg Q$ and then only ever derives $\neg P$, it is really a contrapositive proof in disguise. Write it as a contrapositive; it is shorter and clearer.

A second classic contradiction proof, that there are infinitely many primes, needs one more tool and appears at the end of this section.

0.2.4 Proof by cases

When a hypothesis splits naturally into possibilities, handle each one separately. If $P_1 \vee P_2$ is true, and both $P_1 \Rightarrow Q$ and $P_2 \Rightarrow Q$ are true, then Q is true: whichever of P_1, P_2 holds, its implication delivers Q . The cases must **cover every possibility**. They are allowed to overlap.

Proposition 0.2.4 — $n^2 + n$ is even

For every integer n , the integer $n^2 + n$ is even.

Idea. Nothing about n is given, but every integer is even or odd, and each case gives a formula for n . Factoring first, $n^2 + n = n(n+1)$, makes each case a one-line computation.

Proof. Let n be an integer. Then $n^2 + n = n(n+1)$. Since every integer is even or odd, there are two cases.

Case 1: n is even. By [Definition 0.2.1](#), $n = 2k$ for some integer k . Then $n(n+1) = 2(k(n+1))$, and $k(n+1)$ is an integer.

Case 2: n is odd. By [Definition 0.2.1](#), $n = 2k+1$ for some integer k . Then $n+1 = 2(k+1)$, so $n(n+1) = 2(n(k+1))$, and $n(k+1)$ is an integer.

In both cases $n^2 + n$ is even by [Definition 0.2.1](#). This shows the claim for every integer n . ■

0.2.5 Proving “if and only if”

By the definition of the biconditional in the previous section, $P \Leftrightarrow Q$ means $(P \Rightarrow Q) \wedge (Q \Rightarrow P)$. So a proof of an “iff” has **two directions**, labeled $(\Rightarrow)$ and $(\Leftarrow)$, and each is proved by any method we like. Often one direction is already known.

Proposition 0.2.5 — Parity of a square

For every integer n , n is even if and only if n^2 is even.

Proof. Let n be an integer.

$(\Rightarrow)$ Suppose n is even, so $n = 2k$ for some integer k by [Definition 0.2.1](#). Then $n^2 = 4k^2 = 2(2k^2)$, and $2k^2$ is an integer, so n^2 is even.

$(\Leftarrow)$ This is [Proposition 0.2.2](#).

Both directions hold, so n is even if and only if n^2 is even. ■

Warning. Reversing the steps is not automatic. A common slip is to “prove the converse” by rewriting the same argument backwards without checking that every step reverses. For instance, from $x = 2$ we get $x^2 = 4$; but from $x^2 = 4$ we only get $x = 2$ **or** $x = -2$. Each direction needs its own justification.

0.2.6 Existence and uniqueness

To prove “there exists x with $P(x)$ ”, the most direct method is to **produce a witness** and check it. For example, “there exists an integer n with $n^2 = n + 2$ ” is proved by $n = 2$: indeed $2^2 = 4 = 2 + 2$.

To prove “there is **at most one** x with $P(x)$ ”, the standard move is: *suppose x and x' both satisfy P , and show $x = x'$* . Notice that we do not assume $x \neq x'$; we simply take two objects with the property and prove they are the same. “There exists a **unique** x ” means both existence and at most one, so it takes both arguments.

Here is an example that we will meet again when we study fields and groups. A **binary operation** $*$ on a collection S is a rule that assigns to each ordered pair a, b of elements of S an element $a * b$ of S . Addition and multiplication of integers are binary operations on $\mathbb{Z}$. An **identity element** for $*$ is an element e of S such that $e * a = a$ and $a * e = a$ for **every** a in S .

Proposition 0.2.6 — Uniqueness of the identity

Let $*$ be a binary operation on S . Then $*$ has at most one identity element.

Idea. Suppose e and e' are both identities. The only thing we know about them is how they behave in a product, so look at the one product that involves both, $e * e'$, and evaluate it in two ways.

Proof. Suppose e and e' are both identity elements for $*$. Since e' is an identity, $e * e' = e$, taking $a = e$ in its defining property. Since e is an identity, $e * e' = e'$, taking $a = e'$ in its defining property. Therefore $e = e * e' = e'$. This shows that any two identity elements are equal. ■

So 0 is **the** identity for addition on $\mathbb{Z}$, and 1 is **the** identity for multiplication. The result says nothing about existence. Subtraction on $\mathbb{Z}$ has no identity at all: $a - e = a$ for every a forces $e = 0$, but then $e - a = -a$, which is not a when $a = 1$. Uniqueness proofs of exactly this shape will recur for zero vectors, negatives, inverse matrices and inverse functions.

0.2.7 Counterexamples

By [Definition 0.1.4](#), the negation of “ $\forall x : P(x)$ ” is “ $\exists x : \neg P(x)$ ”. So to **refute** a universal statement, one witness suffices: a single x where $P(x)$ fails. Such an x is a **counterexample**. Conversely, no number of examples can prove a universal statement over an infinite range.

Example. (Refuting “every prime is odd”)

Decide whether every prime number is odd. Justify your answer.

Solution. False. The statement is $\forall p \text{ prime} : p \text{ is odd}$, so by [Definition 0.1.4](#) one prime that is not odd refutes it. Take $p = 2$. It is prime, since $2 \geq 2$ and a positive divisor of 2 is at most 2, so it is 1 or 2. It is even, since $2 = 2 \cdot 1$, and hence not odd. So 2 is a

counterexample.

Warning. Many examples prove nothing. The number $n^2 + n + 41$ is prime for every integer n from 0 to 39. Yet the statement “ $n^2 + n + 41$ is prime for every $n \geq 0$ ” is false: at $n = 40$ we get $40^2 + 40 + 41 = 1681 = 41^2$, which is not prime.

0.2.8 Mathematical induction

Some statements are really infinitely many statements, one for each natural number: “for every $n \geq 1$, $1 + 2 + \cdots + n = n(n+1)/2$ ”. Checking cases one at a time never finishes. Induction replaces the infinite list of checks by two finite tasks, like knocking over a row of dominoes: push the first one, and make sure each domino knocks over the next.

We take this as a basic property of the integers, not something to prove.

Theorem 0.2.7 — Principle of Mathematical Induction

Let n_0 be an integer, and for each integer $n \geq n_0$ let $P(n)$ be a statement. Suppose

- (i) (**base case**) $P(n_0)$ is true, and
- (ii) (**inductive step**) for **every** integer $n \geq n_0$, if $P(n)$ is true then $P(n+1)$ is true.

Then $P(n)$ is true for every integer $n \geq n_0$.

In the inductive step, the assumption “ $P(n)$ is true” is called the **induction hypothesis**. It is not circular: we do not assume the whole conclusion, only that the statement holds at one arbitrary n , and we prove that it then holds at $n+1$.

Proposition 0.2.8 — Sum of the first n integers

For every integer $n \geq 1$,

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2}.$$

Idea. Passing from n to $n+1$ adds exactly one term, $n+1$, to the left side. So the step reduces to the algebra identity $\frac{n(n+1)}{2} + (n+1) = \frac{(n+1)(n+2)}{2}$, which holds after taking out the factor $n+1$.

Proof. We prove this by induction on n , using Theorem 0.2.7 with $n_0 = 1$. Let $P(n)$ be the statement $1 + 2 + \cdots + n = n(n+1)/2$.

Base case. For $n = 1$ the left side is 1 and the right side is $1 \cdot 2/2 = 1$. So $P(1)$ is true.

Inductive step. Let $n \geq 1$ and suppose $P(n)$ is true. Then

$$1 + 2 + \cdots + n + (n+1) = \frac{n(n+1)}{2} + (n+1) = (n+1)\left(\frac{n}{2} + 1\right) = \frac{(n+1)(n+2)}{2},$$

where the first equality uses the induction hypothesis $P(n)$. This is $P(n+1)$.

By [Theorem 0.2.7](#), $P(n)$ is true for every integer $n \geq 1$, as claimed. ■

The base case does not have to be 1, and the step does not have to be an equation. In this book the **natural numbers** are $\mathbb{N} = \{0, 1, 2, \dots\}$: they **include** 0. (Some books start at 1; the next section fixes our convention formally.)

Proposition 0.2.9 — $2^n > n$

For every natural number n , we have $2^n > n$.

Proof. We prove this by induction on n , with $n_0 = 0$. For $n = 0$, $2^0 = 1 > 0$. Now let $n \geq 0$ and suppose $2^n > n$. Then

$$2^{n+1} = 2^n + 2^n > n + 2^n \geq n + 1,$$

where the strict inequality uses the induction hypothesis and the last step uses $2^n \geq 1$. Hence $2^{n+1} > n + 1$. By [Theorem 0.2.7](#), $2^n > n$ for every $n \in \mathbb{N}$. ■

The inductive step must work for **every** $n \geq n_0$, including the smallest ones. The following fake proof fails exactly there.

Warning. The “**all horses are the same color**” fallacy. *Claim:* in every group of $n \geq 1$ horses, all horses have the same color. *“Proof”:* for $n = 1$ there is nothing to show. Suppose the claim holds for groups of n horses, and take a group of $n + 1$ horses $h_1, \dots, h_{n+1}$. The group $h_1, \dots, h_n$ is single-colored by the induction hypothesis, and so is $h_2, \dots, h_{n+1}$. The two groups share a horse, so all $n + 1$ horses have its color. The claim is plainly false, so this “proof” has a gap. Before opening the check below, try to find the exact value of n where the step breaks.

Quick check. Where does the inductive step in the horse argument fail?

Solution. It fails when passing from $n = 1$ to $n + 1 = 2$. The two groups are then $\{h_1\}$ and $\{h_2\}$, which share **no** horse, so nothing links the color of h_1 to that of h_2 . The sentence “the two groups share a horse” is true only when $n \geq 2$. Since the step $P(1) \Rightarrow P(2)$ is not proved, the chain of dominoes breaks at the very first push, and the base case alone gives nothing.

0.2.9 Strong induction and well-ordering

In some arguments, knowing $P(n)$ alone is not enough to reach $P(n + 1)$: we need the statement at some **earlier** value, and we do not know in advance which one. Strong induction allows the hypothesis at all earlier values at once.

Theorem 0.2.10 — Strong induction

Let n_0 be an integer, and for each integer $n \geq n_0$ let $P(n)$ be a statement. Suppose that for **every** integer $n \geq n_0$,

if $P(k)$ is true for all integers k with $n_0 \leq k < n$, then $P(n)$ is true.

Then $P(n)$ is true for every integer $n \geq n_0$.

Idea. Ordinary induction applies to the stronger statement $Q(n)$: “ $P(k)$ holds for all k from n_0 to n ”. The hypothesis is exactly what pushes $Q(n)$ to $Q(n+1)$. At $n = n_0$ the condition “for all k with $n_0 \leq k < n_0$ ” is vacuous, so the hypothesis already contains the base case.

Proof. For each integer $n \geq n_0$, let $Q(n)$ be the statement “ $P(k)$ is true for all integers k with $n_0 \leq k \leq n$ ”. We prove $Q(n)$ for all $n \geq n_0$ by [Theorem 0.2.7](#).

Base case. There is no integer k with $n_0 \leq k < n_0$, so the condition “ $P(k)$ is true for all such k ” is vacuously true. By the hypothesis with $n = n_0$, $P(n_0)$ is true. Hence $Q(n_0)$ is true.

Inductive step. Let $n \geq n_0$ and suppose $Q(n)$ is true, so $P(k)$ holds for all k with $n_0 \leq k < n+1$. By the hypothesis applied to $n+1$, $P(n+1)$ is true. Together with $Q(n)$, this gives $Q(n+1)$.

By [Theorem 0.2.7](#), $Q(n)$ holds for all $n \geq n_0$. In particular $P(n)$ holds for all $n \geq n_0$, since $Q(n)$ includes $k = n$. This proves the theorem. ■

The same property of the natural numbers can be phrased without any statements $P(n)$ at all: there is no endless strictly decreasing sequence of natural numbers, so every search for a smallest example ends.

Theorem 0.2.11 — Well-Ordering Principle

Every non-empty collection of natural numbers has a least element: an element m of the collection with $m \leq s$ for every s in the collection.

We take this as a basic property too. It is equivalent to the Principle of Mathematical Induction, in the sense that each can be proved from the other, but we will not need that. Well-ordering is the natural language for “take a smallest counterexample” arguments, which are contradiction proofs in disguise.

Here is a result that needs strong induction, because the smaller number we reach is not $n-1$.

Proposition 0.2.12 — Existence of a prime factor

Every integer $n \geq 2$ has a prime divisor.

Idea. If n is prime, n itself works. If not, $n = ab$ with $1 < a < n$, and a prime dividing a also divides n . The factor a can be any size below n , so we need the statement for all smaller values, which is strong induction.

Proof. We use [Theorem 0.2.10](#) with $n_0 = 2$. Let $n \geq 2$, and suppose every integer k with $2 \leq k < n$ has a prime divisor.

Case 1: n is prime. Then $n \mid n$, so n is a prime divisor of n .

Case 2: n is not prime. Since $n \geq 2$, [Definition 0.2.2](#) gives a positive divisor a of n with $a \neq 1$ and $a \neq n$. Write $n = ab$ with b an integer. Since a and n are positive, $b \geq 1$, and $b \neq 1$ because $a \neq n$; so $b \geq 2$ and $a = n/b < n$. Thus $2 \leq a < n$, and by the hypothesis a has a prime divisor p , say $a = pc$. Then $n = p(cb)$, so $p \mid n$.

In both cases n has a prime divisor. By [Theorem 0.2.10](#), this holds for every $n \geq 2$. ■

Now the promised contradiction proof.

Theorem 0.2.13 — Infinitely many primes

There are infinitely many prime numbers.

Idea. Suppose the primes form a finite list. Build a number that no prime on the list can divide: multiply them all and add 1. Dividing by any p_i leaves remainder 1. But [Proposition 0.2.12](#) says this number has some prime divisor, and that prime must be on the list.

Proof. Suppose, for a contradiction, that there are only finitely many primes, and list them all as $p_1, p_2, \dots, p_k$. Let

$$N = p_1 p_2 \cdots p_k + 1.$$

The list is non-empty, since 2 is prime, and each $p_i \geq 2$; hence $N \geq 3$. By [Proposition 0.2.12](#), N has a prime divisor p . Since the list contains every prime, $p = p_i$ for some i . Then $p_i \mid N$ and $p_i \mid p_1 p_2 \cdots p_k$, so there are integers s, t with $N = p_i s$ and $p_1 \cdots p_k = p_i t$. Subtracting gives $1 = p_i(s - t)$. Since $s - t$ is an integer and $p_i \geq 2$, this is impossible: $p_i(s - t)$ is 0 when $s = t$, and has absolute value at least 2 otherwise. This contradiction shows that there are infinitely many primes. ■

Remark. Well-ordering also justifies the “lowest terms” fact used in [Theorem 0.2.3](#). Let x be rational. There is at least one way of writing $x = a/b$ with $b \geq 1$: if $x = a/b$ with $b < 0$, then also $x = (-a)/(-b)$ with $-b \geq 1$. Among all such ways, the collection of possible denominators b is a non-empty collection of natural numbers, so by [Theorem 0.2.11](#) it has a least element. Take a representation a/b with that least b . If some $d \geq 2$ divided both, say $a = da'$ and $b = db'$, then $x = a'/b'$ with $1 \leq b' < b$, contradicting minimality. So a/b is in lowest terms.

0.2.10 Writing proofs in this book

The proofs in this book follow a few conventions. Knowing them makes proofs faster to read, and copying them makes your own proofs easier to check.

- **The first sentence does mathematics.** Proofs open with “Let ...”, “Suppose ...”, “Since ...” or “By ...”. The motivation has already been given in the Idea block.
- **Every step names its reason.** Each line cites a hypothesis, a definition, or an earlier result by name, as in “by [Proposition 0.2.2](#)”. When a display hides a reason, a clause such as “where the first equality uses the induction hypothesis” points to it.
- **Structure is marked.** Two directions of an iff are labeled $(\Rightarrow)$ and $(\Leftarrow)$, cases are labeled *Case 1* and *Case 2*, and inductions name their base case and inductive step.
- **No hidden gaps.** Words like “obviously” and “clearly” never stand in for an argument. If a step is short, it is written out; if it is routine and has been done before, the proof says so.
- **The ending restates the result:** “This shows ...”, “This proves ...”, “as claimed”.

How to read a proof here. First read the statement and identify the givens and the goal. Then read the Idea, which says why the argument has the shape it has. Then read the Proof with a pencil. At each line, ask which earlier line or result justifies it; if you cannot answer, that is the line to reread. Finally, ask which hypothesis each step used. If a hypothesis was never used, either it was not needed or you have missed a step.

0.2.11 Exercises

0.2.11.1 A. Check your understanding

Exercise A1.

For each statement, name the proof technique you would use first (direct, contrapositive, contradiction, cases, induction, or counterexample), and give a one-sentence reason. Then carry out the proof or refutation.

- The sum of two odd integers is even.
- For every integer n , if $3n + 2$ is odd, then n is odd.
- For every integer $n \geq 1$, $3 \mid 4^n - 1$.
- For every integer $n \geq 1$, $n^2 \geq 2n$.
- There is no largest integer.

Solution.

- Direct.* The hypothesis gives formulas to add. Let $m = 2a + 1$ and $n = 2b + 1$ with a, b integers, by [Definition 0.2.1](#). Then $m + n = 2(a + b + 1)$, which is even.
- Contrapositive.* The conclusion’s negation “ n even” gives a formula, while “ $3n + 2$ odd” does not. By [Theorem 0.1.1](#) it suffices to show: if n is even then $3n + 2$ is even. If $n = 2k$, then $3n + 2 = 2(3k + 1)$, which is even, hence not odd.
- Induction.* It is a statement for every $n \geq 1$, and $4^{n+1} - 1$ is built from $4^n - 1$. For $n = 1$, $4 - 1 = 3 = 3 \cdot 1$. If $4^n - 1 = 3k$, then $4^{n+1} - 1 = 4(4^n - 1) + 3 = 3(4k + 1)$.

By [Theorem 0.2.7](#) the claim holds for all $n \geq 1$.

- (d) *Counterexample.* The statement is false, so a single witness refutes it. At $n = 1$, $n^2 = 1 < 2 = 2n$.
- (e) *Contradiction.* It says an object does not exist, so assume it does. Suppose m is a largest integer. Then $m + 1$ is an integer with $m + 1 > m$, contradicting the choice of m . Hence there is no largest integer.

0.2.11.2 B. Practice

Exercise B1: Contrapositive.

Prove that for every integer n , if n^2 is odd, then n is odd.

Solution. Let n be an integer. By [Theorem 0.1.1](#), it suffices to prove that if n is not odd, then n^2 is not odd. Suppose n is not odd. Since every integer is even or odd, n is even, so $n = 2k$ for some integer k by [Definition 0.2.1](#). Then $n^2 = 4k^2 = 2(2k^2)$, so n^2 is even. Since no integer is both even and odd, n^2 is not odd. This proves the statement.

Exercise B2: Induction.

Prove that for every integer $n \geq 1$, the sum of the first n odd numbers is n^2 :

$$1 + 3 + 5 + \cdots + (2n - 1) = n^2.$$

Solution. We prove this by induction on n , using [Theorem 0.2.7](#) with $n_0 = 1$. Let $P(n)$ be the displayed statement.

Base case. For $n = 1$ the left side is 1 and the right side is $1^2 = 1$, so $P(1)$ is true.

Inductive step. Let $n \geq 1$ and suppose $P(n)$ is true. The $(n + 1)$ -st odd number is $2(n + 1) - 1 = 2n + 1$. Then

$$1 + 3 + \cdots + (2n - 1) + (2n + 1) = n^2 + 2n + 1 = (n + 1)^2,$$

where the first equality uses the induction hypothesis. This is $P(n + 1)$.

By [Theorem 0.2.7](#), $P(n)$ holds for every $n \geq 1$.

Exercise B3: $\sqrt{3}$ is irrational.

You may use the school fact that every integer has exactly one of the forms $3k$, $3k + 1$, $3k + 2$ with k an integer.

- (a) Prove that for every integer n , if $3 \mid n^2$ then $3 \mid n$.
- (b) Hence prove that there is no rational number whose square is 3.

Solution.

- (a) Let n be an integer. By [Theorem 0.1.1](#), it suffices to show that if $3 \nmid n$ then $3 \nmid n^2$. Suppose $3 \nmid n$. Then $n = 3k + 1$ or $n = 3k + 2$ for some integer k .

Case 1: $n = 3k + 1$. Then $n^2 = 9k^2 + 6k + 1 = 3(3k^2 + 2k) + 1$.

Case 2: $n = 3k + 2$. Then $n^2 = 9k^2 + 12k + 4 = 3(3k^2 + 4k + 1) + 1$.

In both cases n^2 has the form $3j + 1$ with j an integer. Since an integer has only one of the three forms, n^2 is not of the form $3j$, so $3 \nmid n^2$. This proves (a).

- (b) Suppose, for a contradiction, that a rational number x satisfies $x^2 = 3$. Write $x = a/b$ in lowest terms, so $b \geq 1$ and no integer $d \geq 2$ divides both a and b . Multiplying $a^2/b^2 = 3$ by $b^2 \neq 0$ gives $a^2 = 3b^2$. Therefore $3 \mid a^2$, and by (a), $3 \mid a$. Write $a = 3c$. Then $9c^2 = 3b^2$, so $b^2 = 3c^2$ and $3 \mid b^2$. By (a) again, $3 \mid b$. Thus 3 divides both a and b , contradicting lowest terms. Hence no rational number has square 3.

0.2.11.3 C. Going deeper

Exercise C1: Stamps of 4 and 5.

- (a) Show that each of 12, 13, 14, 15 can be written as $4a + 5b$ with $a, b \in \mathbb{N}$.
- (b) Prove by strong induction that every integer $n \geq 12$ can be written as $4a + 5b$ with $a, b \in \mathbb{N}$.
- (c) Show that 11 cannot be written in this form, so the bound 12 in (b) cannot be lowered to 11.

Hint: for $n \geq 16$, look at $n - 4$.

Solution.

- (a) $12 = 4 \cdot 3 + 5 \cdot 0$, $13 = 4 \cdot 2 + 5 \cdot 1$, $14 = 4 \cdot 1 + 5 \cdot 2$, and $15 = 4 \cdot 0 + 5 \cdot 3$.
- (b) We use [Theorem 0.2.10](#) with $n_0 = 12$. Let $n \geq 12$, and suppose every integer k with $12 \leq k < n$ can be written as $4a + 5b$ with $a, b \in \mathbb{N}$.

Case 1: $12 \leq n \leq 15$. Then n has the required form by (a).

Case 2: $n \geq 16$. Then $k = n - 4$ satisfies $12 \leq k < n$, so by the hypothesis $n - 4 = 4a + 5b$ with $a, b \in \mathbb{N}$. Hence $n = 4(a + 1) + 5b$, and $a + 1 \in \mathbb{N}$.

In both cases n has the required form. By [Theorem 0.2.10](#), every integer $n \geq 12$ does.

- (c) Suppose $11 = 4a + 5b$ with $a, b \in \mathbb{N}$. Since $5b \leq 11$, $b \in \{0, 1, 2\}$. If $b = 0$ then $4a = 11$; if $b = 1$ then $4a = 6$; if $b = 2$ then $4a = 1$. None of 11, 6, 1 is a multiple of 4, so no such a exists. Hence 11 is not of the form $4a + 5b$.

Exercise C2: Division with remainder.

Let a and b be integers with $b \geq 1$. Let R be the collection of all **non-negative** integers of the form $a - qb$ with q an integer.

- (a) Show that R is non-empty.
- (b) Let r be the least element of R , which exists by [Theorem 0.2.11](#), and write $r = a - qb$. Prove that $0 \leq r < b$.

- (c) Prove that the pair (q, r) is unique: if $a = qb + r = q'b + r'$ with $0 \leq r < b$ and $0 \leq r' < b$, then $q = q'$ and $r = r'$.

Hint for (a): try $q = -|a|$.

Solution.

- (a) Take $q = -|a|$. Then $a - qb = a + |a|b$. Since $b \geq 1$ and $|a| \geq 0$, we have $|a|b \geq |a|$, so $a + |a|b \geq a + |a| \geq 0$, where the last step uses $|a| \geq -a$. Hence $a + |a|b$ belongs to R , and R is non-empty.
- (b) By (a) and [Theorem 0.2.11](#), R has a least element $r = a - qb$ with $q \in \mathbb{Z}$. Since r belongs to R , $r \geq 0$. Suppose, for a contradiction, that $r \geq b$. Then $r - b = a - (q + 1)b \geq 0$, and $q + 1 \in \mathbb{Z}$, so $r - b$ belongs to R . But $b \geq 1$ gives $r - b < r$, contradicting the choice of r as the least element of R . Hence $0 \leq r < b$.
- (c) Suppose $qb + r = q'b + r'$ with $0 \leq r, r' < b$. Then $r' - r = (q - q')b$, so $b \mid r' - r$. From the bounds, $-b < r' - r < b$. Suppose $q \neq q'$. Then $|q - q'| \geq 1$, so $|r' - r| = |q - q'|b \geq b$, contradicting $-b < r' - r < b$. Hence $q = q'$, and then $r' - r = 0$, so $r = r'$. This proves uniqueness.

Together, (a)–(c) show that for integers a and $b \geq 1$ there are unique integers q, r with $a = qb + r$ and $0 \leq r < b$. With $b = 2$, this is the fact that every integer is even or odd, and not both.

0.3 Sets

Every object in linear algebra is a set with some extra structure. A vector space is a set on which we can add and scale, a subspace is a subset of it, and a linear map sends elements of one set to elements of another. Before the structure arrives, we need the plain language of sets: how to describe them, when two of them are equal, and how to combine them. The most important move in this section is proving that two sets are equal by proving two inclusions, a move we will use for subspaces, spans, kernels and images throughout the book.

0.3.1 Sets, elements and set-builder notation

A **set** is a collection of objects, called its **elements**. We write $x \in A$ for “ x is an element of A ” (also “ x belongs to A ”, “ x lies in A ”), and $x \notin A$ for its negation. A set is determined by its elements and nothing else: it has no order and no repetition. So $\{1, 2\}$, $\{2, 1\}$ and $\{1, 1, 2\}$ all describe the same set, whose elements are 1 and 2. We make this precise below.

Small sets can be listed between braces. Infinite sets we meet constantly have fixed names.

Definition 0.3.1 — Common number sets

We write

- $\mathbb{N} = \{0, 1, 2, 3, \dots\}$ for the **natural numbers**; in this book $\mathbb{N}$ **includes** 0;
- $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$ for the **integers**;

- $\mathbb{Q}$ for the **rational numbers**, the numbers a/b with $a, b \in \mathbb{Z}$ and $b \neq 0$;
- $\mathbb{R}$ for the **real numbers**;
- $\mathbb{C}$ for the **complex numbers**, studied later in this chapter.

Books disagree about whether $0 \in \mathbb{N}$. With our convention, the Well-Ordering Principle ([Theorem 0.2.11](#)) applies to collections that contain 0, and “ $n \in \mathbb{N}$ ” means $n \geq 0$. When we need positive integers only, we say “ $n \geq 1$ ”. For real numbers $a \leq b$ we also use **intervals**: $[a, b]$ is the set of real x with $a \leq x \leq b$, (a, b) the set with $a < x < b$, and $[a, b)$, $(a, b]$ are defined in the same way.

Listing elements fails for most sets we care about, such as “the real numbers whose square is less than 4”. Instead we describe a set by a property, and select from a set we already have.

Definition 0.3.2 — Set-builder notation

Let S be a set and let $P(x)$ be a predicate for $x \in S$. The set

$$\{x \in S : P(x)\}$$

is the set whose elements are **exactly** those $x \in S$ for which $P(x)$ is true.

In words: an object y belongs to $\{x \in S : P(x)\}$ if and only if $y \in S$ **and** $P(y)$ is true. Both parts matter. The ambient set S says where we are looking, and the predicate says what we keep. The colon is read “such that”; some books use a vertical bar instead. When it is plain which sets the elements come from, as in the unions and intersections below, we sometimes leave the ambient set out and write $\{x : P(x)\}$.

A second form describes the elements by a formula: $\{2k : k \in \mathbb{Z}\}$ is the set of all numbers of the form $2k$ with k an integer. An object y belongs to it exactly when $y = 2k$ for **some** $k \in \mathbb{Z}$. This form will describe sets such as “all combinations of given vectors” in Chapter 1.

Example. (Reading set-builder notation)

Describe each set more simply.

- $\{x \in \mathbb{R} : x^2 < 4\}$.
- $\{x \in \mathbb{Z} : x^2 < 4\}$.
- $\{n \in \mathbb{Z} : n \text{ is even}\}$ and $\{2k : k \in \mathbb{Z}\}$.
- $\{(x, y) : x, y \in \mathbb{R}, x + y = 1\}$, a set of ordered pairs (made precise under Cartesian products below).

Solution.

- A real number satisfies $x^2 < 4$ exactly when $-2 < x < 2$, so the set is the interval $(-2, 2)$.
- The same predicate, selected from $\mathbb{Z}$ instead, keeps only integers: the set is $\{-1, 0, 1\}$. Changing the ambient set changes the set.
- Both are the set $\{\dots, -4, -2, 0, 2, 4, \dots\}$ of even integers. The first selects the integers with a property; the second produces each element by the formula $2k$. By the definition of “even” ([Definition 0.2.1](#)), an integer is even exactly when it equals

$2k$ for some $k \in \mathbb{Z}$.

- (d) The pairs of reals whose coordinates add to 1: the line in the plane through $(1, 0)$ and $(0, 1)$.

What if no element of S satisfies the predicate? We still want a set, and it has no elements.

Definition 0.3.3 — Empty set

The **empty set** $\emptyset$ is the set with **no** elements: $x \notin \emptyset$ for every object x .

For example, $\{x \in \mathbb{R} : x^2 = -1\} = \emptyset$, since $x^2 \geq 0$ for every real x (Example (Quantified statements)). Also $\{n \in \mathbb{Z} : 0 < n < 1\} = \emptyset$. We will see shortly that there is only **one** empty set, so these two descriptions give the same set.

Statements about all elements of $\emptyset$ are vacuously true: “every element of $\emptyset$ is a prime number” has no element that could fail, just as in the degenerate case discussed after Example (Quantified statements).

Warning. $\emptyset$, $\{\emptyset\}$ and $\{0\}$ are three different sets. The set $\emptyset$ has no elements. The set $\{\emptyset\}$ has exactly one element, namely the empty set. The set $\{0\}$ has exactly one element, namely the number 0. A box containing an empty box is not an empty box. In Chapter 1 this matters: $\{0\}$ will be a subspace, and $\emptyset$ never is.

0.3.2 Subsets

Most sets in this book live inside a bigger set: the even integers inside $\mathbb{Z}$, a line inside the plane, and later a subspace inside a vector space. The first thing to check about a candidate subspace will be that it sits inside the space at all. We need a word for “sits inside”.

A is a subset of B when everything in A is also in B.

Definition 0.3.4 — Subset

Let A and B be sets. We say A is a **subset** of B , and write $A \subseteq B$, if

$$\forall x : (x \in A \Rightarrow x \in B).$$

If $A \subseteq B$ and $A \neq B$, then A is a **proper** subset of B , written $A \subsetneq B$.

In words: $A \subseteq B$ says that **every** element of A is an element of B . It says nothing about elements of B ; B may contain many more. The definition is an implication inside a “for every”, so the proof template of the previous section applies: to show $A \subseteq B$, **let** $x \in A$ and show $x \in B$.

By Definition 0.1.4 and Theorem 0.1.2, the negation reads

$$A \not\subseteq B \iff \exists x : x \in A \text{ and } x \notin B.$$

So to show A is **not** a subset of B , exhibit one element of A outside B .

Example. (Subsets)

Decide whether each inclusion holds.

- (a) $\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R}$.
- (b) $\{1, 2\} \subseteq \{1, 2, 3\}$ and $\{1, 4\} \subseteq \{1, 2, 3\}$.
- (c) $A \subseteq A$ and $\emptyset \subseteq A$, for any set A .
- (d) $\{x \in \mathbb{R} : x^2 = 1\} \subseteq \mathbb{Z}$.

Solution.

- (a) All hold. Every natural number is an integer; every integer n equals $n/1$, so it is rational; every rational number is real.
- (b) The first holds: each of 1, 2 is an element of $\{1, 2, 3\}$. The second fails, although it differs from the first in only one element: 1 passes, but $4 \in \{1, 4\}$ and $4 \notin \{1, 2, 3\}$, so 4 is a witness for $\{1, 4\} \not\subseteq \{1, 2, 3\}$.
- (c) Both hold. For $A \subseteq A$: let $x \in A$; then $x \in A$. For $\emptyset \subseteq A$: the implication $x \in \emptyset \Rightarrow x \in A$ has a false hypothesis for every x by [Definition 0.3.3](#), so it is vacuously true ([Definition 0.1.2](#)). The empty set is a subset of every set, including itself.
- (d) Holds. Let $x \in \mathbb{R}$ with $x^2 = 1$. Then $(x - 1)(x + 1) = 0$, so $x = 1$ or $x = -1$, and both are integers.

Warning. $\in$ and $\subseteq$ are different relations. $x \in A$ says the object x is one of the elements of A . $A \subseteq B$ compares two sets, element by element. So $1 \in \{1, 2\}$ is true while $1 \subseteq \{1, 2\}$ makes no sense (1 is not a set here); and $\{1\} \subseteq \{1, 2\}$ is true while $\{1\} \in \{1, 2\}$ is false.

Quick check. Let $X = \{\{1, 2\}, 3\}$. Is $\{1, 2\} \subseteq X$? Is $\{1, 2\} \in X$?

Solution. The set X has exactly two elements: the set $\{1, 2\}$ and the number 3. So $\{1, 2\} \in X$ is **true**. But $\{1, 2\} \subseteq X$ is **false**: $1 \in \{1, 2\}$, and 1 is neither the set $\{1, 2\}$ nor the number 3, so $1 \notin X$.

0.3.3 Equality of sets and double inclusion

We said a set is determined by its elements. Here is the precise version.

Definition 0.3.5 — Set equality

Two sets A and B are **equal**, written $A = B$, if they have exactly the same elements:

$$\forall x : (x \in A \Leftrightarrow x \in B).$$

So $\{1, 2\} = \{2, 1\} = \{1, 1, 2\}$: each object belongs to one of these sets exactly when it is 1 or 2. Order and repetition in a list are features of how we **write** a set, not of the set.

Proving a biconditional for every x at once is awkward. It is much easier to prove two implications separately, and that is exactly what the next result allows.

Theorem 0.3.1 — Double inclusion

Let A and B be sets. Then $A = B$ if and only if $A \subseteq B$ and $B \subseteq A$.

Proof. By [Definition 0.3.5](#), $A = B$ means that for every x , both $x \in A \Rightarrow x \in B$ and $x \in B \Rightarrow x \in A$ hold, since a biconditional is the conjunction of an implication and its converse. A statement of the form “for every x , $P(x)$ and $Q(x)$ ” holds exactly when “for every x , $P(x)$ ” and “for every x , $Q(x)$ ” both hold, by [Definition 0.1.3](#). Hence $A = B$ holds exactly when $\forall x (x \in A \Rightarrow x \in B)$ and $\forall x (x \in B \Rightarrow x \in A)$ both hold. By [Definition 0.3.4](#), these are $A \subseteq B$ and $B \subseteq A$. This proves the theorem. ■

This is the move “**prove set equality by double inclusion**”. The proof is written in two labeled halves, $(\subseteq)$ and $(\supseteq)$, each starting “Let $x \in \dots$ ”. Here is a first consequence.

Remark. There is only one empty set. If E and E' are sets with no elements, then $E \subseteq E'$ and $E' \subseteq E$ hold vacuously, as in [Example \(Subsets\)](#) (c), so $E = E'$ by [Theorem 0.3.1](#). This is why we may say **the** empty set.

Example. (A solution set by double inclusion)

Prove that $\{x \in \mathbb{R} : x^2 - 3x + 2 = 0\} = \{1, 2\}$.

Solution. Write $S = \{x \in \mathbb{R} : x^2 - 3x + 2 = 0\}$. By [Theorem 0.3.1](#) it suffices to prove $S \subseteq \{1, 2\}$ and $\{1, 2\} \subseteq S$.

$(\subseteq)$ Let $x \in S$. Then x is real and $x^2 - 3x + 2 = 0$. Factoring, $(x - 1)(x - 2) = 0$. A product of two real numbers is zero only if one factor is zero, so $x = 1$ or $x = 2$. Hence $x \in \{1, 2\}$.

$(\supseteq)$ Let $x \in \{1, 2\}$, so $x = 1$ or $x = 2$. Both are real. If $x = 1$, then $1 - 3 + 2 = 0$; if $x = 2$, then $4 - 6 + 2 = 0$. In either case $x \in S$.

This shows $S = \{1, 2\}$.

Solving an equation “at school” usually produces only the $(\subseteq)$ half: “if x is a solution, then x is 1 or 2”. The $(\supseteq)$ half, checking that the candidates really are solutions, is the part that gets forgotten. It matters whenever a step in the solving is not reversible, such as squaring both sides.

0.3.4 Union, intersection and difference

Given two sets, there are three basic ways to form a new one: keep what is in either, keep what is in both, or keep what is in the first but not the second. Each is a set-builder description whose predicate uses a connective from the section on logic.

Definition 0.3.6 — Union

The **union** of sets A and B is

$$A \cup B = \{x : x \in A \text{ or } x \in B\}.$$

Definition 0.3.7 — Intersection

The **intersection** of sets A and B is

$$A \cap B = \{x : x \in A \text{ and } x \in B\}.$$

We call A and B **disjoint** if $A \cap B = \emptyset$.

Definition 0.3.8 — Set difference and complement

The **difference** of sets A and B is

$$A \setminus B = \{x \in A : x \notin B\}.$$

When all sets under discussion are subsets of one fixed set U , the **complement** of $B \subseteq U$ is $B^c = U \setminus B$.

In words: the union keeps an element that lies in **at least one** of the sets (the “or” is inclusive, as always); the intersection keeps an element that lies in **both**; the difference $A \setminus B$ keeps the elements of A that are **not** in B . The complement depends on the choice of U : the complement of $\mathbb{N}$ is the set of negative integers inside $\mathbb{Z}$, but a much bigger set inside $\mathbb{R}$.

Example. (Operations on finite sets)

Let $A = \{1, 2, 3, 4\}$ and $B = \{3, 4, 5, 6\}$. Compute $A \cup B$, $A \cap B$, $A \setminus B$ and $B \setminus A$.

Solution. Checking each element against the definitions: $A \cup B = \{1, 2, 3, 4, 5, 6\}$; $A \cap B = \{3, 4\}$; $A \setminus B = \{1, 2\}$; and $B \setminus A = \{5, 6\}$. Since $A \setminus B \neq B \setminus A$, the difference is not symmetric in its two arguments, unlike union and intersection.

Example. (Operations on intervals)

Let $A = [0, 2]$ and $B = [1, 3]$. Compute $A \cup B$, $A \cap B$ and $A \setminus B$.

Solution. A real x lies in $A \cup B$ when $0 \leq x \leq 2$ or $1 \leq x \leq 3$, which happens exactly when $0 \leq x \leq 3$; so $A \cup B = [0, 3]$. It lies in $A \cap B$ when both hold, that is, when $1 \leq x \leq 2$; so $A \cap B = [1, 2]$. It lies in $A \setminus B$ when $0 \leq x \leq 2$ and not $1 \leq x \leq 3$. Given $x \leq 2$, “not $1 \leq x \leq 3$ ” means $x < 1$; so $A \setminus B = [0, 1)$. The endpoint 1 is excluded because $1 \in B$.

Unions, intersections and differences interact through laws that mirror those for “or”, “and”

and “not”. The first pair is De Morgan’s laws, the set version of [Theorem 0.1.2](#).

Theorem 0.3.2 — De Morgan’s laws for sets

Let A , B and C be sets. Then

- (a) $A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$;
- (b) $A \setminus (B \cap C) = (A \setminus B) \cup (A \setminus C)$.

In particular, if $B, C \subseteq U$, then $(B \cup C)^c = B^c \cap C^c$ and $(B \cap C)^c = B^c \cup C^c$.

Idea. Unwind membership into connectives. “ $x \notin B \cup C$ ” is “not ($x \in B$ or $x \in C$)”, and De Morgan for statements turns that into “ $x \notin B$ and $x \notin C$ ”. So each law is its logical counterpart, applied one element at a time. We prove (a) by double inclusion; (b) is an exercise.

Proof. We prove (a) using [Theorem 0.3.1](#).

($\subseteq$) Let $x \in A \setminus (B \cup C)$. By [Definition 0.3.8](#), $x \in A$ and $x \notin B \cup C$. By [Definition 0.3.6](#), the second says that “ $x \in B$ or $x \in C$ ” is false, so by [Theorem 0.1.2](#), $x \notin B$ and $x \notin C$. Since $x \in A$ and $x \notin B$, we have $x \in A \setminus B$; since $x \in A$ and $x \notin C$, we have $x \in A \setminus C$. Hence $x \in (A \setminus B) \cap (A \setminus C)$ by [Definition 0.3.7](#).

($\supseteq$) Let $x \in (A \setminus B) \cap (A \setminus C)$. By [Definition 0.3.7](#) and [Definition 0.3.8](#), $x \in A$, $x \notin B$ and $x \notin C$. By [Theorem 0.1.2](#), “ $x \in B$ or $x \in C$ ” is false, so $x \notin B \cup C$. Together with $x \in A$, this gives $x \in A \setminus (B \cup C)$.

This proves (a). For the complement form, take $A = U$ and use [Definition 0.3.8](#): $(B \cup C)^c = U \setminus (B \cup C) = (U \setminus B) \cap (U \setminus C) = B^c \cap C^c$, and similarly for (b). ■

The second pair of laws says each of $\cap$ and $\cup$ distributes over the other, like multiplication over addition, except that here **both** directions hold.

Theorem 0.3.3 — Distributive laws for sets

Let A , B and C be sets. Then

- (a) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$;
- (b) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$.

Idea. Double inclusion again, and in each half the “or” in a union forces a split into cases. For (b) ($\supseteq$), an element of both $A \cup B$ and $A \cup C$ is either in A , and then done, or not in A , and then it must be in B and in C . We prove (b); (a) is an exercise.

Proof. We prove (b) using [Theorem 0.3.1](#).

($\subseteq$) Let $x \in A \cup (B \cap C)$. By [Definition 0.3.6](#), $x \in A$ or $x \in B \cap C$.

Case 1: $x \in A$. Then $x \in A \cup B$ and $x \in A \cup C$ by [Definition 0.3.6](#).

Case 2: $x \in B \cap C$. Then $x \in B$ and $x \in C$ by [Definition 0.3.7](#), so again $x \in A \cup B$ and $x \in A \cup C$.

In both cases $x \in (A \cup B) \cap (A \cup C)$ by [Definition 0.3.7](#).

($\supseteq$) Let $x \in (A \cup B) \cap (A \cup C)$, so $x \in A \cup B$ and $x \in A \cup C$.

Case 1: $x \in A$. Then $x \in A \cup (B \cap C)$.

Case 2: $x \notin A$. Since $x \in A \cup B$ and $x \notin A$, we get $x \in B$. Since $x \in A \cup C$ and $x \notin A$, we get $x \in C$. Hence $x \in B \cap C$, so $x \in A \cup (B \cap C)$.

In both cases $x \in A \cup (B \cap C)$. This proves (b). ■

Remark. In Chapter 1 we will see that the intersection of two subspaces is always a subspace, while their union usually is not: the two axes in the plane are subspaces, but $(1, 0) + (0, 1) = (1, 1)$ lies on neither. The union is replaced there by a different construction, the sum of subspaces.

0.3.5 Indexed families of sets

Two sets are not always enough. In Chapter 1 we will need “the intersection of **all** subspaces that contain a given set”, and there are usually infinitely many of them. So we need unions and intersections of any number of sets at once.

An indexed family is a list of sets with one set for each label; its union keeps what lies in some member, its intersection what lies in every member.

Definition 0.3.9 — Unions and intersections of indexed families

Let I be a set, and for each $i \in I$ let A_i be a set. The collection $(A_i)_{i \in I}$ is an **indexed family** of sets, with **index set** I . Its **union** is

$$\bigcup_{i \in I} A_i = \{x : \exists i \in I : x \in A_i\},$$

and, if I is **non-empty**, its **intersection** is

$$\bigcap_{i \in I} A_i = \{x : \forall i \in I : x \in A_i\}.$$

In words: x lies in the union when it lies in **at least one** A_i , and in the intersection when it lies in **every** A_i . When $I = \{1, 2\}$, these are $A_1 \cup A_2$ and $A_1 \cap A_2$. When $I = \{1, 2, 3, \dots\}$, we also write $\bigcup_{n=1}^{\infty} A_n$ and $\bigcap_{n=1}^{\infty} A_n$.

Why I must be non-empty for the intersection. If $I = \emptyset$, the condition “ $\forall i \in I : x \in A_i$ ” is vacuously true for **every** object x , so the “intersection” would contain everything, which is not a set we want. (When all sets live inside a fixed U , one sometimes defines the empty intersection to be U .) The empty union causes no such trouble: “ $\exists i \in \emptyset$ ” is always false, so $\bigcup_{i \in \emptyset} A_i = \emptyset$.

Example. (Indexed unions and intersections of intervals)

For each integer $n \geq 1$, let $A_n = [1/n, 1]$ and $B_n = (0, 1/n)$. Show that

$$\bigcup_{n=1}^{\infty} A_n = (0, 1] \quad \text{and} \quad \bigcap_{n=1}^{\infty} B_n = \emptyset.$$

You may use the fact that for every real $x > 0$ there is an integer $n \geq 1$ with $1/n < x$.

Solution. The union. ($\subseteq$) Let $x \in \bigcup_n A_n$. By [Definition 0.3.9](#), $x \in A_n$ for some $n \geq 1$, so $0 < 1/n \leq x \leq 1$. Hence $x \in (0, 1]$. ($\supseteq$) Let $x \in (0, 1]$. Since $x > 0$, there is an integer $n \geq 1$ with $1/n < x$. Then $1/n \leq x \leq 1$, so $x \in A_n$, and hence x lies in the union. By [Theorem 0.3.1](#), the union is $(0, 1]$.

The intersection. Suppose, for a contradiction, that some x lies in $\bigcap_n B_n$. Then $x \in B_1 = (0, 1)$, so $x > 0$. Choose an integer $n \geq 1$ with $1/n < x$. Then $x \notin (0, 1/n) = B_n$, contradicting $x \in B_n$ for every n . Hence the intersection has no elements, so it equals $\emptyset$.

Each B_n is non-empty (it contains $1/(2n)$), and each is contained in the one before. Yet their intersection is empty. Infinite intersections can shrink to nothing even when no single step does.

De Morgan's laws hold for indexed families with the same proofs: for a non-empty index set I , $A \setminus \bigcup_i B_i = \bigcap_i (A \setminus B_i)$ and $A \setminus \bigcap_i B_i = \bigcup_i (A \setminus B_i)$. The “or” of the finite version becomes “there exists”, and the “and” becomes “for every”, exactly as in [Definition 0.1.4](#).

0.3.6 Cartesian products and n -tuples

A set forgets order: $\{1, 2\} = \{2, 1\}$. But a point in the plane is not like that. The point $(1, 2)$ is different from $(2, 1)$. We need objects that remember order.

An **ordered pair** (a, b) has a first entry a and a second entry b , and two ordered pairs are equal exactly when their entries agree in order:

$$(a, b) = (a', b') \iff a = a' \text{ and } b = b'.$$

Definition 0.3.10 — Cartesian product

Let A and B be sets. The **Cartesian product** of A and B is the set of ordered pairs

$$A \times B = \{(a, b) : a \in A, b \in B\}.$$

More generally, for sets $A_1, \dots, A_n$, the product $A_1 \times \dots \times A_n$ is the set of all ordered lists $(a_1, \dots, a_n)$ with $a_i \in A_i$ for each i .

For example, $\{1, 2\} \times \{x, y\} = \{(1, x), (1, y), (2, x), (2, y)\}$. The order of the factors matters: $(1, x) \in \{1, 2\} \times \{x, y\}$, but $(1, x) \notin \{x, y\} \times \{1, 2\}$, whose pairs have a letter first. A degenerate case: $A \times \emptyset = \emptyset$ for every set A , since no pair can have a second entry from $\emptyset$.

Definition 0.3.11 — n -tuples

Let A be a set and $n \geq 1$ an integer. An **n -tuple** of elements of A is an ordered list $(a_1, \dots, a_n)$ with each $a_i \in A$. The set of all of them is

$$A^n = \underbrace{A \times A \times \cdots \times A}_{n \text{ factors}}.$$

Two n -tuples are equal exactly when $a_i = b_i$ for **every** $i = 1, \dots, n$.

Example. (Sets of tuples)

Describe some elements of $\mathbb{R}^2$, $\mathbb{Z}^3$ and $\{0, 1\}^3$, and count the elements of $\{0, 1\}^3$.

Solution. The set $\mathbb{R}^2$ is the plane: it contains $(0, 0)$, $(1, -2)$ and $(\sqrt{2}, \pi)$. The set $\mathbb{Z}^3$ contains $(1, 0, -4)$ but not $(1, \frac{1}{2}, 0)$, whose second entry is not an integer. The set $\{0, 1\}^3$ consists of the lists of three bits:

$$\{0, 1\}^3 = \{(0, 0, 0), (0, 0, 1), (0, 1, 0), (0, 1, 1), (1, 0, 0), (1, 0, 1), (1, 1, 0), (1, 1, 1)\}.$$

There are two choices for each of the three entries, giving $2 \cdot 2 \cdot 2 = 8$ elements. In the same way, if A has k elements then A^n has k^n elements.

In Chapter 1, $\mathbb{R}^n$ and more generally F^n become the first examples of vector spaces, with n -tuples added and scaled entry by entry.

0.3.7 Power sets

Sets can themselves be elements of sets, as the warning about $\{\emptyset\}$ showed. One such set collects every subset of a given set.

Definition 0.3.12 — Power set

The **power set** of a set S is the set of all subsets of S :

$$\mathcal{P}(S) = \{X : X \subseteq S\}.$$

So $X \in \mathcal{P}(S)$ means exactly $X \subseteq S$. For example,

$$\mathcal{P}(\{1, 2\}) = \{\emptyset, \{1\}, \{2\}, \{1, 2\}\},$$

which has 4 elements. The degenerate case is instructive: $\mathcal{P}(\emptyset) = \{\emptyset\}$ has **one** element, because the empty set has exactly one subset, itself. For a set with n elements the power set has 2^n elements; the last exercise proves this by induction.

0.3.8 Exercises

0.3.8.1 A. Check your understanding

Exercise A1.

Decide whether each statement is true or false. Give a one-line reason.

- (a) $0 \in \mathbb{N}$.
- (b) $\emptyset \in \emptyset$.
- (c) $\emptyset \subseteq \emptyset$.
- (d) $\{0\} = \emptyset$.
- (e) $\{1, 2\} = \{2, 1, 1\}$ and $(1, 2) = (2, 1)$.
- (f) $\{1\} \in \mathcal{P}(\{1, 2\})$ and $\{1\} \subseteq \mathcal{P}(\{1, 2\})$.

Solution.

- (a) True. By [Definition 0.3.1](#), $\mathbb{N}$ includes 0 in this book.
- (b) False. By [Definition 0.3.3](#), nothing is an element of $\emptyset$, not even $\emptyset$.
- (c) True. The empty set is a subset of every set, vacuously, as in [Example \(Subsets\)](#).
- (d) False. $0 \in \{0\}$ but $0 \notin \emptyset$, so the two sets do not have the same elements ([Definition 0.3.5](#)).
- (e) The first is true: both sets have exactly the elements 1 and 2. The second is false: ordered pairs are equal only when their first entries agree, and $1 \neq 2$.
- (f) The first is true: $\{1\} \subseteq \{1, 2\}$, so $\{1\} \in \mathcal{P}(\{1, 2\})$ by [Definition 0.3.12](#). The second is false: $1 \in \{1\}$, but 1 is a number, not a subset of $\{1, 2\}$, so $1 \notin \mathcal{P}(\{1, 2\})$.

0.3.8.2 B. Practice

Exercise B1: The second De Morgan law.

Let A, B, C be sets. Prove that $A \setminus (B \cap C) = (A \setminus B) \cup (A \setminus C)$, which is part (b) of [Theorem 0.3.2](#).

Solution. We use [Theorem 0.3.1](#).

($\subseteq$) Let $x \in A \setminus (B \cap C)$. By [Definition 0.3.8](#), $x \in A$ and $x \notin B \cap C$. By [Definition 0.3.7](#), “ $x \in B$ and $x \in C$ ” is false, so by [Theorem 0.1.2](#), $x \notin B$ or $x \notin C$. If $x \notin B$, then $x \in A \setminus B$; if $x \notin C$, then $x \in A \setminus C$. In either case $x \in (A \setminus B) \cup (A \setminus C)$ by [Definition 0.3.6](#).

($\supseteq$) Let $x \in (A \setminus B) \cup (A \setminus C)$. By [Definition 0.3.6](#), $x \in A \setminus B$ or $x \in A \setminus C$. In the first case $x \in A$ and $x \notin B$; in the second, $x \in A$ and $x \notin C$. In both cases $x \in A$, and “ $x \notin B$ or $x \notin C$ ” is true, so by [Theorem 0.1.2](#) “ $x \in B$ and $x \in C$ ” is false, that is, $x \notin B \cap C$. Hence $x \in A \setminus (B \cap C)$.

This proves the equality.

Exercise B2: A distributive law.

Let A, B, C be sets. Prove that $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$, which is part (a) of [Theorem 0.3.3](#).

Solution. We use [Theorem 0.3.1](#).

($\subseteq$) Let $x \in A \cap (B \cup C)$. Then $x \in A$ and $x \in B \cup C$, so $x \in B$ or $x \in C$. If $x \in B$, then $x \in A \cap B$; if $x \in C$, then $x \in A \cap C$. In either case $x \in (A \cap B) \cup (A \cap C)$.

($\supseteq$) Let $x \in (A \cap B) \cup (A \cap C)$. If $x \in A \cap B$, then $x \in A$ and $x \in B \subseteq B \cup C$. If $x \in A \cap C$, then $x \in A$ and $x \in C \subseteq B \cup C$. In either case $x \in A$ and $x \in B \cup C$, so $x \in A \cap (B \cup C)$.

This proves the equality.

Exercise B3: An indexed intersection.

Prove that

$$\bigcap_{n=1}^{\infty} \left(-\frac{1}{n}, \frac{1}{n}\right) = \{0\}.$$

You may use the fact that for every real $x > 0$ there is an integer $n \geq 1$ with $1/n < x$.

Solution. Write $B_n = (-1/n, 1/n)$ and $B = \bigcap_{n \geq 1} B_n$. We use [Theorem 0.3.1](#).

($\supseteq$) For every $n \geq 1$, $-1/n < 0 < 1/n$, so $0 \in B_n$. By [Definition 0.3.9](#), $0 \in B$.

($\subseteq$) Let $x \in B$. Suppose, for a contradiction, that $x \neq 0$. Then $|x| > 0$, so there is an integer $n \geq 1$ with $1/n < |x|$. Then $x \geq 1/n$ or $x \leq -1/n$, so $x \notin B_n$. This contradicts $x \in B$, which requires $x \in B_n$ for **every** n . Hence $x = 0$, that is, $x \in \{0\}$.

This shows $B = \{0\}$.

0.3.8.3 C. Going deeper**Exercise C1: When union equals intersection.**

Let A and B be sets. Prove that $A \cup B = A \cap B$ if and only if $A = B$.

Solution. ($\Leftarrow$) Suppose $A = B$. Then $A \cup B = A \cup A$ and $A \cap B = A \cap A$. By [Definition 0.3.6](#) and [Definition 0.3.7](#), “ $x \in A$ or $x \in A$ ” and “ $x \in A$ and $x \in A$ ” are both equivalent to $x \in A$, so $A \cup A = A = A \cap A$. Hence $A \cup B = A \cap B$.

($\Rightarrow$) Suppose $A \cup B = A \cap B$. We show $A = B$ using [Theorem 0.3.1](#).

($\subseteq$) Let $x \in A$. Then $x \in A \cup B$ by [Definition 0.3.6](#). Since $A \cup B = A \cap B$, we get $x \in A \cap B$, and in particular $x \in B$.

($\supseteq$) Swapping the roles of A and B in the previous paragraph, which is allowed since $A \cup B = B \cup A$ and $A \cap B = B \cap A$, gives $B \subseteq A$.

Hence $A = B$. This proves the equivalence.

Exercise C2: Counting subsets.

For a finite set S , write $|S|$ for its number of elements. The aim is to prove that $|\mathcal{P}(S)| = 2^{|S|}$ for every finite set S .

- (a) Check the claim when $S = \emptyset$.
- (b) Let S have $n + 1$ elements, fix $s \in S$, and let $S' = S \setminus \{s\}$. Show that every subset X of S belongs to exactly one of the two collections: the subsets of S with $s \notin X$, and the subsets of S with $s \in X$.
- (c) Show that the subsets of S with $s \notin X$ are exactly the subsets of S' , and that $X \mapsto X \setminus \{s\}$ and $T \mapsto T \cup \{s\}$ pair off the subsets of S containing s with the subsets of S' , each undoing the other.
- (d) Hence prove the claim by induction on $n = |S|$.

Hint: in (c), use double inclusion to check that $(X \setminus \{s\}) \cup \{s\} = X$ when $s \in X$.

Solution.

- (a) The only subset of $\emptyset$ is $\emptyset$ itself, since any $X \subseteq \emptyset$ has no elements and so equals $\emptyset$ by [Theorem 0.3.1](#). So $\mathcal{P}(\emptyset) = \{\emptyset\}$ has $1 = 2^0$ element.
- (b) Let $X \subseteq S$. The statement $s \in X$ is either true or false, and not both. So X lies in exactly one of the two collections.
- (c) *First collection.* If $X \subseteq S$ and $s \notin X$, then every $x \in X$ lies in S and is not s , so $x \in S'$; hence $X \subseteq S'$. Conversely, if $T \subseteq S'$, then $T \subseteq S$ and $s \notin T$, since no element of S' equals s . So the first collection is exactly $\mathcal{P}(S')$.

Second collection. If $X \subseteq S$ with $s \in X$, then $X \setminus \{s\} \subseteq S'$. If $T \subseteq S'$, then $T \cup \{s\} \subseteq S$ contains s . Now we check that the two operations undo each other. For X containing s : an element lies in $(X \setminus \{s\}) \cup \{s\}$ exactly when it is in X and not s , or it is s ; since $s \in X$, this is exactly when it is in X . So $(X \setminus \{s\}) \cup \{s\} = X$. For $T \subseteq S'$: an element lies in $(T \cup \{s\}) \setminus \{s\}$ exactly when it is in T or equal to s , and not equal to s ; since $s \notin T$, this is exactly when it is in T . So $(T \cup \{s\}) \setminus \{s\} = T$. Hence the two operations pair the subsets of S containing s one-to-one with the subsets of S' , and the two collections have the same number of elements.

- (d) Let $P(n)$ be the statement “every set with n elements has exactly 2^n subsets”. We use [Theorem 0.2.7](#) with $n_0 = 0$. The base case $P(0)$ is (a), since the only set with no elements is $\emptyset$. Let $n \geq 0$, suppose $P(n)$, and let S have $n + 1$ elements. Fix $s \in S$, which exists since $n + 1 \geq 1$, and let $S' = S \setminus \{s\}$, a set with n elements. By (b), the number of subsets of S is the size of the first collection plus the size of the second. By (c) and the induction hypothesis, each of these sizes equals $|\mathcal{P}(S')| = 2^n$. Hence S has $2^n + 2^n = 2^{n+1}$ subsets, which is $P(n + 1)$. By [Theorem 0.2.7](#), $P(n)$ holds for every $n \in \mathbb{N}$, as claimed.

0.4 Functions

Sets are the objects of this chapter. The next question is how to move between them, and the tool is a function. Almost everything in linear algebra is a question about a function: a linear map is a function, a matrix will turn out to encode one, and “does $A\mathbf{x} = \mathbf{b}$ have a solution, and is it unique?” is a question about whether a function is surjective and injective. This section fixes the language of functions, composition and inverses, and proves the handful of facts about them that the rest of the book uses without comment.

0.4.1 What a function is

At school a function is usually a formula, such as $x^2 + 1$. A formula alone is not enough. The same formula x^2 behaves differently depending on which numbers we feed it and where we allow the answers to land, and whether an equation $x^2 = c$ can be solved depends on exactly that information. So a function carries three pieces of data, not one.

A function is a machine with a fixed input set and a fixed target set that turns each input into exactly one output.

Definition 0.4.1 — Function

Let X and Y be sets. A **function** (or **map**) $f: X \rightarrow Y$ is a rule that assigns to **every** element $x \in X$ **exactly one** element $f(x) \in Y$. The set X is the **domain** of f , the set Y is its **codomain**, and $f(x)$ is the **value** of f at x . We also write $x \mapsto f(x)$.

Two functions $f: X \rightarrow Y$ and $g: X' \rightarrow Y'$ are **equal** if $X = X'$, $Y = Y'$, and $f(x) = g(x)$ for every $x \in X$.

In words: the first clause says every input gets an output, so nothing in the domain is left out. The second says the output is unique, so the rule never hesitates between two values. The codomain is where outputs are *allowed* to land; nothing says every element of Y must be hit.

If you want a definition that avoids the vague word “rule”, use the Cartesian product ([Definition 0.3.10](#)): a function $X \rightarrow Y$ is a subset $G \subseteq X \times Y$ such that for every $x \in X$ there is exactly one $y \in Y$ with $(x, y) \in G$. The set G is the **graph** of f , and $f(x)$ is that unique y . Nothing in this book depends on which version you prefer.

Example. (First examples)

Check each of the following against the two clauses of [Definition 0.4.1](#).

- (a) $f: \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = 3x - 5$. Every real x gives one real number $3x - 5$.
- (b) $s: \mathbb{R}^2 \rightarrow \mathbb{R}$, $s(x, y) = x + y$. The domain is a Cartesian product; each pair gives one sum.
- (c) $\ell: \{\text{finite strings of the letters } a, b\} \rightarrow \mathbb{N}$, sending a string to its length. The empty string has length 0, which is why it matters that $\mathbb{N}$ contains 0.
- (d) For any set X , the rule $x \mapsto x$ is a function $X \rightarrow X$. It looks too simple to matter, but it will play the role that the number 1 plays for multiplication.

- (e) For any set Y , there is exactly one function $\emptyset \rightarrow Y$, the **empty function**. Both clauses are about elements of the domain, and there are none, so they hold vacuously. It is a degenerate case that makes statements like “for every set X ” true without exceptions.

The failures are just as instructive. The rule “ $x \mapsto$ the real number y with $y^2 = x$ ” is **not** a function $\mathbb{R} \rightarrow \mathbb{R}$. At $x = -1$ there is no such y , so the first clause fails. At $x = 4$ there are two, $y = 2$ and $y = -2$, so the second clause fails. Repairing both problems gives a genuine function $[0, \infty) \rightarrow \mathbb{R}$, $x \mapsto \sqrt{x}$, where $\sqrt{x}$ means the non-negative root.

The equality clause says the codomain is part of the function. The functions $\mathbb{R} \rightarrow \mathbb{R}$, $x \mapsto x^2$ and $\mathbb{R} \rightarrow [0, \infty)$, $x \mapsto x^2$ have the same values but are **different** functions. This is not pedantry: we will see in a moment that one of them is surjective and the other is not.

Quick check. Let $f, g: \mathbb{R} \rightarrow \mathbb{R}$ be $f(x) = |x|$ and $g(x) = \sqrt{x^2}$. Are f and g equal?

Solution. Yes. They have the same domain and the same codomain, and for every real x we have $\sqrt{x^2} = |x|$, because $|x|$ is the non-negative number whose square is x^2 . Different formulas can describe the same function; what counts is the values.

0.4.2 Images and preimages

Often we want to push a whole set through a function, or ask which inputs land in a given region. Both operations get a name.

Definition 0.4.2 — Image and preimage

Let $f: X \rightarrow Y$ be a function, $A \subseteq X$ and $B \subseteq Y$.

- The **image** of A under f is $f(A) := \{f(a) : a \in A\} \subseteq Y$. The set $f(X)$ is called the **image of f** .
- The **preimage** of B under f is $f^{-1}(B) := \{x \in X : f(x) \in B\} \subseteq X$.

In words: $f(A)$ collects the outputs of the inputs in A . The preimage $f^{-1}(B)$ collects every input whose output lies in B . So $y \in f(A)$ means “ $y = f(a)$ for **some** $a \in A$ ”, while $x \in f^{-1}(B)$ means “ $f(x) \in B$ ”, with no existential quantifier at all.

Example. (Images and preimages of the squaring map)

Let $q: \mathbb{R} \rightarrow \mathbb{R}$, $q(x) = x^2$. Compute $q([-1, 2])$, $q^{-1}([1, 4])$ and $q^{-1}([-5, -1])$.

Solution. As x runs over $[-1, 2]$, the square x^2 takes every value in $[0, 4]$: values in $[0, 1]$ come from $[0, 1]$, values in $[1, 4]$ from $[1, 2]$, and no square is negative or exceeds $2^2 = 4$ on this interval. Hence $q([-1, 2]) = [0, 4]$.

Next, $x^2 \in [1, 4]$ means $1 \leq |x| \leq 2$. Hence $q^{-1}([1, 4]) = [-2, -1] \cup [1, 2]$.

Finally, no real square lies in $[-5, -1]$, so $q^{-1}([-5, -1]) = \emptyset$. A preimage may be empty;

that is not an error.

Warning. The symbol $f^{-1}(B)$ is defined for **every** function, invertible or not; it does not mean that an inverse function exists. For the same reason, pushing forward and pulling back need not undo each other. With $q(x) = x^2$ on $\mathbb{R}$: for $B = [-1, 4]$ we get $q^{-1}(B) = [-2, 2]$ and $q(q^{-1}(B)) = [0, 4] \neq B$; for $A = [0, 1]$ we get $q(A) = [0, 1]$ and $q^{-1}(q(A)) = [-1, 1] \neq A$.

What does survive is one inclusion in each direction.

Proposition 0.4.1 — Image and preimage inclusions

Let $f: X \rightarrow Y$, $A \subseteq X$ and $B \subseteq Y$. Then

$$f(f^{-1}(B)) \subseteq B \quad \text{and} \quad A \subseteq f^{-1}(f(A)).$$

Proof. Let $y \in f(f^{-1}(B))$. By [Definition 0.4.2](#), $y = f(x)$ for some $x \in f^{-1}(B)$, and $x \in f^{-1}(B)$ means $f(x) \in B$. Hence $y \in B$.

Let $a \in A$. Then $f(a) \in f(A)$ by definition of the image, so $a \in f^{-1}(f(A))$ by definition of the preimage. This proves both inclusions. ■

The warning shows both inclusions can be strict. In [Exercise \(C3: When images and preimages undo each other\)](#) you will find exactly which property of f turns each one into an equality.

0.4.3 Injective, surjective, bijective

Given $f: X \rightarrow Y$ and a target $y \in Y$, the most basic question is how many solutions the equation $f(x) = y$ has. There are two separate issues: whether there is **at least one** solution, and whether there is **at most one**. In linear algebra these become “does $A\mathbf{x} = \mathbf{b}$ have a solution?” and “is the solution unique?”, and they are answered by different tools. So the two halves deserve separate names.

Injective means no two inputs collide; surjective means every target is hit; bijective means both, so every equation $f(x) = y$ has exactly one solution.

Definition 0.4.3 — Injective, surjective, bijective

Let $f: X \rightarrow Y$ be a function.

- f is **injective** (one-to-one) if **for all** $x_1, x_2 \in X$, $f(x_1) = f(x_2)$ implies $x_1 = x_2$.
- f is **surjective** (onto) if **for every** $y \in Y$ **there exists** $x \in X$ with $f(x) = y$.
Equivalently, $f(X) = Y$.
- f is **bijective** if it is both injective and surjective.

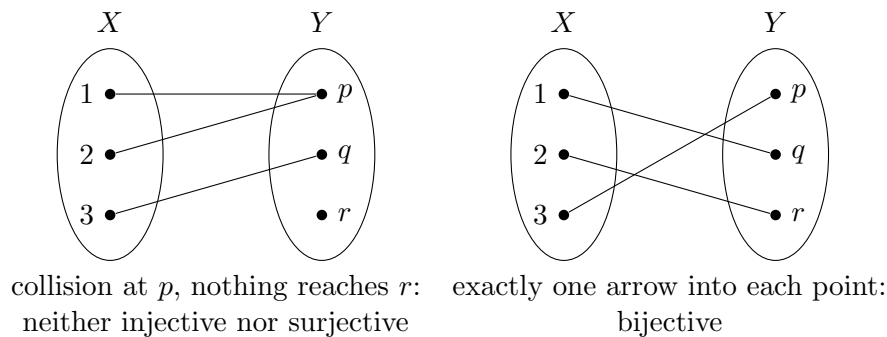
In words: injectivity says that if two inputs give the same output, they were the same input.

Equivalently, by the contrapositive ([Theorem 0.1.1](#)), distinct inputs give distinct outputs. Surjectivity says the image of f fills the whole codomain. The quantifiers matter: in surjectivity the x is allowed to depend on y .

Negating these, as in [Example \(What “not injective” means\)](#) and [Example \(What “not surjective” means\)](#), tells us what a counterexample looks like. The map f is **not injective** when there exist $x_1 \neq x_2$ with $f(x_1) = f(x_2)$: one collision is enough. It is **not surjective** when there exists $y \in Y$ that equals $f(x)$ for no $x \in X$: one missed target is enough.

The standard moves. To prove f injective, write “Suppose $f(x_1) = f(x_2)$ ” and work towards $x_1 = x_2$. To prove f surjective, write “Let $y \in Y$ ” and then produce an x , usually found backwards by solving $f(x) = y$ on scrap paper, and check forwards that $f(x) = y$.

The picture below shows maps between three-element sets, drawn as arrows. A collision is two arrows landing on one point; a missed target is a point with no arrow.



Example. (Deciding injectivity and surjectivity)

Determine which of the following functions are injective, surjective or bijective.

- (a) $f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = 3x - 5$.
- (b) $g: \mathbb{Z} \rightarrow \mathbb{Z}, g(n) = 2n$.
- (c) $s: \mathbb{R}^2 \rightarrow \mathbb{R}, s(x, y) = x + y$.
- (d) $\text{id}_X: X \rightarrow X, x \mapsto x$, for any set X .

Solution.

- (a) Bijective. Suppose $f(x_1) = f(x_2)$. Then $3x_1 - 5 = 3x_2 - 5$, so $3x_1 = 3x_2$, and dividing by $3 \neq 0$ gives $x_1 = x_2$. Hence f is injective. Let $y \in \mathbb{R}$. Solving $3x - 5 = y$ suggests $x = (y + 5)/3$, and indeed $f((y + 5)/3) = (y + 5) - 5 = y$. Hence f is surjective.
- (b) Injective, not surjective. Suppose $2n_1 = 2n_2$; then $n_1 = n_2$. But 1 is not hit: $2n = 1$ has no integer solution, since $2n$ is even. Note that the same formula $x \mapsto 2x$ on $\mathbb{R} \rightarrow \mathbb{R}$ is surjective; the failure comes from the domain and codomain, not the formula.
- (c) Surjective, not injective. Let $t \in \mathbb{R}$; then $s(t, 0) = t$, so s is surjective. But $s(1, 0) = 1 = s(0, 1)$ while $(1, 0) \neq (0, 1)$, a collision, so s is not injective.
- (d) Bijective. If $\text{id}_X(x_1) = \text{id}_X(x_2)$ then $x_1 = x_2$ directly, and every $x \in X$ is hit

by itself. This degenerate case is the model of a bijection and reappears as the identity function below.

The cleanest way to see that domain and codomain are part of the question is to keep the formula $x \mapsto x^2$ fixed and change only the sets. Write $[0, \infty)$ for the non-negative reals.

Function	Injective?	Surjective?
$\mathbb{R} \rightarrow \mathbb{R}$	no: $(-1)^2 = 1^2$	no: -1 is not a square
$\mathbb{R} \rightarrow [0, \infty)$	no: $(-1)^2 = 1^2$	yes: $y = (\sqrt{y})^2$
$[0, \infty) \rightarrow \mathbb{R}$	yes	no: -1 is not a square
$[0, \infty) \rightarrow [0, \infty)$	yes	yes

For the injective rows: if $x_1, x_2 \geq 0$ and $x_1^2 = x_2^2$, then $(x_1 - x_2)(x_1 + x_2) = 0$. Either $x_1 = x_2$, or $x_1 + x_2 = 0$, which for non-negative numbers forces $x_1 = x_2 = 0$. In both cases $x_1 = x_2$.

Read the table as a set of minimal changes. Shrinking the domain from $\mathbb{R}$ to $[0, \infty)$ removes the collision between -1 and 1 , so the injectivity clause now holds. Shrinking the codomain removes the missed targets, so the surjectivity clause now holds. The formula never changed.

Warning. Do not confuse injectivity with the statement “ $x_1 = x_2 \Rightarrow f(x_1) = f(x_2)$ ”. That implication is true for **every** function; it is the second clause of [Definition 0.4.1](#). Injectivity is its converse, “ $f(x_1) = f(x_2) \Rightarrow x_1 = x_2$ ”, and $x \mapsto x^2$ on $\mathbb{R}$ satisfies the first but not the second.

Why these definitions. Surjectivity depends on the codomain, and injectivity on the domain, so neither is a property of a formula alone; this is why [Definition 0.4.1](#) insists on both sets. The names are descriptive: “in-jective” throws the domain *into* the codomain without collisions, and “sur-jective” (French *sur*, onto) lands *onto* all of it. In Chapter 3 we will meet a fast test for linear maps: a linear map is injective exactly when the only input sent to *is*. That test is special to linear maps and fails for general functions, which is on here as on the general definitions come.

0.4.4 Composition

If the outputs of one function can be fed into another, we can run them in sequence. This is the operation that matrix multiplication will describe in Chapter 3.

Definition 0.4.4 — Composition

Let $f: X \rightarrow Y$ and $g: Y \rightarrow Z$ be functions. The **composition** $g \circ f: X \rightarrow Z$ is defined by

$$(g \circ f)(x) := g(f(x)) \quad \text{for every } x \in X.$$

Read $g \circ f$ as “ g after f ”: the function written on the **right** acts **first**. The definition needs the codomain of f to equal the domain of g , so that $g(f(x))$ makes sense.

Example. (Order matters)

(a) Let $f, g: \mathbb{R} \rightarrow \mathbb{R}$ be $f(x) = x + 3$ and $g(x) = 2x$. Compute $g \circ f$ and $f \circ g$.

- (b) Let $q: \mathbb{R} \rightarrow [0, \infty)$, $q(x) = x^2$, and $r: [0, \infty) \rightarrow \mathbb{R}$, $r(y) = \sqrt{y}$. Compute $r \circ q$ and $q \circ r$.

Solution.

- (a) $(g \circ f)(x) = g(x + 3) = 2x + 6$, while $(f \circ g)(x) = f(2x) = 2x + 3$. At $x = 0$ these are $6 \neq 3$, so $g \circ f \neq f \circ g$. Composition is not commutative, even when both orders make sense.
- (b) $r \circ q: \mathbb{R} \rightarrow \mathbb{R}$ is $x \mapsto \sqrt{x^2} = |x|$. In the other order, $q \circ r: [0, \infty) \rightarrow [0, \infty)$ is $y \mapsto (\sqrt{y})^2 = y$. So $q \circ r$ does nothing, while $r \circ q$ forgets the sign of x . The two orders do not even have the same domain. We will return to this pair when we discuss one-sided inverses.

Although the order matters, the grouping does not. This is what lets us write $h \circ g \circ f$ without brackets.

Theorem 0.4.2 — Associativity of composition

Let $f: X \rightarrow Y$, $g: Y \rightarrow Z$ and $h: Z \rightarrow W$ be functions. Then $(h \circ g) \circ f = h \circ (g \circ f)$.

Proof. Both sides are functions $X \rightarrow W$. Let $x \in X$. By Definition 0.4.4 applied twice on each side,

$$((h \circ g) \circ f)(x) = (h \circ g)(f(x)) = h(g(f(x))) = h((g \circ f)(x)) = (h \circ (g \circ f))(x).$$

Since the values agree at every x , the two functions are equal by Definition 0.4.1. ■

Next, the properties of the previous subsection pass through composition.

Theorem 0.4.3 — Composition preserves injectivity and surjectivity

Let $f: X \rightarrow Y$ and $g: Y \rightarrow Z$ be functions.

- (a) If f and g are injective, then $g \circ f$ is injective.
- (b) If f and g are surjective, then $g \circ f$ is surjective.
- (c) If f and g are bijective, then $g \circ f$ is bijective.

Idea. For (a), start from a collision of $g \circ f$ and peel off one function at a time, outermost first: injectivity of g removes g , then injectivity of f removes f . For (b), run backwards: reach z from some y using g , then reach that y from some x using f .

Proof.

- (a) Suppose $(g \circ f)(x_1) = (g \circ f)(x_2)$, that is, $g(f(x_1)) = g(f(x_2))$. Since g is injective, $f(x_1) = f(x_2)$. Since f is injective, $x_1 = x_2$. Hence $g \circ f$ is injective.
- (b) Let $z \in Z$. Since g is surjective, there is $y \in Y$ with $g(y) = z$. Since f is surjective,

- there is $x \in X$ with $f(x) = y$. Then $(g \circ f)(x) = g(y) = z$. Hence $g \circ f$ is surjective.
- (c) If f and g are bijective, then $g \circ f$ is injective by (a) and surjective by (b), so it is bijective. ■

The converse of (a) is false, but half of it survives, and that half is surprisingly useful. Suppose $g \circ f$ is injective. A collision of f would be a collision of $g \circ f$, so f must be injective; but g is only used on the image $f(X)$, so g may still collide outside it. The surjective case is dual.

Theorem 0.4.4 — Partial converse for compositions

Let $f: X \rightarrow Y$ and $g: Y \rightarrow Z$ be functions.

- (a) If $g \circ f$ is injective, then f is injective.
 (b) If $g \circ f$ is surjective, then g is surjective.

Proof.

- (a) Suppose $f(x_1) = f(x_2)$. Applying g to both sides gives $(g \circ f)(x_1) = (g \circ f)(x_2)$. Since $g \circ f$ is injective, $x_1 = x_2$. Hence f is injective.
 (b) Let $z \in Z$. Since $g \circ f$ is surjective, there is $x \in X$ with $g(f(x)) = z$. Put $y = f(x) \in Y$; then $g(y) = z$. Hence g is surjective. ■

Remark. The other halves really fail. Take $X = Z = \{1\}$, $Y = \{1, 2\}$, $f(1) = 1$ and $g(1) = g(2) = 1$. Then $g \circ f = \text{id}_{\{1\}}$ is bijective, yet g is not injective ($g(1) = g(2)$) and f is not surjective (nothing maps to 2).

0.4.5 Identity and one-sided inverses

The function that does nothing deserves a name, because “undoing f ” means “composing with f gives the function that does nothing”.

Definition 0.4.5 — Identity function

For a set X , the **identity function** on X is $\text{id}_X: X \rightarrow X$, $\text{id}_X(x) = x$.

For every $f: X \rightarrow Y$ we have $f \circ \text{id}_X = f$ and $\text{id}_Y \circ f = f$, since both sides send x to $f(x)$. Notice the subscripts: on the right of f the identity lives on the domain, on the left on the codomain.

Now look again at [Example \(Order matters\)](#) (b). There $q \circ r = \text{id}_{[0, \infty)}$, but $r \circ q = |\cdot| \neq \text{id}_{\mathbb{R}}$. So r undoes q from one side only. This one-sided behavior is common enough to need a word.

Definition 0.4.6 — Left and right inverses

Let $f: X \rightarrow Y$. A function $g: Y \rightarrow X$ is a **left inverse** of f if $g \circ f = \text{id}_X$, and a **right inverse** of f if $f \circ g = \text{id}_Y$.

“Left” and “right” refer to where g is written in the composition. A left inverse recovers the input from the output: $g(f(x)) = x$. A right inverse picks, for each target y , an input that reaches it: $f(g(y)) = y$. In [Example \(Order matters\)](#) (b), r is a right inverse of q , and q is a left inverse of r . But r is **not** a left inverse of q , because $r(q(-1)) = 1 \neq -1$.

The descriptions “recovers the input” and “reaches every target” should remind you of injective and surjective. They are exactly that.

Theorem 0.4.5 — Left inverses and injectivity

Let $f: X \rightarrow Y$ with X **non-empty**. Then f has a left inverse if and only if f is injective.

Idea. $(\Rightarrow)$ is [Theorem 0.4.4](#) with $g \circ f = \text{id}_X$, which is injective. For $(\Leftarrow)$ we must define $g(y)$ for every $y \in Y$. If y is in the image, injectivity gives a **unique** x with $f(x) = y$, and $g(y)$ must be that x . If y is not in the image, the condition $g \circ f = \text{id}_X$ says nothing about $g(y)$, so we send y anywhere in X . That is where non-emptiness is used.

Proof. $(\Rightarrow)$ Suppose $g \circ f = \text{id}_X$. The identity id_X is injective, so f is injective by [Theorem 0.4.4](#) (a).

$(\Leftarrow)$ Suppose f is injective. Since $X \neq \emptyset$, we may fix some $x_0 \in X$. Define $g: Y \rightarrow X$ as follows. If $y \in f(X)$, there is some $x \in X$ with $f(x) = y$, and this x is unique because f is injective; set $g(y) = x$. If $y \notin f(X)$, set $g(y) = x_0$. Every y receives exactly one value, so g is a function. For $x \in X$, the element $y = f(x)$ lies in $f(X)$, and the unique input sent to it is x , so $g(f(x)) = x$. Hence $g \circ f = \text{id}_X$, as claimed. ■

The hypothesis $X \neq \emptyset$ is not decoration. The empty function $\emptyset \rightarrow \{1\}$ is injective (there are no two inputs to collide), but there is no function $\{1\} \rightarrow \emptyset$ at all, since 1 would have nowhere to go.

Theorem 0.4.6 — Right inverses and surjectivity

Let $f: X \rightarrow Y$. Then f has a right inverse if and only if f is surjective.

Proof. $(\Rightarrow)$ Suppose $f \circ g = \text{id}_Y$. The identity id_Y is surjective, so f is surjective by [Theorem 0.4.4](#) (b).

$(\Leftarrow)$ Suppose f is surjective. For each $y \in Y$ the set $f^{-1}(\{y\})$ is non-empty, so we may choose one element of it and call it $g(y)$. This defines $g: Y \rightarrow X$ with $f(g(y)) = y$ for every $y \in Y$, that is, $f \circ g = \text{id}_Y$. ■

When Y is infinite, the step “choose one element from each of infinitely many non-empty sets at once” is an axiom of set theory, the **axiom of choice**; we use it freely and never mention it again in this chapter.

Unlike an inverse in the usual sense, one-sided inverses are rarely unique. For $q: \mathbb{R} \rightarrow [0, \infty)$, $x \mapsto x^2$, both $y \mapsto \sqrt{y}$ and $y \mapsto -\sqrt{y}$ are right inverses: each picks one of the two square roots.

Quick check. Give a function $f: \mathbb{N} \rightarrow \mathbb{N}$ that has a left inverse but no right inverse.

Solution. Take $f(n) = n + 1$. It is injective, since $n_1 + 1 = n_2 + 1$ gives $n_1 = n_2$, so by [Theorem 0.4.5](#) it has a left inverse; explicitly, $g(m) = m - 1$ for $m \geq 1$ and $g(0) = 0$ satisfies $g(f(n)) = n$. It is not surjective, since $n + 1 = 0$ has no solution in $\mathbb{N}$. So by [Theorem 0.4.6](#) it has no right inverse.

0.4.6 Inverses

When a function has an inverse on both sides, the two one-sided inverses turn out to coincide, and we get the notion of inverse we know from e^x and $\ln x$.

Definition 0.4.7 — Inverse function

Let $f: X \rightarrow Y$. A function $g: Y \rightarrow X$ is an **inverse** of f if

$$g \circ f = \text{id}_X \quad \text{and} \quad f \circ g = \text{id}_Y.$$

If f has an inverse, f is **invertible**.

Both conditions are needed: [Example \(Order matters\)](#) (b) shows a pair with one but not the other. The next theorem says which functions are invertible and that “the” inverse is justified.

Theorem 0.4.7 — Bijective if and only if invertible

A function $f: X \rightarrow Y$ is invertible if and only if it is bijective. Moreover, an inverse of f , when it exists, is **unique**; we denote it $f^{-1}: Y \rightarrow X$.

Idea. For $(\Leftarrow)$ one could combine the two one-sided theorems, but that route needs $X \neq \emptyset$ and a choice. A bijection makes both unnecessary: every y has **exactly one** preimage, so $f^{-1}(y)$ is forced. For uniqueness, the standard move is to assume two inverses g and h , and sandwich: $g = g \circ (f \circ h) = (g \circ f) \circ h = h$.

Proof. $(\Rightarrow)$ Suppose g is an inverse of f . Since $g \circ f = \text{id}_X$ is injective, f is injective by [Theorem 0.4.4](#) (a). Since $f \circ g = \text{id}_Y$ is surjective, f is surjective by [Theorem 0.4.4](#) (b). Hence f is bijective.

$(\Leftarrow)$ Suppose f is bijective, and let $y \in Y$. Since f is surjective, some $x \in X$ has $f(x) = y$, and since f is injective, this x is unique. Define $g(y)$ to be this x . Then $f(g(y)) = y$ for every $y \in Y$, so $f \circ g = \text{id}_Y$. For $x \in X$, the unique input sent to $f(x)$ is x itself, so $g(f(x)) = x$, and $g \circ f = \text{id}_X$. Hence g is an inverse of f .

(Uniqueness) Suppose g and h are both inverses of f . Then

$$g = g \circ \text{id}_Y = g \circ (f \circ h) = (g \circ f) \circ h = \text{id}_X \circ h = h,$$

where the third equality is [Theorem 0.4.2](#). This proves the theorem. ■

The uniqueness computation used only $g \circ f = \text{id}_X$ and $f \circ h = \text{id}_Y$. So it proves more: **if f has a left inverse g and a right inverse h , then $g = h$** , and this common function is the inverse. The same sandwich will prove that a matrix has at most one inverse.

Warning. The notation f^{-1} now has two meanings. The preimage $f^{-1}(B)$ of a **set** $B \subseteq Y$ exists for every f . The inverse function f^{-1} , applied to an **element** $y \in Y$, exists only for bijective f . Writing $q^{-1}(4)$ for $q(x) = x^2$ on $\mathbb{R}$ is meaningless, while $q^{-1}(\{4\}) = \{-2, 2\}$ is fine. When f is bijective the two agree: $f^{-1}(\{y\}) = \{f^{-1}(y)\}$.

Example. (Finding inverses)

For each function, decide whether it is invertible, and if so find its inverse.

- (a) $f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = 5 - 2x$.
- (b) $t: \mathbb{R}^2 \rightarrow \mathbb{R}^2, t(x, y) = (y, x)$.
- (c) $g: \mathbb{R} \rightarrow \mathbb{R}, g(x) = x^2$.
- (d) $h: [0, \infty) \rightarrow [0, \infty), h(x) = x^2$.

Solution.

- (a) Solving $5 - 2x = y$ gives $x = (5 - y)/2$. Put $k(y) = (5 - y)/2$. Check: $k(f(x)) = (5 - (5 - 2x))/2 = x$ and $f(k(y)) = 5 - (5 - y) = y$. So k is an inverse of f , and $f^{-1}(y) = (5 - y)/2$.
- (b) $t(t(x, y)) = t(y, x) = (x, y)$, so $t \circ t = \text{id}_{\mathbb{R}^2}$. Hence t is its own inverse: $t^{-1} = t$.
- (c) Not invertible. $g(-1) = g(1)$, so g is not injective, hence not bijective, and [Theorem 0.4.7](#) says it has no inverse.
- (d) Invertible. By the table in the previous subsection h is bijective, and the inverse is $h^{-1}(y) = \sqrt{y}$: indeed $\sqrt{x^2} = x$ for $x \geq 0$ and $(\sqrt{y})^2 = y$ for $y \geq 0$.

To undo two steps, undo the last one first. When you dress, the shirt goes on before the jacket; to undress, the jacket comes off first.

Theorem 0.4.8 — Inverse of a composition

Let $f: X \rightarrow Y$ and $g: Y \rightarrow Z$ be bijective. Then $g \circ f$ is bijective and

$$(g \circ f)^{-1} = f^{-1} \circ g^{-1}.$$

Proof. Using [Theorem 0.4.2](#) to regroup,

$$(f^{-1} \circ g^{-1}) \circ (g \circ f) = f^{-1} \circ (g^{-1} \circ g) \circ f = f^{-1} \circ \text{id}_Y \circ f = f^{-1} \circ f = \text{id}_X,$$

and in the same way $(g \circ f) \circ (f^{-1} \circ g^{-1}) = g \circ (f \circ f^{-1}) \circ g^{-1} = g \circ g^{-1} = \text{id}_Z$. Hence $f^{-1} \circ g^{-1}$ is an inverse of $g \circ f$. By [Theorem 0.4.7](#), $g \circ f$ is bijective and its unique inverse is $f^{-1} \circ g^{-1}$, as claimed. ■

The same reversal will appear for matrices as $(AB)^{-1} = B^{-1}A^{-1}$, and for the same reason.

0.4.7 Maps between finite sets

For functions between two finite sets **of the same size**, one of the two conditions comes for free. This is the pigeonhole principle, and it is the ancestor of a move used constantly from Chapter 3 on: instead of checking both injectivity and surjectivity, count, and check only one.

We write $|X|$ for the number of elements of a finite set X . We use two counting facts about lists without proof: a list of n entries with a repeated entry contains at most $n - 1$ distinct elements, and a subset of a finite set Y with $|Y|$ elements is all of Y .

Theorem 0.4.9 — Injective iff surjective for finite sets of equal size

Let X and Y be finite sets with $|X| = |Y|$, and let $f: X \rightarrow Y$. Then f is injective if and only if f is surjective. In that case f is bijective.

Idea. List the elements of X as $x_1, \dots, x_n$; the image $f(X)$ is then listed by $f(x_1), \dots, f(x_n)$. Injective means this list has no repeats, so it has n distinct entries and must fill Y . Surjective means the list fills Y , which has n elements, so there is no room for a repeat.

Proof. Let $n = |X| = |Y|$, and write $X = \{x_1, \dots, x_n\}$ with the x_i distinct. By [Definition 0.4.2](#), $f(X)$ consists of the entries of the list $f(x_1), \dots, f(x_n)$.

($\Rightarrow$) Suppose f is injective. Since the x_i are distinct, the entries $f(x_1), \dots, f(x_n)$ are distinct. Hence $f(X)$ is a subset of Y with $n = |Y|$ elements, so $f(X) = Y$, and f is surjective.

($\Leftarrow$) We prove the contrapositive ([Theorem 0.1.1](#)). Suppose f is not injective. Then $f(x_i) = f(x_j)$ for some $i \neq j$, so the list $f(x_1), \dots, f(x_n)$ has a repeated entry and $f(X)$ has at most $n - 1$ elements. Hence $f(X) \neq Y$, and f is not surjective.

So if f is injective or surjective, the two directions show that it is both, that is, bijective. ■

For a finite set X , this applies to every $f: X \rightarrow X$: such a map is injective if and only if it is surjective. In Chapter 3 the same shape of statement returns for linear maps $V \rightarrow V$ on a finite-dimensional space, with dimension playing the role of the number of elements.

Warning. The theorem needs finiteness. The map $g: \mathbb{N} \rightarrow \mathbb{N}$, $g(n) = 2n$, is injective but misses every odd number, so it is not surjective, even though domain and codomain are the same set. Infinite sets have room for an injection into a proper part of themselves.

0.4.8 Exercises

0.4.8.1 A. Check your understanding

Exercise A1.

- (a) State what it means for $f: X \rightarrow Y$ to be injective, and what it means to be surjective.
- (b) True or false: the functions $\mathbb{R} \rightarrow \mathbb{R}$, $x \mapsto x^3$ and $\mathbb{R} \rightarrow \mathbb{R}$, $x \mapsto x \cdot x \cdot x$ are equal. Give a reason.
- (c) True or false: if $g \circ f$ is injective, then g is injective. Give a reason.
- (d) Write down the first line of a proof that a function f is injective, and the first line of a proof that it is surjective.
- (e) True or false: the preimage $f^{-1}(B)$ is defined only when f is bijective.
- (f) True or false: every injective function $\{1, 2, 3\} \rightarrow \{1, 2, 3\}$ is bijective. Name the result you use.

Solution.

- (a) f is injective if for all $x_1, x_2 \in X$, $f(x_1) = f(x_2)$ implies $x_1 = x_2$. It is surjective if for every $y \in Y$ there exists $x \in X$ with $f(x) = y$.
- (b) True. They have the same domain and codomain, and $x^3 = x \cdot x \cdot x$ for every real x . Equality of functions is about values, not formulas.
- (c) False. In the remark after [Theorem 0.4.4](#), $g \circ f = \text{id}_{\{1\}}$ is injective but $g(1) = g(2)$. Only f is forced to be injective.
- (d) Injective: “Suppose $x_1, x_2 \in X$ satisfy $f(x_1) = f(x_2)$.” Surjective: “Let $y \in Y$.”
- (e) False. $f^{-1}(B) = \{x \in X : f(x) \in B\}$ makes sense for every function; see [Definition 0.4.2](#) and the warning after it.
- (f) True, by [Theorem 0.4.9](#): domain and codomain are finite with the same number of elements, so an injective map is surjective, hence bijective.

0.4.8.2 B. Practice

Exercise B1: Injective or surjective?.

Determine whether each function is injective, and whether it is surjective. Justify your answers.

- (a) $f: \mathbb{Z} \rightarrow \mathbb{Z}$, $f(n) = 3n + 1$.
- (b) $g: \mathbb{R}^2 \rightarrow \mathbb{R}^2$, $g(x, y) = (x + y, x - y)$.
- (c) $h: \mathbb{N} \rightarrow \mathbb{N}$, $h(n) = \lfloor n/2 \rfloor$, the largest integer not exceeding $n/2$.
- (d) $p: \mathbb{R} \rightarrow \mathbb{R}^2$, $p(t) = (t, t^2)$.

Solution.

- (a) Injective, not surjective. Suppose $3n_1 + 1 = 3n_2 + 1$. Then $3n_1 = 3n_2$, so $n_1 = n_2$. But 0 is not hit: $3n + 1 = 0$ would give $n = -1/3 \notin \mathbb{Z}$.

- (b) Bijective. Suppose $g(x_1, y_1) = g(x_2, y_2)$. Then $x_1 + y_1 = x_2 + y_2$ and $x_1 - y_1 = x_2 - y_2$. Adding the two equations gives $2x_1 = 2x_2$, and subtracting gives $2y_1 = 2y_2$, so $(x_1, y_1) = (x_2, y_2)$. Hence g is injective. Let $(u, v) \in \mathbb{R}^2$. Then

$$g\left(\frac{u+v}{2}, \frac{u-v}{2}\right) = \left(\frac{u+v}{2} + \frac{u-v}{2}, \frac{u+v}{2} - \frac{u-v}{2}\right) = (u, v).$$

Hence g is surjective.

- (c) Surjective, not injective. Let $m \in \mathbb{N}$. Then $h(2m) = \lfloor m \rfloor = m$, so h is surjective. But $h(0) = 0 = h(1)$, a collision, so h is not injective.
- (d) Injective, not surjective. Suppose $(t_1, t_1^2) = (t_2, t_2^2)$. Comparing first coordinates gives $t_1 = t_2$. But $(0, 1)$ is not hit: $p(t) = (0, 1)$ forces $t = 0$ and then $t^2 = 0 \neq 1$.

Exercise B2: Inverses.

- (a) Show that $f: \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = 7 - 4x$, is invertible and find f^{-1} .
- (b) Let $D = \mathbb{R} \setminus \{1\}$ and define $k: D \rightarrow D$ by $k(x) = \frac{x+1}{x-1}$. Show that k really takes values in D , and that $k \circ k = \text{id}_D$. Hence deduce that k is bijective and find k^{-1} .
- (c) With g as in B1(b), find g^{-1} . Hence find the inverse of $g \circ g$ without computing $g \circ g$ first.

Solution.

- (a) Put $e(y) = (7 - y)/4$. Then $e(f(x)) = (7 - (7 - 4x))/4 = x$ and $f(e(y)) = 7 - (7 - y) = y$. So e is an inverse of f , and by [Theorem 0.4.7](#) f is invertible with $f^{-1}(y) = (7 - y)/4$.
- (b) For $x \in D$ the denominator $x - 1$ is non-zero, so $k(x)$ is a real number. If $k(x) = 1$, then $x + 1 = x - 1$, that is, $1 = -1$, which is false. Hence $k(x) \in D$. Next, for $x \in D$,

$$k(k(x)) = \frac{\frac{x+1}{x-1} + 1}{\frac{x+1}{x-1} - 1} = \frac{(x+1) + (x-1)}{(x+1) - (x-1)} = \frac{2x}{2} = x,$$

where the second equality multiplies numerator and denominator by $x - 1 \neq 0$. So $k \circ k = \text{id}_D$. This says k is an inverse of itself, so k is invertible, hence bijective by [Theorem 0.4.7](#), and $k^{-1} = k$.

- (c) The solution of B1(b) shows that $e(u, v) = (\frac{u+v}{2}, \frac{u-v}{2})$ satisfies $g \circ e = \text{id}_{\mathbb{R}^2}$. Also $e(g(x, y)) = e(x + y, x - y) = (\frac{2x}{2}, \frac{2y}{2}) = (x, y)$, so $e \circ g = \text{id}_{\mathbb{R}^2}$. Hence $g^{-1} = e$. By [Theorem 0.4.8](#),

$$(g \circ g)^{-1} = g^{-1} \circ g^{-1}, \quad g^{-1}(g^{-1}(u, v)) = g^{-1}\left(\frac{u+v}{2}, \frac{u-v}{2}\right) = \left(\frac{u}{2}, \frac{v}{2}\right).$$

(As a check, $g(g(x, y)) = g(x + y, x - y) = (2x, 2y)$, which $(u, v) \mapsto (u/2, v/2)$ indeed undoes.)

Exercise B3: Images of unions and intersections.

Let $f: X \rightarrow Y$ and $A_1, A_2 \subseteq X$.

- (a) Prove that $f(A_1 \cup A_2) = f(A_1) \cup f(A_2)$.
- (b) Prove that $f(A_1 \cap A_2) \subseteq f(A_1) \cap f(A_2)$.
- (c) Using $q(x) = x^2$ on $\mathbb{R}$, give an example where the inclusion in (b) is strict.

Solution.

- (a) We use double inclusion ([Theorem 0.3.1](#)).

($\subseteq$) Let $y \in f(A_1 \cup A_2)$. Then $y = f(a)$ for some $a \in A_1 \cup A_2$. If $a \in A_1$, then $y \in f(A_1)$; if $a \in A_2$, then $y \in f(A_2)$. In either case $y \in f(A_1) \cup f(A_2)$.

($\supseteq$) Let $y \in f(A_1) \cup f(A_2)$. If $y \in f(A_1)$, then $y = f(a)$ for some $a \in A_1 \subseteq A_1 \cup A_2$, so $y \in f(A_1 \cup A_2)$. The case $y \in f(A_2)$ is the same with the roles of A_1 and A_2 swapped. This proves the equality.

- (b) Let $y \in f(A_1 \cap A_2)$. Then $y = f(a)$ for some $a \in A_1 \cap A_2$. Since $a \in A_1$, $y \in f(A_1)$, and since $a \in A_2$, $y \in f(A_2)$. Hence $y \in f(A_1) \cap f(A_2)$.
- (c) Take $A_1 = \{-1\}$ and $A_2 = \{1\}$. Then $A_1 \cap A_2 = \emptyset$, so $q(A_1 \cap A_2) = \emptyset$, while $q(A_1) \cap q(A_2) = \{1\} \cap \{1\} = \{1\}$. The inclusion is strict. The collision $q(-1) = q(1)$ is what goes wrong; for an injective f , equality holds in (b).

0.4.8.3 C. Going deeper

Exercise C1: Left inverses are not unique.

- (a) Let $f: X \rightarrow Y$ be injective but **not** surjective, with X having at least two elements. Prove that f has at least two different left inverses.
- (b) Find two different left inverses of $f: \mathbb{N} \rightarrow \mathbb{N}$, $f(n) = n + 1$.
- (c) Explain why a bijective function has exactly one left inverse.

Hint for (a): look at the proof of [Theorem 0.4.5](#) and ask where it made a free choice.

Solution.

- (a) Since f is not surjective, there is $y_0 \in Y \setminus f(X)$. Since X has at least two elements, there are $x_0 \neq x_1$ in X . As in the proof of [Theorem 0.4.5](#), for $y \in f(X)$ let $\bar{y}$ be the unique $x \in X$ with $f(x) = y$. Define $g_0, g_1: Y \rightarrow X$ by $g_i(y) = \bar{y}$ for $y \in f(X)$, and $g_i(y) = x_i$ for $y \notin f(X)$. The proof of [Theorem 0.4.5](#) shows that each g_i is a left inverse of f , since $g_i(f(x)) = x$ uses only the values on $f(X)$. But $g_0(y_0) = x_0 \neq x_1 = g_1(y_0)$, so $g_0 \neq g_1$.
- (b) The only element missed by f is 0. Define $g_0(m) = m - 1$ for $m \geq 1$ and $g_0(0) = 0$; define $g_1(m) = m - 1$ for $m \geq 1$ and $g_1(0) = 5$. Both satisfy $g_i(n + 1) = n$ for every $n \in \mathbb{N}$, and $g_0(0) \neq g_1(0)$.
- (c) Let f be bijective, and let g be any left inverse. By [Theorem 0.4.7](#), f has an inverse f^{-1} , which in particular is a right inverse. By the observation following that theorem, a left inverse and a right inverse of the same function are equal, so $g = f^{-1}$. Hence every left inverse equals f^{-1} . Conversely, f^{-1} is itself a left inverse, since $f^{-1} \circ f = \text{id}_X$. So f has exactly one left inverse.

Exercise C2: Self-maps of finite and infinite sets.

- (a) Let X be a finite set and $f: X \rightarrow X$ injective. Prove that f is bijective.
- (b) Show that (a) fails for $X = \mathbb{N}$: give an injective $f: \mathbb{N} \rightarrow \mathbb{N}$ that is not bijective.
- (c) Give a surjective $h: \mathbb{N} \rightarrow \mathbb{N}$ that is not bijective.
- (d) Let X be finite and $f, g: X \rightarrow X$ with $g \circ f = \text{id}_X$. Prove that $f \circ g = \text{id}_X$ as well.

Hint for (d): first show that f is bijective.

Solution.

- (a) The domain and codomain are the same finite set, so $|X| = |X|$. By [Theorem 0.4.9](#), the injective map f is bijective.
- (b) $f(n) = n + 1$ is injective, since $n_1 + 1 = n_2 + 1$ gives $n_1 = n_2$, but 0 is not of the form $n + 1$ with $n \in \mathbb{N}$. Hence f is not surjective, so not bijective.
- (c) Take $h(n) = \lfloor n/2 \rfloor$. By B1(c) it is surjective but not injective.
- (d) Since $g \circ f = \text{id}_X$ is injective, f is injective by [Theorem 0.4.4](#) (a). By (a), f is bijective, so it has an inverse f^{-1} by [Theorem 0.4.7](#). Then

$$g = g \circ \text{id}_X = g \circ (f \circ f^{-1}) = (g \circ f) \circ f^{-1} = \text{id}_X \circ f^{-1} = f^{-1},$$

using [Theorem 0.4.2](#). Hence $f \circ g = f \circ f^{-1} = \text{id}_X$, as claimed. (For $X = \mathbb{N}$ this fails: with $f(n) = n + 1$ and g_0 from C1(b), $g_0 \circ f = \text{id}_{\mathbb{N}}$ but $f(g_0(0)) = 1 \neq 0$. The matrix version of (d), “ $AB = I$ implies $BA = I$ for square matrices”, is proved in Chapter 2.)

Exercise C3: When images and preimages undo each other.

Let $f: X \rightarrow Y$.

- (a) Prove that $f(f^{-1}(B)) = B$ for **every** $B \subseteq Y$ if and only if f is surjective.
- (b) Prove that $f^{-1}(f(A)) = A$ for **every** $A \subseteq X$ if and only if f is injective.

Hint: for each direction you need only one well-chosen set; try a one-element set.

Solution.

- (a) ($\Rightarrow$) Suppose the equality holds for every B , and let $y \in Y$. Taking $B = \{y\}$ gives $f(f^{-1}(\{y\})) = \{y\}$, so $y \in f(f^{-1}(\{y\}))$. Hence $y = f(x)$ for some x , and f is surjective.
- ($\Leftarrow$) Suppose f is surjective, and let $B \subseteq Y$. The inclusion $f(f^{-1}(B)) \subseteq B$ is [Proposition 0.4.1](#). For the reverse, let $y \in B$. Since f is surjective, $y = f(x)$ for some $x \in X$. Then $f(x) \in B$, so $x \in f^{-1}(B)$, and hence $y = f(x) \in f(f^{-1}(B))$. By [Theorem 0.3.1](#), $f(f^{-1}(B)) = B$.
- (b) ($\Rightarrow$) Suppose the equality holds for every A , and suppose $f(x_1) = f(x_2)$. Taking $A = \{x_1\}$ gives $f^{-1}(\{f(x_1)\}) = \{x_1\}$. Since $f(x_2) = f(x_1)$, we have $x_2 \in f^{-1}(\{f(x_1)\}) = \{x_1\}$, so $x_2 = x_1$. Hence f is injective.

($\Leftarrow$) Suppose f is injective, and let $A \subseteq X$. The inclusion $A \subseteq f^{-1}(f(A))$ is [Proposition 0.4.1](#). For the reverse, let $x \in f^{-1}(f(A))$. Then $f(x) \in f(A)$, so $f(x) = f(a)$ for some $a \in A$. Since f is injective, $x = a \in A$. By [Theorem 0.3.1](#), $f^{-1}(f(A)) = A$.

0.5 Relations, Equivalence Classes and Quotients

Mathematics constantly declares different-looking things to be “the same”: $\frac{1}{2}$ and $\frac{2}{4}$ are the same number, 14:00 and 2 p.m. are the same time, and in Chapter 3 two vectors will count as the same when their difference lies in a fixed subspace. This section makes “the same, for present purposes” precise. The notion is an equivalence relation. Gluing equivalent things together produces a new set, the quotient, and the one real danger of working with quotients is defining a function on them that secretly depends on a choice. We will learn to check for that.

0.5.1 Relations

Before we can say what a good notion of sameness is, we need a word for any way of comparing two elements of a set, good or bad. Examples are $\leq$ on numbers, “divides” on integers, and “has the same birthday as” on people. What such a comparison really amounts to is the list of pairs that are related, and a list of pairs is a subset of the Cartesian product ([Definition 0.3.10](#)).

Definition 0.5.1 — Relation

A **relation** on a set X is a subset $R \subseteq X \times X$. When $(x, y) \in R$ we say x **is related to** y and write $x R y$, or $x \sim y$ when the relation is fixed.

Example. (Relations)

- (a) On $\mathbb{Z}$: $a \leq b$. The pairs $(1, 2)$ and $(2, 2)$ belong to the relation, $(2, 1)$ does not.
- (b) On $\mathbb{Z}$: $a \mid b$ (“ a divides b ”), meaning $b = ka$ for some $k \in \mathbb{Z}$. So $3 \mid 12$ and $3 \mid 0$, but $3 \nmid 5$.
- (c) On $\mathbb{R}$: $x \sim y$ if $x^2 + y^2 = 1$. As a subset of $\mathbb{R} \times \mathbb{R} = \mathbb{R}^2$ this relation is the unit circle, so a relation can literally be a picture.
- (d) On any set X : the **empty relation** $R = \emptyset$, in which nothing is related to anything, and the **full relation** $R = X \times X$, in which everything is related to everything. These two extremes are useful test cases for every property below.

Every subset of $X \times X$ is a relation, so the definition has no non-examples. The interesting question is which properties a given relation has.

0.5.2 Equivalence relations

What must any reasonable notion of “ x is the same as y ” satisfy? Think of “has the same birthday as”. Everyone has the same birthday as themselves. If you share a birthday with me, I share one with you. If x shares a birthday with y , and y with z , then x shares one with z . Those three features are exactly what we ask for.

An equivalence relation is a way of declaring elements “the same” that behaves like equality: everything is the same as itself, sameness goes both ways, and it can be chained.

Definition 0.5.2 — Equivalence relation

A relation $\sim$ on a set X is an **equivalence relation** if it satisfies the following three conditions.

- (E1) **Reflexive:** $x \sim x$ for every $x \in X$.
- (E2) **Symmetric:** for all $x, y \in X$, if $x \sim y$ then $y \sim x$.
- (E3) **Transitive:** for all $x, y, z \in X$, if $x \sim y$ and $y \sim z$ then $x \sim z$.

In words: (E1) asks every single element to be related to itself, with no exceptions. (E2) says the relation cannot be one-way. (E3) says a chain of two links can be shortened to one link, and so, by applying it repeatedly, can any finite chain. When $x \sim y$ we say x and y are **equivalent**.

The standard move for checking an equivalence relation is to write three short paragraphs, one per condition, each starting from its hypothesis (“Let $x \in X$.”, “Suppose $x \sim y$.”, “Suppose $x \sim y$ and $y \sim z$.”).

Example. (Equivalence relations)

Show that each of the following is an equivalence relation.

- (a) Equality on any set X : $x \sim y$ if $x = y$.
- (b) **Congruence modulo n** on $\mathbb{Z}$, for a fixed integer $n \geq 1$: $a \equiv b \pmod{n}$ if $n \mid a - b$.
- (c) “Same length” on the set S of finite strings of the letters a, b : $u \sim v$ if u and v have the same number of letters.
- (d) “Same value”: for a function $f: X \rightarrow Y$, put $x \sim_f x'$ if $f(x) = f(x')$.
- (e) Fractions: on $P = \mathbb{Z} \times (\mathbb{Z} \setminus \{0\})$, put $(a, b) \sim (c, d)$ if $ad = bc$.

Solution.

- (a) Every x equals itself; $x = y$ gives $y = x$; $x = y$ and $y = z$ give $x = z$. Equality is the degenerate case: it is the equivalence relation that identifies nothing new, and the model the other examples imitate.
- (b) *Reflexive.* Let $a \in \mathbb{Z}$. Then $a - a = 0 = 0 \cdot n$, so $n \mid a - a$. *Symmetric.* Suppose $a - b = kn$ with $k \in \mathbb{Z}$. Then $b - a = (-k)n$, so $n \mid b - a$. *Transitive.* Suppose $a - b = kn$ and $b - c = \ell n$ with $k, \ell \in \mathbb{Z}$. Then $a - c = (a - b) + (b - c) = (k + \ell)n$, so $n \mid a - c$. For $n = 2$ this is “same parity”: $a \equiv b \pmod{2}$ says a and b are both even or both odd.

- (c) This is the special case of (d) in which $f: S \rightarrow \mathbb{N}$ sends a string to its length, so it follows from (d), whose proof does not use (c).
- (d) *Reflexive:* $f(x) = f(x)$. *Symmetric:* $f(x) = f(x')$ gives $f(x') = f(x)$. *Transitive:* $f(x) = f(x')$ and $f(x') = f(x'')$ give $f(x) = f(x'')$. Each condition is inherited from the same property of equality in Y . Many equivalence relations are of this form: “same distance from the origin” on $\mathbb{R}^2$ is $\sim_f$ for $f(x, y) = x^2 + y^2$.
- (e) *Reflexive:* $ab = ba$, so $(a, b) \sim (a, b)$. *Symmetric:* $ad = bc$ gives $cb = da$, which says $(c, d) \sim (a, b)$. *Transitive:* suppose $ad = bc$ and $cf = de$. Multiplying the first equation by f and the second by b gives $adf = bcf = bde$. Hence $d(af - be) = 0$. Since $d \neq 0$, we get $af = be$, that is, $(a, b) \sim (e, f)$. Transitivity is exactly where second entries must be non-zero: with $d = 0$ allowed, $(1, 1) \sim (0, 0) \sim (2, 1)$ but $(1, 1) \not\sim (2, 1)$.

Minimal changes to these examples show that each condition can fail on its own.

Example. (Relations that are not equivalence relations)

Decide which of (E1)–(E3) each relation satisfies.

- (a) $\leq$ on $\mathbb{Z}$.
 (b) On $\mathbb{R}$: $x \sim y$ if $|x - y| \leq 1$.
 (c) On $\mathbb{R}$: $x \sim y$ if $xy > 0$.

Solution.

- (a) Change “=” to “ $\leq$ ”. Reflexivity survives ($a \leq a$) and so does transitivity ($a \leq b \leq c$ gives $a \leq c$). **Symmetry fails:** $1 \leq 2$ but $2 \not\leq 1$.
- (b) Change “ $|x - y| = 0$ ” (which is equality) to “ $|x - y| \leq 1$ ”. Reflexivity survives, since $|x - x| = 0$, and so does symmetry, since $|y - x| = |x - y|$. **Transitivity fails:** $0 \sim 1$ and $1 \sim 2$, but $|0 - 2| = 2 > 1$. “Close to” cannot be chained; many small steps make a big one.
- (c) Symmetry holds, since $yx = xy$. Transitivity holds: if $xy > 0$ and $yz > 0$, then $xy^2z > 0$, and since $y^2 > 0$ this gives $xz > 0$. **Reflexivity fails, at exactly one point:** $0 \cdot 0 = 0$, so $0 \not\sim 0$. On $\mathbb{R} \setminus \{0\}$ the same rule is an equivalence relation (“same sign”).

Warning. A tempting argument claims that (E1) follows from (E2) and (E3): “if $x \sim y$, then $y \sim x$ by symmetry, so $x \sim x$ by transitivity.” The gap is the word “if”. The argument needs some y with $x \sim y$, and there may be none. In [Example \(Relations that are not equivalence relations\)](#) (c), the element 0 is related to nothing, which is exactly why it escapes. Reflexivity must always be checked on its own.

Quick check. On $\mathbb{Z}$, define $a \sim b$ if $ab \geq 0$. Is $\sim$ an equivalence relation?

Solution. No. It is reflexive ($a^2 \geq 0$) and symmetric, but not transitive: $1 \sim 0$ and $0 \sim -1$ since both products are 0, yet $1 \cdot (-1) = -1 < 0$, so $1 \not\sim -1$. Compare [Example \(Relations that are not equivalence relations\)](#) (c): moving from $>$ to $\geq$ repairs reflexivity at 0 but lets 0 link the positives to the negatives.

Why these three conditions. They are precisely what the next subsection needs. Reflexivity will guarantee that every element lies in its own class, so no class is empty and nothing is lost. Symmetry and transitivity together will guarantee that two classes are either identical or disjoint. Drop any one of them and the picture of X cut cleanly into pieces falls apart, as the non-examples above show.

Remark. Two equivalence relations on matrices are central later: *row equivalence*, which records that one matrix can be reached from another by row operations, and *similarity*, which records that two square matrices describe the same linear map in different coordinates. Both are introduced once matrices are available, later in this chapter, and studied in Chapters 2 and 3.

0.5.3 Equivalence classes and partitions

Once we have decided which elements count as the same, it is natural to collect everything that is the same as a given element into one set.

Definition 0.5.3 — Equivalence class

Let $\sim$ be an equivalence relation on X , and let $x \in X$. The **equivalence class** of x is

$$[x] := \{y \in X : y \sim x\}.$$

Any element of $[x]$ is called a **representative** of the class.

The word “representative” is chosen carefully. A class is one set, but it usually has many names: $[x] = [y]$ is possible with $x \neq y$.

Example. (Equivalence classes)

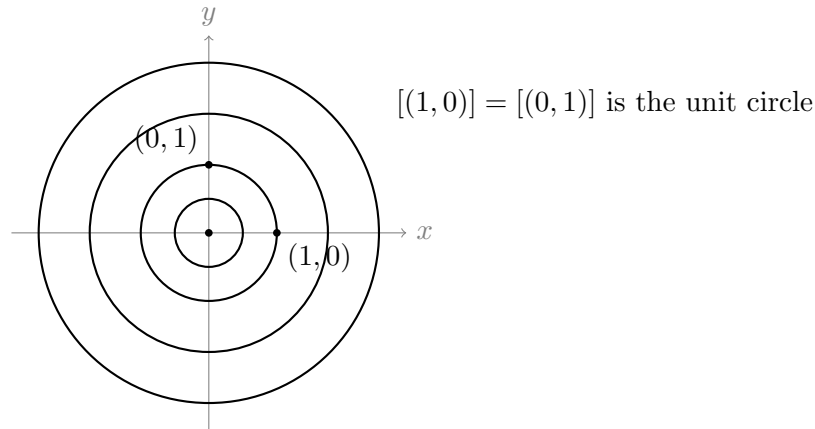
- (a) For equality on X , $[x] = \{x\}$. Every class has one element.
- (b) For congruence modulo 3 on $\mathbb{Z}$:

$$[0] = \{\dots, -6, -3, 0, 3, 6, \dots\}, \quad [1] = \{\dots, -5, -2, 1, 4, 7, \dots\}, \quad [2] = \{\dots, -4, -1, 2, 5, 8, \dots\}.$$

Here $[0] = [3] = [-3]$ and $[2] = [-1]$: one class, many names.

- (c) For “same length” on strings of a, b , the class of ab is $\{aa, ab, ba, bb\}$, and the class of the empty string contains only the empty string.
- (d) For the full relation $X \times X$ on a non-empty set X , there is a single class, $[x] = X$.
- (e) For “same distance from the origin” on $\mathbb{R}^2$, the class of (x, y) is the circle of radius $\sqrt{x^2 + y^2}$ centered at the origin. The class of the origin is the one-point set $\{(0, 0)\}$.

In example (e), the classes look like this. Each circle is one class; the origin on its own is another.



The picture suggests two facts: every point lies on exactly one circle, and two different circles never meet. Both hold for every equivalence relation, and the proof is where the three conditions do their work.

Lemma 0.5.1 — Classes are equal or disjoint

Let $\sim$ be an equivalence relation on X , and let $a, b \in X$.

- (a) $[a] = [b]$ if and only if $a \sim b$.
- (b) Either $[a] = [b]$ or $[a] \cap [b] = \emptyset$.

Proof.

(a) ($\Rightarrow$) Suppose $[a] = [b]$. By reflexivity $a \sim a$, so $a \in [a] = [b]$, which means $a \sim b$.

($\Leftarrow$) Suppose $a \sim b$. Let $x \in [a]$, so $x \sim a$. Since $a \sim b$, transitivity gives $x \sim b$, so $x \in [b]$. Hence $[a] \subseteq [b]$. By symmetry $b \sim a$, and swapping the roles of a and b in the argument gives $[b] \subseteq [a]$. By [Theorem 0.3.1](#), $[a] = [b]$.

(b) Suppose $[a] \cap [b] \neq \emptyset$, and let $c \in [a] \cap [b]$. Then $c \sim a$ and $c \sim b$. By symmetry $a \sim c$, and by transitivity $a \sim b$. By (a), $[a] = [b]$.

■

Part (a) is the working form of the lemma: to show two classes are equal, show one representative is equivalent to the other. The picture of non-overlapping pieces that cover everything has its own name.

Definition 0.5.4 — Partition

A **partition** of a set X is a collection $\mathcal{P}$ of subsets of X , called **blocks**, such that:

- (P1) every block is **non-empty**;
- (P2) **distinct** blocks are disjoint: if $B, B' \in \mathcal{P}$ and $B \neq B'$, then $B \cap B' = \emptyset$;
- (P3) the blocks cover X : every $x \in X$ lies in some block.

By (P2) and (P3), every element of X lies in **exactly one** block. For example, $\{\{1, 3\}, \{2\}\}$ is a partition of $\{1, 2, 3\}$, while $\{\{1, 2\}, \{2, 3\}\}$ is not, because the distinct blocks share 2 and (P2) fails.

Quick check. Is $\{\{1, 2, 3\}, \emptyset\}$ a partition of $\{1, 2, 3\}$? Is $\{\{1\}, \{2\}\}$?

Solution. Neither. The first contains the empty block, so (P1) fails, even though (P2) and (P3) hold. The second does not cover 3, so (P3) fails.

The main theorem says that equivalence relations and partitions are two descriptions of the same thing: “which elements are the same” and “how X is cut into pieces”.

Theorem 0.5.2 — Equivalence relations and partitions

Let X be a set.

- (a) If $\sim$ is an equivalence relation on X , then the set of its equivalence classes is a partition of X .
- (b) If $\mathcal{P}$ is a partition of X , then the relation “ $x \sim_{\mathcal{P}} y$ if x and y lie in the same block” is an equivalence relation on X , and its equivalence classes are exactly the blocks of $\mathcal{P}$.
- (c) The two constructions undo each other: starting from $\sim$, the relation defined in (b) by the partition in (a) is $\sim$ again.

Idea. Part (a) is the lemma plus reflexivity: (P1) and (P3) come from $x \in [x]$, and (P2) is Lemma 0.5.1 (b). Part (b) runs the other way, and each block condition pays for one equivalence condition: covering gives reflexivity, and disjointness gives transitivity, because a middle element y can lie in only one block. Part (c) is Lemma 0.5.1 (a) read as a sentence about blocks.

Proof.

- (a) Let $\mathcal{P} = \{[x] : x \in X\}$. For every $x \in X$, reflexivity gives $x \in [x]$. Hence every class is non-empty, which is (P1), and every element lies in some class, which is (P3). If $[a] \neq [b]$, then $[a] \cap [b] = \emptyset$ by Lemma 0.5.1 (b), which is (P2). Hence $\mathcal{P}$ is a partition of X .
- (b) *Reflexive.* Let $x \in X$. By (P3), x lies in some block, so $x \sim_{\mathcal{P}} x$. *Symmetric.* “ x and y lie in a common block” does not depend on the order of x and y . *Transitive.* Suppose $x, y \in B$ and $y, z \in B'$ for blocks B, B' . Then $y \in B \cap B'$, so $B = B'$ by (P2). Hence $x, z \in B$, and $x \sim_{\mathcal{P}} z$.

Now let $B \in \mathcal{P}$ and $x \in B$; we show $[x] = B$, where the class is taken for $\sim_{\mathcal{P}}$. ($\subseteq$) If $y \in [x]$, then y and x lie in a common block B' ; since $x \in B \cap B'$, (P2) gives $B' = B$, so $y \in B$. ($\supseteq$) If $y \in B$, then y and x both lie in B , so $y \sim_{\mathcal{P}} x$ and $y \in [x]$. Hence $[x] = B$ by Theorem 0.3.1. Every class $[x]$ is therefore the block containing x , which exists by (P3); and every block B is a class, namely $[x]$ for any $x \in B$, which exists by (P1). So the classes are exactly the blocks.

- (c) Let $\sim$ be an equivalence relation, and let $\approx$ be the relation built by (b) from the classes of $\sim$. If $x \sim y$, then $x, y \in [y]$, so $x \approx y$. Conversely, if $x \approx y$, then $x, y \in [z]$ for some z ; thus $x \sim z$ and $y \sim z$, and symmetry and transitivity give

$x \sim y$. Hence $\approx$ and $\sim$ are the same relation. This proves the theorem. ■

So whenever you meet an equivalence relation, you may think of it as a way of cutting the set into non-overlapping pieces, and vice versa. Exercise C2 uses this to count equivalence relations by counting partitions instead.

0.5.4 Quotient sets

Now we take the decisive step: treat each piece as a single new element. The time of day is the classic case. Hours counted from some starting moment are integers, but a clock only records the class of the hour modulo 12.

Definition 0.5.5 — Quotient set

Let $\sim$ be an equivalence relation on X . The **quotient set** of X by $\sim$ is the set of all equivalence classes,

$$X/\sim := \{[x] : x \in X\}.$$

The function $\pi: X \rightarrow X/\sim$, $\pi(x) = [x]$, is the **quotient map**.

The elements of $X/\sim$ are sets, but once the quotient is formed we stop looking inside them and treat each class as a single point. The quotient map is surjective, since every class is $[x] = \pi(x)$ for some x . By Lemma 0.5.1 (a), $\pi(x) = \pi(y)$ exactly when $x \sim y$: the quotient map makes equivalent elements literally equal.

Example. (Quotient sets)

- (a) For an integer $n \geq 1$, the quotient of $\mathbb{Z}$ by congruence modulo n is written $\mathbb{Z}/n\mathbb{Z}$. Show that it has exactly n elements, namely $[0], [1], \dots, [n-1]$.
- (b) Describe the quotient of the set S of strings in a, b by “same length”, and of $\mathbb{R}^2$ by “same distance from the origin”.

Solution.

- (a) Let $a \in \mathbb{Z}$. By division with remainder (Exercise C2: Division with remainder), $a = qn + r$ with $q, r \in \mathbb{Z}$ and $0 \leq r < n$. Then $a - r = qn$, so $a \equiv r \pmod{n}$, and $[a] = [r]$ by Lemma 0.5.1 (a). Hence every class is one of $[0], \dots, [n-1]$. These are distinct: if $0 \leq r < s < n$, then $0 < s - r < n$, so $s - r$ is not a multiple of n , hence $r \not\equiv s$ and $[r] \neq [s]$ by Lemma 0.5.1 (a). So $\mathbb{Z}/n\mathbb{Z} = \{[0], [1], \dots, [n-1]\}$ has exactly n elements.
- (b) Two strings are identified exactly when they have the same length, so there is one class for each length $k \in \mathbb{N}$: the set of all 2^k strings of length k . Similarly, $\mathbb{R}^2$ modulo “same distance” has one class for each radius $r \in [0, \infty)$: the circle of radius r , or the origin when $r = 0$. In both cases the quotient is “the same as” the set of values $\mathbb{N}$ or $[0, \infty)$, and the next subsection makes that precise.

0.5.5 Functions on a quotient

We want to compute with classes. On a clock, “three hours after 11 o’clock” is 2 o’clock, which is really addition on $\mathbb{Z}/12\mathbb{Z}$. The natural way to define such an operation is through representatives: to add two classes, pick an element of each, add them, and take the class of the result. The danger is that a class has many representatives, and different choices might give different answers. When they do, the “definition” does not define a function at all.

Here is that failure in its simplest form.

Example. (The numerator is not well defined)

Let $P = \mathbb{Z} \times (\mathbb{Z} \setminus \{0\})$ with $(a, b) \sim (c, d)$ if $ad = bc$, as in [Example \(Equivalence relations\)](#) (e). Its classes are the rational numbers: the class of (a, b) is the fraction a/b , with all its other names. Is “ $[(a, b)] \mapsto a$ ” a function $P/\sim \rightarrow \mathbb{Z}$?

Solution. No. Since $1 \cdot 4 = 2 \cdot 2$, we have $(1, 2) \sim (2, 4)$, so $[(1, 2)] = [(2, 4)]$ by [Lemma 0.5.1](#) (a). The rule assigns to this one class the value 1 through one name and 2 through the other. So it does not assign **exactly one** value to each element of the domain, and [Definition 0.4.1](#) fails. “The numerator of $\frac{1}{2}$ ” is a statement about how a fraction is written, not about the number.

By contrast, “ $[(a, b)] \mapsto 1$ if $a = 0$, and 0 otherwise” is fine. If $(a, b) \sim (c, d)$ and $a = 0$, then $bc = ad = 0$, so $c = 0$ because $b \neq 0$. By symmetry, $c = 0$ forces $a = 0$. So equivalent pairs always get the same value.

The contrast in that example is the whole story, and the following theorem turns it into a test. A function f on X passes to the quotient exactly when it cannot tell equivalent elements apart.

Theorem 0.5.3 — Functions on a quotient

Let $\sim$ be an equivalence relation on X , let $\pi: X \rightarrow X/\sim$ be the quotient map, and let $f: X \rightarrow Y$ be a function.

- (a) If $f(a) = f(b)$ **whenever** $a \sim b$, then there is **exactly one** function $\bar{f}: X/\sim \rightarrow Y$ with $\bar{f}([x]) = f(x)$ for every $x \in X$, that is, $\bar{f} \circ \pi = f$.
- (b) Conversely, if some function $\bar{f}: X/\sim \rightarrow Y$ satisfies $\bar{f} \circ \pi = f$, then $f(a) = f(b)$ whenever $a \sim b$.

Idea. The formula $\bar{f}([x]) = f(x)$ leaves no freedom: if $\bar{f}$ exists, this is its value. The only question is whether the formula makes sense, since the class $C = [x]$ may also be called $[x']$. So look at **all** the values f takes on C at once, the image $f(C)$. A function needs exactly one value; $f(C)$ is non-empty because C is, and has at most one element because all elements of C are equivalent.

Proof.

- (a) Let $C \in X/\sim$. By [Definition 0.5.5](#), $C = [x]$ for some $x \in X$, so C is non-empty, and hence the image $f(C)$ is non-empty. If $x', x'' \in C$, then $[x'] = C = [x'']$ by [Lemma 0.5.1](#) (a) applied to $x' \sim x$ and $x'' \sim x$; hence $x' \sim x''$ by the same lemma,

and $f(x') = f(x'')$ by hypothesis. Therefore $f(C)$ has exactly one element, and we define $\bar{f}(C)$ to be that element. This assigns exactly one value to each class, so $\bar{f}$ is a function. For $x \in X$, we have $x \in [x]$ by reflexivity, so $\bar{f}([x]) \in f([x]) = \{f(x)\}$, that is, $\bar{f}([x]) = f(x)$.

For uniqueness, suppose $g: X/\sim \rightarrow Y$ also satisfies $g([x]) = f(x)$ for every x . Every class is $[x]$ for some x , and there $g([x]) = f(x) = \bar{f}([x])$. Hence $g = \bar{f}$.

- (b) Suppose $\bar{f} \circ \pi = f$ and $a \sim b$. By [Lemma 0.5.1](#) (a), $[a] = [b]$. Hence $f(a) = \bar{f}([a]) = \bar{f}([b]) = f(b)$, as claimed. ■

In practice we rarely name f and $\bar{f}$ separately. We write down a rule “ $[x] \mapsto (\text{expression in } x)$ ” and check that equivalent representatives give the same value. That check is what people mean when they say a function is **well defined**. For example, “same value” in [Example \(Equivalence relations\)](#) (d) satisfies the hypothesis by its very definition, so every $f: X \rightarrow Y$ induces a function $X/\sim_f \rightarrow Y$. This $\bar{f}$ is injective: $\bar{f}([x]) = \bar{f}([x'])$ means $f(x) = f(x')$, which is $x \sim_f x'$, so $[x] = [x']$. This makes precise the remark that $\mathbb{R}^2$ modulo “same distance” is “the same as” $[0, \infty)$: the induced map $[(x, y)] \mapsto \sqrt{x^2 + y^2}$ is a bijection onto $[0, \infty)$.

The most important application in this chapter is arithmetic modulo n . The operation takes two classes, so we check both representatives at once.

Example. (Addition modulo n is well defined)

Fix an integer $n \geq 1$. Show that the rule $[a] + [b] := [a + b]$ defines a function $\mathbb{Z}/n\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z} \rightarrow \mathbb{Z}/n\mathbb{Z}$. Then compute $[3] + [2]$ in $\mathbb{Z}/4\mathbb{Z}$.

Solution. We must show that the value $[a + b]$ depends only on the classes $[a]$ and $[b]$, not on the representatives a and b . Suppose $[a] = [a']$ and $[b] = [b']$. By [Lemma 0.5.1](#) (a), $a \equiv a' \pmod{n}$ and $b \equiv b' \pmod{n}$, so $a - a' = kn$ and $b - b' = \ell n$ for some $k, \ell \in \mathbb{Z}$. Then

$$(a + b) - (a' + b') = (a - a') + (b - b') = (k + \ell)n,$$

so $a + b \equiv a' + b' \pmod{n}$, and $[a + b] = [a' + b']$ by the same lemma. Hence every choice of representatives gives the same class, and the rule defines a function.

In $\mathbb{Z}/4\mathbb{Z}$, $[3] + [2] = [5] = [1]$, since $5 - 1 = 4$. Choosing other names gives the same answer, as it must: $[7] + [-2] = [5] = [1]$.

This is the same principle as [Theorem 0.5.3](#), applied to a function of two variables. The same check ([Exercise \(B3: Multiplication modulo \$n\$ \)](#)) shows multiplication $[a][b] := [ab]$ is well defined. With these operations, $\mathbb{Z}/p\mathbb{Z}$ for a prime p will become our first finite field later in this chapter, and the identical pattern, now with “ $\mathbf{v} \sim \mathbf{w}$ if $\mathbf{v} - \mathbf{w} \in U$ ”, will build quotient vector spaces in Chapter 3.

Warning. A rule written in terms of a representative is not automatically a function on classes. Define it, then check it. For example, “ $[a] \mapsto a$ ” from $\mathbb{Z}/5\mathbb{Z}$ to $\mathbb{Z}$ is not a function, because $[0] = [5]$ would be sent to both 0 and 5. The check is not a formality: forgetting it is the most common error in any argument that uses quotients.

Quick check. Which of these rules defines a function? (i) $\mathbb{Z}/6\mathbb{Z} \rightarrow \mathbb{Z}/3\mathbb{Z}$, $[a]_6 \mapsto [a]_3$.
(ii) $\mathbb{Z}/6\mathbb{Z} \rightarrow \mathbb{Z}/4\mathbb{Z}$, $[a]_6 \mapsto [a]_4$. Here the subscript records the modulus.

Solution.

- (i) Yes. If $[a]_6 = [a']_6$, then $a - a' = 6k = 3(2k)$, so $[a]_3 = [a']_3$.
- (ii) No. $[0]_6 = [6]_6$, but $[0]_4 \neq [6]_4 = [2]_4$, since $6 - 0 = 6$ is not a multiple of 4.
The first works because $3 \mid 6$; the second fails because $4 \nmid 6$.

0.5.6 Exercises

0.5.6.1 A. Check your understanding

Exercise A1.

- (a) State the three conditions for a relation $\sim$ on X to be an equivalence relation.
- (b) True or false: a symmetric and transitive relation is automatically reflexive. Give a reason.
- (c) True or false: if two equivalence classes $[a]$ and $[b]$ have an element in common, then $a \sim b$.
- (d) You want to define a function on $X/\sim$ by a formula in a representative x . What must you check?
- (e) How many elements does $\mathbb{Z}/5\mathbb{Z}$ have? Is $[12] = [-3]$ in $\mathbb{Z}/5\mathbb{Z}$?

Solution.

- (a) Reflexive: $x \sim x$ for every $x \in X$. Symmetric: $x \sim y$ implies $y \sim x$. Transitive: $x \sim y$ and $y \sim z$ imply $x \sim z$.
- (b) False. On $\mathbb{R}$, $x \sim y$ if $xy > 0$ is symmetric and transitive, but $0 \not\sim 0$; see [Example \(Relations that are not equivalence relations\)](#) (c) and the warning after it.
- (c) True. By [Lemma 0.5.1](#) (b) the classes are equal, and by part (a) of the same lemma $a \sim b$.
- (d) That equivalent representatives give the same value: if $x \sim x'$, the formula gives the same result for x and x' . By [Theorem 0.5.3](#) this is exactly what is needed.
- (e) Five, by [Example \(Quotient sets\)](#) (a). Yes: $12 - (-3) = 15 = 3 \cdot 5$, so $12 \equiv -3 \pmod{5}$, and both classes equal $[2]$.

0.5.6.2 B. Practice

Exercise B1: Which are equivalence relations?.

Determine which of the following are equivalence relations. For those that are, describe the equivalence classes. For those that are not, name a condition that fails with a specific witness.

- (a) On $\mathbb{Z}$: $a \sim b$ if $a + b$ is even.

- (b) On $\mathbb{Z}$: $a \sim b$ if $a - b$ is odd.
- (c) On $\mathbb{R}$: $x \sim y$ if $x - y \in \mathbb{Z}$.
- (d) On $\{1, 2, 3, \dots\}$: $m \sim n$ if m and n have a common divisor greater than 1.
- (e) On the set of all subsets of $\{1, 2, 3\}$: $A \sim B$ if A and B have the same number of elements.

Solution.

- (a) Equivalence relation. *Reflexive*: $a + a = 2a$ is even. *Symmetric*: $b + a = a + b$. *Transitive*: if $a + b$ and $b + c$ are even, then $a + c = (a + b) + (b + c) - 2b$ is a sum of even integers, hence even. There are two classes: the even integers and the odd integers. (In fact $a + b$ is even exactly when $a - b = (a + b) - 2b$ is even, so this is congruence modulo 2.)
- (b) Not an equivalence relation. Reflexivity fails: $0 - 0 = 0$ is even, so $0 \not\sim 0$. (Transitivity also fails: $0 \sim 1$ and $1 \sim 2$, but $0 - 2$ is even.)
- (c) Equivalence relation. *Reflexive*: $x - x = 0 \in \mathbb{Z}$. *Symmetric*: if $x - y = k \in \mathbb{Z}$, then $y - x = -k \in \mathbb{Z}$. *Transitive*: if $x - y = k$ and $y - z = \ell$ with $k, \ell \in \mathbb{Z}$, then $x - z = k + \ell \in \mathbb{Z}$. The class of x is $\{x + k : k \in \mathbb{Z}\}$, the points at integer distance from x in either direction. Each class contains exactly one number in $[0, 1)$, namely $x - \lfloor x \rfloor$.
- (d) Not an equivalence relation. Reflexivity fails at 1: the only positive divisor of 1 is 1. Transitivity also fails: $2 \sim 6$ (common divisor 2) and $6 \sim 3$ (common divisor 3), but 2 and 3 have no common divisor greater than 1.
- (e) Equivalence relation, since it is “same value” (Example (Equivalence relations) (d)) for the function $A \mapsto |A|$. There are four classes: $\{\emptyset\}$; $\{\{1\}, \{2\}, \{3\}\}$; $\{\{1, 2\}, \{1, 3\}, \{2, 3\}\}$; $\{\{1, 2, 3\}\}$.

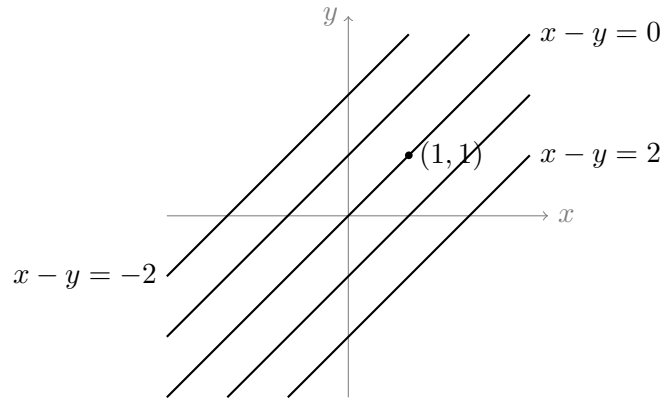
Exercise B2: Parallel lines as classes.

On $\mathbb{R}^2$, define $(x, y) \sim (x', y')$ if $x - y = x' - y'$.

- (a) Prove that $\sim$ is an equivalence relation.
- (b) Describe the class of (x_0, y_0) geometrically, and draw several classes.
- (c) Show that $[(x, y)] \mapsto x - y$ is a well-defined function $\mathbb{R}^2/\sim \rightarrow \mathbb{R}$, and that it is a bijection.

Solution.

- (a) This is “same value” for $f: \mathbb{R}^2 \rightarrow \mathbb{R}$, $f(x, y) = x - y$, so it is an equivalence relation by Example (Equivalence relations) (d). Directly: $x - y = x - y$; equality of the numbers $x - y$ and $x' - y'$ is symmetric and transitive.
- (b) Put $c = x_0 - y_0$. Then $[(x_0, y_0)] = \{(x, y) \in \mathbb{R}^2 : x - y = c\} = \{(x, y) : y = x - c\}$, the line of slope 1 through (x_0, y_0) , which meets the y -axis at $(0, -c)$. The classes are all the lines parallel to $y = x$: each point of the plane lies on exactly one of them, as Theorem 0.5.2 predicts.



- (c) Let $f(x, y) = x - y$. If $(x, y) \sim (x', y')$, then $f(x, y) = f(x', y')$ by the definition of $\sim$. By [Theorem 0.5.3](#) (a), there is a function $\bar{f}: \mathbb{R}^2/\sim \rightarrow \mathbb{R}$ with $\bar{f}([(x, y)]) = x - y$.

Injective. Suppose $\bar{f}([(x, y)]) = \bar{f}([(x', y')])$. Then $x - y = x' - y'$, so $(x, y) \sim (x', y')$, and $[(x, y)] = [(x', y')]$ by [Lemma 0.5.1](#) (a).

Surjective. Let $c \in \mathbb{R}$. Then $\bar{f}([(c, 0)]) = c - 0 = c$.

Hence $\bar{f}$ is a bijection: the set of parallel lines is labeled, one to one, by the real number c .

Exercise B3: Multiplication modulo n .

Fix an integer $n \geq 1$.

- Prove that $[a][b] := [ab]$ is a well-defined operation on $\mathbb{Z}/n\mathbb{Z}$.
- Write out the multiplication table of $\mathbb{Z}/4\mathbb{Z}$, using the representatives 0, 1, 2, 3.
- Hence find a class $[a] \neq [0]$ in $\mathbb{Z}/4\mathbb{Z}$ with $[a][a] = [0]$.

Solution.

- (a) Suppose $[a] = [a']$ and $[b] = [b']$. By [Lemma 0.5.1](#) (a), $a - a' = kn$ and $b - b' = \ell n$ for some $k, \ell \in \mathbb{Z}$. Then

$$ab - a'b' = ab - a'b + a'b - a'b' = (a - a')b + a'(b - b') = (kb + a'\ell)n,$$

so $ab \equiv a'b' \pmod{n}$ and $[ab] = [a'b']$ by the same lemma. Hence the value does not depend on the representatives, and the operation is well defined.

- (b) Reducing each product modulo 4:

$\cdot$	[0]	[1]	[2]	[3]
[0]	[0]	[0]	[0]	[0]
[1]	[0]	[1]	[2]	[3]
[2]	[0]	[2]	[0]	[2]
[3]	[0]	[3]	[2]	[1]

For instance $[2][3] = [6] = [2]$ and $[3][3] = [9] = [1]$.

- (c) From the table, $[2][2] = [4] = [0]$, while $[2] \neq [0]$ since $4 \nmid 2$. So a product of non-zero classes can be zero in $\mathbb{Z}/4\mathbb{Z}$, something that never happens with ordinary numbers.

0.5.6.3 C. Going deeper

Exercise C1: Squares and powers modulo n .

- (a) Let $n \geq 1$. Prove that $[x] \mapsto [x^2]$ is a well-defined function $\mathbb{Z}/n\mathbb{Z} \rightarrow \mathbb{Z}/n\mathbb{Z}$.
- (b) On $\mathbb{N}$, let $\equiv_3$ be congruence modulo 3. Show that the rule $[x] \mapsto [2^x]_3$ does **not** define a function from $\mathbb{N}/\equiv_3$ to $\mathbb{Z}/3\mathbb{Z}$.
- (c) On $\mathbb{N}$, let $\equiv_2$ be congruence modulo 2. Prove that $[x] \mapsto [2^x]_3$ **does** define a function from $\mathbb{N}/\equiv_2$ to $\mathbb{Z}/3\mathbb{Z}$.

Hint for (c): compute 2^2 modulo 3.

Solution.

- (a) Suppose $[x] = [x']$. By [Exercise \(B3: Multiplication modulo \$n\$ \)](#) (a) with $a = b = x$ and $a' = b' = x'$, we get $[x][x] = [x'][x']$, that is, $[x^2] = [x'^2]$. Hence the value depends only on the class, and the function is well defined.
- (b) In $\mathbb{N}$, $0 \equiv 3 \pmod{3}$, so $[0] = [3]$. The rule sends this class to $[2^0]_3 = [1]_3$ through the representative 0, and to $[2^3]_3 = [8]_3 = [2]_3$ through the representative 3. Since $8 - 1 = 7$ is not a multiple of 3, $[1]_3 \neq [2]_3$. So one class receives two different values, and the rule is not a function.
- (c) By [Theorem 0.5.3](#) (a), it suffices to show that $x \equiv x' \pmod{2}$ implies $2^x \equiv 2^{x'} \pmod{3}$. By symmetry of both relations we may assume $x \leq x'$, so $x' = x + 2k$ with $k \in \mathbb{N}$. First, $4^k \equiv 1 \pmod{3}$ for every $k \in \mathbb{N}$, by induction on k ([Theorem 0.2.7](#)): $4^0 = 1$, and if $4^k - 1 = 3m$, then $4^{k+1} - 1 = 4(4^k - 1) + 3 = 3(4m + 1)$. Hence

$$2^{x'} - 2^x = 2^x(4^k - 1)$$

is a multiple of 3, so $[2^{x'}]_3 = [2^x]_3$, as claimed. The difference between (b) and (c) is that the powers of 2 modulo 3 repeat with period 2, and 3 is not a multiple of 2.

Exercise C2: Counting equivalence relations.

- (a) List all partitions of $\{1, 2, 3\}$.
- (b) Hence deduce that there are exactly 5 equivalence relations on $\{1, 2, 3\}$.
- (c) Write the equivalence relation corresponding to the partition $\{\{1, 3\}, \{2\}\}$ as a subset of $\{1, 2, 3\} \times \{1, 2, 3\}$.

Solution.

- (a) Sort by the number of blocks. One block: $\{\{1, 2, 3\}\}$. Two blocks: one block is a single element and the other the remaining pair, giving $\{\{1\}, \{2, 3\}\}$, $\{\{2\}, \{1, 3\}\}$, $\{\{3\}, \{1, 2\}\}$. Three blocks: $\{\{1\}, \{2\}, \{3\}\}$. Four or more non-empty disjoint

blocks are impossible in a three-element set. That is 5 partitions.

- (b) By [Theorem 0.5.2](#) (a), every equivalence relation gives a partition, its set of classes. By [Theorem 0.5.2](#) (c), the relation can be recovered from that partition, so distinct equivalence relations give distinct partitions. By [Theorem 0.5.2](#) (b), every partition arises this way. Hence equivalence relations on $\{1, 2, 3\}$ correspond one to one with partitions, and there are exactly 5 of them.
- (c) Two elements are related exactly when they lie in a common block:

$$\{(1, 1), (2, 2), (3, 3), (1, 3), (3, 1)\}.$$

Exercise C3: Intersections and unions of equivalence relations.

Let R_1 and R_2 be equivalence relations on a set X , viewed as subsets of $X \times X$.

- (a) Prove that $R_1 \cap R_2$ is an equivalence relation on X .
- (b) Show by an example that $R_1 \cup R_2$ need not be an equivalence relation. Which condition fails?
- (c) For congruence modulo 2 and modulo 3 on $\mathbb{Z}$, identify $R_1 \cap R_2$ as a familiar relation.

Hint for (b): try congruence modulo 2 and modulo 3 on $\mathbb{Z}$.

Solution.

- (a) *Reflexive.* Let $x \in X$. Then $(x, x) \in R_1$ and $(x, x) \in R_2$, since both are reflexive, so $(x, x) \in R_1 \cap R_2$. *Symmetric.* Suppose $(x, y) \in R_1 \cap R_2$. Since R_1 and R_2 are symmetric, $(y, x) \in R_1$ and $(y, x) \in R_2$, so $(y, x) \in R_1 \cap R_2$. *Transitive.* Suppose $(x, y), (y, z) \in R_1 \cap R_2$. Both pairs lie in R_1 , which is transitive, so $(x, z) \in R_1$; likewise $(x, z) \in R_2$. Hence $(x, z) \in R_1 \cap R_2$.
- (b) Let R_1 be congruence modulo 2 and R_2 congruence modulo 3 on $\mathbb{Z}$. The union is still reflexive and symmetric, since each of R_1, R_2 is. But transitivity fails: $(0, 2) \in R_1$ since $2 \mid 2$, and $(2, 5) \in R_2$ since $3 \mid 3$, yet $(0, 5)$ lies in neither, because 5 is not a multiple of 2 or of 3.
- (c) $(a, b) \in R_1 \cap R_2$ means $2 \mid a - b$ and $3 \mid a - b$. This holds exactly when $6 \mid a - b$. Indeed, if $a - b = 2k$ and $a - b = 3m$ with $k, m \in \mathbb{Z}$, then $a - b = 3(a - b) - 2(a - b) = 3 \cdot 2k - 2 \cdot 3m = 6(k - m)$. Conversely, if $a - b = 6j$, then $a - b = 2(3j) = 3(2j)$. So $R_1 \cap R_2$ is congruence modulo 6. The same phenomenon, where intersections behave well and unions do not, will reappear for subspaces in Chapter 1.

0.6 Complex Numbers

The equation $x^2 + 1 = 0$ has no real solution, because the square of a real number is never negative. For most of school mathematics this gap is harmless. In linear algebra it is not: later in the book we meet real matrices, such as a rotation of the plane, whose most useful data cannot be found among the real numbers. This section builds the complex numbers, the number system that closes the gap, and collects the facts about them that the rest of the book uses.

0.6.1 The complex numbers as pairs of reals

The quick way to introduce complex numbers is to say “invent a symbol i with $i^2 = -1$ and compute as usual”. That raises a fair worry. How do we know that such a symbol does not lead to a contradiction, the way “invent a number z with $0 \cdot z = 1$ ” does? The answer is to build the new numbers out of objects we already trust: pairs of real numbers.

Here is the recurring expression that suggests the definition. If we pretend $i^2 = -1$ and expand, we get

$$(a + bi)(c + di) = ac + adi + bci + bd i^2 = (ac - bd) + (ad + bc)i.$$

So a “number $a + bi$ ” is really the pair (a, b) , and the product should be the pair $(ac - bd, ad + bc)$. We take this formula as the definition, so nothing is pretended.

A complex number is a pair of real numbers, multiplied as if $i^2 = -1$.

Definition 0.6.1 — Complex Numbers

The set of **complex numbers** is $\mathbb{C} := \mathbb{R}^2 = \{(a, b) : a, b \in \mathbb{R}\}$, with addition and multiplication defined **for every** $(a, b), (c, d) \in \mathbb{C}$ by

$$(a, b) + (c, d) := (a + c, b + d), \quad (a, b)(c, d) := (ac - bd, ad + bc).$$

We write a for $(a, 0)$ and i for $(0, 1)$, so that $(a, b) = a + bi$. For $z = a + bi$ with $a, b \in \mathbb{R}$, the **real part** of z is $\operatorname{Re} z := a$ and the **imaginary part** of z is $\operatorname{Im} z := b$.

In words: addition is coordinate by coordinate, as for points of the plane. Multiplication is the formula found by expanding. The last sentence is a naming convention: the real number a is identified with the point $(a, 0)$ on the horizontal axis, and i is the point one unit up. Note that the imaginary part b is a **real** number; it is not bi .

Two things need checking before this convention is safe.

- **The notation $a + bi$ is honest.** By the definitions, $a + bi = (a, 0) + (b, 0)(0, 1) = (a, 0) + (0, b) = (a, b)$. Two complex numbers are equal exactly when they are equal as pairs, so $a + bi = c + di$ with $a, b, c, d \in \mathbb{R}$ means $a = c$ **and** $b = d$. This is how we “compare real and imaginary parts”.
- **Identifying a with $(a, 0)$ does not change real arithmetic.** We have $(a, 0) + (c, 0) = (a + c, 0)$ and $(a, 0)(c, 0) = (ac - 0, 0 + 0) = (ac, 0)$. So adding or multiplying two real numbers inside $\mathbb{C}$ gives the same answer as in $\mathbb{R}$.

With these in place, the equation that motivated everything is now a computation: $i^2 = (0, 1)(0, 1) = (0 \cdot 0 - 1 \cdot 1, 0 \cdot 1 + 1 \cdot 0) = (-1, 0) = -1$.

Example. (First Computations in $\mathbb{C}$)

Compute $(2 + 3i)(1 - i)$, the powers $i^0, i^1, \dots, i^7$, and the product $(a + bi)(a - bi)$ for real a, b .

Solution. By the multiplication rule with $a = 2, b = 3, c = 1, d = -1$,

$$(2 + 3i)(1 - i) = (2 \cdot 1 - 3 \cdot (-1)) + (2 \cdot (-1) + 3 \cdot 1)i = 5 + i.$$

For the powers, $i^0 = 1$, $i^1 = i$, $i^2 = -1$, $i^3 = i^2 \cdot i = -i$ and $i^4 = (-i)i = -i^2 = 1$. From then on the pattern repeats with period 4: $i^5 = i$, $i^6 = -1$, $i^7 = -i$.

Finally, $(a + bi)(a - bi) = (a^2 - b(-b)) + (a(-b) + ba)i = a^2 + b^2$. This product is a real number, and it is **non-negative**. We will meet it again as $|z|^2$.

The degenerate case deserves a look too. A real number $a = a + 0i$ is a complex number whose imaginary part is zero, and the number $0 = 0 + 0i$ is the only complex number with both parts zero. Every computation with real numbers is also a computation in $\mathbb{C}$, by the second check above.

A non-example by minimal change. Change one sign in the multiplication rule: define $(a, b) * (c, d) := (ac + bd, ad + bc)$, which is what expanding gives if we pretend $j^2 = +1$ instead. Addition is unchanged, and this product is still commutative and has $(1, 0)$ as a unit. But

$$(1, 1) * (1, -1) = (1 - 1, -1 + 1) = (0, 0),$$

so two non-zero elements multiply to zero. Division also breaks down: $(1, 1) * (c, d) = (c + d, c + d)$ can never equal the unit $(1, 0)$, because that would need $c + d = 1$ and $c + d = 0$ at once. So the non-zero element $(1, 1)$ has no inverse. The single sign in $ac - bd$ is what makes the complex numbers a number system in which we can divide, as we are about to see.

Quick check. Using the definition, compute i^{10} and $(1 + i)^2$. Is $\text{Im}(3 - 4i)$ equal to $-4i$?

Solution. Since the powers of i repeat with period 4 and $10 = 8 + 2$, we get $i^{10} = i^2 = -1$. Next, $(1 + i)^2 = (1 - 1) + (1 + 1)i = 2i$. No: the imaginary part is the real number -4 , not $-4i$.

The usual rules of arithmetic hold. Before computing freely, we should confirm that rearranging, regrouping and expanding brackets are legal in $\mathbb{C}$. The definition does not promise this; it has to be checked.

Proposition 0.6.1 — Arithmetic Laws in $\mathbb{C}$

For all $z, w, u \in \mathbb{C}$:

- (a) $z + w = w + z$ and $zw = wz$;
- (b) $(z + w) + u = z + (w + u)$ and $(zw)u = z(wu)$;
- (c) $z(w + u) = zw + zu$;
- (d) $z + 0 = z$, $z \cdot 1 = z$, and $z + (-z) = 0$, where $-(a + bi) := (-a) + (-b)i$.

Idea. Every law reduces to the corresponding law for real numbers once both sides are written as pairs. Addition is coordinate by coordinate, so its laws are immediate. For multiplication the only real work is associativity: expand both sides and see that they produce the same four terms in each coordinate.

Proof. Let $z = a + bi$, $w = c + di$ and $u = e + fi$ with $a, b, c, d, e, f \in \mathbb{R}$. Since addition acts coordinate by coordinate, the laws for addition in (a), (b) and (d) follow from the same laws in $\mathbb{R}$, applied to each coordinate.

- (a) By definition $zw = (ac - bd) + (ad + bc)i$ and $wz = (ca - db) + (cb + da)i$. These are equal because real multiplication and addition are commutative.

(b) Expanding with the definition twice gives

$$(zw)u = ((ac - bd)e - (ad + bc)f) + ((ac - bd)f + (ad + bc)e)i$$

and

$$z(wu) = (a(ce - df) - b(cf + de)) + (a(cf + de) + b(ce - df))i.$$

Multiplying out, both real parts equal $ace - adf - bcf - bde$ and both imaginary parts equal $acf + ade + bce - bdf$. Hence $(zw)u = z(wu)$.

(c) We have $w + u = (c + e) + (d + f)i$, so

$$z(w+u) = (a(c+e) - b(d+f)) + (a(d+f) + b(c+e))i = ((ac - bd) + (ae - bf)) + ((ad + bc) + (af + be))i,$$

which is $zw + zu$ by the definitions of the product and the sum.

(d) With $1 = 1 + 0i$, we get $z \cdot 1 = (a \cdot 1 - b \cdot 0) + (a \cdot 0 + b \cdot 1)i = a + bi = z$. Also $z + (-z) = (a - a) + (b - b)i = 0$. This proves the proposition. ■

From now on we compute in $\mathbb{C}$ exactly as in $\mathbb{R}$: expand brackets, collect terms, and replace i^2 by -1 . The proposition is what makes that habit legitimate.

0.6.2 Conjugate and modulus

Division is still missing. To divide by $3 + 4i$ we want some number to multiply by that lands in the real numbers, where division is familiar. The computation $(a + bi)(a - bi) = a^2 + b^2$ in [Example \(First Computations in \$\mathbb{C}\$ \)](#) shows exactly which number does the job. Both ingredients get names.

Conjugating reflects a point in the real axis; the modulus is its distance from the origin.

Definition 0.6.2 — Conjugate and Modulus

Let $z = a + bi \in \mathbb{C}$ with $a, b \in \mathbb{R}$. The **complex conjugate** of z is $\bar{z} := a - bi$. The **modulus** (or absolute value) of z is the real number $|z| := \sqrt{a^2 + b^2} \geq 0$, the non-negative square root.

In words: the conjugate keeps the real part and flips the sign of the imaginary part. The modulus is the length of the segment from 0 to the point (a, b) , by Pythagoras. For a real number $a = a + 0i$, we get $\bar{a} = a$ and $|a| = \sqrt{a^2}$, the usual absolute value, so the notation does not clash with its real meaning. For example, $\overline{3 + 4i} = 3 - 4i$, $|3 + 4i| = 5$, $\bar{i} = -i$ and $|i| = 1$.

The following rules are used constantly, in this section and much later for complex inner products.

Theorem 0.6.2 — Properties of Conjugation and Modulus

For all $z, w \in \mathbb{C}$:

(a) $\overline{z + w} = \bar{z} + \bar{w}$ and $\bar{\bar{z}} = z$;

- (b) $\overline{zw} = \overline{z}\overline{w}$;
- (c) $z\overline{z} = |z|^2$ and $z + \overline{z} = 2\operatorname{Re} z$;
- (d) $|zw| = |z||w|$ and $|\overline{z}| = |z|$.

Idea. Parts (a)–(c) are direct computations with real and imaginary parts. For (d), expanding $|zw|$ directly is legitimate but messy. It is cleaner to square: by (c) a squared modulus is a product with a conjugate, and (b) lets the conjugate split across the product.

Proof. Let $z = a + bi$ and $w = c + di$ with $a, b, c, d \in \mathbb{R}$.

- (a) We have $\overline{z + w} = (a + c) - (b + d)i = (a - bi) + (c - di) = \overline{z} + \overline{w}$, and $\overline{\overline{z}} = \overline{a - bi} = a + bi = z$.
- (b) Since $zw = (ac - bd) + (ad + bc)i$, we get $\overline{zw} = (ac - bd) - (ad + bc)i$. On the other hand,

$$\overline{z}\overline{w} = (a - bi)(c - di) = (ac - (-b)(-d)) + (a(-d) + (-b)c)i = (ac - bd) - (ad + bc)i.$$
- (c) By the computation in [Example \(First Computations in \$\mathbb{C}\$ \)](#), $z\overline{z} = a^2 + b^2 = |z|^2$. Also $z + \overline{z} = 2a = 2\operatorname{Re} z$.
- (d) By (c), then (b), then [Proposition 0.6.1](#) to reorder the four factors,

$$|zw|^2 = (zw)\overline{zw} = zw\overline{z}\overline{w} = (z\overline{z})(w\overline{w}) = |z|^2|w|^2.$$

Both $|zw|$ and $|z||w|$ are non-negative real numbers with the same square, so they are equal. Finally $|\overline{z}| = \sqrt{a^2 + (-b)^2} = |z|$. This proves the theorem. ■

The first payoff is division. If $z \neq 0$, then $|z|^2 = a^2 + b^2$ is a **strictly** positive real number, so we may divide by it.

Proposition 0.6.3 — Inverse of a Complex Number

Let $z \in \mathbb{C}$ with $z \neq 0$. Then $w := \frac{1}{|z|^2} \overline{z}$ satisfies $zw = 1$. We write z^{-1} or $\frac{1}{z}$ for this w .

Proof. Write $z = a + bi$. Since $z \neq 0$, at least one of a, b is non-zero, so $|z|^2 = a^2 + b^2 > 0$. By [Proposition 0.6.1](#) and [Theorem 0.6.2](#) (c),

$$zw = \frac{1}{|z|^2} (z\overline{z}) = \frac{1}{|z|^2} |z|^2 = 1,$$

as claimed. ■

In practice, “multiply top and bottom by the conjugate”. For instance

$$\frac{1}{3 + 4i} = \frac{3 - 4i}{(3 + 4i)(3 - 4i)} = \frac{3 - 4i}{25}, \quad \frac{5 + i}{1 - i} = \frac{(5 + i)(1 + i)}{(1 - i)(1 + i)} = \frac{4 + 6i}{2} = 2 + 3i,$$

which agrees with $(2 + 3i)(1 - i) = 5 + i$ from [Example \(First Computations in \$\mathbb{C}\$ \)](#). Only one inverse exists; this is a special case of a general fact about fields proved in the next section.

The triangle inequality. The modulus measures distance: $|z - w|$ is the distance between the points z and w of the plane. The side of a triangle is never longer than the other two sides together, and this is the algebraic form of that fact.

Theorem 0.6.4 — Triangle Inequality

For all $z, w \in \mathbb{C}$,

$$|z + w| \leq |z| + |w|.$$

Idea. Moduli are square roots, which are awkward to add. Both sides are non-negative, so it suffices to compare their squares. Expand $|z + w|^2$ using $|u|^2 = u\bar{u}$: the cross terms combine into $2\operatorname{Re}(z\bar{w})$. A real part is never bigger than the modulus, and that single estimate finishes the proof.

Proof. First, for any $u = x + yi$ with $x, y \in \mathbb{R}$, we have $\operatorname{Re} u = x \leq |x| = \sqrt{x^2} \leq \sqrt{x^2 + y^2} = |u|$, since the square root is increasing.

Now let $z, w \in \mathbb{C}$. By [Theorem 0.6.2 \(c\)](#), (a) and [Proposition 0.6.1](#),

$$|z + w|^2 = (z + w)(\bar{z} + \bar{w}) = z\bar{z} + z\bar{w} + w\bar{z} + w\bar{w} = |z|^2 + (z\bar{w} + \overline{z\bar{w}}) + |w|^2,$$

where we used $w\bar{z} = \overline{z\bar{w}}$, which follows from [Theorem 0.6.2 \(b\)](#) and (a). By [Theorem 0.6.2 \(c\)](#) the bracket equals $2\operatorname{Re}(z\bar{w})$. Therefore, using $\operatorname{Re} u \leq |u|$ and then [Theorem 0.6.2 \(d\)](#),

$$|z + w|^2 \leq |z|^2 + 2|z\bar{w}| + |w|^2 = |z|^2 + 2|z||w| + |w|^2 = (|z| + |w|)^2.$$

Both $|z + w|$ and $|z| + |w|$ are non-negative, so taking square roots preserves the inequality. This proves the triangle inequality. ■

The route “square, expand, bound the cross term” is worth remembering: it is exactly the route we will take for norms in inner product spaces. The inequality is often strict. For $z = 1$ and $w = -1$, the left side is 0 and the right side is 2.

0.6.3 Two things that do not carry over from the reals

Complex numbers obey the rules of arithmetic, but not every habit from the real line survives.

The first loss is order. On $\mathbb{R}$ we use two facts about positivity: for every x , exactly one of $x > 0$, $x = 0$, $-x > 0$ holds, and sums and products of positive numbers are positive. Suppose some relation $>$ on $\mathbb{C}$ had both properties. Since $i \neq 0$, either $i > 0$ or $-i > 0$, and in both cases the product rule gives $i^2 = (-i)^2 = -1 > 0$. The same argument applied to $1 \neq 0$ gives $1 = 1^2 = (-1)^2 > 0$. Then both 1 and -1 are positive, so their sum 0 is positive, which contradicts “exactly one”.

Warning. There is no ordering on $\mathbb{C}$ compatible with addition and multiplication, so an inequality such as $z < w$ between complex numbers means nothing. Inequalities in this book compare **real** numbers: moduli $|z|$, real parts, or quantities known to be real. Writing $i > 0$ leads, as shown above, to $0 > 0$.

The second loss is the square-root rule $\sqrt{x}\sqrt{y} = \sqrt{xy}$, which on $\mathbb{R}$ is only claimed for $x, y \geq 0$.

Warning. Every non-zero complex number has **two** square roots, and no choice of “the” square root makes $\sqrt{z}\sqrt{w} = \sqrt{zw}$ true for all z, w . With $\sqrt{-1} = i$: $\sqrt{-1} \cdot \sqrt{-1} = i^2 = -1$, but $\sqrt{(-1)(-1)} = \sqrt{1} = 1$. Avoid the symbol $\sqrt{z}$ for non-real z ; say “a square root of z ” and solve $w^2 = z$ instead.

Solving $w^2 = z$ is a finite computation, done by comparing real and imaginary parts.

Example. (Square Roots of $5 - 12i$)

Find all $w \in \mathbb{C}$ with $w^2 = 5 - 12i$.

Solution. Write $w = x + yi$ with $x, y \in \mathbb{R}$. Then $w^2 = (x^2 - y^2) + 2xyi$, so comparing real and imaginary parts, $w^2 = 5 - 12i$ holds exactly when

$$x^2 - y^2 = 5 \quad \text{and} \quad xy = -6.$$

Taking moduli in $w^2 = 5 - 12i$ and using [Theorem 0.6.2](#) (d) gives $|w|^2 = |5 - 12i| = \sqrt{25 + 144} = 13$, that is $x^2 + y^2 = 13$. Adding and subtracting this and the first equation gives $x^2 = 9$ and $y^2 = 4$, so $x = \pm 3$ and $y = \pm 2$. The condition $xy = -6 < 0$ says x and y have opposite signs. Hence $w = 3 - 2i$ or $w = -3 + 2i$. Both work: $(3 - 2i)^2 = 9 - 12i + 4i^2 = 5 - 12i$, and $(-3 + 2i)^2 = (3 - 2i)^2$.

0.6.4 Polar form: multiplication rotates and scales

Addition in $\mathbb{C}$ is the familiar addition of arrows in the plane. Multiplication has a geometric meaning too, and it only becomes visible when we describe a point by its distance from 0 and its angle, instead of by its coordinates.

Let $z \neq 0$ and put $r = |z| > 0$. The number z/r has modulus 1 by [Theorem 0.6.2](#) (d), so it lies on the unit circle. From trigonometry, every point (x, y) with $x^2 + y^2 = 1$ is $(\cos \theta, \sin \theta)$ for some real θ . We abbreviate

$$e^{i\theta} := \cos \theta + i \sin \theta \quad (\theta \in \mathbb{R}),$$

and then every non-zero complex number can be written as

$$z = re^{i\theta}, \quad r = |z| > 0, \quad \theta \in \mathbb{R}.$$

This is the **polar form** of z , and θ is **an** argument of z . The word “an” matters: θ and $\theta + 2\pi k$ give the same point for every integer k . Conversely, if $re^{i\theta} = se^{i\varphi}$ with $r, s > 0$, then taking moduli gives $r = s$ (as $|e^{i\theta}| = \sqrt{\cos^2 \theta + \sin^2 \theta} = 1$). Comparing real and imaginary parts then gives $\cos \theta = \cos \varphi$ and $\sin \theta = \sin \varphi$, which from trigonometry means $\theta - \varphi$ is an integer multiple of 2π .

For now $e^{i\theta}$ is only a name for $\cos \theta + i \sin \theta$. The next result is the reason the name is a good one: these numbers multiply the way exponentials do.

Theorem 0.6.5 — Multiplication in Polar Form

For all $\theta, \varphi \in \mathbb{R}$,

$$e^{i\theta} e^{i\varphi} = e^{i(\theta+\varphi)}.$$

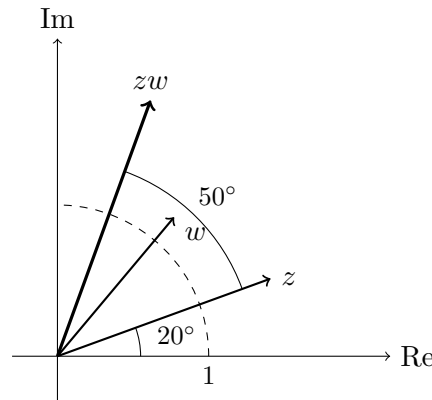
Consequently, if $z = re^{i\theta}$ and $w = se^{i\varphi}$ with $r, s > 0$, then $zw = rs e^{i(\theta+\varphi)}$.

Proof. By the multiplication rule,

$$e^{i\theta} e^{i\varphi} = (\cos \theta \cos \varphi - \sin \theta \sin \varphi) + (\cos \theta \sin \varphi + \sin \theta \cos \varphi)i = \cos(\theta + \varphi) + i \sin(\theta + \varphi),$$

where the last equality uses the addition formulas for cosine and sine. This is $e^{i(\theta+\varphi)}$. For the second statement, $zw = (rs)(e^{i\theta} e^{i\varphi})$ by [Proposition 0.6.1](#), and the first statement finishes the proof. ■

In words: **to multiply two complex numbers, multiply their moduli and add their arguments**. Multiplying by a fixed $w = se^{i\varphi}$ is therefore a rotation of the plane about 0 by the angle φ , combined with a stretch by the factor s . The picture shows z , w and zw for $|z| = 1.5$, $|w| = 1.2$ and arguments 20° and 50° .



The arrow for zw has length $1.5 \times 1.2 = 1.8$ and makes the angle $20^\circ + 50^\circ = 70^\circ$ with the positive real axis: it is the arrow for z , turned by w 's angle and stretched by w 's length. The small arc marks the argument of z , the large arc marks the extra turn by the argument of w , and the dashed arc is the unit circle, where the numbers $e^{i\theta}$ live.

Two special cases are worth isolating. Multiplying by $i = e^{i\pi/2}$ is a quarter turn counterclockwise, which explains $i^2 = -1$: two quarter turns send 1 to -1 . And $e^{i\theta} e^{-i\theta} = e^0 = 1$, so $(e^{i\theta})^{-1} = e^{-i\theta} = \overline{e^{i\theta}}$.

Repeating [Theorem 0.6.5](#) gives powers.

Theorem 0.6.6 — De Moivre's Formula

For every $\theta \in \mathbb{R}$ and every $n \in \mathbb{N}$,

$$(e^{i\theta})^n = e^{in\theta}, \quad \text{that is,} \quad (\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta.$$

Consequently $(re^{i\theta})^n = r^n e^{in\theta}$ for every real $r > 0$.

Proof. We use induction on n ([Theorem 0.2.7](#)). For $n = 0$, both sides equal 1, since $e^{i \cdot 0} = \cos 0 + i \sin 0 = 1$. Suppose $(e^{i\theta})^n = e^{in\theta}$ for some $n \in \mathbb{N}$. Then $(e^{i\theta})^{n+1} = (e^{i\theta})^n e^{i\theta} = e^{in\theta} e^{i\theta} = e^{i(n+1)\theta}$ by the induction hypothesis and [Theorem 0.6.5](#). The consequence follows by the same induction, since $(re^{i\theta})^{n+1} = (r^n e^{in\theta})(re^{i\theta}) = r^{n+1} e^{i(n+1)\theta}$ by [Theorem 0.6.5](#). ■

Powers that are painful to expand become one line in polar form.

Example. (A Power via Polar Form)

Compute $(\sqrt{3} + i)^6$.

Solution. The modulus is $|\sqrt{3} + i| = \sqrt{3 + 1} = 2$, and $\sqrt{3} + i = 2\left(\frac{\sqrt{3}}{2} + \frac{1}{2}i\right) = 2\left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6}\right) = 2e^{i\pi/6}$. By [Theorem 0.6.6](#),

$$(\sqrt{3} + i)^6 = 2^6 e^{i \cdot 6\pi/6} = 64(\cos \pi + i \sin \pi) = -64.$$

Expanding $(\sqrt{3} + i)^6$ with the binomial theorem gives the same answer after seven terms and some cancellation.

Quick check. Use polar form to compute $(1 + i)^8$.

Solution. We have $|1 + i| = \sqrt{2}$ and $1 + i = \sqrt{2}\left(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}i\right) = \sqrt{2}e^{i\pi/4}$. By [Theorem 0.6.6](#), $(1 + i)^8 = (\sqrt{2})^8 e^{i \cdot 8\pi/4} = 16e^{2\pi i} = 16$. As a cross-check, $(1 + i)^2 = 2i$, so $(1 + i)^8 = (2i)^4 = 16i^4 = 16$.

0.6.5 Roots of unity

Polar form also turns the equation $z^n = c$ into a question about angles. The most important case is $c = 1$.

Example. (Roots of Unity)

Let $n \geq 1$ be an integer. Find all $z \in \mathbb{C}$ with $z^n = 1$, and draw them for $n = 6$.

Solution. Since $0^n = 0 \neq 1$, every solution is non-zero, so we may write $z = re^{i\theta}$ with $r > 0$. By [Theorem 0.6.6](#), $z^n = r^n e^{in\theta}$, and $1 = 1 \cdot e^{i0}$. By uniqueness of the polar form up to multiples of 2π , $z^n = 1$ holds exactly when $r^n = 1$ and $n\theta = 2\pi k$ for some integer

k . Since $r > 0$ is real, $r^n = 1$ forces $r = 1$. Hence the solutions are

$$e^{2\pi i k/n}, \quad k \in \mathbb{Z}.$$

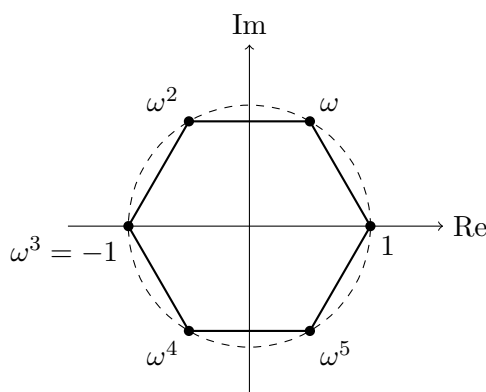
Put $\omega := e^{2\pi i/n}$. By [Theorem 0.6.6](#), $e^{2\pi i k/n} = \omega^k$ for $k \geq 0$. Two integers k, l give the same number exactly when $2\pi(k-l)/n$ is a multiple of 2π , that is, when n divides $k-l$. So there are exactly n distinct solutions,

$$1, \omega, \omega^2, \dots, \omega^{n-1},$$

the **n -th roots of unity**. They sit on the unit circle at equally spaced angles. For $n \geq 3$ they are the corners of a regular polygon with n corners, one of them at 1; for $n = 2$ they are ± 1 .

For $n = 6$, the angles are multiples of 60° , and the roots are

$$1, \quad \frac{1}{2} + \frac{\sqrt{3}}{2}i, \quad -\frac{1}{2} + \frac{\sqrt{3}}{2}i, \quad -1, \quad -\frac{1}{2} - \frac{\sqrt{3}}{2}i, \quad \frac{1}{2} - \frac{\sqrt{3}}{2}i.$$



The six roots sum to 0: since $\omega^3 = e^{i\pi} = -1$, we have $\omega^{k+3} = -\omega^k$, so the roots cancel in the opposite pairs $\{1, \omega^3\}$, $\{\omega, \omega^4\}$, $\{\omega^2, \omega^5\}$. In the picture, the hexagon is symmetric about its center 0. The sum of the n -th roots of unity is 0 for every $n \geq 2$; see [Exercise \(C2: Roots of Unity Sum to Zero\)](#).

0.6.6 The Fundamental Theorem of Algebra

We built $\mathbb{C}$ so that one polynomial equation, $z^2 + 1 = 0$, has a solution. It turns out that nothing more ever needs to be added: every polynomial equation already has a solution in $\mathbb{C}$. Here a **complex polynomial of degree n** means an expression $a_n z^n + \dots + a_1 z + a_0$ with $a_0, \dots, a_n \in \mathbb{C}$ and $a_n \neq 0$; polynomials are studied properly in a later section of this chapter.

Theorem 0.6.7 — Fundamental Theorem of Algebra

Let $n \geq 1$ and let $a_0, a_1, \dots, a_n \in \mathbb{C}$ with $a_n \neq 0$. Then there exists $z \in \mathbb{C}$ with

$$a_n z^n + a_{n-1} z^{n-1} + \dots + a_1 z + a_0 = 0.$$

We do not prove this theorem here. Its proof needs tools we have not built, and it is proved in

Chapter 5. Nothing in Chapters 0–4 depends on it.

Remark. Combined with division of polynomials (also in Chapter 5), the theorem gives the form in which it is usually applied: every complex polynomial of degree $n \geq 1$ **splits into linear factors**, $a_n z^n + \cdots + a_0 = a_n(z - z_1)(z - z_2) \cdots (z - z_n)$ for some $z_1, \dots, z_n \in \mathbb{C}$, not necessarily distinct. For linear algebra this is the fact that every complex square matrix has an eigenvalue, the starting point of the theory of canonical forms.

One companion fact needs no heavy machinery. If the coefficients $a_0, \dots, a_n$ are **real** and $p(z) = a_n z^n + \cdots + a_0$ has $p(z_0) = 0$, then applying [Theorem 0.6.2](#) (a) and (b) term by term, and $\overline{a_k} = a_k$, gives $p(\overline{z_0}) = \overline{p(z_0)} = \overline{0} = 0$. So the non-real roots of a real polynomial come in conjugate pairs. For example, $z^2 - 2z + 5$ has the roots $1 + 2i$ and $1 - 2i$: indeed $(1 + 2i)^2 - 2(1 + 2i) + 5 = (-3 + 4i) - (2 + 4i) + 5 = 0$.

0.6.7 Exercises

0.6.7.1 A. Check your understanding

Exercise A1.

- (a) Define the complex conjugate $\bar{z}$ and the modulus $|z|$ of $z = a + bi$, where $a, b \in \mathbb{R}$.
- (b) State the triangle inequality for complex numbers.
- (c) True or false: $|z + w| = |z| + |w|$ for all $z, w \in \mathbb{C}$. Justify your answer.
- (d) True or false: $z\bar{z}$ is a non-negative real number for every $z \in \mathbb{C}$. Justify your answer.
- (e) True or false: if $z, w \in \mathbb{C}$ and $z^2 = w^2$, then $z = w$. Justify your answer.
- (f) State the Fundamental Theorem of Algebra.

Solution.

- (a) See [Definition 0.6.2](#): $\bar{z} = a - bi$ and $|z| = \sqrt{a^2 + b^2} \geq 0$.
- (b) See [Theorem 0.6.4](#): $|z + w| \leq |z| + |w|$ for all $z, w \in \mathbb{C}$.
- (c) False. For $z = 1$ and $w = -1$, $|z + w| = 0$ but $|z| + |w| = 2$.
- (d) True. By [Theorem 0.6.2](#) (c), $z\bar{z} = |z|^2 = a^2 + b^2$, which is real and non-negative.
- (e) False. For $z = 1$ and $w = -1$, $z^2 = w^2 = 1$ but $z \neq w$. (What is true is that $z^2 = w^2$ implies $z = w$ or $z = -w$.)
- (f) See [Theorem 0.6.7](#): every complex polynomial $a_n z^n + \cdots + a_0$ with $n \geq 1$ and $a_n \neq 0$ has a root in $\mathbb{C}$.

0.6.7.2 B. Practice

Exercise B1: Arithmetic.

Write each of the following in the form $a + bi$ with $a, b \in \mathbb{R}$, or as a real number where indicated.

- (a) $(3 - 2i)(4 + i)$.
 (b) $\frac{1 + 2i}{2 - i}$.
 (c) $|(3 + 4i)(5 - 12i)|$, without expanding the product.
 (d) $\operatorname{Re} \frac{1}{1 + i}$ and $\operatorname{Im} \frac{1}{1 + i}$.

Solution.

- (a) Expanding and using $i^2 = -1$: $(3 - 2i)(4 + i) = 12 + 3i - 8i - 2i^2 = 14 - 5i$.
 (b) Multiplying top and bottom by $\overline{2 - i} = 2 + i$ ([Proposition 0.6.3](#)),

$$\frac{1 + 2i}{2 - i} = \frac{(1 + 2i)(2 + i)}{(2 - i)(2 + i)} = \frac{2 + i + 4i + 2i^2}{4 + 1} = \frac{5i}{5} = i.$$

- (c) By [Theorem 0.6.2](#) (d), $|(3 + 4i)(5 - 12i)| = |3 + 4i| |5 - 12i| = \sqrt{25} \cdot \sqrt{169} = 5 \cdot 13 = 65$.
 (d) By [Proposition 0.6.3](#), $\frac{1}{1 + i} = \frac{1 - i}{|1 + i|^2} = \frac{1 - i}{2} = \frac{1}{2} - \frac{1}{2}i$. Hence the real part is $\frac{1}{2}$ and the imaginary part is $-\frac{1}{2}$.

Exercise B2: A Square Root.

- (a) Find all $w \in \mathbb{C}$ with $w^2 = -3 + 4i$.
 (b) Hence solve $z^2 + 2z + 4 - 4i = 0$.

Hint: for (b), complete the square.

Solution.

- (a) Write $w = x + yi$ with $x, y \in \mathbb{R}$. Comparing real and imaginary parts in $(x^2 - y^2) + 2xyi = -3 + 4i$ gives $x^2 - y^2 = -3$ and $xy = 2$. Taking moduli, by [Theorem 0.6.2](#) (d), $x^2 + y^2 = |w|^2 = |-3 + 4i| = 5$. Adding and subtracting, $x^2 = 1$ and $y^2 = 4$. Since $xy = 2 > 0$, x and y have the same sign. Therefore $w = 1 + 2i$ or $w = -1 - 2i$. Check: $(1 + 2i)^2 = 1 + 4i - 4 = -3 + 4i$, and $(-1 - 2i)^2 = (1 + 2i)^2$.
 (b) Completing the square, $z^2 + 2z + 4 - 4i = (z + 1)^2 - 1 + 4 - 4i = (z + 1)^2 - (-3 + 4i)$. So the equation holds exactly when $(z + 1)^2 = -3 + 4i$. By (a), this means $z + 1 = 1 + 2i$ or $z + 1 = -1 - 2i$. Hence the solutions are $z = 2i$ and $z = -2 - 2i$.

Exercise B3.

Let $z \in \mathbb{C}$ with $z \neq 0$. Prove that $\frac{1}{z} = \bar{z}$ if and only if $|z| = 1$.

Solution. By [Proposition 0.6.3](#), $\frac{1}{z} = \frac{1}{|z|^2} \bar{z}$.

($\Leftarrow$) Suppose $|z| = 1$. Then $\frac{1}{z} = \frac{1}{1} \bar{z} = \bar{z}$.

($\Rightarrow$) Suppose $\frac{1}{z} = \bar{z}$. Then $\frac{1}{|z|^2} \bar{z} = \bar{z}$. Since $z \neq 0$, also $\bar{z} \neq 0$, so multiplying both sides by $(\bar{z})^{-1}$ gives $\frac{1}{|z|^2} = 1$, that is, $|z|^2 = 1$. As $|z| \geq 0$, this forces $|z| = 1$. This proves the equivalence.

Exercise B4: Polar Form.

- (a) Compute $(1 - \sqrt{3}i)^6$.
- (b) Find all $z \in \mathbb{C}$ with $z^3 = -8i$, in the form $a + bi$.

Solution.

- (a) We have $|1 - \sqrt{3}i| = \sqrt{1+3} = 2$ and $1 - \sqrt{3}i = 2(\cos(-\frac{\pi}{3}) + i \sin(-\frac{\pi}{3})) = 2e^{-i\pi/3}$. By [Theorem 0.6.6](#), $(1 - \sqrt{3}i)^6 = 64e^{-2\pi i} = 64(\cos 2\pi - i \sin 2\pi) = 64$.
- (b) Since $0^3 \neq -8i$, write $z = re^{i\theta}$ with $r > 0$. Also $-8i = 8e^{-i\pi/2}$. By [Theorem 0.6.6](#), $z^3 = r^3 e^{3i\theta}$. By uniqueness of polar form up to multiples of 2π , $z^3 = -8i$ holds exactly when $r^3 = 8$ and $3\theta = -\frac{\pi}{2} + 2\pi k$ for some integer k . Therefore $r = 2$ and $\theta = -\frac{\pi}{6} + \frac{2\pi k}{3}$. The values $k = 0, 1, 2$ give $\theta = -\frac{\pi}{6}, \frac{\pi}{2}, \frac{7\pi}{6}$, and every other k differs from one of these by a multiple of 3, which changes θ by a multiple of 2π . Hence the solutions are

$$2e^{-i\pi/6} = \sqrt{3} - i, \quad 2e^{i\pi/2} = 2i, \quad 2e^{7i\pi/6} = -\sqrt{3} - i.$$

Check: $(2i)^3 = 8i^3 = -8i$.

0.6.7.3 C. Going deeper

Exercise C1: Reverse Triangle Inequality.

Prove that $||z| - |w|| \leq |z - w|$ for all $z, w \in \mathbb{C}$.

Hint: write $z = (z - w) + w$.

Solution. Let $z, w \in \mathbb{C}$. Since $z = (z - w) + w$, the triangle inequality ([Theorem 0.6.4](#)) gives $|z| \leq |z - w| + |w|$, so $|z| - |w| \leq |z - w|$. Swapping the roles of z and w gives $|w| - |z| \leq |w - z|$. Moreover $|w - z| = |(-1)(z - w)| = |-1||z - w| = |z - w|$ by [Theorem 0.6.2](#) (d). The real number $||z| - |w||$ equals one of $|z| - |w|$ and $|w| - |z|$, and both are at most $|z - w|$. This proves the inequality.

Exercise C2: Roots of Unity Sum to Zero.

Let $n \geq 2$ and $\omega = e^{2\pi i/n}$.

- (a) Prove that $1 + \omega + \omega^2 + \cdots + \omega^{n-1} = 0$.
- (b) Deduce that $\sum_{k=0}^{n-1} \cos \frac{2\pi k}{n} = 0$ and $\sum_{k=0}^{n-1} \sin \frac{2\pi k}{n} = 0$.

Hint: compare S with ωS , where S is the sum in (a).

Solution.

- (a) Let $S = 1 + \omega + \cdots + \omega^{n-1}$. By Theorem 0.6.6, $\omega^n = e^{2\pi i} = 1$. Therefore

$$\omega S = \omega + \omega^2 + \cdots + \omega^{n-1} + \omega^n = \omega + \omega^2 + \cdots + \omega^{n-1} + 1 = S,$$

so $(\omega - 1)S = 0$. Since $n \geq 2$, we have $0 < 2\pi/n \leq \pi$, so $\operatorname{Re} \omega = \cos(2\pi/n) < 1$ and $\omega \neq 1$. Hence $\omega - 1 \neq 0$, and multiplying $(\omega - 1)S = 0$ by $(\omega - 1)^{-1}$ (Proposition 0.6.3) gives $S = 0$.

- (b) By Theorem 0.6.6, $\omega^k = \cos \frac{2\pi k}{n} + i \sin \frac{2\pi k}{n}$ for each k . Since addition in $\mathbb{C}$ is coordinate by coordinate, the real part of S is $\sum_{k=0}^{n-1} \cos \frac{2\pi k}{n}$ and the imaginary part is $\sum_{k=0}^{n-1} \sin \frac{2\pi k}{n}$. By (a), $S = 0$, so both sums are 0.

Exercise C3: Symmetries of $\mathbb{C}$ Fixing $\mathbb{R}$.

Let $\sigma: \mathbb{C} \rightarrow \mathbb{C}$ be a function with $\sigma(z + w) = \sigma(z) + \sigma(w)$ and $\sigma(zw) = \sigma(z)\sigma(w)$ for all $z, w \in \mathbb{C}$, and $\sigma(a) = a$ for every $a \in \mathbb{R}$.

- (a) Prove that $\sigma(i)^2 = -1$.
- (b) Prove that the only complex numbers u with $u^2 = -1$ are i and $-i$.
- (c) Deduce that either $\sigma(z) = z$ for every $z \in \mathbb{C}$, or $\sigma(z) = \bar{z}$ for every $z \in \mathbb{C}$.
- (d) Show that $z \mapsto \bar{z}$ does satisfy all three conditions on σ .

Hint: for (b), write $u = x + yi$ and compare real and imaginary parts.

Solution.

- (a) Since σ respects products and fixes the real number -1 , we get $\sigma(i)^2 = \sigma(i \cdot i) = \sigma(-1) = -1$.
- (b) Let $u = x + yi$ with $x, y \in \mathbb{R}$ and $u^2 = -1$. Comparing real and imaginary parts in $(x^2 - y^2) + 2xyi = -1$ gives $x^2 - y^2 = -1$ and $xy = 0$. If $y = 0$, then $x^2 = -1$, which is impossible for real x . Hence $x = 0$, so $y^2 = 1$ and $y = \pm 1$. Therefore $u = i$ or $u = -i$, and both satisfy $u^2 = -1$.
- (c) By (a) and (b), $\sigma(i) = i$ or $\sigma(i) = -i$. Let $z = a + bi$ with $a, b \in \mathbb{R}$. Since σ respects sums and products and fixes a and b ,

$$\sigma(z) = \sigma(a) + \sigma(b)\sigma(i) = a + b\sigma(i).$$

If $\sigma(i) = i$, this gives $\sigma(z) = z$ for every z . If $\sigma(i) = -i$, it gives $\sigma(z) = a - bi = \bar{z}$ for every z .

- (d) By Theorem 0.6.2 (a) and (b), conjugation respects sums and products. For $a \in \mathbb{R}$, $\bar{a} = a + 0i = a - 0i = a$. So conjugation satisfies all three conditions.

0.7 Fields

Linear algebra is about equations like $2x + 3y = 7$, and so far we have met several number systems in which such equations make sense: $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$, and the arithmetic of remainders $\mathbb{Z}/n\mathbb{Z}$. Which of them can serve as the numbers of linear algebra? This section answers by isolating the rules that solving linear equations actually uses. A number system obeying those rules is called a field, and fields are the scalars of every later chapter.

0.7.1 What solving linear equations needs

Here is a small system, solved the way you learned at school:

$$\begin{aligned} 2x + 3y &= 7, \\ x - y &= 1. \end{aligned}$$

Subtracting twice the second equation from the first gives $5y = 5$. Dividing by 5 gives $y = 1$, and then $x = 1 + y = 2$. Check: $2 \cdot 2 + 3 \cdot 1 = 7$ and $2 - 1 = 1$.

Look at what we used. We added and subtracted equations, multiplied an equation by a number, and divided by a **non-zero** number. Silently we also rearranged terms, regrouped them and expanded brackets. Every step of Gaussian elimination is of this kind, and every one of them works in $\mathbb{Q}$, $\mathbb{R}$ and $\mathbb{C}$.

Now try the integers. The equation $2x = 1$ involves only integers, yet it has no integer solution, because the step “divide by 2” leaves $\mathbb{Z}$. Elimination stalls as soon as a pivot has no inverse. So what separates $\mathbb{Z}$ from $\mathbb{Q}$ is division, and division is exactly what we must demand.

A field is a number system where you can add, subtract, multiply, and divide by anything non-zero, with the usual rules.

0.7.2 The definition

Definition 0.7.1 — Field

A **field** is a set F together with two functions, **addition** $+: F \times F \rightarrow F$, $(a, b) \mapsto a + b$, and **multiplication** $\cdot: F \times F \rightarrow F$, $(a, b) \mapsto a \cdot b = ab$, satisfying the following axioms.

- (F1) $a + b = b + a$ **for all** $a, b \in F$.
- (F2) $(a + b) + c = a + (b + c)$ **for all** $a, b, c \in F$.
- (F3) There exists an element $0 \in F$ such that $a + 0 = a$ **for every** $a \in F$.
- (F4) **For every** $a \in F$ there exists an element $-a \in F$ such that $a + (-a) = 0$.
- (F5) $(ab)c = a(bc)$ **for all** $a, b, c \in F$.
- (F6) $ab = ba$ **for all** $a, b \in F$.
- (F7) There exists an element $1 \in F$ with $1 \neq 0$ such that $a \cdot 1 = a$ **for every** $a \in F$.
- (F8) **For every non-zero** $a \in F$ there exists an element $a^{-1} \in F$ such that $aa^{-1} = 1$.
- (F9) $a(b + c) = ab + ac$ **for all** $a, b, c \in F$.

Read the axioms in groups.

- **(F1)–(F4)** are about addition alone. It is commutative and associative, it has a neutral element 0, and every element has a negative. Subtraction is then defined by $a - b := a + (-b)$.
- **(F5)–(F8)** say the same things about multiplication, with one change: only **non-zero** elements are required to have inverses. Division is defined by $a/b := ab^{-1}$ for $b \neq 0$. We never divide by 0, and the axioms never ask us to.
- **(F7)** also insists that $1 \neq 0$: the two neutral elements are different.
- **(F9)** is the one rule linking the two operations. It is what lets us expand brackets.

Note the order of quantifiers in (F3): **one** element 0 works **for every** a . This is stronger than “for every a there is some element z_a with $a + z_a = a$ ”. In (F4) the order is reversed, as it must be: the negative $-a$ depends on a .

We often say “ F is a field” when the operations are clear, and write $(F, +, \cdot)$ when they need to be named. Since addition and multiplication are functions into F , sums and products of elements of F stay in F ; when we build a field inside a bigger number system, this “closure” is the first thing to check.

0.7.3 First consequences of the axioms

The notation in the definition hides a question. We wrote “0”, “ $-a$ ” and “ a^{-1} ” as if each were a single well-defined element. The axioms only say that such elements **exist**. If a field could contain two different elements behaving like 0, the symbol 0 would be ambiguous, and so would $a - b$ and a/b . The next theorem removes this worry and proves the familiar rules of algebra from the nine axioms.

Theorem 0.7.1 — Basic Properties of Fields

Let F be a field and let $a, b, c \in F$.

- The element 0 in (F3) is unique, and the element 1 in (F7) is unique.
- (Cancellation) If $a + b = a + c$, then $b = c$. If $ab = ac$ and $a \neq 0$, then $b = c$.
- The negative $-a$ in (F4) is unique, and for $a \neq 0$ the inverse a^{-1} in (F8) is unique.
- $0 \cdot a = 0$.
- $(-1)a = -a$.
- If $ab = 0$, then $a = 0$ or $b = 0$.

Idea. Each part is a short chain of axioms, and the chain is found backwards from the goal. For uniqueness we use the standard move: suppose two elements both have the property, and show they are equal. For (d) there is no axiom about $0 \cdot a$ at all, so we must create a situation where one applies. The trick is to rewrite 0 in a fancy way as $0 + 0$, expand with (F9), and cancel. For (f), the hypothesis $a \neq 0$ is there to be spent on (F8).

We prove the uniqueness of 0, additive cancellation, the uniqueness of negatives, (d) and (f). The remaining statements are [Exercise \(B1: The Remaining Basic Properties\)](#).

Proof.

- (a) Suppose 0 and $0'$ both satisfy (F3). Applying (F3) for $0'$ to the element 0, and (F3) for 0 to the element $0'$, and using (F1),

$$0 = 0 + 0' = 0' + 0 = 0'.$$

So the zero element is unique.

- (b) Suppose $a + b = a + c$. Adding $-a$ on the left of both sides and using (F2),

$$((-a) + a) + b = ((-a) + a) + c.$$

By (F1) and (F4), $(-a) + a = a + (-a) = 0$. Hence $0 + b = 0 + c$, and by (F1) and (F3) this says $b = c$.

- (c) Suppose b and c both satisfy (F4) for a , that is, $a + b = 0$ and $a + c = 0$. Then $a + b = a + c$, so $b = c$ by additive cancellation in (b).
 (d) By (F3), $0 + 0 = 0$. Therefore, by (F9),

$$a \cdot 0 = a(0 + 0) = a \cdot 0 + a \cdot 0.$$

By (F3), also $a \cdot 0 = a \cdot 0 + 0$. Comparing, $a \cdot 0 + 0 = a \cdot 0 + a \cdot 0$, and additive cancellation (b) gives $0 = a \cdot 0$. Finally $0 \cdot a = a \cdot 0$ by (F6), so $0 \cdot a = 0$.

- (e) Suppose $ab = 0$. If $a = 0$ there is nothing to prove, so suppose $a \neq 0$. By (F8), a^{-1} exists. Then, by (F7), (F8), (F6) and (F5),

$$b = b \cdot 1 = b(aa^{-1}) = (ba)a^{-1} = (ab)a^{-1} = 0 \cdot a^{-1} = 0,$$

where the last equality is (d). This shows that $a = 0$ or $b = 0$. ■

Two small consequences are used without comment from now on: $0 + a = a$ and $1 \cdot a = a$, by (F1), (F3), (F6) and (F7). More importantly, every identity of school algebra that is proved from these rules, such as $(a + b)^2 = a^2 + 2ab + b^2$ or $(-a)(-b) = ab$, now holds in every field. Part (f) is the property we will use most: a product of non-zero elements is non-zero.

0.7.4 Examples

We check the standard examples first, then an exotic one, then a degenerate case.

Example. (The Fields $\mathbb{Q}$, $\mathbb{R}$ and $\mathbb{C}$)

Explain why $\mathbb{Q}$, $\mathbb{R}$ and $\mathbb{C}$, with their usual addition and multiplication, are fields.

Solution. For $\mathbb{R}$, the axioms (F1)–(F9) are the rules of arithmetic for real numbers that we take as familiar, and $1 \neq 0$.

For $\mathbb{Q}$, the sum, product and negative of fractions are fractions, and so is the inverse $\frac{q}{p}$ of a non-zero fraction $\frac{p}{q}$. So the operations of $\mathbb{R}$ restrict to functions $\mathbb{Q} \times \mathbb{Q} \rightarrow \mathbb{Q}$, and $0, 1, -a, a^{-1}$ lie in $\mathbb{Q}$ whenever a does. The identities in (F1), (F2), (F5), (F6) and (F9)

hold for all real numbers, so in particular for rational ones. Hence $\mathbb{Q}$ is a field.

For $\mathbb{C}$, the axioms (F1)–(F6) and (F9), together with the existence of $1 = 1 + 0i$, are [Proposition 0.6.1](#). We have $1 = (1, 0) \neq (0, 0) = 0$. Finally (F8) is [Proposition 0.6.3](#): a non-zero z has the inverse $\bar{z}/|z|^2$. Hence $\mathbb{C}$ is a field.

The next example is a field strictly between $\mathbb{Q}$ and $\mathbb{R}$. It shows that the inverse axiom (F8) can take real work to check.

Example. ($\mathbb{Q}(\sqrt{2})$ Is a Field)

Let $\mathbb{Q}(\sqrt{2}) := \{a + b\sqrt{2} : a, b \in \mathbb{Q}\} \subseteq \mathbb{R}$, with the addition and multiplication of $\mathbb{R}$. Show that $\mathbb{Q}(\sqrt{2})$ is a field, and find the inverse of $1 + 2\sqrt{2}$.

Solution. Let $x = a + b\sqrt{2}$ and $y = c + d\sqrt{2}$ with $a, b, c, d \in \mathbb{Q}$.

Closure. We have $x + y = (a + c) + (b + d)\sqrt{2}$ and $xy = (ac + 2bd) + (ad + bc)\sqrt{2}$, using $\sqrt{2} \cdot \sqrt{2} = 2$. The coefficients are rational, so $x + y, xy \in \mathbb{Q}(\sqrt{2})$.

Axioms that are identities. (F1), (F2), (F5), (F6) and (F9) hold for all real numbers, hence for the elements of $\mathbb{Q}(\sqrt{2})$.

Neutral elements and negatives. $0 = 0 + 0\sqrt{2}$ and $1 = 1 + 0\sqrt{2}$ lie in $\mathbb{Q}(\sqrt{2})$, and $1 \neq 0$. The negative $-x = (-a) + (-b)\sqrt{2}$ lies in $\mathbb{Q}(\sqrt{2})$.

Inverses. Suppose $x \neq 0$. We first claim $a - b\sqrt{2} \neq 0$. If $b \neq 0$ and $a - b\sqrt{2} = 0$, then $\sqrt{2} = a/b$ would be rational, contradicting [Theorem 0.2.3](#). If $b = 0$, then $a = x \neq 0$, so $a - b\sqrt{2} = a \neq 0$. Therefore $a - b\sqrt{2} \neq 0$, and by [Theorem 0.7.1](#) (f) in $\mathbb{R}$,

$$N := (a + b\sqrt{2})(a - b\sqrt{2}) = a^2 - 2b^2 \neq 0.$$

Since N is a non-zero rational number, the element

$$x' := \frac{a}{N} + \frac{-b}{N}\sqrt{2} \in \mathbb{Q}(\sqrt{2})$$

satisfies $xx' = \frac{1}{N}(a + b\sqrt{2})(a - b\sqrt{2}) = 1$. So (F8) holds, and $\mathbb{Q}(\sqrt{2})$ is a field.

For $x = 1 + 2\sqrt{2}$ we get $N = 1 - 8 = -7$, so $x^{-1} = -\frac{1}{7} + \frac{2}{7}\sqrt{2} = \frac{-1+2\sqrt{2}}{7}$. Check: $(1 + 2\sqrt{2})(-1 + 2\sqrt{2}) = -1 + 2\sqrt{2} - 2\sqrt{2} + 8 = 7$.

The method is “rationalize the denominator”, the same move as multiplying by the conjugate in $\mathbb{C}$. The irrationality of $\sqrt{2}$ is exactly what guarantees the new denominator is not zero.

Quick check. Is $\{a + b\sqrt{3} : a, b \in \mathbb{Q}\}$, with the operations of $\mathbb{R}$, a field?

Solution. Yes, by the same argument as [Example \(\$\mathbb{Q}\(\sqrt{2}\)\$ Is a Field\)](#). Products stay in the set because $\sqrt{3} \cdot \sqrt{3} = 3$ is rational. For inverses, the number $a^2 - 3b^2$ is non-zero whenever $a + b\sqrt{3} \neq 0$, because $\sqrt{3}$ is irrational ([Exercise \(B3: \$\sqrt{3}\$ is irrational\)](#)). The inverse is $\frac{a-b\sqrt{3}}{a^2-3b^2}$.

The degenerate case. Let $F = \{0\}$ with $0 + 0 = 0$ and $0 \cdot 0 = 0$. Every axiom except one holds: (F1), (F2), (F5), (F6) and (F9) are identities between elements that all equal 0, (F3) and

(F4) hold with $-0 = 0$, and (F8) holds vacuously because there is no non-zero element. The only failure is (F7): the one element that could serve as 1 is 0, and the axiom demands $1 \neq 0$. So $\{0\}$ is not a field, and every field has at least the two elements 0 and 1. This one-element system is the reason the clause $1 \neq 0$ is there.

0.7.5 Finite fields

Every field so far is infinite. The arithmetic of remainders gives finite ones, and they are genuinely useful: computer science and coding theory do linear algebra over them all the time.

Recall from [Definition 0.5.2](#) and [Definition 0.5.5](#) that for an integer $n \geq 1$, congruence modulo n is an equivalence relation on $\mathbb{Z}$. We write $[a]$ for the class of a and $\mathbb{Z}/n\mathbb{Z}$ for the quotient set. By division with remainder ([Exercise \(C2: Division with remainder\)](#)), $\mathbb{Z}/n\mathbb{Z} = \{[0], [1], \dots, [n-1]\}$, and these n classes are distinct. Also $[a] = [b]$ exactly when n divides $a - b$.

Example. (Arithmetic on $\mathbb{Z}/n\mathbb{Z}$ and the Field $\mathbb{F}_2$)

Let $n \geq 1$. Define $[a] + [b] := [a + b]$ and $[a][b] := [ab]$. Show that these operations are well defined, that they satisfy every field axiom except possibly (F8) and the clause $1 \neq 0$, and that $1 \neq 0$ holds exactly when $n \geq 2$. Write out the tables for $n = 2$.

Solution. Well-definedness. The formulas use representatives, so we must check that the answer does not depend on which representatives we pick. For addition this is [Example \(Addition modulo \$n\$ is well defined\)](#). For multiplication, suppose $[a] = [a']$ and $[b] = [b']$, so n divides $a - a'$ and $b - b'$. Then

$$ab - a'b' = a(b - b') + (a - a')b'$$

is divisible by n , so $[ab] = [a'b']$.

Axioms. Every identity in (F1), (F2), (F5), (F6), (F9) transfers from $\mathbb{Z}$, because both sides can be computed on representatives. For instance, for (F9),

$$[a]([b] + [c]) = [a][b + c] = [a(b + c)] = [ab + ac] = [ab] + [ac] = [a][b] + [a][c].$$

The same one-line computation works for the other four. The class $[0]$ satisfies (F3) since $[a] + [0] = [a]$, the class $[-a]$ is a negative of $[a]$ as in (F4), and $[1]$ satisfies $[a][1] = [a]$.

The clause $1 \neq 0$. We have $[1] = [0]$ exactly when n divides 1, that is, when $n = 1$. So $[1] \neq [0]$ exactly when $n \geq 2$.

Tables for $n = 2$. Writing 0, 1 for $[0], [1]$, the set $\mathbb{F}_2 := \mathbb{Z}/2\mathbb{Z} = \{0, 1\}$ has the tables

+	0	1
0	0	1
1	1	0

·	0	1
0	0	0
1	0	1

The only surprise is $1 + 1 = 0$, since 2 is even. Here (F8) holds as well, since the only non-zero element is 1 and $1 \cdot 1 = 1$. Hence $\mathbb{F}_2$ is a field.

So for $n \geq 2$, whether $\mathbb{Z}/n\mathbb{Z}$ is a field comes down to one question: does every non-zero class have an inverse? To find an inverse of $[a]$ we need an integer x with $[ax] = [1]$, that is, $ax + ny = 1$

for some integer y . The following lemma produces such x and y .

Lemma 0.7.2 — Bézout's Lemma for Integers

Let $a, n \in \mathbb{Z}$ with $n \geq 1$, and suppose that the only positive integer dividing both a and n is 1. Then there exist $x, y \in \mathbb{Z}$ with

$$ax + ny = 1.$$

Idea. Consider all positive integers of the form $ax + ny$. There is a smallest one, d , by well-ordering. The plan: ① show d divides a and n ; ② conclude $d = 1$ from the hypothesis. For ①, divide a by d : the remainder is again of the form $ax + ny$ and is smaller than d , so minimality forces it to be 0.

Proof. Let $S = \{ax + ny : x, y \in \mathbb{Z}\} \cap \{1, 2, 3, \dots\}$. Since $n = a \cdot 0 + n \cdot 1 \geq 1$, we have $n \in S$, so S is a non-empty subset of $\mathbb{N}$. By the well-ordering principle (Theorem 0.2.11), S has a smallest element $d = ax_0 + ny_0$.

By division with remainder (Exercise (C2: Division with remainder)), $a = qd + r$ for some integers q, r with $0 \leq r < d$. Then

$$r = a - q(ax_0 + ny_0) = a(1 - qx_0) + n(-qy_0)$$

has the form $ax + ny$. If $r \geq 1$, then $r \in S$ and $r < d$, contradicting the minimality of d . Hence $r = 0$, and d divides a . The same argument with n in place of a shows that d divides n .

Therefore d is a positive integer dividing both a and n . By hypothesis $d = 1$, so $ax_0 + ny_0 = 1$, as claimed. ■

Now we can decide exactly when $\mathbb{Z}/n\mathbb{Z}$ is a field.

Theorem 0.7.3 — $\mathbb{Z}/n\mathbb{Z}$ Is a Field Exactly for Prime n

Let $n \geq 2$ be an integer. Then $\mathbb{Z}/n\mathbb{Z}$ is a field if and only if n is prime.

Idea. By Example (Arithmetic on $\mathbb{Z}/n\mathbb{Z}$ and the Field $\mathbb{F}_2$), only (F8) is in question. If n is prime, a non-zero class $[a]$ has a not divisible by n , so a and n share no factor, and Bézout hands us the inverse. If $n = rs$ factors, then $[r][s] = [n] = [0]$ with both factors non-zero, which no field allows.

Proof. By Example (Arithmetic on $\mathbb{Z}/n\mathbb{Z}$ and the Field $\mathbb{F}_2$), $\mathbb{Z}/n\mathbb{Z}$ satisfies all field axioms except possibly (F8), including $[1] \neq [0]$ since $n \geq 2$.

($\Leftarrow$) Suppose $n = p$ is prime, and let $[a] \neq [0]$. Then p does not divide a . A positive integer dividing p is 1 or p , and p does not divide a , so the only positive common divisor of a and p is 1. By Lemma 0.7.2 there are $x, y \in \mathbb{Z}$ with $ax + py = 1$. Since $ax - 1 = -py$ is divisible by p ,

$$[a][x] = [ax] = [1].$$

So $[x]$ is an inverse of $[a]$, and (F8) holds. Hence $\mathbb{Z}/p\mathbb{Z}$ is a field.

($\Rightarrow$) We prove the contrapositive. Suppose $n \geq 2$ is not prime. Then $n = rs$ for some integers r, s with $1 < r < n$ and $1 < s < n$. Since $0 < r < n$, n does not divide r , so $[r] \neq [0]$; likewise $[s] \neq [0]$. But $[r][s] = [rs] = [n] = [0]$. By [Theorem 0.7.1](#) (f), this cannot happen in a field. Hence $\mathbb{Z}/n\mathbb{Z}$ is not a field. This proves the theorem. ■

For a prime p we write $\mathbb{F}_p := \mathbb{Z}/p\mathbb{Z}$, and usually drop the brackets, writing its elements as $0, 1, \dots, p-1$ and computing “mod p ”. So $\mathbb{F}_2, \mathbb{F}_3, \mathbb{F}_5, \mathbb{F}_7, \dots$ are fields, while $\mathbb{Z}/4\mathbb{Z}$, $\mathbb{Z}/6\mathbb{Z}$ and $\mathbb{Z}/9\mathbb{Z}$ are not.

Quick check. Find the inverse of 3 in $\mathbb{F}_7$.

Solution. We need x with $3x$ leaving remainder 1 on division by 7. Trying $x = 1, 2, \dots$: $3 \cdot 5 = 15 = 2 \cdot 7 + 1$. So $3^{-1} = 5$ in $\mathbb{F}_7$. (By [Theorem 0.7.1](#) (c) this is the only inverse.)

Because $\mathbb{F}_p$ is a field, the elimination from the start of the section runs unchanged in it. Only the arithmetic is new.

Example. (Elimination over $\mathbb{F}_5$)

Solve the system $2x + y = 3$, $3x + 2y = 4$ over $\mathbb{F}_5$.

Solution. In $\mathbb{F}_5$ the inverses are $1^{-1} = 1$, $2^{-1} = 3$, $3^{-1} = 2$, $4^{-1} = 4$, since $2 \cdot 3 = 6 = 5 + 1$ and $4 \cdot 4 = 16 = 15 + 1$. Multiply the first equation by $2^{-1} = 3$: since $3 \cdot 2 = 1$, $3 \cdot 1 = 3$ and $3 \cdot 3 = 9 = 4$, it becomes $x + 3y = 4$. Subtract 3 times this from the second equation: the x -terms cancel, and we get

$$(2 - 9)y = 4 - 12, \quad \text{that is,} \quad 3y = 2,$$

since $-7 = -10 + 3$ and $-8 = -10 + 2$. Multiply by $3^{-1} = 2$: $y = 4$. Then $x = 4 - 3y = 4 - 12 = -8 = 2$. Check: $2 \cdot 2 + 4 = 8 = 3$ and $3 \cdot 2 + 2 \cdot 4 = 14 = 4$ in $\mathbb{F}_5$. Hence $(x, y) = (2, 4)$ is the unique solution.

0.7.6 Non-examples

Each of these fails some part of the slogan, and it pays to name the exact clause that fails.

- $\mathbb{Z}$. Everything but (F8) holds: $\mathbb{Z}$ is closed under $+$ and $\cdot$, contains 0, 1 and negatives, and inherits the identities from $\mathbb{R}$. But $2 \neq 0$ has no inverse in $\mathbb{Z}$, since $2m$ is even and never equals 1. Changing $\mathbb{Z}$ to $\mathbb{Q}$ is the minimal change that repairs it.
- $\mathbb{Z}/6\mathbb{Z}$. This is $\mathbb{F}_5$ with the modulus changed from 5 to 6. All axioms but (F8) still hold ([Example \(Arithmetic on \$\mathbb{Z}/n\mathbb{Z}\$ and the Field \$\mathbb{F}_2\$ \)](#)), but $[2][3] = [0]$ with $[2], [3] \neq [0]$. By [Theorem 0.7.1](#) (f), $[2]$ cannot have an inverse; indeed $[2][x]$ is always one of $[0], [2], [4]$.
- 2×2 **real matrices**, with the matrix operations you know from school (they are treated properly later in this chapter). The identity matrix plays the role of 1. Two axioms fail.

(F6) fails:

$$\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}.$$

(F8) fails too: $E = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ is not the zero matrix, but every product EX has second row zero, so EX is never the identity.

Warning. “[a] $\neq$ [0]” does not mean [a] can be divided by in $\mathbb{Z}/n\mathbb{Z}$. In $\mathbb{Z}/6\mathbb{Z}$ the equation $[2]x = [1]$ has no solution, and $[2]x = [4]$ has two, $x = [2]$ and $x = [5]$. Before dividing modulo n , check that n is prime, or at least that a and n share no factor.

0.7.7 Why these axioms

Each clause is there because something we rely on breaks without it.

- **Without (F8)**, elimination stalls, as $2x = 1$ over $\mathbb{Z}$ showed. Worse, the theorems of the next chapters fail: over $\mathbb{Z}$, the analogues of vector spaces (called modules) need not have bases, and “a set of n independent vectors in an n -dimensional space spans it” becomes false. This book always works over a field.
- **Without (F6)**, scalars would not commute with each other. The matrices above show such systems exist, and there are even systems with division but without commutativity. Linear algebra constantly swaps scalars, as in $a(b\mathbf{v}) = b(a\mathbf{v})$ for a vector $\mathbf{v}$ (Chapter 1), so we keep commutativity.
- **Without $1 \neq 0$** , the one-element system $\{0\}$ would count, and every statement of the form “a field has a non-zero element” would fail. Keeping it costs nothing.
- **The name** “field” is historical and carries no extra meaning.

Remark. Many identities we are used to are consequences of the axioms and hold in **every** field, including $\mathbb{F}_2$. But properties that involve **order** or **size**, such as “squares are non-negative” or “ $1 + 1 + \cdots + 1 \neq 0$ ”, are not among the axioms. The first already fails in $\mathbb{C}$, where $i^2 = -1$, as the warning on ordering in the previous section explains. The second fails in $\mathbb{F}_p$, which is the subject of the last part of this section.

0.7.8 The characteristic of a field

In $\mathbb{Q}$ we can add 1 to itself as often as we like without reaching 0. In $\mathbb{F}_2$ two copies already give 0, and in $\mathbb{F}_p$ it takes p copies. This number is the most basic invariant of a field, and it decides whether familiar arguments such as “ $2x = 0$ implies $x = 0$ ” are valid.

First some notation. For $a \in F$ and $m \in \mathbb{N}$, define the **integer multiple** $m \cdot a$ by $0 \cdot a := 0$ and $(m+1) \cdot a := m \cdot a + a$. So $m \cdot a = a + a + \cdots + a$ with m terms. This agrees with the field product when $m = 0$, by [Theorem 0.7.1](#) (d).

Lemma 0.7.4 — Integer Multiples

Let F be a field, $a \in F$ and $m, k \in \mathbb{N}$. Then

$$(m \cdot 1) + (k \cdot 1) = (m + k) \cdot 1, \quad (m \cdot 1)(k \cdot 1) = (mk) \cdot 1, \quad (m \cdot 1)a = m \cdot a.$$

Proof. All three follow by induction on one variable ([Theorem 0.2.7](#)). For the first, induct on k : the case $k = 0$ is $m \cdot 1 + 0 = m \cdot 1$ by (F3), and if it holds for k , then by (F2) and the definition,

$$m \cdot 1 + (k + 1) \cdot 1 = (m \cdot 1 + k \cdot 1) + 1 = (m + k) \cdot 1 + 1 = (m + k + 1) \cdot 1.$$

For the third, induct on m : the case $m = 0$ is $0 \cdot a = 0$ by [Theorem 0.7.1](#) (d), and if it holds for m , then by (F6), (F9) and (F7),

$$((m + 1) \cdot 1)a = (m \cdot 1 + 1)a = (m \cdot 1)a + 1 \cdot a = m \cdot a + a = (m + 1) \cdot a.$$

The second is the third with $a = k \cdot 1$, combined with $m \cdot (k \cdot 1) = (mk) \cdot 1$, which follows from the first by induction on m . ■

Definition 0.7.2 — Characteristic

Let F be a field. If there is a **positive** integer n with $n \cdot 1 = 0$, the **characteristic** of F is the **smallest** such n . If there is no such n , the characteristic of F is 0.

In words: count how many copies of 1 must be added to reach 0. If it never happens, the convention is to say “characteristic 0”, not “characteristic ∞ ”: in that case the only integer $m \geq 0$ with $m \cdot 1 = 0$ is $m = 0$.

- $\mathbb{Q}, \mathbb{R}, \mathbb{C}$ have characteristic 0, because $n \cdot 1 = n \neq 0$ for every positive integer n .
- $\mathbb{F}_p$ has characteristic p : $n \cdot [1] = [n]$, which is $[0]$ exactly when p divides n , and the smallest positive such n is p .
- $\mathbb{F}_2$ is the smallest case: characteristic 2, because $1 + 1 = 0$.

Can a field have characteristic 6? In $\mathbb{Z}/6\mathbb{Z}$, six copies of $[1]$ give $[0]$, but $\mathbb{Z}/6\mathbb{Z}$ is not a field. This is no accident.

Theorem 0.7.5 — The Characteristic Is 0 or Prime

The characteristic of a field is either 0 or a prime number.

Idea. Suppose the characteristic n factors as rs with smaller factors. Then the elements $r \cdot 1$ and $s \cdot 1$ are non-zero, by minimality of n , but their product is $n \cdot 1 = 0$. A field has no such pairs. This is the same argument as in [Theorem 0.7.3](#), run inside an arbitrary field.

Proof. Let F be a field of characteristic $n \neq 0$. Since the integer multiple $1 \cdot 1 = 0 + 1 = 1$ is non-zero by (F7), we have $n \neq 1$, so $n \geq 2$. Suppose, for a contradiction, that n is not

prime. Then $n = rs$ with integers $1 < r < n$ and $1 < s < n$. By [Lemma 0.7.4](#),

$$(r \cdot 1)(s \cdot 1) = (rs) \cdot 1 = n \cdot 1 = 0.$$

By [Theorem 0.7.1](#) (f), $r \cdot 1 = 0$ or $s \cdot 1 = 0$. Either way we have a positive integer smaller than n whose multiple of 1 is 0, contradicting the minimality of n . Hence n is prime. ■

So the characteristic is 0, 2, 3, 5, 7, and so on. Together with [Exercise \(C2: Finite Fields Have Prime Characteristic\)](#), this says a finite field always has prime characteristic.

Warning. In a field of characteristic 2, such as $\mathbb{F}_2$, we have $1 + 1 = 0$, so $-1 = 1$ and $x = -x$ for **every** x . The familiar argument “ $x = -x$, so $2x = 0$, so $x = 0$ ” divides by $2 = 1 + 1$, which is 0 there. For example, in $\mathbb{F}_2$ the element 1 satisfies $1 = -1$ but is not 0. This is why later results, notably on symmetric bilinear forms in Chapter 13, carry the hypothesis “characteristic $\neq 2$ ”.

With fields in hand, the stage for linear algebra is set: in Chapter 1, a vector space is defined **over a field** F , and every result proved there holds at once for $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$, $\mathbb{Q}(\sqrt{2})$ and $\mathbb{F}_p$.

0.7.9 Exercises

0.7.9.1 A. Check your understanding

Exercise A1.

- (a) Define what it means for a set F with two operations to be a field.
- (b) True or false: in every field, $a^2 = 0$ implies $a = 0$. Justify your answer.
- (c) True or false: $\mathbb{Z}/8\mathbb{Z}$ is a field. Justify your answer.
- (d) True or false: in every field, $1 + 1 \neq 0$. Justify your answer.
- (e) What are the characteristics of $\mathbb{C}$ and of $\mathbb{F}_7$?
- (f) The set $\mathbb{N}$ with the usual addition and multiplication is not a field. Name two axioms that fail, each with a witness.

Solution.

- (a) See [Definition 0.7.1](#): two operations satisfying (F1)–(F9), including $1 \neq 0$.
- (b) True. If $a \cdot a = 0$, then $a = 0$ or $a = 0$ by [Theorem 0.7.1](#) (f).
- (c) False. Since $8 = 2 \cdot 4$ is not prime, $\mathbb{Z}/8\mathbb{Z}$ is not a field by [Theorem 0.7.3](#); concretely $[2][4] = [0]$ with $[2], [4] \neq [0]$.
- (d) False. In $\mathbb{F}_2$, $1 + 1 = 0$.
- (e) $\mathbb{C}$ has characteristic 0, and $\mathbb{F}_7$ has characteristic 7 (see the examples after [Definition 0.7.2](#)).
- (f) (F4) fails: $1 \in \mathbb{N}$, but no $b \in \mathbb{N}$ has $1 + b = 0$. (F8) fails: $2 \neq 0$, but no $b \in \mathbb{N}$ has $2b = 1$.

0.7.9.2 B. Practice

Exercise B1: The Remaining Basic Properties.

Let F be a field and $a, b, c \in F$. Using only the axioms and the parts of [Theorem 0.7.1](#) proved in the text, prove:

- (a) the element 1 in (F7) is unique;
- (b) if $ab = ac$ and $a \neq 0$, then $b = c$;
- (c) for $a \neq 0$, the inverse a^{-1} is unique;
- (d) $(-1)a = -a$.

Hint: for (d), show that $a + (-1)a = 0$.

Solution.

- (a) Suppose 1 and $1'$ both satisfy (F7). Then, by (F7) for $1'$, (F6), and (F7) for 1, $1 = 1 \cdot 1' = 1' \cdot 1 = 1'$.
- (b) Suppose $ab = ac$ and $a \neq 0$. By (F8), a^{-1} exists, and multiplying both sides by it gives $(ab)a^{-1} = (ac)a^{-1}$. By (F6) and (F5), $(ab)a^{-1} = (ba)a^{-1} = b(aa^{-1}) = b \cdot 1 = b$, using (F8) and (F7) at the end. In the same way $(ac)a^{-1} = c$. Hence $b = c$.
- (c) Suppose b and c both satisfy (F8) for $a \neq 0$: $ab = 1 = ac$. By (b), $b = c$.
- (d) By (F7) and (F6), $a = 1 \cdot a$. Therefore, using (F6) and (F9),

$$a + (-1)a = 1 \cdot a + (-1)a = a \cdot 1 + a(-1) = a(1 + (-1)) = a \cdot 0 = 0,$$

where the last two equalities use (F4) and [Theorem 0.7.1](#) (d). So $(-1)a$ satisfies (F4) for a . By uniqueness of negatives ([Theorem 0.7.1](#) (c)), $(-1)a = -a$.

Exercise B2: Solving Equations in $\mathbb{F}_7$.

- (a) Solve $3x + 5 = 2$ in $\mathbb{F}_7$.
- (b) Solve the system $x + 2y = 3$, $3x + y = 1$ over $\mathbb{F}_7$.

Solution.

- (a) Adding -5 to both sides gives $3x = 2 - 5 = -3 = 4$ in $\mathbb{F}_7$. Since $3 \cdot 5 = 15 = 1$ in $\mathbb{F}_7$, we have $3^{-1} = 5$. Multiplying by 5, $x = 20 = 6$. Check: $3 \cdot 6 + 5 = 23 = 2$ in $\mathbb{F}_7$. Hence $x = 6$.
- (b) Subtract 3 times the first equation from the second: $(1 - 6)y = 1 - 9$, that is, $-5y = -8$, or $2y = 6$ in $\mathbb{F}_7$. Since $2 \cdot 4 = 8 = 1$, $2^{-1} = 4$, so $y = 24 = 3$. Then $x = 3 - 2y = 3 - 6 = -3 = 4$. Check: $4 + 2 \cdot 3 = 10 = 3$ and $3 \cdot 4 + 3 = 15 = 1$ in $\mathbb{F}_7$. Hence $(x, y) = (4, 3)$.

Exercise B3: Which Are Fields?

Determine which of the following are fields, with the stated operations. Justify your answer.

- (a) $\mathbb{Z}/9\mathbb{Z}$, with the operations of [Example \(Arithmetic on \$\mathbb{Z}/n\mathbb{Z}\$ and the Field \$\mathbb{F}_2\$ \)](#).
- (b) $\mathbb{Q}(i) := \{a + bi : a, b \in \mathbb{Q}\}$, with the operations of $\mathbb{C}$.
- (c) $\{a + bi : a, b \in \mathbb{Z}\}$, with the operations of $\mathbb{C}$.
- (d) $\mathbb{F}_5$.

Solution.

- (a) Not a field. Since $9 = 3 \cdot 3$ is not prime, [Theorem 0.7.3](#) applies; concretely $[3][3] = [0]$ with $[3] \neq [0]$, which is impossible in a field by [Theorem 0.7.1 \(f\)](#).
- (b) A field. Let $x = a + bi$ and $y = c + di$ with $a, b, c, d \in \mathbb{Q}$. Then $x + y = (a + c) + (b + d)i$, $xy = (ac - bd) + (ad + bc)i$ and $-x = -a - bi$ have rational parts, and $0, 1 \in \mathbb{Q}(i)$ with $1 \neq 0$. The identities (F1), (F2), (F5), (F6), (F9) hold because they hold in $\mathbb{C}$ ([Example \(The Fields \$\mathbb{Q}\$, \$\mathbb{R}\$ and \$\mathbb{C}\$ \)](#)). If $x \neq 0$, then $a^2 + b^2$ is a non-zero rational number, and by [Proposition 0.6.3](#) $x^{-1} = \frac{a}{a^2 + b^2} - \frac{b}{a^2 + b^2}i$, which lies in $\mathbb{Q}(i)$. So (F8) holds.
- (c) Not a field: (F8) fails for 2. Suppose $u = a + bi$ with $a, b \in \mathbb{Z}$ satisfied $2u = 1$. This is also an equation in the field $\mathbb{C}$, where $\frac{1}{2}$ is an inverse of 2; by uniqueness of inverses in $\mathbb{C}$ ([Theorem 0.7.1 \(c\)](#)), $u = \frac{1}{2}$. But $\frac{1}{2}$ has real part $\frac{1}{2} \notin \mathbb{Z}$, a contradiction.
- (d) A field, by [Theorem 0.7.3](#), since 5 is prime.

0.7.9.3 C. Going deeper

Exercise C1: A Field with Four Elements.

Let $F = \{a + b\alpha : a, b \in \mathbb{F}_2\}$, a set of four formal symbols $0, 1, \alpha, \beta$, where $\beta := 1 + \alpha$. Define

$$(a + b\alpha) + (c + d\alpha) := (a + c) + (b + d)\alpha, \quad (a + b\alpha)(c + d\alpha) := (ac + bd) + (ad + bc + bd)\alpha,$$

with all coefficient arithmetic in $\mathbb{F}_2$. (The product is what expanding gives if we replace α^2 by $\alpha + 1$.)

- (a) Write out the addition and multiplication tables of F .
- (b) Prove that F is a field.
- (c) Explain why there is no bijection $\varphi: F \rightarrow \mathbb{Z}/4\mathbb{Z}$ with $\varphi(x + y) = \varphi(x) + \varphi(y)$ and $\varphi(xy) = \varphi(x)\varphi(y)$ for all $x, y \in F$. So this four-element field is not “ $\mathbb{Z}/4\mathbb{Z}$ in disguise”.

Hint: for (c), look for a non-zero element of $\mathbb{Z}/4\mathbb{Z}$ whose square is zero.

Solution.

- (a) Using the formulas, with $1 + 1 = 0$ in $\mathbb{F}_2$: $\begin{array}{c|ccc} & 0 & 1 & \alpha & \beta \\ \hline 0 & 0 & 1 & \alpha & \beta \\ 1 & 1 & 0 & \beta & \alpha \\ \alpha & \alpha & \beta & 0 & 1 \\ \beta & \beta & \alpha & 1 & 0 \end{array}$

$$\begin{array}{c} \bullet \quad \begin{array}{c|ccc} & 0 & 1 & \alpha & \beta \\ \hline 0 & 0 & 1 & \alpha & \beta \\ 1 & 1 & 0 & \beta & \alpha \\ \alpha & \alpha & \beta & 0 & 1 \\ \beta & \beta & \alpha & 1 & 0 \end{array} \end{array}$$

$\cdot$	0	1	α	β
0	0	0	0	0
1	0	1	α	β
α	0	α	β	1
β	0	β	1	α

] For instance, $\alpha \cdot \alpha = (0 + 1\alpha)(0 + 1\alpha) = (0 + 1) + (0 + 0 + 1)\alpha = 1 + \alpha = \beta$, and $\alpha\beta = (0 + 1\alpha)(1 + 1\alpha) = (0 + 1) + (0 + 1 + 1)\alpha = 1$.

- (b) Let $x = a + b\alpha$, $y = c + d\alpha$, $z = e + f\alpha$ with $a, \dots, f \in \mathbb{F}_2$. Since $\mathbb{F}_2$ is a field (Example (Arithmetic on $\mathbb{Z}/n\mathbb{Z}$ and the Field $\mathbb{F}_2$)), we may compute with the coefficients using the rules of algebra.

Addition. It acts coordinate by coordinate, so (F1) and (F2) follow from the same laws in $\mathbb{F}_2$, the element $0 = 0 + 0\alpha$ satisfies (F3), and $-x = x$ satisfies (F4) because $a + a = 0$ and $b + b = 0$ in $\mathbb{F}_2$.

(F6). The formula for xy is unchanged when (a, b) and (c, d) are swapped: $ac + bd = ca + db$ and $ad + bc + bd = cb + da + db$.

(F7). With $1 = 1 + 0\alpha$, $x \cdot 1 = (a + 0) + (0 + b + 0)\alpha = x$, and $1 \neq 0$ since the coefficients differ.

(F9). Since $y + z = (c + e) + (d + f)\alpha$,

$$x(y + z) = (a(c + e) + b(d + f)) + (a(d + f) + b(c + e) + b(d + f))\alpha,$$

and expanding the coefficients shows this is $((ac + bd) + (ae + bf)) + ((ad + bc + bd) + (af + be + bf))\alpha = xy + xz$.

(F5). Write $xy = g + h\alpha$ with $g = ac + bd$, $h = ad + bc + bd$. Then $(xy)z = (ge + hf) + (gf + he + hf)\alpha$. Expanding,

$$ge + hf = ace + adf + bcf + bde + bdf, \quad gf + he + hf = acf + ade + adf + bce + bcf + bde,$$

where in the second we used $bdf + bdf = 0$ in $\mathbb{F}_2$. Similarly $yz = k + l\alpha$ with $k = ce + df$, $l = cf + de + df$, and $x(yz) = (ak + bl) + (al + bk + bl)\alpha$. Expanding gives $ak + bl = ace + adf + bcf + bde + bdf$ and $al + bk + bl = acf + ade + adf + bce + bcf + bde$, again using $bdf + bdf = 0$. Hence $(xy)z = x(yz)$.

(F8). From the table, $1 \cdot 1 = 1$, $\alpha\beta = 1$ and $\beta\alpha = 1$, so every non-zero element has an inverse.

This proves that F is a field.

- (c) Suppose such a bijection φ existed. First, $\varphi(0) = \varphi(0 + 0) = \varphi(0) + \varphi(0)$, so additive cancellation in $\mathbb{Z}/4\mathbb{Z}$ (adding $-\varphi(0)$, which only uses (F1)–(F4), valid there by Example (Arithmetic on $\mathbb{Z}/n\mathbb{Z}$ and the Field $\mathbb{F}_2$)) gives $\varphi(0) = [0]$. Since φ is surjective, there is $x \in F$ with $\varphi(x) = [2]$. As $[2] \neq [0] = \varphi(0)$, we have $x \neq 0$. But

$$\varphi(x \cdot x) = \varphi(x)\varphi(x) = [2][2] = [4] = [0] = \varphi(0),$$

so $x \cdot x = 0$ because φ is injective. This contradicts Theorem 0.7.1 (f) in the field F (from (b)). Hence no such φ exists.

Exercise C2: Finite Fields Have Prime Characteristic.

- (a) Let F be a finite field. Prove that the characteristic of F is a prime p .
- (b) Let F be a field of characteristic $p > 0$. Prove that $p \cdot a = 0$ for every $a \in F$.

Hint: for (a), the elements $1 \cdot 1, 2 \cdot 1, 3 \cdot 1, \dots$ cannot all be different.

Solution.

- (a) The elements $m \cdot 1$ for $m = 1, 2, 3, \dots$ all lie in the finite set F , so they cannot all be distinct. Hence there are positive integers $k < m$ with $m \cdot 1 = k \cdot 1$. By [Lemma 0.7.4](#), $m \cdot 1 = k \cdot 1 + (m - k) \cdot 1$, so $k \cdot 1 + (m - k) \cdot 1 = k \cdot 1 + 0$, and additive cancellation ([Theorem 0.7.1](#) (b)) gives $(m - k) \cdot 1 = 0$ with $m - k \geq 1$. Therefore the characteristic of F is not 0. By [Theorem 0.7.5](#), it is a prime p .
- (b) Let $a \in F$. By [Lemma 0.7.4](#) and the definition of characteristic, $p \cdot a = (p \cdot 1) a = 0 \cdot a = 0$, where the last step is [Theorem 0.7.1](#) (d).

Exercise C3: The Frobenius Identity.

Let p be a prime.

- (a) Let $a, b \in \mathbb{Z}$ with p dividing ab . Prove that p divides a or p divides b .
- (b) Deduce that p divides the binomial coefficient $\binom{p}{k}$ for $0 < k < p$.
- (c) Let F be a field of characteristic p . Prove that $(x + y)^p = x^p + y^p$ for all $x, y \in F$.

You may use the binomial theorem $(x + y)^n = \sum_{k=0}^n \binom{n}{k} \cdot (x^k y^{n-k})$ in any field, with integer multiples as in [Lemma 0.7.4](#); its inductive proof uses only the field axioms.

Hint: for (a), use [Lemma 0.7.2](#). For (b), use $p! = \binom{p}{k} k! (p - k)!$.

Solution.

- (a) Suppose p does not divide a . A positive integer dividing p is 1 or p , so the only positive common divisor of a and p is 1. By [Lemma 0.7.2](#) there are $x, y \in \mathbb{Z}$ with $ax + py = 1$. Multiplying by b , $b = (ab)x + p(by)$. Both terms on the right are divisible by p , the first because p divides ab . Hence p divides b .
- (b) First, by (a) and induction on the number of factors, if p divides a product of integers then it divides one of the factors. The integers $1, 2, \dots, p - 1$ are not divisible by p , so p divides neither $k!$ nor $(p - k)!$ for $0 < k < p$, and hence, by (a) once more, not their product. Now p divides $p! = \binom{p}{k} \cdot (k! (p - k)!)$. By (a), p divides $\binom{p}{k}$.
- (c) Let $x, y \in F$. By the binomial theorem,

$$(x + y)^p = x^p + y^p + \sum_{k=1}^{p-1} \binom{p}{k} \cdot (x^k y^{p-k}),$$

since $\binom{p}{0} = \binom{p}{p} = 1$. Fix $0 < k < p$. By (b), $\binom{p}{k} = pm$ for some $m \in \mathbb{N}$. By

Lemma 0.7.4,

$$\binom{p}{k} \cdot (x^k y^{p-k}) = ((pm) \cdot 1)(x^k y^{p-k}) = (p \cdot 1)(m \cdot 1)(x^k y^{p-k}) = 0,$$

because $p \cdot 1 = 0$ in characteristic p and 0 times anything is 0 (Theorem 0.7.1 (d)). So every middle term vanishes, and $(x + y)^p = x^p + y^p$, as claimed.

0.8 Polynomials

We now have fields: number systems in which we can add, subtract, multiply and divide by anything non-zero. The next objects built on top of a field are polynomials. They will be everywhere in this book: as examples of vectors in Chapter 1, as the characteristic and minimal polynomials of a matrix later on, and as the reason some fields behave differently from others. This section says precisely what a polynomial is, introduces its degree, and explains why a polynomial is **not** the same thing as the function it defines.

0.8.1 What a polynomial is

At school, a polynomial is “an expression like $3x^2 - x + 2$ ”. That is a good picture, but it leaves one question open: what is x ? One tempting answer is that x is a number to be plugged in, so a polynomial is really a function. This answer breaks down as soon as we leave the familiar number systems.

Work over the field $\mathbb{F}_2 = \{0, 1\}$ of Example (Arithmetic on $\mathbb{Z}/n\mathbb{Z}$ and the Field $\mathbb{F}_2$), where $1 + 1 = 0$. The expressions $x^2 + x$ and 0 look different. But if we plug in the two elements of $\mathbb{F}_2$, we get $0^2 + 0 = 0$ and $1^2 + 1 = 1 + 1 = 0$. As functions on $\mathbb{F}_2$, the two are the same. If a polynomial *were* its function, then $x^2 + x$ would be the zero polynomial, and it would have no sensible degree. We want $x^2 + x$ to be a genuine polynomial of degree 2, so we must define polynomials by their coefficients alone.

A polynomial is its list of coefficients, and nothing else.

Definition 0.8.1 — Polynomial

Let F be a field. A **polynomial over F** in the indeterminate x is a sequence $(a_0, a_1, a_2, \dots)$ of elements of F , indexed by $\mathbb{N}$, such that **only finitely many** a_k are non-zero. That is, there is some $N \in \mathbb{N}$ with $a_k = 0$ **for every** $k > N$.

The element a_k is the **coefficient of x^k** . We write the polynomial as

$$p = a_0 + a_1x + a_2x^2 + \cdots + a_Nx^N = \sum_{k=0}^N a_kx^k.$$

Two polynomials $p = \sum a_kx^k$ and $q = \sum b_kx^k$ are **equal** if and only if $a_k = b_k$ **for every** $k \in \mathbb{N}$.

In words: a polynomial is an infinite list of coefficients that is eventually all zero. The expression

$\sum_{k=0}^N a_k x^k$ is only a way of displaying the list: x^k marks the position k , and we stop writing once the remaining coefficients are all zero. Two polynomials are equal exactly when they agree position by position. A sequence indexed by $\mathbb{N}$ is a function $\mathbb{N} \rightarrow F$, so this is just equality of functions in the sense of [Definition 0.4.1](#).

There is one thing to check about the notation. The number N in the display is not unique: if $a_k = 0$ for every $k > N$, the same is true for every $k > N + 1$. Writing $\sum_{k=0}^N a_k x^k$ or $\sum_{k=0}^{N+1} a_k x^k$ only adds a term $0 \cdot x^{N+1}$, which we agree to drop. So the display names the same sequence whichever admissible N we use. We also drop terms with zero coefficient in the middle, write x for $1 \cdot x^1$, and write a_0 for $a_0 x^0$.

Example. (Polynomials over various fields)

Read off the coefficient sequence of each polynomial, and check it against [Definition 0.8.1](#).

$$3x^2 - x + 2 \in \mathbb{R}[x], \quad x^5 + \pi x^3 - \sqrt{2} \in \mathbb{R}[x], \quad (1+i)x - i \in \mathbb{C}[x], \quad x^2 + x \in \mathbb{F}_2[x], \quad 7, \quad 0.$$

(The symbol $F[x]$ for the set of all polynomials over F is defined formally in [Definition 0.8.2](#) below.)

Solution. The first is the sequence $(2, -1, 3, 0, 0, \dots)$ of real numbers. It has finitely many non-zero terms, so it is a polynomial over $\mathbb{R}$.

The second is $(-\sqrt{2}, 0, 0, \pi, 0, 1, 0, \dots)$. Its coefficients are real but irrational; the definition only asks that they lie in the field.

The third is $(-i, 1+i, 0, \dots)$, a polynomial over $\mathbb{C}$.

The fourth is $(0, 1, 1, 0, \dots)$ with entries in $\mathbb{F}_2$. It is **not** the zero sequence, so it is not the zero polynomial, even though it vanishes at both points of $\mathbb{F}_2$.

The constant 7 is the sequence $(7, 0, 0, \dots)$. It is a polynomial over $\mathbb{Q}$, $\mathbb{R}$ or $\mathbb{C}$. Over $\mathbb{F}_2$ it would be $(1, 0, 0, \dots)$, because $7 = 1$ there.

The last, $0 = (0, 0, 0, \dots)$, is the **zero polynomial**. It is the degenerate case: every coefficient is zero, and the finiteness condition holds with $N = 0$. It matters because it is the one polynomial whose degree needs a special convention, as we will see.

Now the non-example, by a minimal change. Take the polynomial $1 + x + x^2$, whose sequence is $(1, 1, 1, 0, 0, \dots)$, and change every later 0 to 1. The result $(1, 1, 1, 1, \dots)$, informally “ $1 + x + x^2 + x^3 + \dots$ ”, is still a sequence of elements of $\mathbb{R}$. What fails is the clause **only finitely many** coefficients are non-zero, so this is **not** a polynomial. (It is a *formal power series*, which we will not need.) Expressions such as $x^{-1} + 1$ and $\sqrt{x}$ are not polynomials either: they are not of the form $\sum a_k x^k$ with $k \in \mathbb{N}$ at all.

Why insist on finitely many non-zero coefficients? Because we want to plug elements of the field into polynomials, which means forming the sum $\sum a_k c^k$, and in a field only finite sums make sense. Finiteness is also what gives every non-zero polynomial a highest power, its degree.

0.8.2 Adding and multiplying polynomials

The coefficient picture tells us how to add and multiply. Adding $3x^2 - x + 2$ and $x + 5$ at school means collecting like powers, so we add coefficients position by position. Multiplying means expanding every product $a_i x^i \cdot b_j x^j = a_i b_j x^{i+j}$ and collecting like powers, so the coefficient of x^k collects all pairs (i, j) with $i + j = k$.

Definition 0.8.2 — The polynomial ring $F[x]$

]

Let F be a field. The set of **all** polynomials over F is denoted $F[x]$. For $p = \sum a_k x^k$ and $q = \sum b_k x^k$ in $F[x]$, define the **sum** $p + q$ and the **product** pq by their coefficients:

$$p + q = \sum_k (a_k + b_k) x^k, \quad pq = \sum_k \left(\sum_{i=0}^k a_i b_{k-i} \right) x^k.$$

We must check that $p + q$ and pq are again polynomials, that is, that their coefficient sequences are eventually zero. Suppose $a_i = 0$ for every $i > m$ and $b_j = 0$ for every $j > n$. If $k > \max(m, n)$, then $a_k + b_k = 0 + 0 = 0$. If $k > m + n$, then in each product $a_i b_{k-i}$ either $i > m$ or $k - i > n$, because otherwise $k = i + (k - i) \leq m + n$. So every term is 0, and the coefficient of x^k in pq is 0. Hence both are polynomials, and addition and multiplication are well defined on $F[x]$.

With this multiplication, $x \cdot x$ has coefficient sequence $(0, 0, 1, 0, \dots)$, and by induction on k the k -th power of the polynomial x is the sequence with a 1 in position k and 0 elsewhere. So the notation x^k in [Definition 0.8.1](#) agrees with the actual power, and $\sum a_k x^k$ is an honest sum of products.

Example. (A product, coefficient by coefficient)

In $\mathbb{R}[x]$, compute $(3x^2 - x + 2)(x + 1)$ from the definition.

Solution. Here $(a_0, a_1, a_2) = (2, -1, 3)$ and $(b_0, b_1) = (1, 1)$, with all other coefficients 0. The coefficient of x^k in the product is $\sum_{i=0}^k a_i b_{k-i}$:

$$\begin{aligned} k = 0 : a_0 b_0 &= 2, \\ k = 1 : a_0 b_1 + a_1 b_0 &= 2 - 1 = 1, \\ k = 2 : a_0 b_2 + a_1 b_1 + a_2 b_0 &= 0 - 1 + 3 = 2, \\ k = 3 : a_0 b_3 + a_1 b_2 + a_2 b_1 + a_3 b_0 &= 0 + 0 + 3 + 0 = 3, \end{aligned}$$

and every coefficient with $k \geq 4$ is 0 by the argument above (here $m + n = 3$). Hence $(3x^2 - x + 2)(x + 1) = 3x^3 + 2x^2 + x + 2$, which is what expanding by hand gives.

The operations obey the rules we expect. We record them once, so that later sections can use them freely.

Theorem 0.8.1 — Arithmetic in $F[x]$

]

Let F be a field and $p, q, r \in F[x]$. Then:

1. $p + q = q + p$ and $(p + q) + r = p + (q + r)$;
2. $p + 0 = p$, and $p + (-p) = 0$, where $-p = \sum (-a_k)x^k$;
3. $pq = qp$ and $(pq)r = p(qr)$;
4. $1 \cdot p = p$, where 1 is the constant polynomial $(1, 0, 0, \dots)$;
5. $p(q + r) = pq + pr$.

Idea. Two polynomials are equal when their coefficients agree, so each identity reduces to an identity between coefficients, which the field axioms supply. For products it helps to write the coefficient of x^k in pq symmetrically, as a sum over all pairs (i, j) of natural numbers with $i + j = k$.

Proof. Let $p = \sum a_k x^k$, $q = \sum b_k x^k$ and $r = \sum c_k x^k$. By [Definition 0.8.1](#), it suffices to compare the coefficient of x^k on both sides, for each $k \in \mathbb{N}$.

Parts 1, 2 and 4 hold because in each position they are the corresponding field axioms of [Definition 0.7.1](#): for example, the coefficient of x^k in $p + q$ is $a_k + b_k = b_k + a_k$, the coefficient in $q + p$. For 4, the coefficient of x^k in $1 \cdot p$ is $\sum_{i+j=k} e_i a_j$ with $e_0 = 1$ and $e_i = 0$ for $i \geq 1$, which equals a_k .

For 3, the coefficient of x^k in pq is $\sum_{i+j=k} a_i b_j$. Since multiplication in F is commutative, this equals $\sum_{i+j=k} b_j a_i$, the coefficient in qp . Next, the coefficient of x^k in $(pq)r$ is

$$\sum_{s+l=k} \left(\sum_{i+j=s} a_i b_j \right) c_l = \sum_{i+j+l=k} a_i b_j c_l,$$

where the equality uses distributivity and associativity in F . The same computation for $p(qr)$ gives $\sum_{i+s=k} a_i \left(\sum_{j+l=s} b_j c_l \right) = \sum_{i+j+l=k} a_i b_j c_l$. Therefore $(pq)r = p(qr)$.

For 5, the coefficient of x^k in $p(q + r)$ is $\sum_{i+j=k} a_i (b_j + c_j) = \sum_{i+j=k} a_i b_j + \sum_{i+j=k} a_i c_j$ by distributivity in F , which is the coefficient in $pq + pr$. This proves the theorem. ■

A set with an addition and a multiplication satisfying these rules is called a **commutative ring**; this is where the name “polynomial ring” comes from. We will not need the word again in this chapter. What matters is that we may now calculate with polynomials exactly as at school.

Quick check. Over $\mathbb{R}$, are $(x + 1)^2$ and $x^2 + 2x + 1$ equal as polynomials? Over $\mathbb{F}_2$, are $(x + 1)^2$ and $x^2 + 1$ equal as polynomials?

Solution. Both answers are yes, but for a reason that depends on the field. Expanding gives $(x + 1)^2 = x^2 + 2x + 1$, with coefficient sequence $(1, 2, 1, 0, \dots)$. Over $\mathbb{R}$ this is exactly the sequence of $x^2 + 2x + 1$. Over $\mathbb{F}_2$ we have $2 = 1 + 1 = 0$, so the sequence is $(1, 0, 1, 0, \dots)$, which is the sequence of $x^2 + 1$.

0.8.3 Degree

The most useful single number attached to a polynomial is the highest power of x that actually occurs. Every non-zero polynomial has one: it has at least one non-zero coefficient and only finitely many, so there is a largest position holding a non-zero coefficient. The zero polynomial has no such position, and we must decide what to do with it.

Definition 0.8.3 — Degree

Let $p = \sum a_k x^k \in F[x]$.

- If $p \neq 0$, the **degree** of p , written $\deg p$, is the **largest** $n \in \mathbb{N}$ with $a_n \neq 0$. The coefficient a_n is the **leading coefficient** of p , and p is **monic** if its leading coefficient is 1.
- The zero polynomial has degree $\deg 0 = -\infty$.

We use the conventions $-\infty < n$ and $-\infty + n = n + (-\infty) = -\infty$ for every $n \in \mathbb{N}$, and $-\infty + (-\infty) = -\infty$.

Examples, simplest first. A non-zero constant c has degree 0; its leading coefficient is c . The polynomial $x + 5 \in \mathbb{R}[x]$ is monic of degree 1. The polynomial $3x^3 + 2x^2 + x + 2$ from [Example \(A product, coefficient by coefficient\)](#) has degree 3 and leading coefficient 3. Over $\mathbb{F}_2$, $x^2 + x$ has degree 2, even though it vanishes at every point of $\mathbb{F}_2$.

Why $-\infty$, and not 0 or -1 ? Look at the formula we expect: the degree of a product is the sum of the degrees. With $\deg 0 = 0$ it fails at once, since $0 \cdot x = 0$ would give $0 = \deg 0 = \deg 0 + \deg x = 1$. With $\deg 0 = -1$ it fails too: $-1 = \deg(0 \cdot x) \neq \deg 0 + \deg x = 0$. The formula needs $\deg 0 + n = \deg 0$ for every n , and $-\infty$ is exactly the symbol with that property. The convention keeps a formula true, and that is its only job.

It turns out that the degree of a sum is controlled by the larger degree, and the degree of a product is the sum of the degrees:

Theorem 0.8.2 — Degree of a sum

Let F be a field and $p, q \in F[x]$. Then

$$\deg(p + q) \leq \max(\deg p, \deg q),$$

with **equality** whenever $\deg p \neq \deg q$.

Proof. Let $p = \sum a_k x^k$, $q = \sum b_k x^k$ and $M = \max(\deg p, \deg q)$. If $M = -\infty$, then $p = q = 0$, so $p + q = 0$ and both sides are $-\infty$. Suppose $M \in \mathbb{N}$. For every $k > M$ we have $k > \deg p$ and $k > \deg q$, so $a_k = b_k = 0$ by [Definition 0.8.3](#). Hence the coefficient $a_k + b_k$ of x^k in $p + q$ is 0 for every $k > M$, and therefore $\deg(p + q) \leq M$.

Now suppose $\deg p \neq \deg q$; by symmetry we may assume $\deg p > \deg q$, so $M = \deg p \in \mathbb{N}$. Then $a_M \neq 0$ and $b_M = 0$, because $M > \deg q$. The coefficient of x^M in $p + q$ is $a_M + 0 = a_M \neq 0$, so $\deg(p + q) \geq M$. Together with the first part, $\deg(p + q) = M$. This proves the theorem.

The inequality can be strict: over any field, $\deg(x + (-x)) = \deg 0 = -\infty < 1$. The leading terms cancel. That is the only way the degree of a sum can drop.

Theorem 0.8.3 — Degree of a product

Let F be a field and $p, q \in F[x]$. Then

$$\deg(pq) = \deg p + \deg q.$$

If p and q are non-zero, the leading coefficient of pq is the product of their leading coefficients.

Idea. Expanding $(a_mx^m + \text{lower terms})(b_nx^n + \text{lower terms})$, the only way to reach x^{m+n} is $a_mx^m \cdot b_nx^n$, and nothing reaches beyond. So the top coefficient is a_mb_n . The one real question is whether a_mb_n could be 0. In a field it cannot, and this is exactly where the field axioms enter.

Proof. If $p = 0$ or $q = 0$, then $pq = 0$ by the definition of the product, so both sides equal $-\infty$ by the conventions in Definition 0.8.3. Suppose now $p = \sum a_kx^k$ and $q = \sum b_kx^k$ are non-zero, with $m = \deg p$ and $n = \deg q$. Then $a_m \neq 0$, $b_n \neq 0$, $a_i = 0$ for $i > m$ and $b_j = 0$ for $j > n$.

We showed after Definition 0.8.2 that the coefficient of x^k in pq is 0 for every $k > m+n$. For $k = m+n$, the coefficient is $\sum_{i+j=m+n} a_ib_j$. A term with $i > m$ has $a_i = 0$, and a term with $i < m$ has $j = m+n-i > n$, so $b_j = 0$. Hence only $i = m$, $j = n$ survives, and the coefficient is a_mb_n .

Since $a_m \neq 0$ and $b_n \neq 0$, Theorem 0.7.1 gives $a_mb_n \neq 0$, because a product of non-zero elements of a field is non-zero. Therefore $\deg(pq) = m+n$ and the leading coefficient of pq is a_mb_n , as claimed.

Warning. The field axioms are doing real work here. The same definitions make sense with coefficients in $\mathbb{Z}/6\mathbb{Z}$, which is **not** a field. There $(2x+1)(3x^2+x) = 6x^3+5x^2+x = 5x^2+x$, because $2 \cdot 3 = 6 = 0$. The factors have degrees 1 and 2, but the product has degree 2, not 3.

The first consequence is that polynomials over a field behave like integers: a product can only be zero if a factor is, and non-zero factors can be canceled.

Corollary 0.8.4 — No zero divisors in $F[x]$

]

Let F be a field and $p, q, r \in F[x]$.

1. If $pq = 0$, then $p = 0$ or $q = 0$.
2. If $pq = pr$ and $p \neq 0$, then $q = r$.

Proof. For 1, we prove the contrapositive. Suppose $p \neq 0$ and $q \neq 0$. Then $\deg p, \deg q \in \mathbb{N}$, so by Theorem 0.8.3, $\deg(pq) = \deg p + \deg q \in \mathbb{N}$. In particular $\deg(pq) \neq -\infty$, so $pq \neq 0$.

For 2, suppose $pq = pr$ and $p \neq 0$. By Theorem 0.8.1, $p(q - r) = pq - pr = 0$. By part 1 and $p \neq 0$, we get $q - r = 0$, that is, $q = r$. This proves the corollary. ■

Quick check. Let $p, q \in \mathbb{R}[x]$ with $\deg p = 3$ and $\deg q = 2$. What are $\deg(pq)$, $\deg(p + q)$ and $\deg(p^2)$? Can you say what $\deg(p - x^3)$ is?

Solution. By Theorem 0.8.3, $\deg(pq) = 5$ and $\deg(p^2) = 6$. Since the degrees differ, Theorem 0.8.2 gives $\deg(p + q) = 3$. The degree of $p - x^3$ is not determined: if the leading coefficient of p is 1, the x^3 terms cancel and the degree is at most 2 (it could even be $-\infty$ if $p = x^3$); otherwise it is 3.

0.8.4 Polynomials of bounded degree

In linear algebra we constantly need “all polynomials up to a given size”. For instance, the quadratics $ax^2 + bx + c$ are described by three numbers, much like points of $\mathbb{R}^3$. So we give such collections names.

Definition 0.8.4 — Polynomials of degree at most n

Let F be a field and $n \in \mathbb{N}$. The set of polynomials over F of degree **at most** n is

$$F[x]_{\leq n} = \{p \in F[x] : \deg p \leq n\} = \{a_0 + a_1x + \cdots + a_nx^n : a_0, \dots, a_n \in F\}.$$

The two descriptions agree: $\deg p \leq n$ says exactly that every coefficient beyond position n is zero, and the coefficients $a_0, \dots, a_n$ are then free, any of them allowed to be zero. In particular the zero polynomial lies in $F[x]_{\leq n}$, since $-\infty \leq n$.

Example. (Inside $\mathbb{R}[x]_{\leq 2}$)

Which of $3x^2 - 2x + 1$, $5x + 7$, -4 , 0 and $x^3 - x^3 + x$ lie in $\mathbb{R}[x]_{\leq 2}$? Which lie in $\mathbb{R}[x]_{\leq 0}$?

Solution. All five lie in $\mathbb{R}[x]_{\leq 2}$. Their degrees are 2, 1, 0, $-\infty$, and 1 (the last is the polynomial x , since $x^3 - x^3 = 0$ as polynomials). Only -4 and 0 lie in $\mathbb{R}[x]_{\leq 0}$, which is the set of constant polynomials, 0 included.

Sometimes we need the degree to be exactly n , for instance when we speak of “a quadratic” and mean that the x^2 term is really there.

Definition 0.8.5 — Polynomials of degree exactly n

Let F be a field and $n \in \mathbb{N}$. The set of polynomials over F of degree **exactly** n is

$$F[x]_{=n} = \{p \in F[x] : \deg p = n\} = \{a_0 + a_1x + \cdots + a_nx^n : a_0, \dots, a_n \in F, a_n \neq 0\}.$$

Example. (Comparing the two notations)

For each of $x^2 + 1$, $3x - 2$ and 0 , decide whether it lies in $\mathbb{R}[x]_{\leq 2}$ and whether it lies in $\mathbb{R}[x]_{=2}$.

Solution. The polynomial $x^2 + 1$ has degree 2, so it lies in both. The polynomial $3x - 2$ has degree $1 \leq 2$, so it lies in $\mathbb{R}[x]_{\leq 2}$ but **not** in $\mathbb{R}[x]_{=2}$. The zero polynomial has degree $-\infty \leq 2$, so it lies in $\mathbb{R}[x]_{\leq 2}$, but $-\infty \neq 2$, so it is **not** in $\mathbb{R}[x]_{=2}$. In general $F[x]_{\leq n} = F[x]_{=n} \cup F[x]_{\leq n-1}$ for $n \geq 1$, and the union is disjoint.

The two sets differ in one small clause, $a_n \neq 0$, and that clause changes their behavior completely.

Warning. $F[x]_{\leq n}$ is closed under addition, but $F[x]_{=n}$ is **not**. For example $x^2 + x$ and $-x^2 + 1$ both lie in $\mathbb{R}[x]_{=2}$, but their sum $x + 1$ has degree 1. The leading terms cancel. Also $F[x]_{=n}$ does not contain 0. In Chapter 1 this is why $F[x]_{\leq n}$ is a subspace and $F[x]_{=n}$ is not.

The positive half is [Theorem 0.8.2](#) at work: if $\deg p \leq n$ and $\deg q \leq n$, then $\deg(p + q) \leq \max(\deg p, \deg q) \leq n$. Multiplying by a constant $c \in F$ also cannot raise the degree, by [Theorem 0.8.3](#).

0.8.5 Evaluation and roots

A polynomial is not a function, but every polynomial *defines* a function: plug an element of the field in for x .

Definition 0.8.6 — Evaluation and roots

Let F be a field, $p = \sum_{k=0}^N a_k x^k \in F[x]$ and $c \in F$. The **value** of p at c is the element

$$p(c) = \sum_{k=0}^N a_k c^k = a_0 + a_1 c + \cdots + a_N c^N \in F,$$

where $c^0 = 1$. An element $c \in F$ is a **root** of p if $p(c) = 0$. The function $F \rightarrow F$, $c \mapsto p(c)$, is the **polynomial function** of p .

The value does not depend on the choice of N : enlarging N adds terms $0 \cdot c^k = 0$ (by [Theorem 0.7.1](#)), which do not change the sum. So $p(c)$ is well defined.

For example, $p = x^2 - 3x + 2 \in \mathbb{R}[x]$ has $p(0) = 2$ and $p(1) = 1 - 3 + 2 = 0$, so 1 is a root. Over $\mathbb{F}_3 = \{0, 1, 2\}$, the polynomial $x^3 + 2$ has $1^3 + 2 = 3 = 0$, so 1 is a root; the other two values are

$0 + 2 = 2$ and $8 + 2 = 10 = 1$, so 1 is its only root in $\mathbb{F}_3$.

Evaluation turns the operations of $F[x]$ into the operations of F :

Theorem 0.8.5 — Evaluation respects sums and products

Let F be a field, $p, q \in F[x]$ and $c \in F$. Then

$$(p + q)(c) = p(c) + q(c), \quad (pq)(c) = p(c)q(c),$$

and the constant polynomial a has value a at c .

Proof. Let $p = \sum_{i=0}^m a_i x^i$ and $q = \sum_{j=0}^n b_j x^j$, where we choose m and n large enough that all non-zero coefficients are included. The statement about constants is immediate from [Definition 0.8.6](#). For sums, taking $N = \max(m, n)$ and padding with zero coefficients, $(p + q)(c) = \sum_{k=0}^N (a_k + b_k) c^k = \sum_{k=0}^N a_k c^k + \sum_{k=0}^N b_k c^k$ by distributivity in F .

For products, all non-zero coefficients of pq occur at positions $k \leq m + n$, so

$$(pq)(c) = \sum_{k=0}^{m+n} \left(\sum_{i+j=k} a_i b_j \right) c^k = \sum_{i=0}^m \sum_{j=0}^n a_i b_j c^i c^j = \sum_{i=0}^m \sum_{j=0}^n (a_i c^i) (b_j c^j) = \left(\sum_{i=0}^m a_i c^i \right) \left(\sum_{j=0}^n b_j c^j \right),$$

where the first equality is [Definition 0.8.2](#), the second uses distributivity and $c^{i+j} = c^i c^j$ and regroups the pairs (i, j) , the third uses **commutativity** of multiplication in F to move c^i past b_j , and the last is distributivity again. This shows $(pq)(c) = p(c)q(c)$. ■

Note which step used commutativity. In the next section we will plug a square matrix A into a polynomial. Products of matrices do not commute in general, but the proof only moves **scalars** past powers of A , and multiplies powers of the same A ; both are still allowed for matrices, so the same identity will hold there.

As a first use, a root of a factor is a root of the product: if $q(c) = 0$, then $(pq)(c) = p(c) \cdot 0 = 0$. Conversely, if $(pq)(c) = 0$ then $p(c)q(c) = 0$, so $p(c) = 0$ or $q(c) = 0$ by [Theorem 0.7.1](#). This is how one solves $x^2 - 3x + 2 = (x - 1)(x - 2) = 0$ at school.

Now we can return to the question from the start of the section.

Example. (Different polynomials, same function)

Over $\mathbb{F}_2$, compare the polynomials $x^2 + x$ and 0, and the polynomials x^2 and x , both as polynomials and as functions $\mathbb{F}_2 \rightarrow \mathbb{F}_2$.

Solution. As polynomials, $x^2 + x$ has coefficient sequence $(0, 1, 1, 0, \dots)$ and 0 has $(0, 0, 0, \dots)$. They differ in position 1, so $x^2 + x \neq 0$ by [Definition 0.8.1](#). Its degree is 2.

As functions, $\mathbb{F}_2 = \{0, 1\}$ has two elements, and

$$0^2 + 0 = 0, \quad 1^2 + 1 = 1 + 1 = 0.$$

So the polynomial function of $x^2 + x$ is the zero function, the same as that of 0. Both elements of $\mathbb{F}_2$ are roots of a polynomial of degree 2 that is not the zero polynomial.

Similarly $x^2 \neq x$ as polynomials, but $0^2 = 0$ and $1^2 = 1$, so they define the same function. Indeed, by [Theorem 0.8.5](#), two polynomials define the same function exactly when their difference is a polynomial whose function is zero, and $x^2 - x = x^2 + x$ in $\mathbb{F}_2[x]$.

Warning. A polynomial is not a function. Two different polynomials can define the same polynomial function, as $x^2 + x$ and 0 do over $\mathbb{F}_2$. So “ $p(c) = 0$ for every $c \in F$ ” does **not** imply $p = 0$ in general, and we may never read off coefficients or a degree from the values alone. This happens over every finite field: if $F = \{c_1, \dots, c_q\}$, then $(x - c_1)(x - c_2) \cdots (x - c_q)$ has degree q by [Theorem 0.8.3](#), yet vanishes at every c_i by [Theorem 0.8.5](#). Over $\mathbb{F}_p$, [Exercise \(C2: A non-zero polynomial vanishing everywhere\)](#) below gives the shorter example $x^p - x$.

Over an infinite field such as $\mathbb{Q}$, $\mathbb{R}$ or $\mathbb{C}$, the problem disappears: a polynomial whose function is zero is the zero polynomial, so polynomials and polynomial functions can be identified. We prove this in Chapter 5, once we can count roots. Until then, whenever we say two polynomials are equal we mean their coefficients agree.

0.8.6 Looking ahead

Two tools will be built on this section later.

Remark. Division with remainder. Over a field, if $f, g \in F[x]$ and $g \neq 0$, there are unique $q, r \in F[x]$ with $f = qg + r$ and $\deg r < \deg g$. This is long division of polynomials, and the convention $\deg 0 = -\infty$ makes “ $\deg r < \deg g$ ” cover the case $r = 0$. We state it here only for orientation; it is proved in Chapter 5 and is not used before then.

Remark. Polynomials in a matrix. In the next section, once matrices and their products are available, we substitute a square matrix A for x , replacing the constant term a_0 by $a_0 I$. This is how the characteristic and minimal polynomials of later chapters act on matrices.

0.8.7 Exercises

0.8.7.1 A. Check your understanding

Exercise A1.

- Define the degree of a polynomial over a field F , including the zero polynomial. State the formula for $\deg(pq)$.
- Is $1 + x + x^2 + x^3 + \cdots$, with every coefficient equal to 1, a polynomial over $\mathbb{R}$? Justify your answer.
- Determine whether the following statement is true: “for all $p, q \in F[x]$, $\deg(p+q) = \max(\deg p, \deg q)$.” Justify your answer.
- Determine whether the following statement is true: “if $p \in \mathbb{F}_2[x]$ and $p(c) = 0$ for

every $c \in \mathbb{F}_2$, then $p = 0$.” Justify your answer.

- (e) Explain why $F[x]_{=2}$ is **not** closed under addition, while $F[x]_{\leq 2}$ is.

Solution.

- (a) For $p = \sum a_k x^k \neq 0$, $\deg p$ is the largest n with $a_n \neq 0$; and $\deg 0 = -\infty$. For all $p, q \in F[x]$, $\deg(pq) = \deg p + \deg q$ ([Theorem 0.8.3](#)).
- (b) No. Its coefficient sequence $(1, 1, 1, \dots)$ has infinitely many non-zero terms, so it fails the clause “only finitely many a_k are non-zero” of [Definition 0.8.1](#).
- (c) False. Take $p = x$ and $q = -x$. Then $p + q = 0$ has degree $-\infty$, while $\max(\deg p, \deg q) = 1$. Only the inequality $\leq$ holds in general ([Theorem 0.8.2](#)).
- (d) False. By [Example \(Different polynomials, same function\)](#), $p = x^2 + x$ satisfies $p(0) = p(1) = 0$, but $p \neq 0$.
- (e) The sum of x^2 and $-x^2 + 1$, both of degree exactly 2, is 1, of degree 0. For $F[x]_{\leq 2}$: if $\deg p \leq 2$ and $\deg q \leq 2$, then $\deg(p + q) \leq \max(\deg p, \deg q) \leq 2$ by [Theorem 0.8.2](#).

0.8.7.2 B. Practice

Exercise B1: Sums, products and degrees.

- (a) In $\mathbb{R}[x]$, let $p = 2x^3 - x + 4$ and $q = -2x^3 + x^2 + 3$. Compute $p + q$ and pq , and give their degrees. Which inequality or formula from this section does each degree illustrate?
- (b) In $\mathbb{F}_3[x]$, compute $(x + 2)(x^2 + x + 1)$. Hence find all roots in $\mathbb{F}_3$ of the product.

Solution.

- (a) Adding coefficients gives $p + q = x^2 - x + 7$, of degree 2. Here $\deg p = \deg q = 3$ and the x^3 terms cancel, so the inequality of [Theorem 0.8.2](#) is strict. Expanding,

$$pq = -4x^6 + 2x^5 + 2x^4 - 3x^3 + 4x^2 - 3x + 12,$$

of degree $6 = 3 + 3$, as [Theorem 0.8.3](#) predicts; its leading coefficient is $2 \cdot (-2) = -4$.

- (b) Expanding, $(x + 2)(x^2 + x + 1) = x^3 + 3x^2 + 3x + 2$. In $\mathbb{F}_3$ we have $3 = 0$, so the product is $x^3 + 2$. By [Theorem 0.8.5](#) and [Theorem 0.7.1](#), c is a root of the product if and only if $c + 2 = 0$ or $c^2 + c + 1 = 0$. The first gives $c = 1$. For the second, the values at 0, 1, 2 are 1, 3 = 0 and 7 = 1. Hence the only root in $\mathbb{F}_3$ is $c = 1$, which occurs in both factors.

Exercise B2: Degree of a composition.

For $p = \sum_{k=0}^m a_k x^k$ and q in $F[x]$, define the **composition** $p \circ q = \sum_{k=0}^m a_k q^k \in F[x]$, where $q^0 = 1$.

- (a) Prove that if p and q are non-constant, then $\deg(p \circ q) = \deg p \cdot \deg q$.
- (b) Give an example showing the formula fails when q is constant.

Hint: first find $\deg(q^k)$.

Solution.

- (a) Let $m = \deg p \geq 1$ and $d = \deg q \geq 1$, so $a_m \neq 0$. By [Theorem 0.8.3](#) and induction on k ([Theorem 0.2.7](#)), $\deg(q^k) = kd$ for every $k \in \mathbb{N}$: the case $k = 0$ is $\deg 1 = 0$, and $\deg(q^{k+1}) = \deg(q^k) + \deg q = kd + d$. Hence, again by [Theorem 0.8.3](#), $\deg(a_m q^m) = 0 + md = md$, and for $k < m$, $\deg(a_k q^k) \leq kd < md$ (it is kd or $-\infty$). Let $s = \sum_{k=0}^{m-1} a_k q^k$. Applying [Theorem 0.8.2](#) $m-1$ times gives $\deg s \leq \max_{k < m} \deg(a_k q^k) < md$. Since $\deg s \neq \deg(a_m q^m)$, the equality case of [Theorem 0.8.2](#) gives $\deg(p \circ q) = \deg(s + a_m q^m) = md$. This proves $\deg(p \circ q) = \deg p \cdot \deg q$.
- (b) Take $p = x - 1$ and $q = 1$. Then $p \circ q = 1 - 1 = 0$ has degree $-\infty$, while $\deg p \cdot \deg q = 1 \cdot 0 = 0$.

Exercise B3: All small polynomials over $\mathbb{F}_2$.

List all polynomials in $\mathbb{F}_2[x]_{\leq 2}$. For each, give its degree and its polynomial function (the values at 0 and 1). Hence determine which of them define the zero function, and explain why every function $\mathbb{F}_2 \rightarrow \mathbb{F}_2$ is the polynomial function of exactly two of them.

Solution. A polynomial $a_0 + a_1x + a_2x^2$ with $a_i \in \mathbb{F}_2$ has value a_0 at 0 and $a_0 + a_1 + a_2$ at 1. With $2^3 = 8$ choices of coefficients:

polynomial	degree	value at 0	value at 1
0	$-\infty$	0	0
1	0	1	1
x	1	0	1
$x + 1$	1	1	0
x^2	2	0	1
$x^2 + 1$	2	1	0
$x^2 + x$	2	0	0
$x^2 + x + 1$	2	1	1

The zero function requires $a_0 = 0$ and $a_0 + a_1 + a_2 = 0$, that is, $a_0 = 0$ and $a_1 = a_2$. Hence exactly 0 and $x^2 + x$ define the zero function.

A function $f: \mathbb{F}_2 \rightarrow \mathbb{F}_2$ is determined by the pair $(f(0), f(1))$. Given such a pair (u, v) , the polynomial $a_0 + a_1x + a_2x^2$ defines f exactly when $a_0 = u$ and $a_1 + a_2 = v - u$. The first fixes a_0 , and for each of the two choices of a_2 the second fixes $a_1 = v - u - a_2$. Therefore exactly two of the eight polynomials define f , and the table confirms this.

0.8.7.3 C. Going deeper

Exercise C1: Roots of a quadratic.

Let F be a field.

- (a) Let $p = ax^2 + bx + c \in F[x]$ and $r \in F$ with $p(r) = 0$. Prove that $p = (x - r)(ax +$

$(b + ar))$.

- (b) Deduce that a non-zero polynomial in $F[x]_{\leq 2}$ has at most 2 roots in F .
- (c) The same definitions make sense with coefficients in $\mathbb{Z}/8\mathbb{Z}$. Show that $x^2 - 1$ has four roots there, and name the step of (b) that breaks.

Hint for (b): use (a) together with Theorem 0.8.5.

Solution.

- (a) Since $p(r) = 0$, we have $ar^2 + br + c = 0$, so $c = -ar^2 - br$. Expanding with Theorem 0.8.1,

$$(x-r)(ax+(b+ar)) = ax^2 + (b+ar)x - arx - r(b+ar) = ax^2 + bx - (br+ar^2) = ax^2 + bx + c = p.$$

- (b) Let $p \neq 0$ with $\deg p \leq 2$. *Case 1.* $\deg p = 0$. Then $p = c \neq 0$ is constant and $p(r) = c \neq 0$ for every r , so p has no roots. *Case 2.* $\deg p = 1$, say $p = bx + c$ with $b \neq 0$. Then $p(r) = 0$ if and only if $r = -b^{-1}c$, so p has exactly one root. *Case 3.* $\deg p = 2$, say $p = ax^2 + bx + c$ with $a \neq 0$. If p has no root we are done. Otherwise let r be a root. By (a), $p = (x - r)q$ with $q = ax + (b + ar)$, of degree 1 since $a \neq 0$. Let s be any root of p . By Theorem 0.8.5, $0 = p(s) = (s - r)q(s)$, so $s = r$ or $q(s) = 0$ by Theorem 0.7.1. By Case 2, q has exactly one root. Hence every root of p is r or the root of q , and p has at most 2 roots. This proves the claim.
- (c) In $\mathbb{Z}/8\mathbb{Z}$, $1^2 = 1$, $3^2 = 9 = 1$, $5^2 = 25 = 1$ and $7^2 = 49 = 1$, so 1, 3, 5, 7 are four roots of $x^2 - 1$. The argument in (b) breaks at “ $(s - r)q(s) = 0$ implies $s - r = 0$ or $q(s) = 0$ ”: for $r = 1$, $s = 3$ we get $(3 - 1)(3 + 1) = 2 \cdot 4 = 0$ with both factors non-zero. That step needs the field property that a product of non-zero elements is non-zero.

Exercise C2: A non-zero polynomial vanishing everywhere.

Let p be a prime and $F = \mathbb{F}_p$.

- (a) Fix $a \in F \setminus \{0\}$. Prove that the map $F \setminus \{0\} \rightarrow F \setminus \{0\}$, $b \mapsto ab$, is a bijection.
- (b) Deduce that $a^{p-1} = 1$ for every $a \in F \setminus \{0\}$.
- (c) Hence show that the polynomial $x^p - x \in \mathbb{F}_p[x]$ is not the zero polynomial but that its polynomial function is the zero function.

Hint for (b): multiply together all the elements of $F \setminus \{0\}$ in two ways.

Solution.

- (a) The map lands in $F \setminus \{0\}$, since a product of non-zero elements of a field is non-zero (Theorem 0.7.1). It is injective: if $ab = ab'$, multiplying by a^{-1} gives $b = b'$. The set $F \setminus \{0\}$ is finite with $p - 1$ elements, so an injective map from it to itself is surjective by Theorem 0.4.9, hence bijective.
- (b) Let P be the product of all $p - 1$ elements of $F \setminus \{0\}$; it is non-zero by Theorem 0.7.1. Since $b \mapsto ab$ is a bijection by (a), the elements ab , as b runs over $F \setminus \{0\}$, are the same elements in a different order. Multiplication in F is commutative and

associative, so the product does not depend on the order, and

$$P = \prod_{b \neq 0} (ab) = a^{p-1} \prod_{b \neq 0} b = a^{p-1} P.$$

Multiplying by P^{-1} gives $a^{p-1} = 1$.

- (c) The coefficient of x^p in $x^p - x$ is $1 \neq 0$, so the polynomial is non-zero, of degree p . For its function: $0^p - 0 = 0$, and for $a \neq 0$, (b) gives $a^p - a = a \cdot a^{p-1} - a = a - a = 0$. Hence every element of $\mathbb{F}_p$ is a root, and the polynomial function of $x^p - x$ is the zero function. For $p = 2$ this is $x^2 - x = x^2 + x$, the polynomial of [Example \(Different polynomials, same function\)](#).

Exercise C3: Which polynomials have inverses?.

- (a) Let F be a field and $p, q \in F[x]$ with $pq = 1$. Prove that p and q are non-zero constants.
- (b) Show that (a) fails with coefficients in $\mathbb{Z}/4\mathbb{Z}$ by computing $(2x + 1)^2$ there.

Solution.

- (a) Since $pq = 1 \neq 0$, neither p nor q is the zero polynomial, so $\deg p, \deg q \in \mathbb{N}$. By [Theorem 0.8.3](#), $\deg p + \deg q = \deg 1 = 0$. Two natural numbers with sum 0 are both 0, so $\deg p = \deg q = 0$. Therefore p and q are non-zero constants.
- (b) In $\mathbb{Z}/4\mathbb{Z}$, $(2x + 1)^2 = 4x^2 + 4x + 1 = 1$, because $4 = 0$. So $2x + 1$ has degree 1 and is its own inverse. The proof of (a) breaks at [Theorem 0.8.3](#), which fails here because the leading coefficient $2 \cdot 2 = 4 = 0$.

0.9 Matrices

You have probably met matrices as tables of numbers that are multiplied by a strange rule. This section rebuilds them over an arbitrary field F and answers the questions a computational course leaves open. Why is the product defined the way it is? Which rules of school algebra survive, and which fail? What does it mean to divide by a matrix? Everything later in the book, from Gaussian elimination in Chapter 2 to the matrix of a linear map in Chapter 3, rests on the answers.

Throughout, F is a field ([Definition 0.7.1](#)). Nothing here uses more about F than the field axioms, so every statement holds over $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$ and $\mathbb{F}_p$ alike.

0.9.1 Matrices and their entries

Definition 0.9.1 — Matrix

Let F be a field and $m, n \geq 1$ integers. An $m \times n$ **matrix over F** is a rectangular array

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}$$

with m rows and n columns, whose **entries** a_{ij} lie in F . Formally, it is a function $\{1, \dots, m\} \times \{1, \dots, n\} \rightarrow F$, $(i, j) \mapsto a_{ij}$. The set of all such matrices is $M_{m \times n}(F)$, and we write $M_n(F) = M_{n \times n}(F)$ for the **square** matrices.

The entry a_{ij} sits in row i and column j : **row first, column second**. We write $A = (a_{ij})$, and $(A)_{ij}$ for the (i, j) -entry when the matrix has no letter of its own. Two matrices are **equal** when they have the **same size** and the same entry in every position; this is equality of functions (Definition 0.4.1).

Two sizes deserve their own names. A matrix in $M_{n \times 1}(F)$ is a **column vector**, and we write

$$F^n = M_{n \times 1}(F), \quad \mathbf{x} = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}.$$

In running text we write $\mathbf{x} = (x_1, \dots, x_n)$ and still mean a column. Columns are printed in bold. A matrix in $M_{1 \times n}(F)$ is a **row vector**. For $j = 1, \dots, n$, the column $\mathbf{e}_j \in F^n$ has 1 in position j and 0 elsewhere.

We will often cut a matrix into its columns. If $\mathbf{a}_j = (a_{1j}, \dots, a_{mj}) \in F^m$ is the j -th column of A , we write

$$A = (\mathbf{a}_1 \quad \mathbf{a}_2 \quad \cdots \quad \mathbf{a}_n).$$

The i -th row of A is the row vector $(a_{i1} \quad \cdots \quad a_{in}) \in M_{1 \times n}(F)$.

Example. (Matrices over different fields)

Identify the size, the field and a few entries of each matrix:

$$A = \begin{pmatrix} 1 & -2 & 0 \\ 3 & 1 & 4 \end{pmatrix}, \quad B = \begin{pmatrix} i & 1+i \\ 0 & 2-i \end{pmatrix}, \quad C = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}, \quad D = (5).$$

Here C is meant over $\mathbb{F}_2$. Also write out the 2×2 real matrix E with $e_{ij} = i + 2j$.

Solution. $A \in M_{2 \times 3}(\mathbb{R})$, with $a_{12} = -2$ and $a_{23} = 4$; its second column is $\mathbf{a}_2 = (-2, 1)$. $B \in M_2(\mathbb{C})$, with $b_{12} = 1 + i$ and $b_{21} = 0$. $C \in M_3(\mathbb{F}_2)$; every entry is 0 or 1. $D \in M_1(\mathbb{R})$ is the degenerate case, a 1×1 matrix. It behaves exactly like the number 5, and 1×1 matrices are how numbers will appear inside matrix formulas. Finally $e_{11} = 3$, $e_{12} = 5$, $e_{21} = 4$, $e_{22} = 6$, so $E = \begin{pmatrix} 3 & 5 \\ 4 & 6 \end{pmatrix}$. Swapping the roles of i and j would give a different matrix, which is why the order “row, column” matters.

0.9.2 Special matrices

A few shapes of matrix occur so often that they have names. The **main diagonal** of a square matrix $A \in M_n(F)$ consists of the entries $a_{11}, a_{22}, \dots, a_{nn}$.

Definition 0.9.2 — Zero matrix

The **zero matrix** $0 = 0_{m \times n} \in M_{m \times n}(F)$ is the matrix all of whose entries are 0.

Definition 0.9.3 — Identity matrix

The **identity matrix** $I_n \in M_n(F)$ has entries

$$(I_n)_{ij} = \delta_{ij} = \begin{cases} 1 & \text{if } i = j, \\ 0 & \text{if } i \neq j. \end{cases}$$

That is, 1 on the main diagonal and 0 elsewhere. We write I when the size is clear. The symbol δ_{ij} is the **Kronecker delta**.

The columns of I_n are exactly $\mathbf{e}_1, \dots, \mathbf{e}_n$.

Definition 0.9.4 — Diagonal matrix

A square matrix $A \in M_n(F)$ is **diagonal** if $a_{ij} = 0$ **whenever** $i \neq j$. We write $\text{diag}(d_1, \dots, d_n)$ for the diagonal matrix with diagonal entries $d_1, \dots, d_n$.

Definition 0.9.5 — Upper triangular matrix

A square matrix $A \in M_n(F)$ is **upper triangular** if $a_{ij} = 0$ **whenever** $i > j$, that is, every entry **strictly below** the main diagonal is zero. It is **lower triangular** if $a_{ij} = 0$ whenever $i < j$.

Example. (Zero and identity matrices)

Write out $0_{2 \times 3}$, I_2 and I_3 . Which of them are diagonal?

Solution.

$$0_{2 \times 3} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad I_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad I_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

$I_2 = \text{diag}(1, 1)$ and $I_3 = \text{diag}(1, 1, 1)$ are diagonal. The matrix $0_{2 \times 3}$ is **not** diagonal, because the definition only applies to square matrices; the square zero matrix $0_{n \times n} = \text{diag}(0, \dots, 0)$ is diagonal.

Example. (Diagonal and triangular)

Classify each matrix as diagonal, upper triangular, lower triangular, or none of these:

$$P = \begin{pmatrix} 3 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad Q = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{pmatrix}, \quad R = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 7 & 6 \end{pmatrix}.$$

Solution. $P = \text{diag}(3, -1, 0)$ is diagonal. A zero on the diagonal is allowed: the definition only constrains off-diagonal entries. P is also both upper and lower triangular, and in general a matrix is diagonal exactly when it is both. Q is upper triangular but not lower triangular, since $q_{12} = 2 \neq 0$. R differs from Q in one entry, $r_{32} = 7$. All other entries below the diagonal are still zero, but the clause “ $a_{ij} = 0$ whenever $i > j$ ” fails at $(i, j) = (3, 2)$, so R is not upper triangular. It is not lower triangular either.

0.9.3 Addition and scalar multiplication

Matrices of the same size are added, and multiplied by scalars, one entry at a time.

Definition 0.9.6 — Matrix addition

Let $A = (a_{ij})$ and $B = (b_{ij})$ be in $M_{m \times n}(F)$, of the **same size**. Their **sum** is $A + B \in M_{m \times n}(F)$ with $(A + B)_{ij} = a_{ij} + b_{ij}$.

Definition 0.9.7 — Scalar multiplication

Let $A = (a_{ij}) \in M_{m \times n}(F)$ and $c \in F$. The **scalar multiple** $cA \in M_{m \times n}(F)$ has entries $(cA)_{ij} = c a_{ij}$.

Definition 0.9.8 — Additive inverse

The **additive inverse** of $A \in M_{m \times n}(F)$ is $-A = (-1)A$, with entries $-a_{ij}$. We write $A - B = A + (-B)$.

Matrices of different sizes are never added: $\begin{pmatrix} 1 & 2 \end{pmatrix} + \begin{pmatrix} 1 \\ 2 \end{pmatrix}$ is not defined.

Example. (Adding and scaling)

Let $A = \begin{pmatrix} 1 & -2 & 0 \\ 3 & 1 & 4 \end{pmatrix}$ and $B = \begin{pmatrix} 2 & 2 & -1 \\ 0 & -5 & 1 \end{pmatrix}$ in $M_{2 \times 3}(\mathbb{R})$. Compute $A + B$, $3A$ and $A - 2B$.

Solution. Entry by entry,

$$A+B = \begin{pmatrix} 3 & 0 & -1 \\ 3 & -4 & 5 \end{pmatrix}, \quad 3A = \begin{pmatrix} 3 & -6 & 0 \\ 9 & 3 & 12 \end{pmatrix}, \quad A-2B = \begin{pmatrix} 1-4 & -2-4 & 0+2 \\ 3-0 & 1+10 & 4-2 \end{pmatrix} = \begin{pmatrix} -3 & -6 & 2 \\ 3 & 11 & 2 \end{pmatrix}.$$

Because both operations work entry by entry, they inherit the rules of F :

Theorem 0.9.1 — Properties of addition and scalar multiplication

Let $A, B, C \in M_{m \times n}(F)$ and $c, d \in F$. Then:

1. $A + B = B + A$;
2. $(A + B) + C = A + (B + C)$;
3. $A + 0 = A$;
4. $A + (-A) = 0$;
5. $c(A + B) = cA + cB$ and $(c + d)A = cA + dA$;
6. $(cd)A = c(dA)$ and $1A = A$.

Proof. Two matrices of the same size are equal when their entries agree, so it suffices to compare (i, j) -entries. For 1, $(A + B)_{ij} = a_{ij} + b_{ij} = b_{ij} + a_{ij} = (B + A)_{ij}$, where the middle equality is commutativity of addition in F . For 5, $(c(A + B))_{ij} = c(a_{ij} + b_{ij}) = ca_{ij} + cb_{ij} = (cA + cB)_{ij}$ by distributivity in F . Each remaining identity reduces in the same way to the corresponding field axiom of Definition 0.7.1 in each entry. This proves the theorem. ■

These eight rules will reappear in Chapter 1 as the axioms of a vector space: $M_{m \times n}(F)$ is one of our standard examples.

0.9.4 Matrix multiplication

Addition was entrywise. The obvious guess for multiplication is also entrywise, but that product is almost useless. The product that matters is the one that records **substitution**, and it looks very different.

Suppose new variables y_1, y_2 are given in terms of x_1, x_2 , and z_1, z_2 in terms of y_1, y_2 :

$$\begin{aligned} y_1 &= 2x_1 + x_2 & z_1 &= y_1 + 2y_2 \\ y_2 &= x_1 + 4x_2 & z_2 &= 3y_1 - y_2 \end{aligned}$$

Record each substitution by its table of coefficients: $B = \begin{pmatrix} 2 & 1 \\ 1 & 4 \end{pmatrix}$ for $\mathbf{y}$ in terms of $\mathbf{x}$, and

$A = \begin{pmatrix} 1 & 2 \\ 3 & -1 \end{pmatrix}$ for $\mathbf{z}$ in terms of $\mathbf{y}$. Substituting,

$$\begin{aligned} z_1 &= (2x_1 + x_2) + 2(x_1 + 4x_2) = (1 \cdot 2 + 2 \cdot 1)x_1 + (1 \cdot 1 + 2 \cdot 4)x_2 = 4x_1 + 9x_2, \\ z_2 &= 3(2x_1 + x_2) - (x_1 + 4x_2) = (3 \cdot 2 + (-1) \cdot 1)x_1 + (3 \cdot 1 + (-1) \cdot 4)x_2 = 5x_1 - x_2. \end{aligned}$$

So $\mathbf{z}$ in terms of $\mathbf{x}$ has coefficient table $\begin{pmatrix} 4 & 9 \\ 5 & -1 \end{pmatrix}$. Look at how each entry arose: the coefficient

of x_k in z_i is $a_{i1}b_{1k} + a_{i2}b_{2k}$, row i of A against column k of B . The entrywise product $\begin{pmatrix} 2 & 2 \\ 3 & -4 \end{pmatrix}$ has nothing to do with the substitution. This forces the definition.

To multiply matrices, run each row of the first against each column of the second: the product of two matrices is the table of the combined substitution.

Definition 0.9.9 — Matrix multiplication

Let $A = (a_{ij}) \in M_{m \times n}(F)$ and $B = (b_{jk}) \in M_{n \times p}(F)$, so that **the number of columns of A equals the number of rows of B** . Their **product** is the matrix $AB \in M_{m \times p}(F)$ with entries

$$(AB)_{ik} = \sum_{j=1}^n a_{ij}b_{jk} = a_{i1}b_{1k} + a_{i2}b_{2k} + \cdots + a_{in}b_{nk}$$

for $1 \leq i \leq m$ and $1 \leq k \leq p$. If the sizes do not match, AB is **not defined**.

In words: to get the (i, k) -entry of AB , walk along row i of A and down column k of B at the same time, multiply the entries you meet in pairs, and add. The size condition says that the row and the column have the same length n , so the pairing makes sense. The product has as many rows as A and as many columns as B :

$$\begin{matrix} A & B \\ m \times n & n \times p \end{matrix} = \begin{matrix} AB \\ m \times p \end{matrix}.$$

In particular, for $A \in M_{m \times n}(F)$ and a column $\mathbf{x} \in F^n$, the product $A\mathbf{x} \in F^m$ is defined, with i -th entry $a_{i1}x_1 + \cdots + a_{in}x_n$. A system of m linear equations in n unknowns is therefore a single equation $A\mathbf{x} = \mathbf{b}$.

Example. (A product of square matrices)

Let $A = \begin{pmatrix} 2 & -1 \\ 0 & 3 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 4 \\ -2 & 1 \end{pmatrix}$ in $M_2(\mathbb{R})$. Compute AB and BA .

Solution. Row i of A against column k of B :

$$AB = \begin{pmatrix} 2 \cdot 1 + (-1)(-2) & 2 \cdot 4 + (-1) \cdot 1 \\ 0 \cdot 1 + 3 \cdot (-2) & 0 \cdot 4 + 3 \cdot 1 \end{pmatrix} = \begin{pmatrix} 4 & 7 \\ -6 & 3 \end{pmatrix}, \quad BA = \begin{pmatrix} 1 \cdot 2 + 4 \cdot 0 & 1 \cdot (-1) + 4 \cdot 3 \\ -2 \cdot 2 + 1 \cdot 0 & (-2)(-1) + 1 \cdot 3 \end{pmatrix} = \begin{pmatrix} 2 & 11 \\ -4 & 5 \end{pmatrix}$$

Both products are defined, and they are different.

Example. (Products of rectangular matrices)

Let $A = \begin{pmatrix} 1 & 0 & 2 \\ -1 & 3 & 1 \end{pmatrix} \in M_{2 \times 3}(\mathbb{R})$ and $B = \begin{pmatrix} 3 & 1 \\ 2 & 1 \\ 1 & 0 \end{pmatrix} \in M_{3 \times 2}(\mathbb{R})$. Determine the sizes of

AB and BA and compute them. Then compute $\begin{pmatrix} 1 & 2 \end{pmatrix} \begin{pmatrix} 3 \\ 4 \end{pmatrix}$ and $\begin{pmatrix} 3 \\ 4 \end{pmatrix} \begin{pmatrix} 1 & 2 \end{pmatrix}$.

Solution. AB is $(2 \times 3)(3 \times 2)$, so it is 2×2 ; BA is $(3 \times 2)(2 \times 3)$, so it is 3×3 . Computing row against column,

$$AB = \begin{pmatrix} 3+0+2 & 1+0+0 \\ -3+6+1 & -1+3+0 \end{pmatrix} = \begin{pmatrix} 5 & 1 \\ 4 & 2 \end{pmatrix}, \quad BA = \begin{pmatrix} 2 & 3 & 7 \\ 1 & 3 & 5 \\ 1 & 0 & 2 \end{pmatrix}.$$

For instance $(BA)_{13} = 3 \cdot 2 + 1 \cdot 1 = 7$. The last two products show the extreme cases: a row times a column is 1×1 , namely $(1 \cdot 3 + 2 \cdot 4) = (11)$, while a column times a row is 2×2 , namely $\begin{pmatrix} 3 & 6 \\ 4 & 8 \end{pmatrix}$.

Here is a non-example by minimal change. Take the same six entries as B , but arrange them in two rows: $C = \begin{pmatrix} 3 & 2 & 1 \\ 1 & 1 & 0 \end{pmatrix} \in M_{2 \times 3}(\mathbb{R})$. The entries are still real numbers and A is unchanged. But A has 3 columns and C has 2 rows, so forming AC would pair a row of length 3 with a column of length 2. The clause “**the number of columns of A equals the number of rows of B** ” fails, and AC is **not defined**.

Why this definition, and not the entrywise one? Because it is the only rule under which “the table of a combined substitution is the product of the tables”. In Chapter 3 the same fact becomes: the matrix of a composition of linear maps is the product of their matrices. Every property of matrix multiplication, good or bad, comes from this.

0.9.5 Three views of a product

The entry formula is how one computes a product by hand, but it hides the structure. Two other descriptions, by columns and by rows, are what we will actually think with. It turns out that multiplying by A on the left acts on each column of B separately, and multiplying by B on the right acts on each row of A separately:

Theorem 0.9.2 — Three views of a product

Let $A \in M_{m \times n}(F)$ and $B \in M_{n \times p}(F)$, and write $B = (\mathbf{b}_1 \ \cdots \ \mathbf{b}_p)$ with columns $\mathbf{b}_k \in F^n$. Then:

1. **(Entries)** $(AB)_{ik} = \sum_{j=1}^n a_{ij}b_{jk}$ for all i, k ;
2. **(Columns)** the k -th column of AB is $A\mathbf{b}_k$, that is,

$$AB = (A\mathbf{b}_1 \ A\mathbf{b}_2 \ \cdots \ A\mathbf{b}_p);$$

3. **(Rows)** the i -th row of AB is (row i of A) B .

Proof. Part 1 is [Definition 0.9.9](#). For part 2, fix k . The column $\mathbf{b}_k$ has entries $b_{1k}, \dots, b_{nk}$, so by [Definition 0.9.9](#) the i -th entry of $A\mathbf{b}_k \in F^m$ is $\sum_{j=1}^n a_{ij}b_{jk}$. By part 1 this is $(AB)_{ik}$, the i -th entry of the k -th column of AB . Hence the k -th column of AB is $A\mathbf{b}_k$.

For part 3, fix i , and let $\mathbf{r} = (a_{i1} \ \cdots \ a_{in})$ be row i of A . Then $\mathbf{r}B$ is a $1 \times p$ matrix whose k -th entry is $\sum_{j=1}^n a_{ij}b_{jk} = (AB)_{ik}$, again by [Definition 0.9.9](#). This is the k -th entry of row i of AB . This proves the theorem.

The column view reduces every matrix product to products $A\mathbf{x}$ of a matrix with a single column. So it pays to understand $A\mathbf{x}$ well, and here is the fact to understand:

Theorem 0.9.3 — $A\mathbf{x}$ is a combination of the columns of A

Let $A = (\mathbf{a}_1 \ \cdots \ \mathbf{a}_n) \in M_{m \times n}(F)$ with columns $\mathbf{a}_j \in F^m$, and let $\mathbf{x} = (x_1, \dots, x_n) \in F^n$. Then

$$A\mathbf{x} = x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n.$$

In particular $A\mathbf{e}_j = \mathbf{a}_j$ for each j . Similarly, if $\mathbf{r}_1, \dots, \mathbf{r}_m$ are the rows of A and $\mathbf{y} = (y_1 \ \cdots \ y_m)$ is a row vector, then $\mathbf{y}A = y_1\mathbf{r}_1 + \cdots + y_m\mathbf{r}_m$.

Proof. Both sides are in F^m , so we compare i -th entries. By Definition 0.9.9, the i -th entry of $A\mathbf{x}$ is $\sum_{j=1}^n a_{ij}x_j$. On the right, the i -th entry of $x_j\mathbf{a}_j$ is $x_j a_{ij}$ by Definition 0.9.7, so by Definition 0.9.6 the i -th entry of $\sum_j x_j\mathbf{a}_j$ is $\sum_{j=1}^n x_j a_{ij}$. These agree because multiplication in F is commutative. Taking $\mathbf{x} = \mathbf{e}_j$, whose only non-zero entry is $x_j = 1$, gives $A\mathbf{e}_j = \mathbf{a}_j$. The statement for rows follows in the same way, comparing the k -th entries $\sum_i y_i a_{ik}$. This proves the theorem. ■

Read the formula as a sentence: **the entries of $\mathbf{x}$ are the weights, and the columns of A are what gets weighted.** So $A\mathbf{x} = \mathbf{b}$ has a solution exactly when $\mathbf{b}$ can be built from the columns of A . This single observation drives much of Chapters 2 and 3. Combined with Theorem 0.9.2, it also says that **every column of AB is a combination of the columns of A , with weights taken from the corresponding column of B , and every row of AB is a combination of the rows of B .**

Example. (One product, three ways)

Let

$$A = \begin{pmatrix} 1 & 2 \\ 3 & -1 \\ 0 & 4 \end{pmatrix} \in M_{3 \times 2}(\mathbb{R}), \quad B = \begin{pmatrix} 2 & 1 \\ -1 & 1 \end{pmatrix} \in M_2(\mathbb{R}), \quad \mathbf{x} = \begin{pmatrix} 2 \\ -1 \end{pmatrix}.$$

Compute $A\mathbf{x}$ as a combination of the columns of A . Then compute AB by columns, check one row by the row view, and check one entry by the entry formula.

Solution. The columns of A are $\mathbf{a}_1 = (1, 3, 0)$ and $\mathbf{a}_2 = (2, -1, 4)$. By Theorem 0.9.3,

$$A\mathbf{x} = 2\mathbf{a}_1 - \mathbf{a}_2 = \begin{pmatrix} 2 \\ 6 \\ 0 \end{pmatrix} - \begin{pmatrix} 2 \\ -1 \\ 4 \end{pmatrix} = \begin{pmatrix} 0 \\ 7 \\ -4 \end{pmatrix}.$$

Columns. The first column of B is $\mathbf{b}_1 = (2, -1) = \mathbf{x}$, so the first column of AB is $A\mathbf{b}_1 = (0, 7, -4)$, just computed. The second column is $\mathbf{b}_2 = (1, 1)$, so $A\mathbf{b}_2 = \mathbf{a}_1 + \mathbf{a}_2 =$

$(3, 2, 4)$. Hence

$$AB = \begin{pmatrix} 0 & 3 \\ 7 & 2 \\ -4 & 4 \end{pmatrix}.$$

Rows. Row 2 of A is $(3 \ -1)$, and by [Theorem 0.9.3](#) (rows version) $(3 \ -1)B = 3(2 \ 1) - (-1 \ 1) = (7 \ 2)$, which is row 2 of AB .

Entries. $(AB)_{32} = a_{31}b_{12} + a_{32}b_{22} = 0 \cdot 1 + 4 \cdot 1 = 4$, as in the answer.

The column view needed nothing beyond scaling and adding columns of A . When the columns of B are simple, it is usually the fastest route.

Quick check. Without multiplying out, what is the second column of $\begin{pmatrix} 5 & -1 & 7 \\ 2 & 0 & 3 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 0 \\ 4 & 1 \end{pmatrix}$?

Solution. The second column of the right factor is $\mathbf{e}_3 = (0, 0, 1)$. By [Theorem 0.9.2](#) and [Theorem 0.9.3](#), the second column of the product is $A\mathbf{e}_3$, the third column of the left factor: $(7, 3)$.

0.9.6 The rules of multiplication

Matrix multiplication keeps the most important rules of arithmetic: associativity, both distributive laws, and an identity.

Theorem 0.9.4 — Properties of matrix multiplication

Let $c \in F$, and let $A, A' \in M_{m \times n}(F)$, $B, B' \in M_{n \times p}(F)$ and $C \in M_{p \times q}(F)$. Then:

1. **(Associativity)** $(AB)C = A(BC)$;
2. **(Distributivity)** $A(B + B') = AB + AB'$ and $(A + A')B = AB + A'B$;
3. **(Identity)** $I_m A = A = A I_n$;
4. **(Scalars)** $c(AB) = (cA)B = A(cB)$;
5. **(Zero)** $0_{\ell \times m} A = 0_{\ell \times n}$ and $A 0_{n \times r} = 0_{m \times r}$ for all $\ell, r \geq 1$.

Idea. Each identity is checked entry by entry. The only one with content is associativity. Both sides have (i, l) -entry equal to the double sum $\sum_j \sum_k a_{ij} b_{jk} c_{kl}$, added up in two different orders, so the proof is “expand, swap the order of the finite sums, and fold back up”. In the substitution picture it is no surprise: substituting in two stages gives the same result whichever pair of stages we combine first.

Proof. For 1, both sides lie in $M_{m \times q}(F)$. Fix i and l . By [Definition 0.9.9](#) applied twice,

$$((AB)C)_{il} = \sum_{k=1}^p (AB)_{ik} c_{kl} = \sum_{k=1}^p \left(\sum_{j=1}^n a_{ij} b_{jk} \right) c_{kl} = \sum_{k=1}^p \sum_{j=1}^n a_{ij} b_{jk} c_{kl},$$

where the last step uses distributivity in F . The same computation gives

$$(A(BC))_{il} = \sum_{j=1}^n a_{ij}(BC)_{jl} = \sum_{j=1}^n a_{ij} \left(\sum_{k=1}^p b_{jk}c_{kl} \right) = \sum_{j=1}^n \sum_{k=1}^p a_{ij}b_{jk}c_{kl}.$$

The two double sums add the same np terms in different orders, so they are equal, by commutativity and associativity of addition in F . Hence $(AB)C = A(BC)$.

For 2, $(A(B + B'))_{ik} = \sum_j a_{ij}(b_{jk} + b'_{jk}) = \sum_j a_{ij}b_{jk} + \sum_j a_{ij}b'_{jk} = (AB + AB')_{ik}$ by distributivity in F ; the other law is the same with the roles of the factors swapped.

For 3, $(I_m A)_{ik} = \sum_{j=1}^m \delta_{ij}a_{jk} = a_{ik}$, because by [Definition 0.9.3](#) only the term $j = i$ survives. Likewise $(AI_n)_{ik} = \sum_j a_{ij}\delta_{jk} = a_{ik}$.

For 4, $(c(AB))_{ik} = c \sum_j a_{ij}b_{jk} = \sum_j (ca_{ij})b_{jk} = ((cA)B)_{ik}$, and moving c next to b_{jk} instead, using commutativity in F , gives $(A(cB))_{ik}$. For 5, every term $0 \cdot a_{jk}$ or $a_{ij} \cdot 0$ is 0 by [Theorem 0.7.1](#). This proves the theorem. ■

Associativity means we may write ABC without brackets, and more generally a product of any number of matrices, as long as consecutive sizes match.

Just as important is what is **missing** from the list. There is no commutative law, and no cancellation law. Each of the following failures will cost you a wrong proof if you forget it.

Warning. $AB \neq BA$ in general. Take $A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$. Then $AB = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ but $BA = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$. Before swapping two factors in a proof, you must know that they commute. (And if A is 2×3 and B is 3×2 , then AB and BA do not even have the same size.)

Warning. A product can be zero with both factors non-zero. In $M_2(\mathbb{R})$,

$$\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

So $M_n(F)$ is not a field for $n \geq 2$, and “ $AB = 0$ implies $A = 0$ or $B = 0$ ” is false.

Warning. Cancellation fails: $AC = BC$ does not imply $A = B$. With

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 2 \\ 3 & 9 \end{pmatrix}, \quad C = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix},$$

we get $AC = BC = \begin{pmatrix} 1 & 1 \\ 3 & 3 \end{pmatrix}$ although $A \neq B$. By the column view, both columns of C are $\mathbf{e}_1$, so AC and BC only see the first columns of A and B , which agree. Canceling C is legal only when C is invertible, as we will see below.

Quick check. Find a non-zero $A \in M_2(\mathbb{R})$ with $A^2 = AA = 0$.

Solution. Take $A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$. By the column view, $A\mathbf{e}_1 = \mathbf{0}$ and $A\mathbf{e}_2 = \mathbf{e}_1$, so the columns of A^2 are $A\mathbf{0} = \mathbf{0}$ and $A\mathbf{e}_1 = \mathbf{0}$. Hence $A^2 = \mathbf{0}$ although $A \neq \mathbf{0}$. This small matrix is the source of a remarkable number of counterexamples in this book; it will reappear often.

0.9.7 The transpose

Rows and columns play symmetric roles in a matrix, and the transpose exchanges them.

Definition 0.9.10 — Transpose

Let $A = (a_{ij}) \in M_{m \times n}(F)$. The **transpose** of A is the matrix $A^T \in M_{n \times m}(F)$ with entries

$$(A^T)_{ij} = a_{ji}.$$

In words: row i of A becomes column i of A^T , and the size flips from $m \times n$ to $n \times m$. A square matrix is reflected in its main diagonal.

Definition 0.9.11 — Symmetric and skew-symmetric matrices

A square matrix $A \in M_n(F)$ is **symmetric** if $A^T = A$, that is, $a_{ij} = a_{ji}$ for all i, j . It is **skew-symmetric** if $A^T = -A$, that is, $a_{ij} = -a_{ji}$ for all i, j .

Example. (Transposes, symmetric and skew-symmetric matrices)

Compute the transpose of $A = \begin{pmatrix} 1 & -2 & 0 \\ 3 & 1 & 4 \end{pmatrix}$. Decide whether

$$S = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{pmatrix}, \quad K = \begin{pmatrix} 0 & 2 & -1 \\ -2 & 0 & 3 \\ 1 & -3 & 0 \end{pmatrix}, \quad S' = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 6 & 6 \end{pmatrix}$$

over $\mathbb{R}$ are symmetric or skew-symmetric.

Solution. The rows of A become columns: $A^T = \begin{pmatrix} 1 & 3 \\ -2 & 1 \\ 0 & 4 \end{pmatrix} \in M_{3 \times 2}(\mathbb{R})$.

S is symmetric: $s_{12} = s_{21} = 2$, $s_{13} = s_{31} = 3$, $s_{23} = s_{32} = 5$. K is skew-symmetric: $k_{12} = 2 = -k_{21}$, $k_{13} = -1 = -k_{31}$, $k_{23} = 3 = -k_{32}$, and each diagonal entry satisfies $k_{ii} = -k_{ii}$, so it is 0. The matrix S' differs from S only in the entry $(3, 2)$. Every other symmetric pair still matches, but $s'_{23} = 5 \neq 6 = s'_{32}$, so S' is **not** symmetric. A 1×1 matrix is always symmetric, and the identity I_n and every diagonal matrix are symmetric.

Over $\mathbb{R}$, a skew-symmetric matrix has zero diagonal, since $a_{ii} = -a_{ii}$ forces $2a_{ii} = 0$. Over

$\mathbb{F}_2$, where $-1 = 1$, “skew-symmetric” and “symmetric” mean the same thing, and I_2 is skew-symmetric. The characteristic of the field (Definition 0.7.2) matters.

The transpose is compatible with the operations, except that it **reverses** products:

Theorem 0.9.5 — Properties of the transpose

Let $c \in F$. For matrices of sizes for which each expression is defined:

1. $(A^\top)^\top = A$;
2. $(A + B)^\top = A^\top + B^\top$;
3. $(cA)^\top = cA^\top$;
4. $(AB)^\top = B^\top A^\top$ for $A \in M_{m \times n}(F)$ and $B \in M_{n \times p}(F)$.

Idea. The order reverses for a size reason before any computation: $(AB)^\top$ is $p \times m$, and $B^\top A^\top$ is $(p \times n)(n \times m) = p \times m$, while $A^\top B^\top$ is $(n \times m)(p \times n)$, usually not even defined. Then compare entries.

Proof. For 1, $((A^\top)^\top)_{ij} = (A^\top)_{ji} = a_{ij}$ by Definition 0.9.10. For 2 and 3, $((A + B)^\top)_{ij} = a_{ji} + b_{ji} = (A^\top + B^\top)_{ij}$ and $((cA)^\top)_{ij} = ca_{ji} = (cA^\top)_{ij}$.

For 4, both sides lie in $M_{p \times m}(F)$. Fix $1 \leq i \leq p$ and $1 \leq k \leq m$. By Definition 0.9.10 and Definition 0.9.9,

$$((AB)^\top)_{ik} = (AB)_{ki} = \sum_{j=1}^n a_{kj} b_{ji}, \quad (B^\top A^\top)_{ik} = \sum_{j=1}^n (B^\top)_{ij} (A^\top)_{jk} = \sum_{j=1}^n b_{ji} a_{kj}.$$

The two sums agree term by term because multiplication in F is commutative. Hence $(AB)^\top = B^\top A^\top$. This proves the theorem. ■

For example, for every $A \in M_{m \times n}(F)$, the square matrix $A^\top A$ is symmetric: by parts 4 and 1, $(A^\top A)^\top = A^\top (A^\top)^\top = A^\top A$. Matrices of this form are central to least squares and the singular value decomposition much later on.

0.9.8 The trace

The trace is the simplest number attached to a square matrix, and it has one surprising property.

Definition 0.9.12 — Trace

The **trace** of a square matrix $A = (a_{ij}) \in M_n(F)$ is the sum of its diagonal entries:

$$\operatorname{tr} A = \sum_{i=1}^n a_{ii} = a_{11} + a_{22} + \cdots + a_{nn} \in F.$$

Example. (Computing traces)

Compute $\text{tr } A$ for $A = \begin{pmatrix} 2 & 7 & -1 \\ 0 & -3 & 5 \\ 4 & 1 & 6 \end{pmatrix} \in M_3(\mathbb{R})$, and $\text{tr } I_n$ over $\mathbb{R}$ and over $\mathbb{F}_2$.

Solution. $\text{tr } A = 2 + (-3) + 6 = 5$; the off-diagonal entries play no role. $\text{tr } I_n = 1 + \cdots + 1 = n \cdot 1$. Over $\mathbb{R}$ this is n . Over $\mathbb{F}_2$ it is 0 when n is even and 1 when n is odd. The trace is **not** defined for a non-square matrix.

Theorem 0.9.6 — Properties of the trace

Let $A, B \in M_n(F)$ and $c \in F$. Then:

1. $\text{tr}(A + B) = \text{tr } A + \text{tr } B$ and $\text{tr}(cA) = c \text{tr } A$;
2. $\text{tr}(A^T) = \text{tr } A$;
3. $\text{tr}(AB) = \text{tr}(BA)$. More generally, this holds for all $A \in M_{m \times n}(F)$ and $B \in M_{n \times m}(F)$, where AB is $m \times m$ and BA is $n \times n$.

Idea. Part 3 is surprising because $AB \neq BA$ in general. But the trace only sees the diagonal, and $\text{tr}(AB) = \sum_i \sum_j a_{ij} b_{ji}$ is a sum over all pairs (i, j) of the product of a_{ij} with the “mirror” entry b_{ji} . That description does not care which matrix is written first.

Proof. Parts 1 and 2 follow from the definitions: the diagonal entries of $A + B$, cA and A^T are $a_{ii} + b_{ii}$, ca_{ii} and a_{ii} .

For 3, let $A \in M_{m \times n}(F)$ and $B \in M_{n \times m}(F)$. By [Definition 0.9.12](#) and [Definition 0.9.9](#),

$$\text{tr}(AB) = \sum_{i=1}^m (AB)_{ii} = \sum_{i=1}^m \sum_{j=1}^n a_{ij} b_{ji} = \sum_{j=1}^n \sum_{i=1}^m b_{ji} a_{ij} = \sum_{j=1}^n (BA)_{jj} = \text{tr}(BA),$$

where the middle equality swaps the order of the finite double sum and uses commutativity of multiplication in F . This proves the theorem. ■

A first payoff: for all $A, B \in M_n(F)$, the matrix $AB - BA$ has trace $\text{tr}(AB) - \text{tr}(BA) = 0$. So over $\mathbb{R}$ it can never equal I_n , whose trace is n ([Exercise \(C1: \$AB - BA = I\$ is impossible, or is it?\)](#)). The trace will also turn out to be one of the quantities that does not change when we change coordinates.

It is tempting to conclude that the trace of a product ignores the order of **all** the factors. It does not.

Warning. $\text{tr}(ABC) \neq \text{tr}(BAC)$ **in general**. Take $A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$, $B = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ and $C = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$. Then $ABC = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ has trace 1, but $BA = 0$, so $BAC = 0$ has trace 0. What **is** true is that a **cyclic** rotation preserves the trace: $\text{tr}(ABC) = \text{tr}((AB)C) = \text{tr}(C(AB)) = \text{tr}(CAB)$ by part 3.

0.9.9 Inverse matrices

In a field we solve $ax = b$ by multiplying by a^{-1} , and this works exactly when $a \neq 0$. We would like to solve a system $A\mathbf{x} = \mathbf{b}$ the same way, by “dividing by A ”. What should that mean, and for which A is it possible?

The naive answer, “every non-zero matrix”, fails. Let $N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \neq 0$. For **any** $B \in M_2(F)$, the first column of BN is B times the first column of N , that is $B\mathbf{0}$, by [Theorem 0.9.2](#). The first column of I_2 is $\mathbf{e}_1 \neq \mathbf{0}$, so $BN \neq I_2$: no matrix undoes N from the left. Dividing needs a genuine two-sided inverse, and not every non-zero matrix has one.

An invertible matrix is a square matrix whose effect can be undone by multiplying, on either side.

Definition 0.9.13 — Invertible matrix

A **square** matrix $A \in M_n(F)$ is **invertible** if there exists $B \in M_n(F)$ such that

$$AB = I_n \quad \text{and} \quad BA = I_n.$$

Such a B is called **an inverse** of A . A square matrix that is not invertible is called **singular**.

In words: A is invertible when some matrix B of the same size cancels it **from the right** ($AB = I_n$) **and** from the left ($BA = I_n$). The definition asks for one B doing both jobs, and it only speaks about square matrices.

Is “an inverse” really “the inverse”? Yes: an inverse, if it exists, is unique. This is part 1 of the next theorem, and it lets us write A^{-1} . The remaining parts are what inverses are for.

Theorem 0.9.7 — Properties of inverses

Let $A, B \in M_n(F)$.

1. **(Uniqueness)** If B and B' are both inverses of A , then $B = B'$. We write A^{-1} for **the** inverse of an invertible A .
2. If A is invertible, then so is A^{-1} , and $(A^{-1})^{-1} = A$.
3. If A and B are invertible, then so is AB , and $(AB)^{-1} = B^{-1}A^{-1}$.
4. If A is invertible, then so is A^T , and $(A^T)^{-1} = (A^{-1})^T$.
5. If A is invertible and $c \in F$ is **non-zero**, then cA is invertible, and $(cA)^{-1} = c^{-1}A^{-1}$.
6. **(Cancellation)** If A is invertible, then $AX = AY$ implies $X = Y$ for all $X, Y \in M_{n \times p}(F)$, and $XA = YA$ implies $X = Y$ for all $X, Y \in M_{p \times n}(F)$.
7. **(Solving)** If A is invertible, then for every $\mathbf{b} \in F^n$ the equation $A\mathbf{x} = \mathbf{b}$ has **exactly one** solution $\mathbf{x} \in F^n$, namely $\mathbf{x} = A^{-1}\mathbf{b}$.

Idea. Uniqueness is the standard trick for “prove unique by assuming two”: sandwich B' between B and A , and use associativity to collapse the product both ways. For the other parts the candidate inverse is written in the statement, so we only multiply it out on **both** sides, as the definition demands. The order in part 3 is the “socks and shoes”

rule: to undo “first B , then A ”, undo A first.

Proof. For 1, suppose $AB = BA = I_n$ and $AB' = B'A = I_n$. By [Theorem 0.9.4](#) (identity and associativity),

$$B = BI_n = B(AB') = (BA)B' = I_n B' = B'.$$

For 2, the equations $AA^{-1} = I_n$ and $A^{-1}A = I_n$ say precisely that A is an inverse of A^{-1} . By part 1 it is the inverse.

For 3, by associativity,

$$(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = AI_n A^{-1} = AA^{-1} = I_n,$$

and similarly $(B^{-1}A^{-1})(AB) = B^{-1}(A^{-1}A)B = B^{-1}B = I_n$. Hence $B^{-1}A^{-1}$ is an inverse of AB , and by part 1 it is $(AB)^{-1}$.

For 4, by [Theorem 0.9.5](#), $A^T(A^{-1})^T = (A^{-1}A)^T = I_n^T = I_n$ and $(A^{-1})^T A^T = (AA^{-1})^T = I_n$, where $I_n^T = I_n$ since I_n is symmetric. Hence $(A^{-1})^T$ is the inverse of A^T .

For 5, since $c \neq 0$, the inverse c^{-1} exists in F . By [Theorem 0.9.4](#) (scalars), $(cA)(c^{-1}A^{-1}) = (cc^{-1})(AA^{-1}) = I_n$, and similarly in the other order.

For 6, suppose $AX = AY$. Multiplying on the left by A^{-1} and using associativity, $X = I_n X = (A^{-1}A)X = A^{-1}(AX) = A^{-1}(AY) = (A^{-1}A)Y = Y$. The statement for $XA = YA$ follows by multiplying on the right by A^{-1} .

For 7, $\mathbf{x} = A^{-1}\mathbf{b}$ is a solution, since $A(A^{-1}\mathbf{b}) = (AA^{-1})\mathbf{b} = \mathbf{b}$. If $\mathbf{x}$ is any solution, then $A\mathbf{x} = \mathbf{b} = A(A^{-1}\mathbf{b})$, and part 6 (with $p = 1$) gives $\mathbf{x} = A^{-1}\mathbf{b}$. This proves the theorem. ■

Part 6 is the repair of the failed cancellation law from the warnings above: **you may cancel an invertible matrix**, on the side where it stands, and nothing else.

Example. (Some invertible matrices)

Show that each of the following is invertible and find its inverse: I_n ; a diagonal matrix $D = \text{diag}(d_1, \dots, d_n)$ with **all** $d_i \neq 0$; $A = \begin{pmatrix} 2 & 1 \\ 5 & 3 \end{pmatrix} \in M_2(\mathbb{R})$; and $U = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \in M_2(\mathbb{F}_2)$.

Solution. $I_n I_n = I_n$, so I_n is its own inverse. This is the degenerate case: I_n plays the role that 1 plays among numbers.

For D , let $E = \text{diag}(d_1^{-1}, \dots, d_n^{-1})$, which exists since every $d_i \neq 0$. Then $(DE)_{ik} = \sum_j d_{ij} e_{jk}$, and only $j = i = k$ can contribute, so $DE = \text{diag}(d_1 d_1^{-1}, \dots, d_n d_n^{-1}) = I_n$; likewise $ED = I_n$. Hence $D^{-1} = E$. For $n = 1$ this says the 1×1 matrix (a) is invertible when $a \neq 0$, with inverse (a^{-1}) , exactly as in the field.

For A , let $B = \begin{pmatrix} 3 & -1 \\ -5 & 2 \end{pmatrix}$. Then

$$AB = \begin{pmatrix} 6-5 & -2+2 \\ 15-15 & -5+6 \end{pmatrix} = I_2, \quad BA = \begin{pmatrix} 6-5 & 3-3 \\ -10+10 & -5+6 \end{pmatrix} = I_2,$$

so $A^{-1} = B$. (Where B came from is explained by [Theorem 0.9.8](#) below.)

For U over $\mathbb{F}_2$, $U^2 = \begin{pmatrix} 1 & 1+1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, since $1+1=0$. So U is its own inverse.

Over $\mathbb{R}$ the same array would have inverse $\begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}$: the answer depends on the field.

Now a non-example by minimal change. The matrix $\text{diag}(1, 2)$ is invertible, with inverse $\text{diag}(1, \frac{1}{2})$. Change the entry 2 to 0: $A = \text{diag}(1, 0)$ is still diagonal and still non-zero. But for any $B \in M_2(F)$, the second column of BA is B times the second column of A , which is $B\mathbf{0} = \mathbf{0}$, while the second column of I_2 is $\mathbf{e}_2$. So $BA \neq I_2$ for every B : the clause “**there exists** B with $\dots BA = I_n$ ” fails, and $\text{diag}(1, 0)$ is singular. The same argument shows that any square matrix with a zero column is singular, and in particular so is every zero matrix $0_{n \times n}$.

Why does the definition insist on a **square** matrix and on **both** equations? For non-square matrices one-sided inverses can exist without the other side: with $C = \begin{pmatrix} 1 & 0 \end{pmatrix} \in M_{1 \times 2}(\mathbb{R})$ and $D = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, we get $CD = (1) = I_1$ but $DC = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \neq I_2$. Demanding both equations makes the inverse unique (part 1) and makes both cancellation laws (part 6) hold.

Warning. For square matrices, $AB = I_n$ **already implies** $BA = I_n$. **This is true, but it is not obvious, and we do not know it yet.** Nothing in this chapter proves it: the entries of AB and BA are different sums. It is proved in Chapter 2, using row reduction. Until then, to show that B is the inverse of A , check **both** $AB = I_n$ and $BA = I_n$. (The non-square example above shows that some argument is needed.)

For 2×2 matrices, invertibility is decided by one number.

Theorem 0.9.8 — Inverse of a 2×2 matrix

Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in M_2(F)$. Then A is invertible **if and only if** $ad - bc \neq 0$. In that case,

$$A^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

Idea. Where does the formula come from? Try to kill the off-diagonal entries of A by multiplying with a matrix built from its own entries. Swapping a and d and negating b and c gives $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} = (ad - bc)I_2$. If $ad - bc \neq 0$, divide by it. If $ad - bc = 0$, the same computation shows that A sends a non-zero column to $\mathbf{0}$, and an invertible matrix cannot do that, since its inverse would have to send back to a non-zero column.

Proof. Write $\Delta = ad - bc$ and $\tilde{A} = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$. By [Definition 0.9.9](#) and commutativity

of multiplication in F ,

$$A\tilde{A} = \begin{pmatrix} ad - bc & -ab + ba \\ cd - dc & -cb + da \end{pmatrix} = \Delta I_2, \quad \tilde{A}A = \begin{pmatrix} da - bc & db - bd \\ -ca + ac & -cb + ad \end{pmatrix} = \Delta I_2.$$

($\Leftarrow$) Suppose $\Delta \neq 0$, so Δ^{-1} exists in F . By [Theorem 0.9.4](#) (scalars), $A(\Delta^{-1}\tilde{A}) = \Delta^{-1}(A\tilde{A}) = \Delta^{-1}\Delta I_2 = I_2$, and likewise $(\Delta^{-1}\tilde{A})A = I_2$. Hence A is invertible with $A^{-1} = \Delta^{-1}\tilde{A}$, which is the stated formula.

($\Rightarrow$) We prove the contrapositive. Suppose $\Delta = 0$. We first find $\mathbf{x} \in F^2$ with $\mathbf{x} \neq \mathbf{0}$ and $A\mathbf{x} = \mathbf{0}$.

Case 1. $A = 0$. Take $\mathbf{x} = \mathbf{e}_1$.

Case 2. $c \neq 0$ or $d \neq 0$. Take $\mathbf{x} = (d, -c) \neq \mathbf{0}$. Then $A\mathbf{x} = (ad - bc, cd - dc) = (\Delta, 0) = \mathbf{0}$.

Case 3. $c = d = 0$ but $a \neq 0$ or $b \neq 0$. Take $\mathbf{x} = (b, -a) \neq \mathbf{0}$. Then $A\mathbf{x} = (ab - ba, cb - da) = (0, 0) = \mathbf{0}$, since $c = d = 0$.

These cases cover every A . Now suppose, for a contradiction, that A is invertible. Then by [Theorem 0.9.4](#),

$$\mathbf{x} = I_2\mathbf{x} = (A^{-1}A)\mathbf{x} = A^{-1}(A\mathbf{x}) = A^{-1}\mathbf{0} = \mathbf{0},$$

contradicting $\mathbf{x} \neq \mathbf{0}$. Hence A is not invertible. This proves the theorem. ■

The number $ad - bc$ will return in Chapter 6 as the determinant of A . Here it arose from a single multiplication, and the proof used nothing beyond the field axioms, so the theorem holds over every field. For instance, $\begin{pmatrix} 2 & 1 \\ 5 & 3 \end{pmatrix}$ has $ad - bc = 6 - 5 = 1$, which explains the inverse found in [Example \(Some invertible matrices\)](#); and $\begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$ has $ad - bc = 0$, so it is singular, with $A(2, -1) = \mathbf{0}$.

0.9.10 Powers and polynomials of a matrix

A square matrix can be multiplied by itself, so we can form powers, and then plug the matrix into a polynomial ([Definition 0.8.1](#)). This is how polynomials act on matrices throughout the book.

Definition 0.9.14 — Powers and polynomials of a matrix

Let $A \in M_n(F)$. The **powers** of A are defined recursively by

$$A^0 = I_n, \quad A^{k+1} = A^k A \quad (k \in \mathbb{N}).$$

For a polynomial $p = a_0 + a_1x + \cdots + a_Nx^N \in F[x]$, the matrix $p(A)$ is

$$p(A) = a_0I_n + a_1A + a_2A^2 + \cdots + a_NA^N \in M_n(F).$$

Two points in this definition deserve attention. The **constant term** a_0 becomes $a_0 I_n$, because a_0 alone is a number and cannot be added to matrices; this matches $x^0 = 1$ with $A^0 = I_n$. And $p(A)$ is well defined, because enlarging N only adds terms $0 \cdot A^k = 0$, exactly as for evaluation at a number (Definition 0.8.6).

The next theorem says that plugging in a matrix respects sums and products, just as plugging in a number does (Theorem 0.8.5).

Theorem 0.9.9 — Polynomials in one matrix

Let $A \in M_n(F)$ and $p, q \in F[x]$. Then:

1. $A^j A^k = A^{j+k}$ for all $j, k \in \mathbb{N}$;
2. $(p + q)(A) = p(A) + q(A)$ and $(pq)(A) = p(A)q(A)$;
3. $p(A)q(A) = q(A)p(A)$.

Idea. In the proof of Theorem 0.8.5, the only delicate step moved c^i past a coefficient b_j . Here the coefficients are scalars, and scalars can be moved freely through matrix products. The other step, $c^i c^j = c^{i+j}$, becomes part 1, which is associativity plus induction. Part 3 is then free, because $pq = qp$ in $F[x]$.

Proof. For 1, fix j and use induction on k (Theorem 0.2.7). For $k = 0$, $A^j A^0 = A^j I_n = A^j$. If $A^j A^k = A^{j+k}$, then by Definition 0.9.14 and associativity (Theorem 0.9.4),

$$A^j A^{k+1} = A^j (A^k A) = (A^j A^k) A = A^{j+k} A = A^{j+k+1}.$$

For 2, let $p = \sum_{i=0}^m a_i x^i$ and $q = \sum_{j=0}^l b_j x^j$. The statement for sums follows from Theorem 0.9.1, since $(a_i + b_i)A^i = a_i A^i + b_i A^i$. For products, by Definition 0.8.2 the coefficient of x^k in pq is $\sum_{i+j=k} a_i b_j$, so

$$(pq)(A) = \sum_k \left(\sum_{i+j=k} a_i b_j \right) A^k = \sum_{i=0}^m \sum_{j=0}^l (a_i b_j) A^i A^j = \left(\sum_{i=0}^m a_i A^i \right) \left(\sum_{j=0}^l b_j A^j \right) = p(A) q(A),$$

where the second equality uses part 1 and Theorem 0.9.1, and the third uses distributivity and the scalar rule $(a_i A^i)(b_j A^j) = (a_i b_j) A^i A^j$ from Theorem 0.9.4.

For 3, $pq = qp$ in $F[x]$ by Theorem 0.8.1. Hence by part 2, $p(A)q(A) = (pq)(A) = (qp)(A) = q(A)p(A)$. This proves the theorem. ■

Example. (Plugging a matrix into a polynomial)

Let $A = \begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix} \in M_2(\mathbb{R})$ and $p = x^2 - 4x + 3$. Compute $p(A)$ directly, and again using the factorization $p = (x - 1)(x - 3)$.

Solution. First $A^2 = \begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix} = \begin{pmatrix} 1 & 8 \\ 0 & 9 \end{pmatrix}$. Hence

$$p(A) = A^2 - 4A + 3I_2 = \begin{pmatrix} 1 & 8 \\ 0 & 9 \end{pmatrix} - \begin{pmatrix} 4 & 8 \\ 0 & 12 \end{pmatrix} + \begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

The constant term contributed $3I_2$, not “3”. By [Theorem 0.9.9](#), $p(A) = (A - I_2)(A - 3I_2)$, and indeed

$$(A - I_2)(A - 3I_2) = \begin{pmatrix} 0 & 2 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} -2 & 2 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

Note that neither factor is zero; this is the second warning of the multiplication part in action. That the diagonal entries 1, 3 of A are the roots of p is no accident, as we will see with eigenvalues.

[Theorem 0.9.9](#) is about polynomials in **one** matrix. Identities with **two** different matrices are another matter.

Warning. School identities fail for matrices that do not commute. Expanding with distributivity, $(A + B)^2 = A^2 + AB + BA + B^2$, and this equals $A^2 + 2AB + B^2$ only when $AB = BA$. With $A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$: $(A + B)^2 = I_2$, but $A^2 + 2AB + B^2 = \begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix}$. Before using a polynomial identity valid for numbers, check that all the matrices involved commute; polynomials in a single A always do.

0.9.11 Two equivalence relations on matrices

We close with two relations on matrices that the rest of the book studies in depth. Both are equivalence relations ([Definition 0.5.2](#)), and the proofs are short exercises in using [Theorem 0.9.7](#).

Example. (Similarity)

For $A, B \in M_n(F)$, say A is **similar** to B , written $A \sim B$, if there is an **invertible** $P \in M_n(F)$ with $B = P^{-1}AP$. Prove that similarity is an equivalence relation on $M_n(F)$. Then check that $A = \begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix}$ is similar to $\text{diag}(1, 3)$ using $P = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, and show that $\text{diag}(1, 2)$ is **not** similar to I_2 over $\mathbb{R}$.

Solution. We check the three conditions of [Definition 0.5.2](#).

Reflexive. Let $A \in M_n(F)$. The identity I_n is invertible with $I_n^{-1} = I_n$ ([Example \(Some invertible matrices\)](#)), and $I_n^{-1}AI_n = A$. Hence $A \sim A$.

Symmetric. Suppose $A \sim B$, so $B = P^{-1}AP$ with P invertible. Multiplying on the left by P and on the right by P^{-1} , and using associativity, $PBP^{-1} = (PP^{-1})A(PP^{-1}) = A$. By [Theorem 0.9.7](#) (part 2), $Q = P^{-1}$ is invertible with $Q^{-1} = P$, so $A = Q^{-1}BQ$. Hence $B \sim A$.

Transitive. Suppose $A \sim B$ and $B \sim C$, so $B = P^{-1}AP$ and $C = Q^{-1}BQ$ with P, Q

invertible. Then

$$C = Q^{-1}(P^{-1}AP)Q = (Q^{-1}P^{-1})A(PQ) = (PQ)^{-1}A(PQ),$$

where the last step is [Theorem 0.9.7](#) (part 3), which also says PQ is invertible. Hence $A \sim C$. This shows that similarity is an equivalence relation.

For the numerical part, $P^{-1} = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}$ by [Theorem 0.9.8](#) (here $ad - bc = 1$), and

$$P^{-1}AP = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & -1 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 3 \end{pmatrix}.$$

Finally, if $P^{-1}I_2P = \text{diag}(1, 2)$ for some invertible P , then $\text{diag}(1, 2) = P^{-1}P = I_2$, which is false. (Alternatively: similar matrices have equal traces by [Theorem 0.9.6](#), and $\text{tr } \text{diag}(1, 2) = 3 \neq 2 = \text{tr } I_2$.) Hence $\text{diag}(1, 2) \not\sim I_2$.

In Chapter 3 we will see that similar matrices describe the same linear map in two different coordinate systems. That is why a quantity like the trace, which similar matrices share, is really a property of the map and not of the matrix. Deciding when two matrices are similar is one of the main problems of this book.

Example. (Row equivalence)

For $A, B \in M_{m \times n}(F)$, say A is **row equivalent** to B if there is an **invertible** $E \in M_m(F)$ with $B = EA$. Prove that row equivalence is an equivalence relation on $M_{m \times n}(F)$. Then, with $E = \begin{pmatrix} 1 & 0 \\ -2 & 1 \end{pmatrix}$, compute EA for $A = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 5 & 0 \end{pmatrix}$ and describe what E did to the rows of A .

Solution. Reflexive. $A = I_m A$ and I_m is invertible, so A is row equivalent to itself.

Symmetric. Suppose $B = EA$ with E invertible. Multiplying on the left by E^{-1} , $E^{-1}B = (E^{-1}E)A = A$, and E^{-1} is invertible by [Theorem 0.9.7](#) (part 2). So B is row equivalent to A .

Transitive. Suppose $B = EA$ and $C = E'B$ with E, E' invertible. Then $C = E'(EA) = (E'E)A$ by associativity, and $E'E$ is invertible by [Theorem 0.9.7](#) (part 3). So A is row equivalent to C . This shows that row equivalence is an equivalence relation.

For the computation, E is invertible, with inverse $\begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}$ (check both products, or use [Theorem 0.9.8](#) with $ad - bc = 1$). By the row view of [Theorem 0.9.2](#) and [Theorem 0.9.3](#), row 1 of EA is $1 \cdot (\text{row 1 of } A)$ and row 2 is $-2 \cdot (\text{row 1}) + 1 \cdot (\text{row 2})$:

$$EA = \begin{pmatrix} 1 & 2 & 1 \\ 2-2 & 5-4 & 0-2 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & -2 \end{pmatrix}.$$

So E subtracted twice row 1 from row 2, which is the first step of Gaussian elimination on A .

In Chapter 2 we will see that every elementary row operation is multiplication on the left by an invertible matrix, and that row equivalence is exactly “reachable by a sequence of row operations”. This is why Gaussian elimination never changes the solutions of a linear system.

0.9.12 Exercises

0.9.12.1 A. Check your understanding

Exercise A1.

- Let $A \in M_{2 \times 3}(F)$ and $B \in M_{3 \times 4}(F)$. Which of AB , BA , $A^\top A$ and $B^\top A^\top$ are defined? Give the size of each one that is.
- Define what it means for $A \in M_n(F)$ to be invertible.
- Determine whether the following statement is true: “if $A, B \in M_n(F)$ and $AB = 0$, then $A = 0$ or $B = 0$.” Justify your answer.
- Determine whether the following statement is true: “ $(A + B)(A - B) = A^2 - B^2$ for all $A, B \in M_2(\mathbb{R})$.” Justify your answer.
- Let $A \in M_3(F)$ have columns $\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3$. Express $A(2, 0, -1)$ and the second column of A^2 in terms of the columns of A and A itself.

Solution.

- AB is $(2 \times 3)(3 \times 4)$, defined, of size 2×4 . BA would pair rows of length 4 with columns of length 2, so it is not defined. $A^\top A$ is $(3 \times 2)(2 \times 3)$, defined, of size 3×3 . $B^\top A^\top$ is $(4 \times 3)(3 \times 2)$, defined, of size 4×2 ; it equals $(AB)^\top$ by [Theorem 0.9.5](#).
- A is invertible if there exists $B \in M_n(F)$ with $AB = I_n$ **and** $BA = I_n$.
- False. $\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} = 0$ with both factors non-zero.
- False. Expanding gives $(A + B)(A - B) = A^2 - AB + BA - B^2$, which differs from $A^2 - B^2$ unless $AB = BA$. For $A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$, we have $A^2 = B^2 = 0$, but $(A + B)(A - B) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \neq 0$.
- By [Theorem 0.9.3](#), $A(2, 0, -1) = 2\mathbf{a}_1 - \mathbf{a}_3$. By [Theorem 0.9.2](#), the second column of $A^2 = AA$ is $A\mathbf{a}_2$.

0.9.12.2 B. Practice

Exercise B1: Products and sizes.

Let

$$A = \begin{pmatrix} 1 & -1 & 2 \\ 0 & 3 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 2 & 1 \\ 1 & 0 \\ 0 & 4 \end{pmatrix}, \quad C = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$$

over $\mathbb{R}$.

- Compute AB and BA .
- Determine which of AC and CA is defined, and compute it.
- Compute $A\mathbf{x}$ for $\mathbf{x} = (2, 0, -1)$ as a combination of the columns of A .

Solution.

- (a) AB is 2×2 and BA is 3×3 . Row against column,

$$AB = \begin{pmatrix} 2-1+0 & 1+0+8 \\ 0+3+0 & 0+0+4 \end{pmatrix} = \begin{pmatrix} 1 & 9 \\ 3 & 4 \end{pmatrix}, \quad BA = \begin{pmatrix} 2 & 1 & 5 \\ 1 & -1 & 2 \\ 0 & 12 & 4 \end{pmatrix}.$$

For instance $(BA)_{12} = 2 \cdot (-1) + 1 \cdot 3 = 1$ and $(BA)_{32} = 0 \cdot (-1) + 4 \cdot 3 = 12$.

- (b) AC would be $(2 \times 3)(2 \times 2)$, and $3 \neq 2$, so it is not defined. CA is $(2 \times 2)(2 \times 3)$, of size 2×3 :

$$CA = \begin{pmatrix} 1 & -1+6 & 2+2 \\ 3 & -3+12 & 6+4 \end{pmatrix} = \begin{pmatrix} 1 & 5 & 4 \\ 3 & 9 & 10 \end{pmatrix}.$$

- (c) By [Theorem 0.9.3](#), $A\mathbf{x} = 2\mathbf{a}_1 + 0\mathbf{a}_2 - \mathbf{a}_3 = 2(1, 0) - (2, 1) = (0, -1)$.

Exercise B2: Traces of products.

- (a) With A and B as in [Exercise \(B1: Products and sizes\)](#), verify that $\text{tr}(AB) = \text{tr}(BA)$, although AB and BA have different sizes.
- (b) Let $A = \text{diag}(1, 2)$, $B = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ and $C = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$. Show that $\text{tr}(ABC) \neq \text{tr}(BAC)$, and verify that $\text{tr}(ABC) = \text{tr}(CAB)$.

Solution.

- (a) From [Exercise \(B1: Products and sizes\)](#), $\text{tr}(AB) = 1 + 4 = 5$ and $\text{tr}(BA) = 2 + (-1) + 4 = 5$, as [Theorem 0.9.6](#) predicts.

- (b) By the column view, BC has columns $B\mathbf{e}_2 = \mathbf{e}_1$ and $B\mathbf{0} = \mathbf{0}$, so $BC = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$, and $ABC = A(BC) = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$, with trace 1. Next, $BA = \begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix}$, so $BAC = \begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix}$, with trace 2. Hence $\text{tr}(ABC) = 1 \neq 2 = \text{tr}(BAC)$. Finally $AB = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ and $CAB = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$, with trace 1 = $\text{tr}(ABC)$, as the cyclic rule predicts.

Exercise B3: Inverses of 2×2 matrices.

- (a) Find the inverse of $\begin{pmatrix} 3 & 4 \\ 2 & 3 \end{pmatrix} \in M_2(\mathbb{R})$, and check both products.
- (b) Show that $A = \begin{pmatrix} 2 & 6 \\ 1 & 3 \end{pmatrix}$ is not invertible by finding a non-zero $\mathbf{x}$ with $A\mathbf{x} = \mathbf{0}$.
- (c) Find the inverse of $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \in M_2(\mathbb{F}_5)$.
- (d) For which $t \in \mathbb{R}$ is $\begin{pmatrix} t & 2 \\ 2 & t \end{pmatrix}$ invertible?

Solution.

- (a) Here $ad - bc = 9 - 8 = 1 \neq 0$, so by [Theorem 0.9.8](#) the inverse is $B = \begin{pmatrix} 3 & -4 \\ -2 & 3 \end{pmatrix}$.
 Check: $\begin{pmatrix} 3 & 4 \\ 2 & 3 \end{pmatrix} B = \begin{pmatrix} 9-8 & -12+12 \\ 6-6 & -8+9 \end{pmatrix} = I_2$ and $B \begin{pmatrix} 3 & 4 \\ 2 & 3 \end{pmatrix} = \begin{pmatrix} 9-8 & 12-12 \\ -6+6 & -8+9 \end{pmatrix} = I_2$.
- (b) Here $ad - bc = 6 - 6 = 0$. Following the proof of [Theorem 0.9.8](#), take $\mathbf{x} = (d, -c) = (3, -1) \neq \mathbf{0}$. Then $A\mathbf{x} = (6-6, 3-3) = \mathbf{0}$. If A were invertible, then $\mathbf{x} = A^{-1}(A\mathbf{x}) = \mathbf{0}$, a contradiction. Hence A is not invertible.
- (c) In $\mathbb{F}_5$, $ad - bc = 4 - 6 = -2 = 3 \neq 0$, and $3^{-1} = 2$ since $3 \cdot 2 = 6 = 1$. By [Theorem 0.9.8](#) the inverse is $2 \begin{pmatrix} 4 & -2 \\ -3 & 1 \end{pmatrix} = \begin{pmatrix} 8 & -4 \\ -6 & 2 \end{pmatrix} = \begin{pmatrix} 3 & 1 \\ 4 & 2 \end{pmatrix}$. Check:
 $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 3 & 1 \\ 4 & 2 \end{pmatrix} = \begin{pmatrix} 11 & 5 \\ 25 & 11 \end{pmatrix} = I_2$ and $\begin{pmatrix} 3 & 1 \\ 4 & 2 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} = \begin{pmatrix} 6 & 10 \\ 10 & 16 \end{pmatrix} = I_2$ in $\mathbb{F}_5$.
- (d) By [Theorem 0.9.8](#), it is invertible if and only if $t^2 - 4 \neq 0$, that is, $t \neq 2$ and $t \neq -2$.

Exercise B4: Transposes of products.

- (a) Prove that $(A_1 A_2 \cdots A_k)^\top = A_k^\top \cdots A_2^\top A_1^\top$ for all $k \geq 1$ and matrices $A_1, \dots, A_k$ whose sizes allow the product.
- (b) Let $A \in M_n(F)$. Prove that $A + A^\top$ is symmetric and $A - A^\top$ is skew-symmetric.

Solution.

- (a) We use induction on k ([Theorem 0.2.7](#)). For $k = 1$ there is nothing to prove. Suppose the formula holds for products of k matrices, and let $A_1, \dots, A_{k+1}$ be given. By associativity, $A_1 \cdots A_{k+1} = (A_1 \cdots A_k) A_{k+1}$. By [Theorem 0.9.5](#) (part 4) and then the induction hypothesis,

$$(A_1 \cdots A_{k+1})^\top = A_{k+1}^\top (A_1 \cdots A_k)^\top = A_{k+1}^\top A_k^\top \cdots A_1^\top.$$

This proves the formula for all $k \geq 1$.

- (b) By [Theorem 0.9.5](#) (parts 1–3), $(A + A^\top)^\top = A^\top + (A^\top)^\top = A^\top + A = A + A^\top$, so $A + A^\top$ is symmetric. Similarly $(A - A^\top)^\top = A^\top - A = -(A - A^\top)$, so $A - A^\top$ is skew-symmetric.

0.9.12.3 C. Going deeper

Exercise C1: $AB - BA = I$ is impossible, or is it?.

- (a) Prove that there are no $A, B \in M_n(\mathbb{R})$, for any $n \geq 1$, with $AB - BA = I_n$.
- (b) Show that the argument of (a) breaks down in $M_2(\mathbb{F}_2)$, and that $A = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$, $B = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ satisfy $AB - BA = I_2$ over $\mathbb{F}_2$.

Hint: consider the trace of both sides.

Solution.

- (a) Suppose, for a contradiction, that $AB - BA = I_n$. Taking traces and using [Theorem 0.9.6](#) (parts 1 and 3),

$$n = \operatorname{tr} I_n = \operatorname{tr}(AB - BA) = \operatorname{tr}(AB) - \operatorname{tr}(BA) = 0.$$

This contradicts $n \geq 1$. Hence no such A, B exist.

- (b) Over $\mathbb{F}_2$ the same computation gives $\operatorname{tr} I_2 = 1 + 1 = 0$, so taking traces yields $0 = 0$ and no contradiction. For the given matrices, the column view gives $AB = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$ and $BA = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$, so $AB - BA = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I_2$, because $-1 = 1$ in $\mathbb{F}_2$.

Exercise C2: Triangular matrices with non-zero diagonal.

Let F be a field.

- (a) Let $U = \begin{pmatrix} a & b \\ 0 & d \end{pmatrix}$ with $a, d \neq 0$. Show that U is invertible and that U^{-1} is upper triangular.
- (b) Let $U = \begin{pmatrix} a & b & c \\ 0 & d & e \\ 0 & 0 & f \end{pmatrix}$ with $a, d, f \neq 0$. Show that

$$V = \begin{pmatrix} a^{-1} & -b(ad)^{-1} & (be - cd)(adf)^{-1} \\ 0 & d^{-1} & -e(df)^{-1} \\ 0 & 0 & f^{-1} \end{pmatrix}$$

is the inverse of U .

- (c) Explain how the entries of V can be found, column by column, by solving $U\mathbf{v}_k = \mathbf{e}_k$.

The general case, for all n , follows from the methods of Chapter 2.

Solution.

- (a) Here $ad - b \cdot 0 = ad \neq 0$, since a product of non-zero elements of a field is non-zero ([Theorem 0.7.1](#)). By [Theorem 0.9.8](#), U is invertible with $U^{-1} = (ad)^{-1} \begin{pmatrix} d & -b \\ 0 & a \end{pmatrix} = \begin{pmatrix} a^{-1} & -b(ad)^{-1} \\ 0 & d^{-1} \end{pmatrix}$, which is upper triangular.
- (b) The inverses a^{-1}, d^{-1}, f^{-1} exist, and $ad, df, adf \neq 0$ by [Theorem 0.7.1](#), so V is defined. We check $UV = I_3$ entry by entry. The diagonal entries are $a \cdot a^{-1} = 1$, $d \cdot d^{-1} = 1$, $f \cdot f^{-1} = 1$. Below the diagonal both matrices are zero, so $(UV)_{21} =$

$(UV)_{31} = (UV)_{32} = 0$. Above the diagonal,

$$(UV)_{12} = a \cdot (-b(ad)^{-1}) + bd^{-1} = -bd^{-1} + bd^{-1} = 0,$$

$$(UV)_{23} = d \cdot (-e(df)^{-1}) + ef^{-1} = -ef^{-1} + ef^{-1} = 0,$$

$$(UV)_{13} = a(be - cd)(adf)^{-1} + b(-e(df)^{-1}) + cf^{-1} = (be - cd)(df)^{-1} - be(df)^{-1} + cd(df)^{-1} = 0.$$

Hence $UV = I_3$. For VU , again the diagonal entries are 1 and the entries below the diagonal are 0, and

$$(VU)_{12} = a^{-1}b - b(ad)^{-1}d = a^{-1}b - a^{-1}b = 0,$$

$$(VU)_{23} = d^{-1}e - e(df)^{-1}f = 0,$$

$$(VU)_{13} = a^{-1}c - b(ad)^{-1}e + (be - cd)(adf)^{-1}f = a^{-1}c - be(ad)^{-1} + be(ad)^{-1} - cd(ad)^{-1} = 0,$$

using $cd(ad)^{-1} = ca^{-1}$. Hence $VU = I_3$, and $V = U^{-1}$. Both products had to be checked, since we may not yet use that one implies the other.

- (c) By [Theorem 0.9.2](#), $UV = I_3$ says exactly that the columns $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ of V satisfy $U\mathbf{v}_k = \mathbf{e}_k$. Because U is upper triangular, each system can be solved from the bottom row up. For example, $U\mathbf{v}_3 = \mathbf{e}_3$ reads $fx_3 = 1$, $dx_2 + ex_3 = 0$, $ax_1 + bx_2 + cx_3 = 0$, giving $x_3 = f^{-1}$, then $x_2 = -e(df)^{-1}$, then $x_1 = (be - cd)(adf)^{-1}$, which is the third column of V . (This only finds a candidate satisfying $UV = I_3$; the check $VU = I_3$ in (b) is still needed.)

Exercise C3: Matrices that commute.

Let $N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \in M_2(\mathbb{R})$.

- Find all $X \in M_2(\mathbb{R})$ with $XN = NX$.
- Deduce that the only matrices $X \in M_2(\mathbb{R})$ with $XY = YX$ **for every** $Y \in M_2(\mathbb{R})$ are the scalar multiples cI_2 , $c \in \mathbb{R}$.

Solution.

- Let $X = \begin{pmatrix} p & q \\ r & s \end{pmatrix}$. By the column view, $XN = (X\mathbf{0} \quad X\mathbf{e}_1) = \begin{pmatrix} 0 & p \\ 0 & r \end{pmatrix}$, and by the row view, NX has rows (row 2 of X) and : $NX = \begin{pmatrix} r & s \\ 0 & 0 \end{pmatrix}$. Comparing entries, $XN = NX$ if and only if $r = 0$ and $p = s$, that is, $X = \begin{pmatrix} p & q \\ 0 & p \end{pmatrix} = pI_2 + qN$ with $p, q \in \mathbb{R}$ arbitrary.
- Every cI_2 commutes with every Y , by [Theorem 0.9.4](#): $(cI_2)Y = cY = Y(cI_2)$. Conversely, suppose $XY = YX$ for every Y . Taking $Y = N$, part (a) gives $X = \begin{pmatrix} p & q \\ 0 & p \end{pmatrix}$. Taking $Y = N^T = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$, we compute $XN^T = \begin{pmatrix} q & 0 \\ p & 0 \end{pmatrix}$ and $N^TX = \begin{pmatrix} 0 & 0 \\ p & q \end{pmatrix}$. Equality forces $q = 0$. Hence $X = pI_2$, as claimed.

0.10 Groups and Permutations

Several structures in this chapter have quietly behaved alike. The invertible $n \times n$ matrices, the non-zero elements of a field, and the bijections of a set can each be combined, have an element that does nothing, and can be undone. This short section gives that common structure a name, proves its basic rules once, and introduces the permutations that the determinant in Chapter 6 is built from. We keep it brief: groups are a language for us here, not a subject.

0.10.1 Groups

Look at four situations side by side. In $\text{GL}_n(F)$, the set of invertible matrices in $M_n(F)$, we multiply matrices, I_n does nothing, and each A is undone by A^{-1} (Theorem 0.9.7). In a field F , we multiply non-zero elements, 1 does nothing, and each a is undone by a^{-1} . The four rotations of a square about its center (by $0^\circ, 90^\circ, 180^\circ, 270^\circ$) can be performed one after another, the rotation by 0° does nothing, and each rotation is undone by rotating back. The bijections $X \rightarrow X$ of a set are composed, id_X does nothing, and each is undone by its inverse function (Theorem 0.4.7). The objects differ completely, but the rules we use are the same three. So we give the pattern a name.

A group is a set with one way of combining elements, in which combining is associative, some element does nothing, and every element can be undone.

Definition 0.10.1 — Group

A **group** is a set G together with a **binary operation** $G \times G \rightarrow G$, $(a, b) \mapsto ab$, such that:

- (G1) **(Associativity)** $(ab)c = a(bc)$ for all $a, b, c \in G$;
- (G2) **(Identity)** there is an element $e \in G$ with $ea = ae = a$ for every $a \in G$;
- (G3) **(Inverses)** for every $a \in G$ there is $b \in G$ with $ab = ba = e$, where e is the element of (G2).

The group is **abelian** if moreover $ab = ba$ for all $a, b \in G$.

In words: the operation is a function on pairs, so the combination of two elements of G lies in G again; this “closure” is built into the phrase “binary operation $G \times G \rightarrow G$ ”. (G1) lets us drop brackets. (G2) asks for an element that changes nothing, from **both** sides. (G3) asks that each element can be undone, again from both sides, back to that same e . When the operation is addition we write $a + b$, 0 and $-a$ instead of ab , e and a^{-1} ; this notation is only used for abelian groups.

Example. (Groups)

Check that each of the following is a group, and decide which are abelian: $(\mathbb{Z}, +)$; $(F, +)$ and $(F \setminus \{0\}, \cdot)$ for a field F ; $\text{GL}_n(F)$ under matrix multiplication; the set S_n of bijections $\{1, \dots, n\} \rightarrow \{1, \dots, n\}$ under composition; and a one-element set $\{e\}$ with $ee = e$.

Solution. $(\mathbb{Z}, +)$: the sum of integers is an integer, addition is associative, 0 is an identity, and $-a$ undoes a . It is abelian. $(F, +)$ is a group for the same reasons, by the field axioms (Definition 0.7.1); it is abelian.

$(F \setminus \{0\}, \cdot)$: the product of two non-zero elements is non-zero by Theorem 0.7.1, so the operation lands in $F \setminus \{0\}$. Associativity and the identity $1 \neq 0$ come from the field axioms, and each $a \neq 0$ has an inverse a^{-1} , which is non-zero since $aa^{-1} = 1 \neq 0$. It is abelian.

$\text{GL}_n(F)$: by Theorem 0.9.7 (part 3), a product of invertible matrices is invertible, so the operation lands in $\text{GL}_n(F)$. Associativity holds by Theorem 0.9.4, I_n is invertible and is an identity, and $A^{-1} \in \text{GL}_n(F)$ undoes A by part 2. For $n \geq 2$ it is **not** abelian: for $n = 2$, $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$, while the product in the other order is $\begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$. (Over $\mathbb{F}_2$ read 2 as 0; the two products still differ.) For $n \geq 3$, put these blocks in the top-left corner and 1's further down the diagonal. $\text{GL}_1(F)$ is just $F \setminus \{0\}$, and is abelian.

S_n : a composition of bijections is a bijection (Theorem 0.4.3), composition is associative (Theorem 0.4.2), id is an identity, and each bijection has an inverse function (Theorem 0.4.7), which is again a bijection. We will see below that S_n is not abelian for $n \geq 3$.

$\{e\}$: all three axioms reduce to $ee = e$. This **trivial group** is the degenerate case; it reappears as the smallest subgroup of every group.

Now non-examples, each a minimal change of an example. Replace addition on $\mathbb{Z}$ by multiplication. The operation still lands in $\mathbb{Z}$, it is associative, and 1 is an identity. But (G3) fails: there is no integer b with $2b = 1$. So $(\mathbb{Z}, \cdot)$ is **not** a group. Similarly, replace $\text{GL}_n(F)$ by all of $M_n(F)$ under multiplication: (G1) and (G2) still hold with I_n , but the zero matrix has no inverse, so (G3) fails. Keep $\mathbb{N}$ under $+$: the identity 0 is there, but 1 has no inverse in $\mathbb{N}$.

Warning. The operation must land in the set. $\text{GL}_n(F)$ under matrix **addition** is not a group: I_n and $-I_n$ are invertible, but $I_n + (-I_n) = 0$ is not, so addition is not even a binary operation on $\text{GL}_n(F)$. Before checking (G1)–(G3), check that combining two elements of the set gives an element of the set.

The rules we proved separately for fields and for matrices follow from the axioms alone. It is worth proving them once:

Theorem 0.10.1 — Basic properties of groups

Let G be a group and $a, b, c \in G$.

1. The identity element of G is unique.
2. The inverse of a is unique. We denote it by a^{-1} .
3. $(a^{-1})^{-1} = a$ and $(ab)^{-1} = b^{-1}a^{-1}$.
4. **(Cancellation)** If $ab = ac$, then $b = c$. If $ba = ca$, then $b = c$.

Proof. For 1, suppose e and e' both satisfy (G2). Then $e = ee'$, using that e' is an identity, and $ee' = e'$, using that e is one. Hence $e = e'$.

For 2, suppose b and b' both satisfy $ab = ba = e$ and $ab' = b'a = e$. By (G2) and (G1),

$$b = be = b(ab') = (ba)b' = eb' = b'.$$

For 3, the equations $aa^{-1} = a^{-1}a = e$ say that a is an inverse of a^{-1} , so $(a^{-1})^{-1} = a$ by part 2. Next, by (G1),

$$(ab)(b^{-1}a^{-1}) = a(bb^{-1})a^{-1} = aea^{-1} = aa^{-1} = e,$$

and similarly $(b^{-1}a^{-1})(ab) = b^{-1}(a^{-1}a)b = e$. By part 2, $(ab)^{-1} = b^{-1}a^{-1}$.

For 4, suppose $ab = ac$. Multiplying on the left by a^{-1} and using (G1), $b = (a^{-1}a)b = a^{-1}(ab) = a^{-1}(ac) = (a^{-1}a)c = c$. The second statement follows by multiplying on the right by a^{-1} . This proves the theorem. ■

These are the arguments of [Theorem 0.9.7](#) with matrices erased. That is the point of the definition: a proof that uses only (G1)–(G3) holds in every group at once.

0.10.2 Subgroups

After a structure comes the substructure. The even integers sit inside $(\mathbb{Z}, +)$ and form a group on their own, with the same operation.

Definition 0.10.2 — Subgroup

Let G be a group. A subset $H \subseteq G$ is a **subgroup** of G if $e \in H$, and $ab \in H$ and $a^{-1} \in H$ for all $a, b \in H$.

Then the operation of G restricts to a binary operation on H , and H is a group with it: (G1) holds in H because it holds in G , and (G2), (G3) hold because e and the inverses lie in H . In practice one checks a single condition:

Theorem 0.10.2 — Subgroup test

Let G be a group and $H \subseteq G$. Then H is a subgroup of G if and only if $H \neq \emptyset$ and $ab^{-1} \in H$ for all $a, b \in H$.

Proof. ($\Rightarrow$) If H is a subgroup, then $e \in H$, so $H \neq \emptyset$. For $a, b \in H$ we have $b^{-1} \in H$, and then $ab^{-1} \in H$.

($\Leftarrow$) Suppose $H \neq \emptyset$ and $ab^{-1} \in H$ for all $a, b \in H$. Pick $h \in H$. Taking $a = b = h$ gives $e = hh^{-1} \in H$. For $b \in H$, taking $a = e$ gives $b^{-1} = eb^{-1} \in H$. For $a, b \in H$, we now know $b^{-1} \in H$, so $ab = a(b^{-1})^{-1} \in H$ by [Theorem 0.10.1](#). Hence H is a subgroup. This proves the theorem. ■

For example, $2\mathbb{Z} = \{2k : k \in \mathbb{Z}\}$ is a subgroup of $(\mathbb{Z}, +)$. In additive notation the test reads “ $a - b \in H$ ”, and $2k - 2l = 2(k - l) \in 2\mathbb{Z}$, while $0 \in 2\mathbb{Z}$. By contrast $\mathbb{N} \subseteq \mathbb{Z}$ is non-empty and closed under $+$, but $0 - 1 = -1 \notin \mathbb{N}$: the test fails, because $\mathbb{N}$ lacks inverses. The set of odd integers fails more badly, since it does not contain 0. In $\mathrm{GL}_n(F)$, the invertible diagonal matrices form a subgroup: I_n is one, and by [Example \(Some invertible matrices\)](#) the inverse of $\mathrm{diag}(d_1, \dots, d_n)$ is $\mathrm{diag}(d_1^{-1}, \dots, d_n^{-1})$, so DE^{-1} is again diagonal with non-zero diagonal entries. Every group G has the subgroups $\{e\}$ and G .

0.10.3 Homomorphisms

After objects come the maps between them that respect the structure.

Definition 0.10.3 — Group homomorphism

Let G and H be groups. A function $\varphi: G \rightarrow H$ is a **homomorphism** if $\varphi(ab) = \varphi(a)\varphi(b)$ for all $a, b \in G$. A **bijective** homomorphism is an **isomorphism**. The **kernel** of φ is $\ker \varphi = \{g \in G : \varphi(g) = e_H\}$, where e_H is the identity of H .

The product ab on the left is taken in G , and $\varphi(a)\varphi(b)$ on the right in H . For example, $F \setminus \{0\} \rightarrow \mathrm{GL}_2(F)$, $a \mapsto \mathrm{diag}(a, 1)$, is a homomorphism, since $\mathrm{diag}(ab, 1) = \mathrm{diag}(a, 1)\mathrm{diag}(b, 1)$; its kernel is $\{1\}$. The map $\mathbb{Z} \rightarrow \mathbb{Z}/n\mathbb{Z}$, $a \mapsto [a]$, is a homomorphism of additive groups, because $[a + b] = [a] + [b]$ is exactly how addition on $\mathbb{Z}/n\mathbb{Z}$ was defined ([Example \(Addition modulo \$n\$ is well defined\)](#)); its kernel is $n\mathbb{Z}$. On the other hand, $x \mapsto x + 1$ from $(\mathbb{R}, +)$ to itself is not a homomorphism: writing $\varphi(x) = x + 1$, we get $\varphi(0 + 0) = 1$ but $\varphi(0) + \varphi(0) = 2$.

Theorem 0.10.3 — Basic properties of homomorphisms

Let $\varphi: G \rightarrow H$ be a homomorphism of groups. Then $\varphi(e_G) = e_H$, $\varphi(a^{-1}) = \varphi(a)^{-1}$ for every $a \in G$, and $\ker \varphi$ is a subgroup of G .

Proof. Since $\varphi(e_G)\varphi(e_G) = \varphi(e_G e_G) = \varphi(e_G) = \varphi(e_G)e_H$, cancellation ([Theorem 0.10.1](#)) gives $\varphi(e_G) = e_H$. Next, $\varphi(a)\varphi(a^{-1}) = \varphi(aa^{-1}) = \varphi(e_G) = e_H$, and similarly $\varphi(a^{-1})\varphi(a) = e_H$, so $\varphi(a^{-1}) = \varphi(a)^{-1}$ by uniqueness of inverses. Finally, $e_G \in \ker \varphi$, and for $a, b \in \ker \varphi$,

$$\varphi(ab^{-1}) = \varphi(a)\varphi(b)^{-1} = e_H e_H^{-1} = e_H,$$

so $ab^{-1} \in \ker \varphi$. By [Theorem 0.10.2](#), $\ker \varphi$ is a subgroup. This proves the theorem. ■

Two homomorphisms will matter later: the determinant $\mathrm{GL}_n(F) \rightarrow F \setminus \{0\}$ and the sign $S_n \rightarrow \{1, -1\}$, both in Chapter 6. Linear maps in Chapter 3 are, among other things, homomorphisms of additive groups, and their kernels will be one of our main tools.

0.10.4 Permutations

A **permutation** of $\{1, \dots, n\}$ is a bijection $\{1, \dots, n\} \rightarrow \{1, \dots, n\}$, and S_n is the group of all of them, the **symmetric group**. It has $n!$ elements: there are n choices for $\sigma(1)$, then $n-1$ for $\sigma(2)$, and so on.

A permutation σ can be written in **two-line notation**, with $\sigma(i)$ below i . A more compact notation lists what happens along a loop. For **distinct** $i_1, \dots, i_k \in \{1, \dots, n\}$, the **cycle** $(i_1 i_2 \cdots i_k)$ is the permutation sending

$$i_1 \mapsto i_2 \mapsto \cdots \mapsto i_k \mapsto i_1$$

and fixing every other element. A cycle of length 2 is a **transposition**: $(i j)$ swaps i and j . The same cycle can be written starting at any of its entries: $(1 2 3) = (2 3 1) = (3 1 2)$, but $(1 3 2)$ is different.

The product $\sigma\tau$ in S_n means the composition $\sigma \circ \tau$: **first apply τ , then σ** . So products of cycles are read **from right to left**.

Example. (Computing with cycles)

In S_3 , compute $(1 3)(1 2)$ and $(1 2)(1 3)$. In S_5 , write $\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 5 & 1 & 2 & 4 \end{pmatrix}$ (two-line notation) as a product of cycles.

Solution. For $(1 3)(1 2)$, follow each element through $(1 2)$ first, then $(1 3)$: $1 \mapsto 2 \mapsto 2$, $2 \mapsto 1 \mapsto 3$, $3 \mapsto 3 \mapsto 1$. So $(1 3)(1 2) = (1 2 3)$. For $(1 2)(1 3)$: $1 \mapsto 3 \mapsto 3$, $3 \mapsto 1 \mapsto 2$, $2 \mapsto 2 \mapsto 1$. So $(1 2)(1 3) = (1 3 2)$. The two products differ, so S_3 is not abelian; the same two transpositions show that S_n is not abelian for every $n \geq 3$.

For σ , start at 1 and follow: $1 \mapsto 3 \mapsto 1$, a loop $(1 3)$. The smallest element not yet used is 2: $2 \mapsto 5 \mapsto 4 \mapsto 2$, a loop $(2 5 4)$. Every element is now used, so $\sigma = (1 3)(2 5 4)$. These two cycles move disjoint sets of elements, so they can be applied in either order.

Warning. Mind the order. $\sigma\tau$ applies τ first, exactly as for functions, and as for matrices: $(AB)\mathbf{x} = A(B\mathbf{x})$ applies B first. Reading a product of cycles from left to right gives the wrong answer, as the two different products $(1 3)(1 2) \neq (1 2)(1 3)$ above show.

Quick check. Compute $(1 2 3)(1 2)$ in S_3 .

Solution. Apply $(1 2)$ first: $1 \mapsto 2 \mapsto 3$, $2 \mapsto 1 \mapsto 2$, $3 \mapsto 3 \mapsto 1$. So 1 and 3 are swapped and 2 is fixed: $(1 2 3)(1 2) = (1 3)$.

The example showed $(1 2 3) = (1 3)(1 2)$: a 3-cycle is a product of transpositions. Every permutation is, and the proof is a model induction on n .

Theorem 0.10.4 — Transpositions generate S_n

Let $n \geq 2$. Every permutation in S_n is a product of transpositions.

Idea. Induction on n : what is the smaller object inside a permutation of $n+1$ elements? If σ fixes $n+1$, it is really a permutation of $\{1, \dots, n\}$. If not, one transposition repairs that: swap $\sigma(n+1)$ back to $n+1$. The result fixes $n+1$, so the induction hypothesis applies, and we undo the swap.

Proof. We use induction on $n \geq 2$ ([Theorem 0.2.7](#)). For $n = 2$, $S_2 = \{\text{id}, (1\ 2)\}$, and $\text{id} = (1\ 2)(1\ 2)$.

Suppose every permutation in S_n is a product of transpositions, and let $\sigma \in S_{n+1}$.

Case 1. $\sigma(n+1) = n+1$. Since σ is injective, $\sigma(i) \neq n+1$ for $i \leq n$, so σ restricts to a function $\sigma': \{1, \dots, n\} \rightarrow \{1, \dots, n\}$. It is injective because σ is, and hence bijective by [Theorem 0.4.9](#). By the induction hypothesis, $\sigma' = \tau_1 \cdots \tau_r$ with transpositions $\tau_j \in S_n$. Regard each τ_j as a transposition in S_{n+1} fixing $n+1$. Then $\tau_1 \cdots \tau_r$ agrees with $\sigma' = \sigma$ on $\{1, \dots, n\}$ and fixes $n+1$, as σ does. Hence $\sigma = \tau_1 \cdots \tau_r$.

Case 2. $\sigma(n+1) = k \neq n+1$. Let $\tau = (k\ n+1)$. Then $(\tau\sigma)(n+1) = \tau(k) = n+1$, so by Case 1, $\tau\sigma = \tau_1 \cdots \tau_r$ for some transpositions τ_j . Since $\tau\tau = \text{id}$, multiplying on the left by τ gives $\sigma = \tau\tau_1 \cdots \tau_r$, a product of transpositions.

This completes the induction, and proves the theorem. ■

The product is not unique: $(1\ 2\ 3) = (1\ 3)(1\ 2) = (1\ 2)(2\ 3)$, and one can always insert $(1\ 2)(1\ 2) = \text{id}$. What turns out to be unique is whether the number of transpositions is even or odd. That fact defines the **sign** of a permutation, which we do not define here; it is the subject of Chapter 6, where it builds the determinant.

0.10.5 Exercises

0.10.5.1 A. Check your understanding

Exercise A1.

- (a) State the three axioms of a group.
- (b) Is $\mathbb{Z}$ with the operation $(a, b) \mapsto a - b$ a group? Justify your answer.
- (c) Determine whether the following statement is true: “in every group, $(ab)^{-1} = a^{-1}b^{-1}$.” Justify your answer.
- (d) Compute $(1\ 3)(1\ 2\ 3)$ in S_3 .
- (e) Explain why $\{A \in M_2(\mathbb{R}) : A \text{ is not invertible}\}$ is **not** a subgroup of $\text{GL}_2(\mathbb{R})$.

Solution.

- (a) (G1) $(ab)c = a(bc)$ for all a, b, c ; (G2) there is e with $ea = ae = a$ for all a ; (G3) for every a there is b with $ab = ba = e$. The operation must be a binary operation $G \times G \rightarrow G$.
- (b) No. Associativity fails: $(1 - 1) - 1 = -1$, but $1 - (1 - 1) = 1$.
- (c) False. By [Theorem 0.10.1](#), $(ab)^{-1} = b^{-1}a^{-1}$, and this can differ from $a^{-1}b^{-1}$. In S_3

take $a = (1\ 2)$ and $b = (1\ 3)$, which are their own inverses. By [Example \(Computing with cycles\)](#), $ab = (1\ 3\ 2)$, whose inverse is $(1\ 2\ 3)$, while $a^{-1}b^{-1} = ab = (1\ 3\ 2)$. The statement is true in abelian groups.

- (d) Apply $(1\ 2\ 3)$ first: $1 \mapsto 2 \mapsto 2$, $2 \mapsto 3 \mapsto 1$, $3 \mapsto 1 \mapsto 3$. Hence $(1\ 3)(1\ 2\ 3) = (1\ 2)$.
- (e) It is not even a subset of $\text{GL}_2(\mathbb{R})$, since its elements are not invertible; for instance it does not contain the identity I_2 .

0.10.5.2 B. Practice

Exercise B1: The table of S_3 .

List the six elements of S_3 in cycle notation and write out the table whose entry in row x and column y is xy . Hence find all elements $x \in S_3$ with $x^2 = \text{id}$, and all x that commute with every element of S_3 .

Solution. The elements are $\text{id}, (1\ 2), (1\ 3), (2\ 3), (1\ 2\ 3), (1\ 3\ 2)$. Computing each product right to left, as in [Example \(Computing with cycles\)](#), the entry in row x , column y is xy :

xy	id	$(1\ 2)$	$(1\ 3)$	$(2\ 3)$	$(1\ 2\ 3)$	$(1\ 3\ 2)$
id	id	$(1\ 2)$	$(1\ 3)$	$(2\ 3)$	$(1\ 2\ 3)$	$(1\ 3\ 2)$
$(1\ 2)$	$(1\ 2)$	id	$(1\ 3\ 2)$	$(1\ 2\ 3)$	$(2\ 3)$	$(1\ 3)$
$(1\ 3)$	$(1\ 3)$	$(1\ 2\ 3)$	id	$(1\ 3\ 2)$	$(1\ 2)$	$(2\ 3)$
$(2\ 3)$	$(2\ 3)$	$(1\ 3\ 2)$	$(1\ 2\ 3)$	id	$(1\ 3)$	$(1\ 2)$
$(1\ 2\ 3)$	$(1\ 2\ 3)$	$(1\ 3)$	$(2\ 3)$	$(1\ 2)$	$(1\ 3\ 2)$	id
$(1\ 3\ 2)$	$(1\ 3\ 2)$	$(2\ 3)$	$(1\ 2)$	$(1\ 3)$	id	$(1\ 2\ 3)$

For example, the entry in row $(1\ 2)$, column $(2\ 3)$ is $(1\ 2)(2\ 3)$: $1 \mapsto 1 \mapsto 2$, $2 \mapsto 3 \mapsto 3$, $3 \mapsto 2 \mapsto 1$, which is $(1\ 2\ 3)$.

Reading the diagonal, $x^2 = \text{id}$ exactly for $x = \text{id}, (1\ 2), (1\ 3), (2\ 3)$. An element x commutes with every element exactly when row x equals column x . This holds for id . It fails for every other element: $(1\ 2)(1\ 3) = (1\ 3\ 2) \neq (1\ 2\ 3) = (1\ 3)(1\ 2)$ handles $(1\ 2)$ and $(1\ 3)$; $(2\ 3)(1\ 2) = (1\ 3\ 2) \neq (1\ 2\ 3) = (1\ 2)(2\ 3)$ handles $(2\ 3)$; and $(1\ 2\ 3)(1\ 2) = (1\ 3) \neq (2\ 3) = (1\ 2)(1\ 2\ 3)$ handles $(1\ 2\ 3)$ and, since $(1\ 3\ 2)(1\ 2) = (2\ 3) \neq (1\ 3) = (1\ 2)(1\ 3\ 2)$, also $(1\ 3\ 2)$. Hence only id commutes with everything.

Exercise B2: Invertible upper triangular matrices.

Let F be a field and $T = \left\{ \begin{pmatrix} a & b \\ 0 & d \end{pmatrix} : a, b, d \in F, a \neq 0, d \neq 0 \right\}$.

- (a) Prove that T is a subgroup of $\text{GL}_2(F)$.
- (b) For which fields F is T abelian? Justify your answer.

Solution.

- (a) Let $X = \begin{pmatrix} a & b \\ 0 & d \end{pmatrix} \in T$. Since $a, d \neq 0$, $ad - b \cdot 0 = ad \neq 0$ by [Theorem 0.7.1](#), so

X is invertible by [Theorem 0.9.8](#). Hence $T \subseteq \text{GL}_2(F)$, and $T \neq \emptyset$ since $I_2 \in T$. Now let $Y = \begin{pmatrix} a' & b' \\ 0 & d' \end{pmatrix} \in T$. By [Theorem 0.9.8](#), $Y^{-1} = (a'd')^{-1} \begin{pmatrix} d' & -b' \\ 0 & a' \end{pmatrix} = \begin{pmatrix} a'^{-1} & -b'(a'd')^{-1} \\ 0 & d'^{-1} \end{pmatrix}$. Therefore

$$XY^{-1} = \begin{pmatrix} aa'^{-1} & -ab'(a'd')^{-1} + bd'^{-1} \\ 0 & dd'^{-1} \end{pmatrix},$$

whose $(2,1)$ -entry is 0 and whose diagonal entries aa'^{-1} , dd'^{-1} are non-zero by [Theorem 0.7.1](#). So $XY^{-1} \in T$, and T is a subgroup by [Theorem 0.10.2](#).

- (b) It depends on the field. *Case 1:* F has an element c with $c \neq 0$ and $c \neq 1$ (for example $c = 2$ in $\mathbb{R}$ or $\mathbb{F}_3$). Let $X = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ and $Y = \text{diag}(1, c)$, both in T . Then $XY = \begin{pmatrix} 1 & c \\ 0 & c \end{pmatrix}$ and $YX = \begin{pmatrix} 1 & 1 \\ 0 & c \end{pmatrix}$, which differ in the $(1,2)$ -entry since $c \neq 1$. So T is **not** abelian. *Case 2:* $F = \{0, 1\}$, that is, $F = \mathbb{F}_2$. Then $a = d = 1$ is forced, so $T = \left\{ \begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix} : b \in \mathbb{F}_2 \right\}$, and $\begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & b' \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & b+b' \\ 0 & 1 \end{pmatrix}$ is symmetric in b, b' . So T is abelian over $\mathbb{F}_2$, and over every other field it is not.

Exercise B3: The exponential is an isomorphism.

Let $\mathbb{R}_{>0} = \{x \in \mathbb{R} : x > 0\}$. Show that $(\mathbb{R}_{>0}, \cdot)$ is a group, and that $\exp: (\mathbb{R}, +) \rightarrow (\mathbb{R}_{>0}, \cdot)$, $x \mapsto e^x$, is an isomorphism. What is its kernel? You may use the familiar facts $e^{x+y} = e^x e^y$, $e^x > 0$, and that $\ln: \mathbb{R}_{>0} \rightarrow \mathbb{R}$ satisfies $\ln(e^x) = x$ and $e^{\ln y} = y$.

Solution. A product of positive reals is positive, so multiplication is a binary operation on $\mathbb{R}_{>0}$. It is associative, $1 \in \mathbb{R}_{>0}$ is an identity, and for $y > 0$ the inverse $1/y$ is positive. Hence $(\mathbb{R}_{>0}, \cdot)$ is a group.

Since $e^x > 0$, $\exp$ maps $\mathbb{R}$ into $\mathbb{R}_{>0}$, and $e^{x+y} = e^x e^y$ says exactly that it is a homomorphism from $(\mathbb{R}, +)$ to $(\mathbb{R}_{>0}, \cdot)$. The function $\ln$ satisfies $\ln \circ \exp = \text{id}_{\mathbb{R}}$ and $\exp \circ \ln = \text{id}_{\mathbb{R}_{>0}}$, so $\exp$ is bijective by [Theorem 0.4.7](#). Hence $\exp$ is an isomorphism. Its kernel is $\{x \in \mathbb{R} : e^x = 1\} = \{0\}$, since $\exp$ is injective and $e^0 = 1$.

0.10.5.3 C. Going deeper

Exercise C1: Every element its own inverse.

- (a) Let G be a group in which $g^2 = gg = e$ for every $g \in G$. Prove that G is abelian.
 (b) Show that $(M_2(\mathbb{F}_2), +)$ is an example of such a group, written additively.

Solution.

- (a) Let $a, b \in G$. The hypothesis $gg = e$ says that every g is its own inverse, by uniqueness of inverses ([Theorem 0.10.1](#)). Applying this to ab , a and b , and using [Theorem 0.10.1](#),

$$ab = (ab)^{-1} = b^{-1}a^{-1} = ba.$$

Hence G is abelian.

- (b) $(M_2(\mathbb{F}_2), +)$ is a group by [Theorem 0.9.1](#): addition is associative, 0 is an identity, and $-A$ is an inverse. In additive notation, “ $g^2 = e$ ” reads $A + A = 0$, and indeed $(A + A)_{ij} = a_{ij} + a_{ij} = (1 + 1)a_{ij} = 0$ in $\mathbb{F}_2$. (This group is abelian anyway, consistent with (a).)

Exercise C2: The order of a product of disjoint cycles.

The **order** of $\sigma \in S_n$ is the smallest integer $m \geq 1$ with $\sigma^m = \text{id}$, where σ^m is the product of m copies of σ . Two cycles are **disjoint** if no element appears in both.

- (a) Let $\sigma = (i_1 \cdots i_k)$ be a k -cycle. Prove that $\sigma^m = \text{id}$ if and only if k divides m . Deduce that σ has order k .
- (b) Prove that disjoint cycles σ and τ commute, and deduce $(\sigma\tau)^m = \sigma^m\tau^m$ for all $m \geq 1$.
- (c) Let σ be a k -cycle and τ an l -cycle, disjoint. Prove that $\sigma\tau$ has order $\text{lcm}(k, l)$.
- (d) Find the order of $(1\ 2)(3\ 4\ 5) \in S_5$.

Hint for (c): σ^m moves only elements that appear in σ .

Solution.

- (a) Index the entries of the cycle modulo k , so $\sigma(i_j) = i_{j+1}$ with $i_{k+1} = i_1$. By induction on m , $\sigma^m(i_j) = i_{j+m}$, indices taken modulo k , and σ^m fixes every element not in the cycle. Hence $\sigma^m = \text{id}$ if and only if $i_{j+m} = i_j$ for all j . Since $i_1, \dots, i_k$ are distinct, this holds if and only if $j + m \equiv j \pmod{k}$, that is, $k \mid m$. The smallest $m \geq 1$ with $k \mid m$ is k , so σ has order k .
- (b) Let A and B be the disjoint sets of elements appearing in σ and τ . Note that σ maps A to A and fixes everything outside A , and τ does the same with B . Let $i \in \{1, \dots, n\}$. If $i \in A$, then $i \notin B$ and $\sigma(i) \in A$, so $\sigma(i) \notin B$; hence $\sigma\tau(i) = \sigma(i) = \tau\sigma(i)$. The case $i \in B$ is the same with the roles swapped. If $i \notin A \cup B$, both sides give i . Hence $\sigma\tau = \tau\sigma$. Then $(\sigma\tau)^m = \sigma^m\tau^m$ follows by induction on m : $(\sigma\tau)^{m+1} = \sigma^m\tau^m\sigma\tau = \sigma^m\sigma\tau^m\tau = \sigma^{m+1}\tau^{m+1}$, moving σ past τ one factor at a time, which is allowed because they commute.
- (c) By (b), $(\sigma\tau)^m = \sigma^m\tau^m$. We claim this is id if and only if $\sigma^m = \text{id}$ and $\tau^m = \text{id}$. One direction is clear. Conversely, suppose $\sigma^m\tau^m = \text{id}$, and let $i \in A$. Then $i \notin B$, so $\tau^m(i) = i$ by the hint, and $\sigma^m(i) = \sigma^m\tau^m(i) = i$. Since σ^m also fixes every $i \notin A$, $\sigma^m = \text{id}$, and then $\tau^m = \sigma^m\tau^m = \text{id}$ as well. By (a), $(\sigma\tau)^m = \text{id}$ if and only if $k \mid m$ and $l \mid m$. The smallest such $m \geq 1$ is $\text{lcm}(k, l)$, which is therefore the order of $\sigma\tau$.
- (d) $(1\ 2)$ and $(3\ 4\ 5)$ are disjoint cycles of lengths 2 and 3, so by (c) the order is $\text{lcm}(2, 3) = 6$.

Chapter 1

Vector Spaces and Dimensions

In this chapter, we introduce the fundamental concept of a vector space and discuss its properties, including basis and dimension.

1.1 Vector Spaces

Why do we bother with a list of eight axioms? To a student first encountering Linear Algebra, this can feel like a bureaucratic exercise in “checking boxes.” However, there is a profound beauty hidden in this formality.

In our preliminary chapter, we saw three very different worlds: the world of geometric arrows ($\mathbb{R}^n$), the world of functions ($F[x]$), and the world of matrices ($M_{m \times n}$). On the surface, an arrow is not a polynomial, and a polynomial is not a matrix. But if you squint, they all behave the same: you can add them, and you can scale them.

By defining a **Vector Space** through these axioms, we are choosing to ignore what the objects *are* and focus entirely on how they *act*. If we prove a theorem using only these eight axioms, that theorem becomes a “universal law” that applies to arrows, functions, and matrices all at once. This is the power of abstraction: solve the problem once, and you solve it for every universe that obeys these rules.

1.1.1 Vector Spaces and Vectors

Definition 1.1.1 — Vector Space

A set V over a field F with two operations: addition $+$ and scalar multiplication $\cdot$.

$$+ : V \times V \rightarrow V, \quad \cdot : F \times V \rightarrow V$$

such that satisfy the following axioms for all $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3 \in V$ and $\alpha, \beta \in F$:

(VS1) $\mathbf{v}_1 + \mathbf{v}_2 = \mathbf{v}_2 + \mathbf{v}_1$

(VS2) $(\mathbf{v}_1 + \mathbf{v}_2) + \mathbf{v}_3 = \mathbf{v}_1 + (\mathbf{v}_2 + \mathbf{v}_3)$

- (VS3) There exists $\mathbf{0} \in V$ such that $\mathbf{v}_1 + \mathbf{0} = \mathbf{v}_1$
- (VS4) There exists $\mathbf{v}'_1 \in V$ such that $\mathbf{v}_1 + \mathbf{v}'_1 = \mathbf{0}$
- (VS5) $\alpha(\mathbf{v}_1 + \mathbf{v}_2) = \alpha\mathbf{v}_1 + \alpha\mathbf{v}_2$
- (VS6) $(\alpha + \beta)\mathbf{v}_1 = \alpha\mathbf{v}_1 + \beta\mathbf{v}_1$
- (VS7) $\alpha(\beta\mathbf{v}_1) = (\alpha\beta)\mathbf{v}_1$
- (VS8) $1\mathbf{v}_1 = \mathbf{v}_1$

Remark. Instead of writing $(V, +, \cdot)$ is a vector space over F , we usually simplify it to V is a vector space over F , or even just V is a vector space if the context is clear.

Remark. We sometimes will abuse the name and refer to $\mathbf{0} = \mathbf{0}_V \in V$ as the “zero vector” of the vector space V . Readers should not confuse $\mathbf{0}_V$ with the zero scalar $0 = 0_F \in F$ from the field F .

Example. (Examples of Vector Spaces)

The following are classic examples of vector spaces:

1. The standard example: F^n with the usual operations of $+$ and $\cdot$ is a vector space over F .
2. Working with polynomials is common too: $F[x]$ with the usual operations of $+$ and $\cdot$ is a vector space over F .
3. Let $\mathcal{D}$ be any open interval. Let

$$C(\mathcal{D}) := \{f : \mathcal{D} \rightarrow \mathbb{R} : f \text{ is continuous}\}.$$

Then $C(\mathcal{D})$ is a vector space over $\mathbb{R}$. The zero vector of this vector space is given by the zero function ($x \mapsto 0$).

4. Let F be any field and fix $n \in \mathbb{N}$. Then $M_n(F)$ is a vector space over F .
5. The set $V = \mathbb{R}_{>0}$ of positive real numbers forms a vector space over $F = \mathbb{R}$ under the following operations: for $x, y \in V$ and $\alpha \in F$, define

$$x \oplus y := xy, \quad \alpha \odot x := x^\alpha = e^{\alpha \log x}.$$

Under these operations, the zero vector is $\mathbf{0} = 1$, and the additive inverse of x is x^{-1} .

You can check if addition and scalar multiplication make sense and follow all axioms of vector spaces.

Example. ($\mathbb{Q}$ Adjoin $\sqrt{2}$)

Show that $\mathbb{Q}(\sqrt{2}) := \{a + b\sqrt{2} : a, b \in \mathbb{Q}\}$ is a vector space over $\mathbb{Q}$ by checking all eight axioms.

Solution. Let $\mathbf{v}_1 = a_1 + b_1\sqrt{2}$, $\mathbf{v}_2 = a_2 + b_2\sqrt{2}$, $\mathbf{v}_3 = a_3 + b_3\sqrt{2} \in \mathbb{Q}(\sqrt{2})$ and $\alpha, \beta \in \mathbb{Q}$.

(VS1)

$$\begin{aligned}
\mathbf{v}_1 + \mathbf{v}_2 &= (a_1 + a_2) + (b_1 + b_2)\sqrt{2} \\
&= (a_2 + a_1) + (b_2 + b_1)\sqrt{2} \\
&= \mathbf{v}_2 + \mathbf{v}_1
\end{aligned}$$

(VS2)

$$\begin{aligned}
(\mathbf{v}_1 + \mathbf{v}_2) + \mathbf{v}_3 &= (a_1 + a_2 + a_3) + (b_1 + b_2 + b_3)\sqrt{2} \\
&= \mathbf{v}_1 + (\mathbf{v}_2 + \mathbf{v}_3)
\end{aligned}$$

(VS3) The element $\mathbf{0} = 0 + 0\sqrt{2} = 0 \in \mathbb{Q}(\sqrt{2})$ satisfies

$$\begin{aligned}
\mathbf{v}_1 + \mathbf{0} &= (a_1 + 0) + (b_1 + 0)\sqrt{2} \\
&= \mathbf{v}_1
\end{aligned}$$

(VS4) For $\mathbf{v}_1 = a_1 + b_1\sqrt{2}$, define $-\mathbf{v}_1 := (-a_1) + (-b_1)\sqrt{2} \in \mathbb{Q}(\sqrt{2})$. Then

$$\begin{aligned}
\mathbf{v}_1 + (-\mathbf{v}_1) &= (a_1 - a_1) + (b_1 - b_1)\sqrt{2} \\
&= \mathbf{0}
\end{aligned}$$

(VS5)

$$\begin{aligned}
\alpha(\mathbf{v}_1 + \mathbf{v}_2) &= \alpha(a_1 + a_2) + \alpha(b_1 + b_2)\sqrt{2} \\
&= (\alpha a_1 + \alpha a_2) + (\alpha b_1 + \alpha b_2)\sqrt{2} \\
&= \alpha\mathbf{v}_1 + \alpha\mathbf{v}_2
\end{aligned}$$

(VS6)

$$\begin{aligned}
(\alpha + \beta)\mathbf{v}_1 &= (\alpha + \beta)a_1 + (\alpha + \beta)b_1\sqrt{2} \\
&= (\alpha a_1 + \beta a_1) + (\alpha b_1 + \beta b_1)\sqrt{2} \\
&= \alpha\mathbf{v}_1 + \beta\mathbf{v}_1
\end{aligned}$$

(VS7)

$$\begin{aligned}
\alpha(\beta\mathbf{v}_1) &= \alpha(\beta a_1 + \beta b_1\sqrt{2}) \\
&= \alpha\beta a_1 + \alpha\beta b_1\sqrt{2} \\
&= (\alpha\beta)\mathbf{v}_1
\end{aligned}$$

(VS8)

$$\begin{aligned}
1 \cdot \mathbf{v}_1 &= 1 \cdot a_1 + 1 \cdot b_1\sqrt{2} \\
&= a_1 + b_1\sqrt{2} \\
&= \mathbf{v}_1
\end{aligned}$$

Thus $\mathbb{Q}(\sqrt{2})$ is a vector space over $\mathbb{Q}$.**Definition 1.1.2 — Vectors and Scalars**

Let V be a vector space over F . Then the elements of V are called **vectors**, and the elements of F are called **scalars**.

Additionally, $\mathbf{0} \in V$ is called the **zero vector**, and $(\mathbf{v}'_1)$ in (VS4) is called the **inverse element** of $\mathbf{v}_1$.

1.1.2 Vector Space Theorems

Theorem 1.1.1 — Left Cancellation Law

Let V be a vector space over F , let $\mathbf{u}, \mathbf{v}_1, \mathbf{v}_2 \in V$. If $\mathbf{u} + \mathbf{v}_1 = \mathbf{u} + \mathbf{v}_2$, then $\mathbf{v}_1 = \mathbf{v}_2$.

Proof. Let $\mathbf{u}'$ be an inverse of $\mathbf{u}$.

$$\begin{array}{ll}
 \mathbf{u} + \mathbf{v}_1 = \mathbf{u} + \mathbf{v}_2 & \text{(given)} \\
 \mathbf{u}' + (\mathbf{u} + \mathbf{v}_1) = \mathbf{u}' + (\mathbf{u} + \mathbf{v}_2) & \text{(add } \mathbf{u}' \text{ to both sides)} \\
 (\mathbf{u}' + \mathbf{u}) + \mathbf{v}_1 = (\mathbf{u}' + \mathbf{u}) + \mathbf{v}_2 & \text{(by VS2)} \\
 \mathbf{0} + \mathbf{v}_1 = \mathbf{0} + \mathbf{v}_2 & \text{(by VS4)} \\
 \mathbf{v}_1 = \mathbf{v}_2 & \text{(by VS3)}
 \end{array}$$

■

Theorem 1.1.2 — Right Cancellation Law

Let V be a vector space over F , let $\mathbf{u}, \mathbf{v}_1, \mathbf{v}_2 \in V$. If $\mathbf{v}_1 + \mathbf{u} = \mathbf{v}_2 + \mathbf{u}$, then $\mathbf{v}_1 = \mathbf{v}_2$.

Proof. Since we have $\mathbf{u} + \mathbf{v}_1 = \mathbf{u} + \mathbf{v}_2 \implies \mathbf{v}_1 = \mathbf{v}_2$, applying (VS1) to it gives $\mathbf{v}_1 + \mathbf{u} = \mathbf{v}_2 + \mathbf{u} \implies \mathbf{v}_1 = \mathbf{v}_2$.

■

Theorem 1.1.3 — Zero Vector is Unique

Let V be a vector space over F . The zero vector $\mathbf{0} \in V$ is unique.

Proof. Suppose that there are two vectors $\mathbf{0}_1, \mathbf{0}_2$.

$$\begin{array}{ll}
 \mathbf{0}_1 + \mathbf{0}_2 = \mathbf{0}_2 + \mathbf{0}_1 & \text{(by VS1)} \\
 \mathbf{0}_1 = \mathbf{0}_2 & \text{(by VS3)}
 \end{array}$$

■

Theorem 1.1.4 — Additive Inverse is Unique

Let V be a vector space over F . Then for every $\mathbf{v} \in V$, its additive inverse described in (VS4), which is $\mathbf{v}'$, is unique.

Proof. Let $\mathbf{v}'_1, \mathbf{v}'_2$ both be inverses of $\mathbf{v}$. Then

$$\begin{aligned}
 \mathbf{v}'_1 &= \mathbf{v}'_1 + \mathbf{0} && \text{(by VS3)} \\
 &= \mathbf{v}'_1 + (\mathbf{v} + \mathbf{v}'_2) && \text{(by VS4)} \\
 &= (\mathbf{v}'_1 + \mathbf{v}) + \mathbf{v}'_2 && \text{(by VS2)} \\
 &= \mathbf{0} + \mathbf{v}'_2 && \text{(by VS4)} \\
 &= \mathbf{v}'_2 && \text{(by VS3)}
 \end{aligned}$$

■

Definition 1.1.3 — Notation of Additive Inverse

The unique inverse of a vector $\mathbf{v} \in V$, for a vector space V over F , will be denoted as $-\mathbf{v}$.

Theorem 1.1.5 — Zero Scalar Annihilates

Let $\mathbf{v} \in V$ for a vector space V over F . Then $0 \cdot \mathbf{v} = \mathbf{0}$.

Proof. Let $\mathbf{w} = 0 \cdot \mathbf{v}$. We show that $\mathbf{w} = \mathbf{0}$.

$$\begin{aligned}
 \mathbf{w} + \mathbf{w} &= 0 \cdot \mathbf{v} + 0 \cdot \mathbf{v} \\
 &= (0 + 0) \cdot \mathbf{v} && \text{(by VS6)} \\
 &= 0 \cdot \mathbf{v} && \text{(arithmetic in } F) \\
 &= \mathbf{w} \\
 (\mathbf{w} + \mathbf{w}) + (-\mathbf{w}) &= \mathbf{w} + (-\mathbf{w}) && \text{(add } -\mathbf{w} \text{ to both sides)} \\
 \mathbf{w} + (\mathbf{w} + (-\mathbf{w})) &= \mathbf{0} && \text{(by VS2, VS4)} \\
 \mathbf{w} + \mathbf{0} &= \mathbf{0} && \text{(by VS4)} \\
 \mathbf{w} &= \mathbf{0} && \text{(by VS3)}
 \end{aligned}$$

■

Theorem 1.1.6 — Negation as Scalar Multiplication

Let V be a vector space over F and $\mathbf{v} \in V$ be any vector. Then $-\mathbf{v} = (-1) \cdot \mathbf{v}$.

Proof. We show that $(-1) \cdot \mathbf{v}$ is an additive inverse of $\mathbf{v}$:

$$\begin{aligned}
 \mathbf{v} + (-1) \cdot \mathbf{v} &= 1 \cdot \mathbf{v} + (-1) \cdot \mathbf{v} && \text{(by VS8)} \\
 &= (1 + (-1)) \cdot \mathbf{v} && \text{(by VS6)} \\
 &= 0 \cdot \mathbf{v} && \text{(arithmetic in } F) \\
 &= \mathbf{0} && (*)
 \end{aligned}$$

By $(*)$, we use [Theorem 1.1.5](#).

Since $(-1) \cdot \mathbf{v}$ is an additive inverse of $\mathbf{v}$, by uniqueness we have $-\mathbf{v} = (-1) \cdot \mathbf{v}$. ■

Theorem 1.1.7 — Negative Scalar Distribution

Let V be a vector space over F . For any $\alpha \in F$ and $\mathbf{v} \in V$, we have $(-\alpha)\mathbf{v} = -(\alpha\mathbf{v})$.

Proof. We show that $(-\alpha)\mathbf{v}$ is an additive inverse of $\alpha\mathbf{v}$:

$$\begin{aligned}
 \alpha\mathbf{v} + (-\alpha)\mathbf{v} &= (\alpha + (-\alpha))\mathbf{v} && \text{(by VS6)} \\
 &= 0 \cdot \mathbf{v} && \text{(arithmetic in } F) \\
 &= \mathbf{0} && (*)
 \end{aligned}$$

By $(*)$, we use [Theorem 1.1.5](#).

Since $(-\alpha)\mathbf{v}$ is an additive inverse of $\alpha\mathbf{v}$, by uniqueness we have $(-\alpha)\mathbf{v} = -(\alpha\mathbf{v})$. ■

Theorem 1.1.8 — Scalar Multiplication by Zero Vector

Let V be a vector space over F . For any scalar $\alpha \in F$, we have $\alpha\mathbf{0} = \mathbf{0}$.

Proof. Let $\mathbf{w} = \alpha\mathbf{0}$. We show that $\mathbf{w} = \mathbf{0}$.

$$\begin{aligned}
 \mathbf{w} &= \alpha\mathbf{0} \\
 &= \alpha(\mathbf{0} + \mathbf{0}) && \text{(by VS3)} \\
 &= \alpha\mathbf{0} + \alpha\mathbf{0} && \text{(by VS5)} \\
 &= \mathbf{w} + \mathbf{w} \\
 \mathbf{w} + (-\mathbf{w}) &= (\mathbf{w} + \mathbf{w}) + (-\mathbf{w}) && \text{(add } -\mathbf{w} \text{ to both sides)} \\
 \mathbf{0} &= \mathbf{w} + (\mathbf{w} + (-\mathbf{w})) && \text{(by VS4, VS2)} \\
 \mathbf{0} &= \mathbf{w} + \mathbf{0} && \text{(by VS4)} \\
 \mathbf{0} &= \mathbf{w} && \text{(by VS3)}
 \end{aligned}$$
■

Theorem 1.1.9 — Zero Product Law

Let V be a vector space over F , $\mathbf{v} \in V$, and $\alpha \in F$. If $\alpha\mathbf{v} = \mathbf{0}$, then either $\alpha = 0$ or $\mathbf{v} = \mathbf{0}$.

Proof. Suppose $\alpha\mathbf{v} = \mathbf{0}$. We consider two cases:

- **Case $\alpha = 0$:** Then the statement holds.
- **Case $\alpha \neq 0$:** Since F is a field, α has a multiplicative inverse α^{-1} .

$$\begin{array}{ll}
 \alpha\mathbf{v} = \mathbf{0} & \text{(given)} \\
 \alpha^{-1}(\alpha\mathbf{v}) = \alpha^{-1}\mathbf{0} & \text{(multiply by } \alpha^{-1}\text{)} \\
 (\alpha^{-1}\alpha)\mathbf{v} = \mathbf{0} & \text{(by VS7 and previous Theorem)} \\
 1 \cdot \mathbf{v} = \mathbf{0} & \text{(field inverse property)} \\
 \mathbf{v} = \mathbf{0} & \text{(by VS8)}
 \end{array}$$

Thus, if $\alpha \neq 0$, we must have $\mathbf{v} = \mathbf{0}$. ■

1.2 Subspaces

Some vector spaces might be “too big” and be unwieldy to deal with. Therefore, it is often convenient to look at “smaller vector spaces” inside a given vector space. These are called **subspaces**.

1.2.1 Definition and the Subspace Test

Definition 1.2.1 — Subspace

Let V be a vector space over F . A subset W of V is called a **subspace** of V , if W is also a vector space over F with addition and scalar multiplication inherited from V .

Remark. Let $(V, +, \cdot)$ be a vector space over F , and $W \subseteq V$. If $(W, +, \cdot)$ is also a vector space over F then W is a subspace of V .

Example. (Trivial Subspaces)

Given a vector space V , then

- V is a subspace of V itself;
- $\{\mathbf{0}\}$ is a subspace of V (also known as the “zero subspace”);
- $\emptyset$ is **not** a subspace of V .

Theorem 1.2.1 — Subspace Test

Let V be a vector space over F and $W \subseteq V$. Then W is a subspace of V if and only if the following are satisfied:

- W is non-empty;
- W is closed under addition: $\mathbf{w}_1 + \mathbf{w}_2 \in W$ for all $\mathbf{w}_1, \mathbf{w}_2 \in W$;
- W is closed under scalar multiplication: $\alpha \mathbf{w} \in W$ for all $\alpha \in F$ and $\mathbf{w} \in W$.

Proof. ($\Rightarrow$) If W is a subspace, then W is a vector space, so it contains the zero vector (hence non-empty) and is closed under addition and scalar multiplication by definition.

($\Leftarrow$) Assume the three conditions hold. Since $W \subseteq V$, the operations on W are inherited from V , so associativity, commutativity, and distributivity are automatically satisfied. It remains to verify the existence of identity elements and inverses.

- **Zero vector:** Since $W \neq \emptyset$, there exists $\mathbf{w} \in W$. By closure under scalar multiplication, $0 \cdot \mathbf{w} = \mathbf{0} \in W$.
- **Additive inverse:** For any $\mathbf{w} \in W$, closure under scalar multiplication gives $(-1) \cdot \mathbf{w} = -\mathbf{w} \in W$.

Thus W is a vector space under the inherited operations, i.e., a subspace of V . ■

Remark. Note that from the proof, the first condition of the Subspace Test can be replaced by the requirement that W contains the zero vector of V .

Remark. In practice, to test whether a given subset of a vector space is a subspace, we use Theorem 1.2.1 instead of checking all eight vector space axioms from scratch, which is much less tedious. Moreover, some introductory textbooks use Theorem 1.2.1 as the definition of a subspace. While this is logically equivalent, it can be somewhat less intuitive than defining a subspace as a subset that is itself a vector space.

1.2.2 Examples of Subspaces

Example. (Line as Subspace)

$W = \{(x, y) \in \mathbb{R}^2 : x + 2y = 0\}$ is a subspace of $\mathbb{R}^2$. Indeed:

- **Zero vector:** $(0, 0) \in W$ since $0 + 2(0) = 0$.
- **Addition:** If $(x_1, y_1), (x_2, y_2) \in W$, then $(x_1 + x_2) + 2(y_1 + y_2) = (x_1 + 2y_1) + (x_2 + 2y_2) = 0 + 0 = 0$, so their sum is in W .
- **Scalar multiplication:** If $(x, y) \in W$ and $c \in \mathbb{R}$, then $(cx) + 2(cy) = c(x + 2y) = c(0) = 0$, so $c(x, y) \in W$.

Example. (Translated Line is Not a Subspace)

$W = \{(x, y) \in \mathbb{R}^2 : x + 2y = 3\}$ is **not** a subspace of $\mathbb{R}^2$. This is because it does not contain the zero vector: $0 + 2(0) = 0 \neq 3$, so $(0, 0) \notin W$.

Example. (Polynomials of Bounded Degree)

Let $F[x]$ be the vector space of all polynomials over a field F . For a fixed $n \in \mathbb{N}$, the set $F[x]_{\leq n}$ of polynomials of degree at most n is a subspace of $F[x]$.

- **Zero vector:** The zero polynomial has degree $-\infty$, so $0 \in F[x]_{\leq n}$.
- **Addition:** If $\deg p \leq n$ and $\deg q \leq n$, then $\deg(p + q) \leq \max(\deg p, \deg q) \leq n$.
- **Scalar multiplication:** If $\deg p \leq n$ and $c \in F$, then $\deg(cp) \leq \deg p \leq n$.

Example. (Polynomials of Exact Degree)

The set $F[x]_{=n}$ of polynomials of degree **exactly** n is **not** a subspace of $F[x]$.

- It does not contain the zero vector (since $\deg 0 = -\infty \neq n$).
- It is not closed under addition. For example, if $p(x) = x^n + x$ and $q(x) = -x^n$, both have degree n , but $p(x) + q(x) = x$, which has degree $1 \neq n$ (assuming $n > 1$).

Example. (Differentiable Functions)

Let $C(\mathbb{R}) = \{f : \mathbb{R} \rightarrow \mathbb{R} \mid f \text{ is continuous}\}$ be the set of all continuous functions on $\mathbb{R}$. We consider the set of differentiable functions:

$$D(\mathbb{R}) = \{f : \mathbb{R} \rightarrow \mathbb{R} \mid f \text{ is differentiable}\}$$

To verify that $D(\mathbb{R})$ is a subspace of $C(\mathbb{R})$:

- **Subset:** Since differentiability implies continuity, we have $D(\mathbb{R}) \subseteq C(\mathbb{R})$.
- **Zero vector:** The zero function $f(x) = 0$ is differentiable, so $\mathbf{0} \in D(\mathbb{R})$.
- **Closure under addition:** If $f, g \in D(\mathbb{R})$, then $f + g$ is differentiable with $(f + g)' = f' + g'$, so $f + g \in D(\mathbb{R})$.
- **Closure under scalar multiplication:** If $f \in D(\mathbb{R})$ and $c \in \mathbb{R}$, then cf is differentiable with $(cf)' = cf'$, so $cf \in D(\mathbb{R})$.

Thus, $D(\mathbb{R})$ is a subspace of $C(\mathbb{R})$.

Example. (Symmetric Matrices)

The set of all symmetric matrices in $M_n(F)$, denoted by $S_n(F) = \{\mathbf{A} \in M_n(F) : \mathbf{A} = \mathbf{A}^\top\}$, is a subspace of $M_n(F)$.

- **Zero vector:** The zero matrix $\mathbf{O}$ is symmetric since $\mathbf{O}^\top = \mathbf{O}$, so $\mathbf{O} \in S_n(F)$.
- **Addition:** If $\mathbf{A}, \mathbf{B} \in S_n(F)$, then $(\mathbf{A} + \mathbf{B})^\top = \mathbf{A}^\top + \mathbf{B}^\top = \mathbf{A} + \mathbf{B}$, so $\mathbf{A} + \mathbf{B} \in S_n(F)$.
- **Scalar multiplication:** If $\mathbf{A} \in S_n(F)$ and $c \in F$, then $(c\mathbf{A})^\top = c\mathbf{A}^\top = c\mathbf{A}$, so $c\mathbf{A} \in S_n(F)$.

Example. (Trace-free Matrices)

The set of all trace-free matrices in $M_n(F)$, denoted by $W = \{\mathbf{A} \in M_n(F) : \text{tr}(\mathbf{A}) = 0\}$, is a subspace of $M_n(F)$.

- **Zero vector:** $\text{tr}(\mathbf{O}) = 0 + 0 + \cdots + 0 = 0$, so $\mathbf{O} \in W$.
- **Addition:** If $\mathbf{A}, \mathbf{B} \in W$, then $\text{tr}(\mathbf{A} + \mathbf{B}) = \text{tr}(\mathbf{A}) + \text{tr}(\mathbf{B}) = 0 + 0 = 0$, so $\mathbf{A} + \mathbf{B} \in W$.
- **Scalar multiplication:** If $\mathbf{A} \in W$ and $c \in F$, then $\text{tr}(c\mathbf{A}) = c \text{tr}(\mathbf{A}) = c(0) = 0$, so $c\mathbf{A} \in W$.

1.2.3 Intersections and Unions

Theorem 1.2.2 — Intersection of Subspaces

Let W_1 and W_2 be subspaces of a vector space V . Then the intersection $W_1 \cap W_2$ is also a subspace of V .

Proof. We use the Subspace Test (Theorem 1.2.1) on $W = W_1 \cap W_2$:

- **Zero vector:** Since W_1 and W_2 are subspaces, $\mathbf{0} \in W_1$ and $\mathbf{0} \in W_2$. Thus $\mathbf{0} \in W_1 \cap W_2$.
- **Closure under addition:** Let $\mathbf{w}, \mathbf{z} \in W_1 \cap W_2$. Then $\mathbf{w}, \mathbf{z} \in W_1$ and $\mathbf{w}, \mathbf{z} \in W_2$. Since W_1 and W_2 are subspaces, they are closed under addition, so $\mathbf{w} + \mathbf{z} \in W_1$ and $\mathbf{w} + \mathbf{z} \in W_2$. Hence $\mathbf{w} + \mathbf{z} \in W_1 \cap W_2$.
- **Closure under scalar multiplication:** Let $\mathbf{w} \in W_1 \cap W_2$ and $c \in F$. Then $\mathbf{w} \in W_1$ and $\mathbf{w} \in W_2$. Since W_1 and W_2 are closed under scalar multiplication, $c\mathbf{w} \in W_1$ and $c\mathbf{w} \in W_2$. Hence $c\mathbf{w} \in W_1 \cap W_2$.

Since all three conditions of the Subspace Test are satisfied, $W_1 \cap W_2$ is a subspace of V . ■

Remark. Unlike the intersection, the **union** of two subspaces is not necessarily a subspace. For $W_1 \cup W_2$ to be a subspace, one subspace must be contained within the other (i.e., $W_1 \subseteq W_2$ or $W_2 \subseteq W_1$).

Example. (Union of Axes)

Consider the vector space $V = \mathbb{R}^2$. Let W_1 be the x -axis and W_2 be the y -axis:

$$W_1 = \{(x, 0) : x \in \mathbb{R}\}, \quad W_2 = \{(0, y) : y \in \mathbb{R}\}.$$

Both W_1 and W_2 are subspaces of $\mathbb{R}^2$. However, their union $W_1 \cup W_2$ is **not** a subspace because it is not closed under addition.

Indeed, $(1, 0) \in W_1 \subseteq W_1 \cup W_2$ and $(0, 1) \in W_2 \subseteq W_1 \cup W_2$, but their sum:

$$(1, 0) + (0, 1) = (1, 1)$$

is not in $W_1 \cup W_2$ since $(1, 1)$ is neither on the x -axis nor the y -axis.

Proposition 1.2.3 — Union of Subspaces

Let W_1, W_2 be subspaces of V . Then $W_1 \cup W_2$ is a subspace of V if and only if $W_1 \subseteq W_2$ or $W_2 \subseteq W_1$.

Proof. ($\Leftarrow$) If $W_1 \subseteq W_2$, then $W_1 \cup W_2 = W_2$, which is a subspace. Similarly, if $W_2 \subseteq W_1$, the union is W_1 , which is also a subspace.

($\Rightarrow$) We prove the contrapositive: if neither is contained in the other, then the union is not a subspace. Assume $W_1 \not\subseteq W_2$ and $W_2 \not\subseteq W_1$.

- Since $W_1 \not\subseteq W_2$, there exists a vector $\mathbf{u} \in W_1$ such that $\mathbf{u} \notin W_2$.
- Since $W_2 \not\subseteq W_1$, there exists a vector $\mathbf{v} \in W_2$ such that $\mathbf{v} \notin W_1$.

Consider the sum $\mathbf{w} = \mathbf{u} + \mathbf{v}$. We show that $\mathbf{w} \notin W_1 \cup W_2$:

- If $\mathbf{w} \in W_1$, then $\mathbf{v} = \mathbf{w} - \mathbf{u}$. Since $\mathbf{w} \in W_1$ and $\mathbf{u} \in W_1$, their difference $\mathbf{v}$ must be in W_1 (by closure). But we chose $\mathbf{v} \notin W_1$, which is a contradiction.
- If $\mathbf{w} \in W_2$, then $\mathbf{u} = \mathbf{w} - \mathbf{v}$. Since $\mathbf{w} \in W_2$ and $\mathbf{v} \in W_2$, their difference $\mathbf{u}$ must be in W_2 . But we chose $\mathbf{u} \notin W_2$, which is a contradiction.

Therefore, $\mathbf{u} + \mathbf{v}$ is in neither W_1 nor W_2 , so it is not in $W_1 \cup W_2$. Thus, the union is not closed under addition and is therefore not a subspace. ■

1.3 Linear Combinations and Span

In the previous section, we studied subspaces. A natural question arises: given a set of vectors, how can we describe the “smallest” subspace that contains them? This leads us to the concept of linear combinations and the span.

1.3.1 Linear Combinations

Definition 1.3.1 — Linear Combination

Let V be a vector space over F , $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m\}$ be a subset of V . Then the expression

$$a_1 \mathbf{v}_1 + a_2 \mathbf{v}_2 + \dots + a_m \mathbf{v}_m$$

is called a **linear combination** of S . The scalars a_i are called the **coefficients** of the

linear combination.

Example. (Examples of Linear Combinations)

Below are some examples of linear combinations in different vector spaces:

1. The vector $(2, 1, 0)$ in $\mathbb{R}^3$ is a linear combination of $S = \{(1, 2, 3), (4, 5, 6)\}$ because

$$(2, 1, 0) = -2 \cdot (1, 2, 3) + 1 \cdot (4, 5, 6).$$

2. The polynomial $f(x) = x^3 + 2x + 1$ in $\mathbb{R}[x]$ is a linear combination of $S = \{1, x, x^2, x^3\}$ because

$$f(x) = 1 \cdot 1 + 2 \cdot x + 0 \cdot x^2 + 1 \cdot x^3.$$

1.3.2 Span and Spanning Sets

Definition 1.3.2 — Span

Let S be a subset of a vector space V over F . The **span** of S , denoted by $\text{span}_F(S)$, is the set of all linear combinations of finite subsets of elements of S .

If $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$, then $\text{span}_F(S) = \{a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + \dots + a_n\mathbf{v}_n : a_i \in F\}$. By convention, $\text{span}_F(\emptyset) = \{\mathbf{0}\}$.

Remark. When the underlying field is clear, we abuse the notation of span_F and write it as span .

Theorem 1.3.1 — Spanning Set as Subspace

The span of any subset $S \subseteq V$ is a subspace of V . Moreover, it is the smallest subspace of V containing S .

Proof. Let $W = \text{span}(S)$.

- **Zero vector:** Since $\text{span}(\emptyset) = \{\mathbf{0}\}$, and for non-empty S , taking all coefficients as zero gives $\mathbf{0}$, we have $\mathbf{0} \in W$.
- **Addition:** Let $\mathbf{x}, \mathbf{y} \in W$. Then $\mathbf{x} = \sum a_i \mathbf{v}_i$ and $\mathbf{y} = \sum b_i \mathbf{v}_i$. Their sum $\mathbf{x} + \mathbf{y} = \sum (a_i + b_i) \mathbf{v}_i$ is also a linear combination of elements in S , so $\mathbf{x} + \mathbf{y} \in W$.
- **Scalar multiplication:** For any $c \in F$, $c\mathbf{x} = \sum (ca_i) \mathbf{v}_i \in W$.

To see it is the smallest subspace, note that any subspace containing S must be closed under addition and scalar multiplication, and thus must contain all linear combinations of elements in S .

■

Definition 1.3.3 — Spanning Set

If $\text{span}(S) = V$, we say that S **spans** V , or that S is a **spanning set** for V .

Example. (Spanning Set for Polynomials)

The set $\{1, x, x^2\}$ spans $\mathbb{R}[x]_{\leq 2}$, the space of polynomials of degree at most 2.

1.4 Linear Independence

A spanning set is like a toolbox that is guaranteed to have every tool you need. But a messy toolbox might have three different hammers that all do the same thing. In mathematics, as in engineering, we value **efficiency**.

If one of the vectors in your spanning set can already be built using the others, then that vector is “dead weight”—it isn’t helping you reach any new territory. We call such a set *linearly dependent*.

In this section, we search for the “cleanest” possible sets. We want to know: “Is every vector in this collection actually contributing something unique, or is someone just coasting on the work of the others?”

1.4.1 Definition and Examples

Definition 1.4.1 — Linear Independence

A subset S of a vector space V is said to be **linearly independent** if for any distinct vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n \in S$, the only solution to the equation:

$$a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + \cdots + a_n\mathbf{v}_n = \mathbf{0}$$

is the trivial solution $a_1 = a_2 = \cdots = a_n = 0$.

If there exists a non-trivial solution (where at least one $a_i \neq 0$), then S is said to be **linearly dependent**.

Remark. Geometrically, in $\mathbb{R}^2$, two vectors are linearly dependent if and only if they lie on the same line through the origin (one is a multiple of the other). In $\mathbb{R}^3$, three vectors are linearly dependent if and only if they lie on the same plane through the origin.

Example. (Examples of Linear Independence)

The following examples illustrate how to check for linear independence in various settings:

1. Let $S_1 = \{(1, 2, 3), (4, 5, 6)\} \subseteq \mathbb{R}^3$. To check linear independence, suppose

$$a(1, 2, 3) + b(4, 5, 6) = (0, 0, 0) \implies \begin{cases} a + 4b = 0 \\ 2a + 5b = 0 \\ 3a + 6b = 0 \end{cases}$$

The first two equations give $a = -4b$ and $2(-4b) + 5b = 0 \implies -3b = 0$. Thus $a = b = 0$, so S_1 is **linearly independent**.

2. Let $S_2 = \{(1, 2, 3), (4, 5, 6), (7, 8, 9)\} \subseteq \mathbb{R}^3$. Note that

$$(1, 2, 3) - 2(4, 5, 6) + (7, 8, 9) = (1 - 8 + 7, 2 - 10 + 8, 3 - 12 + 9) = (0, 0, 0).$$

Since there exists a non-trivial solution, S_2 is **linearly dependent**.

3. Consider $\{2x + 1, x^2 + 3\}$ in $\mathbb{R}[x]_{\leq 2}$. Suppose

$$a(2x + 1) + b(x^2 + 3) = 0 \implies bx^2 + 2ax + (a + 3b) = 0.$$

Comparing coefficients of $x^2, x, 1$, we have $b = 0$, $2a = 0$, and $a + 3b = 0$. This implies $a = b = 0$, so the set is **linearly independent**.

4. Consider $\{\sin x, \cos x, e^x\}$ in $D(\mathbb{R})$. Suppose for all $x \in \mathbb{R}$:

$$a \sin x + b \cos x + ce^x = 0.$$

Setting $x = 0$ gives $b + c = 0$. Setting $x = \pi$ gives $-b + ce^\pi = 0$. Adding these gives $c(1 + e^\pi) = 0$, so $c = 0$ and hence $b = 0$. Finally, setting $x = \pi/2$ gives $a + ce^{\pi/2} = 0 \implies a = 0$. Thus, the set is **linearly independent**.

Theorem 1.4.1 — Linear Dependence Lemma

$\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m\}$ is linearly dependent if and only if either $\mathbf{v}_1 = \mathbf{0}$ or for some r , $\mathbf{v}_r$ is a linear combination of $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_{r-1}\}$.

Proof. ($\implies$) Suppose $S = \{\mathbf{v}_1, \dots, \mathbf{v}_m\}$ is linearly dependent. Then there exist $a_1, \dots, a_m \in F$, not all zero, such that

$$a_1 \mathbf{v}_1 + \dots + a_m \mathbf{v}_m = \mathbf{0}.$$

Let r be the largest index such that $a_r \neq 0$. Then

$$a_1 \mathbf{v}_1 + \dots + a_{r-1} \mathbf{v}_{r-1} + a_r \mathbf{v}_r = \mathbf{0}.$$

So

$$a_r \mathbf{v}_r = -(a_1 \mathbf{v}_1 + \dots + a_{r-1} \mathbf{v}_{r-1}),$$

and dividing by a_r gives

$$\mathbf{v}_r = -\frac{a_1}{a_r} \mathbf{v}_1 - \dots - \frac{a_{r-1}}{a_r} \mathbf{v}_{r-1}.$$

Hence $\mathbf{v}_r$ is a linear combination of $\{\mathbf{v}_1, \dots, \mathbf{v}_{r-1}\}$. (If $r = 1$, this says $a_1 \mathbf{v}_1 = \mathbf{0}$ with $a_1 \neq 0$, so $\mathbf{v}_1 = \mathbf{0}$.)

($\impliedby$) We prove by cases:

- If $\mathbf{v}_1 = \mathbf{0}$, then

$$1 \cdot \mathbf{v}_1 + 0 \cdot \mathbf{v}_2 + \cdots + 0 \cdot \mathbf{v}_m = \mathbf{0},$$

with coefficients not all zero, so S is linearly dependent.

- If for some r we have $\mathbf{v}_r$ is a linear combination of $\{\mathbf{v}_1, \dots, \mathbf{v}_{r-1}\}$, then there exist scalars $c_1, \dots, c_{r-1} \in F$ such that

$$\mathbf{v}_r = c_1 \mathbf{v}_1 + \cdots + c_{r-1} \mathbf{v}_{r-1}.$$

Move everything to one side:

$$c_1 \mathbf{v}_1 + \cdots + c_{r-1} \mathbf{v}_{r-1} - 1 \cdot \mathbf{v}_r = \mathbf{0}.$$

Extending coefficients by zeros for $\mathbf{v}_{r+1}, \dots, \mathbf{v}_m$, we get a nontrivial linear combination of $\mathbf{v}_1, \dots, \mathbf{v}_m$ equal to $\mathbf{0}$, so S is linearly dependent. ■

Remark. The above theorem can be paraphrased to: A set $S = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ with $n \geq 2$ is linearly dependent if and only if at least one vector in S can be written as a linear combination of the others.

1.4.2 The Redundant Members

Having the last remark in mind, we can develop an intuition that says linear independence of a set is equivalent to requiring the set containing no “redundant members.” Next we want to develop the intuition saying that throwing away these “redundant members” will not change the linear span.

Theorem 1.4.2 — Span Preservation

Let $S = \{\mathbf{v}_1, \dots, \mathbf{v}_m, \mathbf{w}\} \subseteq V$. If $\mathbf{w} \in \text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_m\}$, then

$$\text{span}(S) = \text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_m\}.$$

Proof. Let $W = \text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_m\}$ and $W' = \text{span}(S)$. Since $\{\mathbf{v}_1, \dots, \mathbf{v}_m\} \subseteq S$, it is clear that $W \subseteq W'$.

For the reverse inclusion, let $\mathbf{v} \in W'$. Then $\mathbf{v}$ is a linear combination of vectors in S :

$$\mathbf{v} = a_1 \mathbf{v}_1 + \cdots + a_m \mathbf{v}_m + b \mathbf{w}$$

for some scalars a_i, b . Since $\mathbf{w} \in W$, there exist scalars $c_1, \dots, c_m$ such that $\mathbf{w} = c_1 \mathbf{v}_1 + \cdots + c_m \mathbf{v}_m$. Substituting this into the expression for $\mathbf{v}$:

$$\mathbf{v} = a_1 \mathbf{v}_1 + \cdots + a_m \mathbf{v}_m + b(c_1 \mathbf{v}_1 + \cdots + c_m \mathbf{v}_m) = (a_1 + bc_1) \mathbf{v}_1 + \cdots + (a_m + bc_m) \mathbf{v}_m.$$

Thus $\mathbf{v}$ is a linear combination of $\mathbf{v}_1, \dots, \mathbf{v}_m$, so $\mathbf{v} \in W$. This shows $W' \subseteq W$. Hence $W' = W$. ■

1.5 Bases

We have reached what many consider the “Goldilocks Zone” of Linear Algebra.

If a set is too small, it won’t span the whole space—you’ll be left with points you can’t reach. If a set is too large, it will be linearly dependent—you’ll have redundant vectors cluttering your workspace. A **basis** is a set that is “just right.” It is large enough to build everything, but small enough that every piece is essential.

Because a basis has no redundancy, it gives us something incredible: a **unique address system**. In an abstract vector space, it’s hard to tell someone where a vector is. But once we pick a basis, every vector can be described by a unique list of numbers—its coordinates. A basis is the bridge that allows us to turn abstract geometry into concrete arithmetic.

1.5.1 Definition and Standard Bases

Definition 1.5.1 — Basis

A subset B of a vector space V is called a **basis** for V if:

1. B is linearly independent;
2. B spans V .

Example. (Standard Bases)

Commonly used standard bases for familiar vector spaces are listed below:

- The **standard basis** for F^n is $\{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n\}$, where $\mathbf{e}_i$ has a 1 in the i -th position and 0 elsewhere.
- The standard basis for $F[x]_{\leq n}$ is $\{1, x, x^2, \dots, x^n\}$.
- The standard basis for $M_{m \times n}(F)$ is the set of matrices $\mathbf{E}_{ij}$ having a 1 at entry (i, j) and 0 elsewhere.

1.5.2 Unique Representation and Coordinates

The most important property of a basis is that every vector in the space has a unique “address” relative to it.

Theorem 1.5.1 — Unique Representation

Let $B = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ be a basis for V . Then every vector $\mathbf{v} \in V$ can be written as a linear combination of elements in B in **exactly one way**.

Proof. Since B spans V , there exist scalars a_i such that $\mathbf{v} = \sum a_i \mathbf{v}_i$. Suppose there is

another representation $\mathbf{v} = \sum b_i \mathbf{v}_i$. Then:

$$\mathbf{0} = \mathbf{v} - \mathbf{v} = \sum a_i \mathbf{v}_i - \sum b_i \mathbf{v}_i = \sum (a_i - b_i) \mathbf{v}_i.$$

Since B is linearly independent, we must have $a_i - b_i = 0$ for all i , meaning $a_i = b_i$. ■

Definition 1.5.2 — Coordinates

Let $B = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ be an **ordered** basis for V . For any $\mathbf{v} \in V$, the unique scalars $c_1, \dots, c_n$ such that $\mathbf{v} = \sum c_i \mathbf{v}_i$ are called the **coordinates** of $\mathbf{v}$ with respect to B . We write this as a column vector:

$$[\mathbf{v}]_B = \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix}.$$

Remark. Notice the subtle but important distinction: a **basis as a set** determines which vectors can be built (i.e., it determines the space), but to define **coordinates** we need an **ordered basis**—a basis with a fixed ordering of its elements. For example, if $B = \{\mathbf{v}_1, \mathbf{v}_2\}$, then the coordinate vector $[\mathbf{v}]_B = \begin{bmatrix} 3 \\ 5 \end{bmatrix}$ means $\mathbf{v} = 3\mathbf{v}_1 + 5\mathbf{v}_2$. If we had listed the basis in the opposite order $B' = \{\mathbf{v}_2, \mathbf{v}_1\}$, the same vector $\mathbf{v}$ would have coordinates $[\mathbf{v}]_{B'} = \begin{bmatrix} 5 \\ 3 \end{bmatrix}$. The underlying space is the same, but the coordinate representation depends on the ordering.

Remark. The map $\mathbf{v} \mapsto [\mathbf{v}]_B$ provides a way to treat any abstract n -dimensional vector space as if it were simply F^n . This is the power of a basis.

1.6 Sifting and the Replacement Theorem

How “big” is a vector space? We intuitively know that a plane is “bigger” than a line, and 3D space is “bigger” than a plane. But in the abstract world of vector spaces, we need a rigorous way to measure this.

The answer lies in the number of vectors in a basis. But there’s a potential problem: what if one person finds a basis with 3 vectors, and another person finds a basis for the same space with 4 vectors? If that were possible, our entire concept of “size” would collapse.

In this section, we prove the most important technical result of the chapter: the **Steinitz Replacement Theorem**. This theorem is the “engine” that guarantees every basis of a space has the exact same number of vectors. Once we have this, we can finally define the **Dimension** of a space—a single number that captures its fundamental complexity.

1.6.1 The Sifting Method

Theorem 1.6.1 — Sifting Method

Let $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m\}$ be a subset of V . Then there exists a subset $S' \subseteq S$ such that:

1. S' is linearly independent, and
2. $\text{span}(S') = \text{span}(S)$.

Proof. For each $r = 1, 2, \dots, m$, we check whether $\mathbf{v}_r$ is a linear combination of the previous terms $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_{r-1}$. We:

1. Keep $\mathbf{v}_r$ in S' if $\mathbf{v}_r$ is not a linear combination of $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_{r-1}$ (not redundant);
2. Remove $\mathbf{v}_r$ from S' if $\mathbf{v}_r$ is a linear combination of $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_{r-1}$ (redundant).

After this process, we obtain a subset S' in which no element is a linear combination of the preceding members anymore. Therefore, by [Theorem 1.4.1](#), S' is linearly independent. Moreover, according to [Theorem 1.4.2](#), removing “redundant members” from S does not change its linear span. Therefore, we have $\text{span}(S') = \text{span}(S)$. ■

Remark. The sifting method is a powerful algorithmic tool that allows us to “clean up” any set of vectors. Given any collection of vectors, we can systematically remove the redundant ones while preserving the span. This gives us a “cleaned” version that is linearly independent and spans the same space.

This method becomes our “new weapon” in linear algebra proofs. Whenever we encounter a set that might have redundant vectors, we can apply the sifting method to obtain a linearly independent subset with the same span. This technique will be particularly useful in proving results about dimension and in constructing bases from spanning sets.

Example. (Sifting Example)

Consider $S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4, \mathbf{v}_5, \mathbf{v}_6\}$ in $\mathbb{R}^3$ where:

$$\begin{array}{lll} \mathbf{v}_1 = (1, 1, 1), & \mathbf{v}_2 = (2, 2, 2), & \mathbf{v}_3 = (1, 0, 0), \\ \mathbf{v}_4 = (3, 2, 2), & \mathbf{v}_5 = (1, 1, 0), & \mathbf{v}_6 = (0, 0, 1). \end{array}$$

We apply the sifting method ([Theorem 1.6.1](#)):

- $\mathbf{v}_1 \neq \mathbf{0}$, so $\mathbf{v}_1$ is kept.
- $\mathbf{v}_2 = 2\mathbf{v}_1$, so $\mathbf{v}_2$ is removed.
- $\mathbf{v}_3$ is not a linear combination of $\mathbf{v}_1$, so $\mathbf{v}_3$ is kept.
- $\mathbf{v}_4 = 2\mathbf{v}_1 + 1\mathbf{v}_3$, so $\mathbf{v}_4$ is removed.
- Suppose $\mathbf{v}_5 = a\mathbf{v}_1 + b\mathbf{v}_3$. Then $(1, 1, 0) = (a + b, a, a)$, which gives $a = 0$ and $a + b = 1$, so $b = 1$. But then the second coordinate gives $1 = a = 0$, which is a contradiction. So $\mathbf{v}_5$ is not a linear combination of $\mathbf{v}_1, \mathbf{v}_3$. Hence, $\mathbf{v}_5$ is kept.
- $\mathbf{v}_6 = (0, 0, 1) = (1, 1, 1) - (1, 1, 0) = \mathbf{v}_1 - \mathbf{v}_5$, so $\mathbf{v}_6$ is removed.

The sifted set is $S' = \{\mathbf{v}_1, \mathbf{v}_3, \mathbf{v}_5\} = \{(1, 1, 1), (1, 0, 0), (1, 1, 0)\}$, which is linearly independent and spans the same space as S .

Example. (Sifting in $\mathbb{R}^2$)

Consider $S = \{(1, 0), (0, 1), (1, 1)\}$ in $\mathbb{R}^2$.

- $(1, 0) \neq \mathbf{0}$, so $(1, 0)$ is kept.
- $(0, 1) \notin \text{span}\{(1, 0)\}$, so $(0, 1)$ is kept.
- $(1, 1) = (1, 0) + (0, 1) \in \text{span}\{(1, 0), (0, 1)\}$, so $(1, 1)$ is removed.

The sifted set $\{(1, 0), (0, 1)\}$ is linearly independent and still spans $\mathbb{R}^2$.

1.6.2 The Steinitz Replacement Theorem

Theorem 1.6.2 — Steinitz Exchange/Replacement Theorem

Let V be a vector space over F . Suppose:

- $G = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ spans V ;
- $L = \{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_m\}$ is linearly independent.

Then $m \leq n$.

Proof. The strategy is to insert the vectors from L one-by-one into G and “sift” the resulting set to maintain the spanning property. We rely on the fact that if we add a vector $\mathbf{w}$ to a spanning set S , the set $\{\mathbf{w}\} \cup S$ becomes linearly dependent, allowing us to remove a redundant vector (by Theorem 1.4.2).

Consider the first step. We form the set B_1 by inserting $\mathbf{u}_1$ into G :

$$B_1 = \{\mathbf{u}_1, \mathbf{v}_1, \dots, \mathbf{v}_n\}.$$

Since G spans V , B_1 is linearly dependent. By applying the sifting process (Theorem 1.6.1), we can remove a vector to obtain a new spanning set G_1 . Since L is linearly independent, $\mathbf{u}_1 \neq \mathbf{0}$, so the removed vector must be some $\mathbf{v}_i$. Thus:

$$G_1 = \{\mathbf{u}_1\} \cup S_1,$$

where S_1 is a subset of G with size $n - 1$.

Now, consider the general step. Suppose we have successfully exchanged k vectors such that the set

$$G_k = \{\mathbf{u}_1, \dots, \mathbf{u}_k\} \cup S_k$$

spans V , where $S_k \subset G$ contains $n - k$ vectors. We insert the next vector $\mathbf{u}_{k+1}$ to form:

$$B_{k+1} = \{\mathbf{u}_{k+1}\} \cup G_k.$$

Since G_k is a spanning set, $\mathbf{u}_{k+1}$ depends on the vectors in G_k , making B_{k+1} dependent. We must remove a vector to restore the spanning property. We cannot remove any of

the $\mathbf{u}_i$'s, as this would imply a linear dependence relation exclusively among elements of L , contradicting the hypothesis that L is linearly independent. Therefore, we must remove some $\mathbf{v}_j$ from S_k .

This process generates a sequence of subsets of remaining $\mathbf{v}$'s:

$$n = |G| > |S_1| > |S_2| > \cdots > |S_m| \geq 0.$$

Since we are able to perform this exchange m times (once for each $\mathbf{u} \in L$), we must have started with enough $\mathbf{v}$'s to accommodate every removal. Therefore, $n \geq m$. ■

1.6.3 Consequences of the Replacement Theorem

Having proved the Steinitz Replacement Theorem, the major conclusion we can draw from it is that any (finite) basis of the same vector space will have the same size.

Theorem 1.6.3 — Basis Theorem

If a vector space V over F has a finite basis, then every basis of V has the same number of vectors.

Proof. Let B_1 and B_2 be two bases with n and m vectors respectively.

- Since B_1 is linearly independent and B_2 spans, by Replacement Theorem, $n \leq m$.
- Since B_2 is linearly independent and B_1 spans, by Replacement Theorem, $m \leq n$.

Thus $n = m$. ■

Definition 1.6.1 — Dimension

A vector space V over F is called **finite-dimensional** if it has a finite basis. The **dimension** of V , denoted $\dim_F(V)$, is the number of vectors in any basis for V . By convention, $\dim_F(\{\mathbf{0}\}) = 0$.

Remark. The convention $\dim(\{\mathbf{0}\}) = 0$ is consistent with the fact that the **only basis for the zero space is the empty set** $\emptyset$. Indeed, $\emptyset$ is vacuously linearly independent (there is no nontrivial linear combination to consider), and $\text{span}(\emptyset) = \{\mathbf{0}\}$ by our earlier convention. Since $|\emptyset| = 0$, we have $\dim(\{\mathbf{0}\}) = 0$.

Remark. When the underlying field is clear, we abuse the notation of $\dim_F$ and write it as $\dim$.

Example. (Examples of Dimensions)

$\dim(F^n) = n$, $\dim(F[x]_{\leq n}) = n + 1$, and $\dim(M_{m \times n}(F)) = mn$.

Remark. The notation $\dim_F(V)$ reminds us that **dimension depends on the choice of field**. For instance, recall from [Example \(Q Adjoin \$\sqrt{2}\$ \)](#) that $\mathbb{Q}(\sqrt{2}) = \{a + b\sqrt{2} : a, b \in \mathbb{Q}\}$ is a vector space over $\mathbb{Q}$ with basis $\{1, \sqrt{2}\}$, so $\dim_{\mathbb{Q}}(\mathbb{Q}(\sqrt{2})) = 2$.

As a more striking example, consider the complex numbers $\mathbb{C}$. As a vector space over itself, $\mathbb{C}$ has basis $\{1\}$, so $\dim_{\mathbb{C}}(\mathbb{C}) = 1$. However, as a vector space over $\mathbb{R}$, $\mathbb{C}$ has basis $\{1, i\}$, so $\dim_{\mathbb{R}}(\mathbb{C}) = 2$. The same set can have different dimensions depending on which field we use for scalar multiplication!

Theorem 1.6.4 — Size Bounds for Independent and Spanning Sets

Let V be a vector space of dimension n over F , and $S \subseteq V$ be a subset.

1. If $|S| > n$, then S cannot be linearly independent.
2. If $|S| < n$, then S cannot span V .

Proof.

1. Given $|S| > n$, and suppose for the sake of contradiction that S is linearly independent. Let B be a basis of V , then we have $|B| = n$ and B is a spanning set of V . By Replacement Theorem, we have

$$\underbrace{|S|}_{\text{size of lin. indep. set}} \leq \underbrace{|B|}_{\text{size of spanning set}} = n.$$

Giving a contradiction, hence S cannot be linearly independent.

2. Given $|S| < n$, and suppose for the sake of contradiction that S spans V . Let B be a basis of V , then we have $|B| = n$ and B is linearly independent. By Replacement Theorem, we have

$$n = \underbrace{|B|}_{\text{size of lin. indep. set}} \leq \underbrace{|S|}_{\text{size of spanning set}}.$$

This implies $n \leq |S|$, which contradicts the assumption that $|S| < n$. Hence S cannot span V . ■

Remark.

$$\text{Size of lin. indep. set} \leq \text{Size of basis} \leq \text{Size of spanning set}.$$

Theorem 1.6.5 — Basis Extension Theorem

Let V be a finite-dimensional vector space and S be a linearly independent subset of V .

Then S can be extended into a basis of V .

Proof. Since V is finite-dimensional, it has a finite basis $B = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$. Consider the set $S \cup B$. Since B spans V , $S \cup B$ also spans V . We apply the Sifting Method (Theorem 1.6.1) to the ordered set formed by listing the elements of S first, followed by the elements of B .

The Sifting Method will keep all elements of S because S is linearly independent. It then proceeds to examine each element of B , keeping those that are not in the span of the vectors already kept. The resulting subset S' spans $\text{span}(S \cup B) = V$ and is linearly independent by construction. Thus, S' is a basis for V that contains S . ■

Corollary 1.6.6 — Existence of Basis

Every finite spanning set of a vector space V contains a basis.

Proof. Apply the Sifting Method (Theorem 1.6.1) to the spanning set. The resulting subset S' is linearly independent by construction and still spans V . Therefore, S' is a basis for V . ■

Theorem 1.6.7 — Dimension Implies Equality

Let V and W be finite-dimensional vector spaces over F . If

1. $W \subseteq V$;
2. $\dim_F(W) = \dim_F(V)$,

then $W = V$.

Proof. If we show $V \subseteq W$ then we are done. Suppose for contradiction that $V \not\subseteq W$. That means there exists some $\mathbf{v} \in V$ that $\mathbf{v} \notin W$.

Let $S = \{\mathbf{w}_1, \dots, \mathbf{w}_n\}$ be a basis of W , (consequently, $\dim_F(W) = n$). Since $\mathbf{v} \notin W$, we have $\mathbf{v} \notin \text{span}(S)$.

Consider $S' = \{\mathbf{w}_1, \dots, \mathbf{w}_n, \mathbf{v}\}$. Since S is linearly independent, no $\mathbf{w}_i$ is a linear combination of $\mathbf{w}_1, \dots, \mathbf{w}_{i-1}$; and $\mathbf{v}$ is not a linear combination of $\mathbf{w}_1, \dots, \mathbf{w}_n$ because $\mathbf{v} \notin \text{span}(S)$. By Theorem 1.4.1, S' is linearly independent in V . By Replacement Theorem, we have

$$\underbrace{|S'|}_{\text{size of lin. indep. set}} \leq \dim_F(V),$$

which implies $n + 1 \leq n$, which is a contradiction, therefore $V \subseteq W$. ■

Remark. This theorem makes proving $W = V$ somewhat easier. As if we wanted to prove equivalence before, we can only do it by proving $W \subseteq V$ and $V \subseteq W$. Now we can just prove one side and then state that their dimension is the same.

An immediate consequence of the above theorem is the following:

Theorem 1.6.8 — Maximal Independent Subset is a Basis

Let V be a vector space of dimension n over F . Then any n independent vectors form a basis of V . (We say that “any maximal linearly independent subset forms a basis”).

Proof. Let S be a linearly independent set of size n . We want to show that $\text{span}(S) = V$.

Let $W = \text{span}(S)$, this implies that W is a subspace of V , which implies $W \subseteq V$.

Notice that S spans W and S is linearly independent, therefore S is a basis of W . So $\dim_F(W) = |S| = n = \dim_F(V)$. By Theorem 1.6.7, we have $W = V$, then $\text{span}(S) = V$. ■

1.7 Sums of Subspaces

In the previous sections, we studied intersections and unions of subspaces. While the intersection of two subspaces is always a subspace, the union typically is not. This raises a natural question: what is the “right” way to combine two subspaces into a larger one?

The answer is the **sum** of subspaces. Given two subspaces W_1 and W_2 , their sum $W_1 + W_2$ consists of all vectors that can be written as $\mathbf{w}_1 + \mathbf{w}_2$ for some $\mathbf{w}_1 \in W_1$ and $\mathbf{w}_2 \in W_2$. This construction always yields a subspace—in fact, it is the smallest subspace containing both W_1 and W_2 .

A particularly nice situation occurs when W_1 and W_2 have only the zero vector in common. In this case, their sum is called a **direct sum**, and every vector in the sum can be decomposed uniquely into a piece from W_1 and a piece from W_2 . Direct sums are fundamental in decomposing vector spaces into simpler, independent components.

1.7.1 Definition and Properties

Definition 1.7.1 — Sum of Subspaces

Let W_1, W_2 be subspaces of a vector space V . Then we define

$$W_1 + W_2 := \{\mathbf{w}_1 + \mathbf{w}_2 : \mathbf{w}_1 \in W_1 \text{ and } \mathbf{w}_2 \in W_2\}.$$

Theorem 1.7.1 — Sum of Subspaces Theorem

$W_1 + W_2$ is a subspace of V . Furthermore, this is the smallest subspace containing both W_1 and W_2 .

Proof. To prove that $W_1 + W_2$ is a subspace of V , we verify the three conditions of the Subspace Test (Theorem 1.2.1):

- Non-emptiness: As W_1, W_2 are subspaces, they both contain $\mathbf{0}$. And since $\mathbf{0} = \mathbf{0}_{W_1} + \mathbf{0}_{W_2}$, therefore $W_1 + W_2$ is non-empty.
- Closed under addition: Let $\mathbf{a}, \mathbf{b} \in W_1 + W_2$. Then we have

$$\begin{cases} \mathbf{a} = \mathbf{w}_1 + \mathbf{w}_2, & \text{for some } \mathbf{w}_1 \in W_1, \mathbf{w}_2 \in W_2 \\ \mathbf{b} = \mathbf{w}'_1 + \mathbf{w}'_2, & \text{for some } \mathbf{w}'_1 \in W_1, \mathbf{w}'_2 \in W_2 \end{cases}$$

Then $\mathbf{a} + \mathbf{b} = \underbrace{(\mathbf{w}_1 + \mathbf{w}'_1)}_{\in W_1} + \underbrace{(\mathbf{w}_2 + \mathbf{w}'_2)}_{\in W_2} \in W_1 + W_2$. So $W_1 + W_2$ is closed under addition.

- Closed under scalar multiplication: Let $\mathbf{a} \in W_1 + W_2$, and $\lambda \in F$. Then we have $\mathbf{a} = \mathbf{w}_1 + \mathbf{w}_2$ for some $\mathbf{w}_1 \in W_1, \mathbf{w}_2 \in W_2$. Then $\lambda \mathbf{a} = \underbrace{\lambda \mathbf{w}_1}_{\in W_1} + \underbrace{\lambda \mathbf{w}_2}_{\in W_2} \in W_1 + W_2$. So $W_1 + W_2$ is closed under scalar multiplication.

This concludes that $W_1 + W_2$ is a subspace.

To prove that $W_1 + W_2$ is the smallest subspace containing both W_1 and W_2 , we first prove that it actually contains W_1 and W_2 , then we prove that it is the smallest subspace of V containing so.

- Containment: Since W_1 and W_2 are subspaces, they both contain the zero vector $\mathbf{0}$. Since for any $\mathbf{w}_1 \in W_1$, we can write $\mathbf{w}_1 = \mathbf{w}_1 + \mathbf{0}$, by $\mathbf{0} \in W_2$, we have $\mathbf{w}_1 \in W_1 + W_2$. Thus $W_1 \subseteq W_1 + W_2$. The same logic applies to W_2 therefore $W_2 \subseteq W_1 + W_2$.
- Minimality: Let U be any subspace of V such that $W_1 \subseteq U$ and $W_2 \subseteq U$. We want to show that $W_1 + W_2 \subseteq U$. Take any element $\mathbf{v} \in W_1 + W_2$. By definition $\mathbf{v} = \mathbf{w}_1 + \mathbf{w}_2$ for some $\mathbf{w}_1 \in W_1$ and $\mathbf{w}_2 \in W_2$. Because $W_1 \subseteq U$, then $\mathbf{w}_1 \in U$. Similarly, because $W_2 \subseteq U$, then $\mathbf{w}_2 \in U$. At last, since U is a subspace, it is closed under addition. Therefore

$$\mathbf{w}_1 + \mathbf{w}_2 \in U \implies \mathbf{v} \in U.$$

This concludes that $W_1 + W_2$ is the smallest subspace of V containing W_1 and W_2 . ■

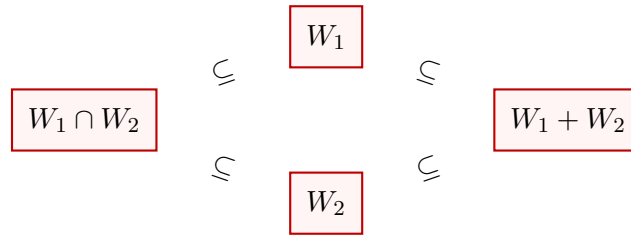
1.7.2 Dimension of $W_1 + W_2$

Theorem 1.7.2 — Dimension Formula of Subspace Sum

Let V be a finite-dimensional vector space, and let W_1, W_2 subspaces of V . Then

$$\dim(W_1 + W_2) = \dim(W_1) + \dim(W_2) - \dim(W_1 \cap W_2).$$

Proof.



Inclusion diagram for subspaces W_1 and W_2

We use the above diagram to build intuition for the proof.

Let $n = \dim(W_1 \cap W_2)$. Let $B = \{\mathbf{b}_1, \dots, \mathbf{b}_n\}$ be a basis of the intersection $W_1 \cap W_2$.

Since B is a linearly independent subset of both W_1 and W_2 , by the Basis Extension Theorem (Theorem 1.6.5), we can extend it to form bases for both subspaces:

1. **For W_1 :** Extend B with $\{\mathbf{s}_1, \dots, \mathbf{s}_p\}$ such that $B_1 = \{\mathbf{b}_1, \dots, \mathbf{b}_n, \mathbf{s}_1, \dots, \mathbf{s}_p\}$ is a basis of W_1 . Note that $\dim(W_1) = n + p$.
2. **For W_2 :** Extend B with $\{\mathbf{t}_1, \dots, \mathbf{t}_q\}$ such that $B_2 = \{\mathbf{b}_1, \dots, \mathbf{b}_n, \mathbf{t}_1, \dots, \mathbf{t}_q\}$ is a basis of W_2 . Note that $\dim(W_2) = n + q$.

Claim. The set $S = \{\mathbf{b}_1, \dots, \mathbf{b}_n, \mathbf{s}_1, \dots, \mathbf{s}_p, \mathbf{t}_1, \dots, \mathbf{t}_q\}$ is a basis of $W_1 + W_2$.

Proof.

- S spans $W_1 + W_2$

Let $\mathbf{w} \in W_1 + W_2$. By definition, $\mathbf{w} = \mathbf{w}_1 + \mathbf{w}_2$ for some $\mathbf{w}_1 \in W_1, \mathbf{w}_2 \in W_2$. Since B_1 spans W_1 and B_2 spans W_2 , both $\mathbf{w}_1$ and $\mathbf{w}_2$ can be written as linear combinations of vectors in S . Thus, their sum $\mathbf{w}$ is also in $\text{span}(S)$.

- S is Linearly Independent

$$\sum_{i=1}^n \alpha_i \mathbf{b}_i + \sum_{j=1}^p \beta_j \mathbf{s}_j + \sum_{k=1}^q \gamma_k \mathbf{t}_k = \mathbf{0} \quad (1.7.1)$$

Rearrange the equation to isolate the terms involving $\mathbf{t}_k$:

$$\mathbf{v} := \sum_{i=1}^n \alpha_i \mathbf{b}_i + \sum_{j=1}^p \beta_j \mathbf{s}_j = - \sum_{k=1}^q \gamma_k \mathbf{t}_k$$

Observe the vector $\mathbf{v}$. From the LHS, we have $\mathbf{v} \in W_1$. From the RHS, we have $\mathbf{v} \in W_2$. Therefore, $\mathbf{v} \in W_1 \cap W_2$.

Since $\mathbf{v} \in W_1 \cap W_2$, it can be written uniquely as a linear combination of the basis B of the intersection:

$$\mathbf{v} = \sum_{i=1}^n \delta_i \mathbf{b}_i$$

Equating this with the RHS expression of $\mathbf{v}$:

$$\sum_{i=1}^n \delta_i \mathbf{b}_i = - \sum_{k=1}^q \gamma_k \mathbf{t}_k \implies \sum_{i=1}^n \delta_i \mathbf{b}_i + \sum_{k=1}^q \gamma_k \mathbf{t}_k = \mathbf{0}.$$

The set $B_2 = \{\mathbf{b}_1, \dots, \mathbf{b}_n, \mathbf{t}_1, \dots, \mathbf{t}_q\}$ is a basis of W_2 and is therefore linearly independent. This forces all coefficients to be zero. Specifically, all $\gamma_k = 0$.

Substitute $\gamma_k = 0$ back into Eq. 1.7.1:

$$\sum_{i=1}^n \alpha_i \mathbf{b}_i + \sum_{j=1}^p \beta_j \mathbf{s}_j = \mathbf{0}$$

Since $B_1 = \{\mathbf{b}_1, \dots, \mathbf{b}_n, \mathbf{s}_1, \dots, \mathbf{s}_p\}$ is a basis of W_1 , it is linearly independent. This forces all $\alpha_i = 0$ and all $\beta_j = 0$.

Since all α, β, γ are zero, S is linearly independent. ■

Use dimension counting, since S is a basis for $W_1 + W_2$:

$$\begin{aligned} \dim(W_1 + W_2) &= |S| \\ &= n + p + q \\ &= (n + p) + (n + q) - n \\ &= \dim(W_1) + \dim(W_2) - \dim(W_1 \cap W_2). \end{aligned}$$



Remark. **Counter-example for 3 Subspaces**

It is a coincidence that this formula resembles the “Inclusion-Exclusion Principle” from set theory. This intuition fails for three or more subspaces. The formula:

$$\dim(W_1 + W_2 + W_3) \neq \sum \dim(W_i) - \sum \dim(W_i \cap W_j) + \dim(W_1 \cap W_2 \cap W_3)$$

Counter-example: Consider three distinct lines through the origin in $\mathbb{R}^2$ (e.g., x-axis, y-axis, and $y = x$).

- LHS: The sum of three lines fills the plane, so dimension is **2**.
- RHS: $1 + 1 + 1 - (0 + 0 + 0) + 0 = 3$.

Since $2 \neq 3$, the formula is false.

1.7.3 Direct Sum of Two Subspaces

Definition 1.7.2 — Direct Sum of Two Subspaces

Let W_1, W_2 be subspaces of a vector space V . If $W_1 \cap W_2 = \{\mathbf{0}\}$, then we write $W_1 + W_2$ as

$$W_1 \oplus W_2$$

and call it the **direct sum** of W_1 and W_2 .

Example.

Consider the vector space $V = M_n(\mathbb{C})$, and two of its subspaces:

- $W_1 = \{\mathbf{A} \in M_n(\mathbb{C}) : \mathbf{A}^\top = \mathbf{A}\}$ (the subspace of symmetric matrices);
- $W_2 = \{\mathbf{A} \in M_n(\mathbb{C}) : \mathbf{A}^\top = -\mathbf{A}\}$ (the subspace of skew-symmetric matrices).

We can derive that $V = W_1 \oplus W_2$.

Proof. We first show that $V = W_1 + W_2$. That is, any square matrix is a sum of a symmetric matrix and a skew-symmetric matrix. Let $\mathbf{A} \in M_n(\mathbb{C})$, notice that we have the identity:

$$\mathbf{A} = \frac{1}{2}(\mathbf{A} + \mathbf{A}^\top) + \frac{1}{2}(\mathbf{A} - \mathbf{A}^\top)$$

where the first term is symmetric and the second term is skew-symmetric. Therefore, any square matrix is a sum of a symmetric matrix and a skew-symmetric matrix.

To establish that $W_1 \cap W_2 = \{\mathbf{O}\}$, let $\mathbf{A} \in W_1 \cap W_2$. By definition, this implies $\mathbf{A} \in W_1$ and $\mathbf{A} \in W_2$. Consequently, the identities $\mathbf{A} = \mathbf{A}^\top$ and $\mathbf{A} = -\mathbf{A}^\top$ must hold simultaneously. Substituting the former into the latter yields $\mathbf{A} = -\mathbf{A}$. This implies $2\mathbf{A} = \mathbf{O}$, and thus $\mathbf{A} = \mathbf{O}$.

Chapter 2

Linear Transformations

Vector spaces are only half of the story; the other half is the maps between them that respect addition and scalar multiplication. This chapter studies these *linear transformations*.

2.0.1 Overview

In this part, we will explore:

- **Linear transformations:** the definition and basic properties, examples and non-examples, and the fact that a linear transformation is determined by its values on a basis.
- **Kernel and image:** the two subspaces attached to every linear transformation, with examples and their basic theorems.
- **The rank–nullity theorem:** nullity and rank, and the identity $\text{nullity}(f) + \text{rank}(f) = \dim(V)$ for a finite-dimensional V .

2.1 Linear Transformations

In Chapter 1, we studied vector spaces as static objects—sets equipped with addition and scalar multiplication. Now we turn to the study of **structure-preserving maps** between vector spaces. Just as group theory studies homomorphisms and topology studies continuous functions, linear algebra studies **linear transformations**.

A linear transformation is a function that “respects” the vector space structure: it preserves addition and scalar multiplication. This simple requirement has profound consequences. Linear transformations are the morphisms in the category of vector spaces, and understanding them is central to all of linear algebra.

2.1.1 Definition and Basic Properties

Definition 2.1.1 — Linear Transformation

Let V, W be vector spaces over F . A function $T : V \rightarrow W$ is called a **linear transfor-**

mation (or **linear map**), if

(LT1) $T(\mathbf{v}_1 + \mathbf{v}_2) = T(\mathbf{v}_1) + T(\mathbf{v}_2)$, for all $\mathbf{v}_1, \mathbf{v}_2 \in V$;

(LT2) $T(\lambda \mathbf{v}) = \lambda T(\mathbf{v})$, for all $\lambda \in F$ and $\mathbf{v} \in V$.

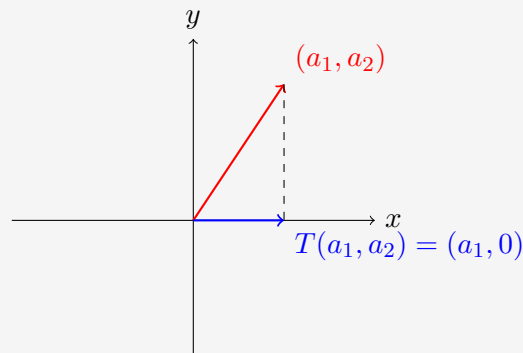
2.1.2 Examples of Linear Transformations

We now present a rich collection of examples. For each, we determine whether the given function is a linear transformation by verifying the two axioms (LT1) and (LT2).

2.1.2.1 Geometric Transformations

Example. (Projection onto a Coordinate Axis)

Determine whether $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by $T(a_1, a_2) = (a_1, 0)$ is a linear transformation.



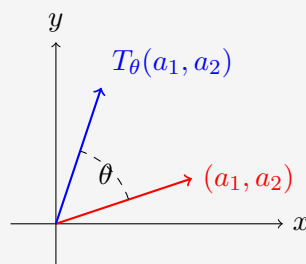
Solution. Let $\mathbf{v} = (a_1, a_2)$ and $\mathbf{w} = (b_1, b_2)$ be vectors in $\mathbb{R}^2$, and let $\lambda \in \mathbb{R}$.

- **Additivity:** $T(\mathbf{v} + \mathbf{w}) = T(a_1 + b_1, a_2 + b_2) = (a_1 + b_1, 0) = (a_1, 0) + (b_1, 0) = T(\mathbf{v}) + T(\mathbf{w})$.
- **Homogeneity:** $T(\lambda \mathbf{v}) = T(\lambda a_1, \lambda a_2) = (\lambda a_1, 0) = \lambda(a_1, 0) = \lambda T(\mathbf{v})$.

Thus T is a linear transformation.

Example. (Rotation in the Plane)

Determine whether rotation by angle θ (counterclockwise) in the plane is a linear transformation. That is, let $T_\theta : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the rotation map.



Solution. Write (a_1, a_2) in polar coordinates: $a_1 = r \cos \phi$ and $a_2 = r \sin \phi$. Then:

$$\begin{aligned} T_\theta(a_1, a_2) &= (r \cos(\theta + \phi), r \sin(\theta + \phi)) \\ &= (r \cos \theta \cos \phi - r \sin \theta \sin \phi, r \sin \theta \cos \phi + r \cos \theta \sin \phi) \\ &= \cos \theta (r \cos \phi, r \sin \phi) + \sin \theta (-r \sin \phi, r \cos \phi) \\ &= \cos \theta (a_1, a_2) + \sin \theta (-a_2, a_1). \end{aligned}$$

Thus $T_\theta(a_1, a_2) = (a_1 \cos \theta - a_2 \sin \theta, a_1 \sin \theta + a_2 \cos \theta)$.

Linearity follows from this explicit formula: for $\mathbf{v}, \mathbf{w} \in \mathbb{R}^2$ and $\lambda \in \mathbb{R}$,

$$\begin{aligned} T_\theta(\mathbf{v} + \mathbf{w}) &= \cos \theta (\mathbf{v} + \mathbf{w}) + \sin \theta \cdot J(\mathbf{v} + \mathbf{w}) = T_\theta(\mathbf{v}) + T_\theta(\mathbf{w}), \\ T_\theta(\lambda \mathbf{v}) &= \cos \theta (\lambda \mathbf{v}) + \sin \theta \cdot J(\lambda \mathbf{v}) = \lambda T_\theta(\mathbf{v}), \end{aligned}$$

where $J(a_1, a_2) = (-a_2, a_1)$ is itself linear.

2.1.2.2 Matrix-Induced Transformations

Example. (Left Multiplication by a Matrix)

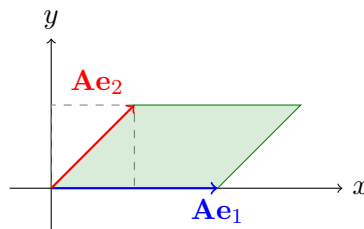
Let $\mathbf{A} \in M_{m \times n}(F)$ be an $m \times n$ matrix. Determine whether $f_{\mathbf{A}} : F^n \rightarrow F^m$ defined by $f_{\mathbf{A}}(\mathbf{v}) = \mathbf{A}\mathbf{v}$ (where $\mathbf{v}$ is viewed as a column vector) is a linear transformation.

Solution. For $\mathbf{v}, \mathbf{w} \in F^n$ and $\lambda \in F$:

- **Additivity:** $f_{\mathbf{A}}(\mathbf{v} + \mathbf{w}) = \mathbf{A}(\mathbf{v} + \mathbf{w}) = \mathbf{A}\mathbf{v} + \mathbf{A}\mathbf{w} = f_{\mathbf{A}}(\mathbf{v}) + f_{\mathbf{A}}(\mathbf{w})$.
- **Homogeneity:** $f_{\mathbf{A}}(\lambda \mathbf{v}) = \mathbf{A}(\lambda \mathbf{v}) = \lambda(\mathbf{A}\mathbf{v}) = \lambda f_{\mathbf{A}}(\mathbf{v})$.

These follow from the distributive and scalar properties of matrix multiplication ([Theorem 0.9.4](#)).

For $F = \mathbb{R}$ and $m = n = 2$, the map $f_{\mathbf{A}}$ sends the unit square to the parallelogram spanned by $\mathbf{A}\mathbf{e}_1$ and $\mathbf{A}\mathbf{e}_2$, the columns of $\mathbf{A}$.



The unit square (dashed) and its image under $\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}$.

Example. (A Specific Matrix Transformation)

Determine whether $T : F^3 \rightarrow F^2$ defined by $T(a_1, a_2, a_3) = (2a_1 + a_3, a_2 - a_1)$ is a linear transformation.

Solution. This is the matrix transformation induced by $\mathbf{A} = \begin{bmatrix} 2 & 0 & 1 \\ -1 & 1 & 0 \end{bmatrix}$, since

$$\begin{bmatrix} 2 & 0 & 1 \\ -1 & 1 & 0 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} = \begin{bmatrix} 2a_1 + a_3 \\ a_2 - a_1 \end{bmatrix}.$$

By the previous example, T is linear.

Example. (Trace)

Determine whether the **trace** function $\text{tr} : M_n(F) \rightarrow F$ defined by $\text{tr}(\mathbf{A}) = \sum_{i=1}^n a_{ii}$ is a linear transformation.

Solution. For matrices $\mathbf{A}, \mathbf{B} \in M_n(F)$ and $\lambda \in F$:

- **Additivity:** $\text{tr}(\mathbf{A} + \mathbf{B}) = \sum_{i=1}^n (a_{ii} + b_{ii}) = \sum_{i=1}^n a_{ii} + \sum_{i=1}^n b_{ii} = \text{tr}(\mathbf{A}) + \text{tr}(\mathbf{B})$.
- **Homogeneity:** $\text{tr}(\lambda \mathbf{A}) = \sum_{i=1}^n \lambda a_{ii} = \lambda \sum_{i=1}^n a_{ii} = \lambda \text{tr}(\mathbf{A})$.

Thus trace is a linear transformation from the vector space of matrices to the scalar field.

2.1.2.3 Polynomial and Function Space Transformations

Example. (Multiplication by x)

Determine whether $T : F[x] \rightarrow F[x]$ defined by $T(f(x)) = x \cdot f(x)$ is a linear transformation.

Solution. For polynomials $f, g \in F[x]$ and $\lambda \in F$:

- **Additivity:** $T(f + g) = x(f + g) = xf + xg = T(f) + T(g)$.
- **Homogeneity:** $T(\lambda f) = x(\lambda f) = \lambda(xf) = \lambda T(f)$.

Thus multiplication by x is a linear transformation.

Example. (Differentiation)

Determine whether $D : \mathbb{R}[x] \rightarrow \mathbb{R}[x]$ defined by $D(f(x)) = f'(x)$ (the derivative) is a linear transformation.

Solution. For polynomials $f, g \in \mathbb{R}[x]$ and $\lambda \in \mathbb{R}$:

- **Additivity:** $D(f + g) = (f + g)' = f' + g' = D(f) + D(g)$.
- **Homogeneity:** $D(\lambda f) = (\lambda f)' = \lambda f' = \lambda D(f)$.

These are the standard sum rule and constant multiple rule from calculus.

Remark. More precisely, $D : \mathbb{R}[x]_{\leq n} \rightarrow \mathbb{R}[x]_{\leq n-1}$ maps polynomials of degree at most n to polynomials of degree at most $n - 1$.

Example. (Integration)

Determine whether $T : \mathbb{R}[x] \rightarrow \mathbb{R}[x]$ defined by $T(f(x)) = \int_0^x f(t) dt$ is a linear transformation.

Solution. For polynomials $f, g \in \mathbb{R}[x]$ and $\lambda \in \mathbb{R}$:

- **Additivity:** $T(f+g) = \int_0^x (f(t)+g(t)) dt = \int_0^x f(t) dt + \int_0^x g(t) dt = T(f) + T(g)$.
- **Homogeneity:** $T(\lambda f) = \int_0^x \lambda f(t) dt = \lambda \int_0^x f(t) dt = \lambda T(f)$.

These follow from the linearity of the integral.

Example. (Shift Operator)

Fix $\theta \in \mathbb{R}$. Determine whether the **shift operator** $S_\theta : C(\mathbb{R}) \rightarrow C(\mathbb{R})$ defined by $S_\theta(f)(x) = f(x - \theta)$ is a linear transformation. (This operator translates the graph of f by θ units to the right.)

Solution. For $f, g \in C(\mathbb{R})$ and $\lambda \in \mathbb{R}$:

- **Additivity:** $S_\theta(f+g)(x) = (f+g)(x-\theta) = f(x-\theta) + g(x-\theta) = S_\theta(f)(x) + S_\theta(g)(x)$.
- **Homogeneity:** $S_\theta(\lambda f)(x) = (\lambda f)(x-\theta) = \lambda f(x-\theta) = \lambda S_\theta(f)(x)$.

Thus the shift operator is linear.

Example. (Composition Operator)

Fix a continuous function $g : \mathbb{R} \rightarrow \mathbb{R}$. Determine whether the **composition operator** $C_g : C(\mathbb{R}) \rightarrow C(\mathbb{R})$ defined by $C_g(f) = f \circ g$, i.e., $C_g(f)(x) = f(g(x))$, is a linear transformation.

Solution. For $f, h \in C(\mathbb{R})$ and $\lambda \in \mathbb{R}$:

- **Additivity:** $C_g(f+h)(x) = (f+h)(g(x)) = f(g(x)) + h(g(x)) = C_g(f)(x) + C_g(h)(x)$.
- **Homogeneity:** $C_g(\lambda f)(x) = (\lambda f)(g(x)) = \lambda f(g(x)) = \lambda C_g(f)(x)$.

The shift operator is the special case where $g(x) = x - \theta$.

2.1.2.4 Identity and Inclusion Maps**Example. (Identity Map)**

Let V be a vector space. Determine whether the **identity map** $\text{id}_V : V \rightarrow V$ defined by $\text{id}_V(\mathbf{v}) = \mathbf{v}$ is a linear transformation.

Solution. For $\mathbf{v}, \mathbf{w} \in V$ and $\lambda \in F$:

- **Additivity:** $\text{id}_V(\mathbf{v} + \mathbf{w}) = \mathbf{v} + \mathbf{w} = \text{id}_V(\mathbf{v}) + \text{id}_V(\mathbf{w})$.
- **Homogeneity:** $\text{id}_V(\lambda\mathbf{v}) = \lambda\mathbf{v} = \lambda\text{id}_V(\mathbf{v})$.

The identity map is trivially linear.

Example. (Inclusion Map)

Let $W \subseteq V$ be a subspace. Determine whether the **inclusion map** $\iota : W \hookrightarrow V$ defined by $\iota(\mathbf{w}) = \mathbf{w}$ is a linear transformation.

Solution. The inclusion map is simply the identity map restricted to W , viewed as a map into V . The verification is identical to the identity map.

Example. (Zero Map)

For any vector spaces V and W , determine whether the **zero map** $0 : V \rightarrow W$ defined by $0(\mathbf{v}) = \mathbf{0}_W$ for all $\mathbf{v} \in V$ is a linear transformation.

Solution. For $\mathbf{v}_1, \mathbf{v}_2 \in V$ and $\lambda \in F$:

- **Additivity:** $0(\mathbf{v}_1 + \mathbf{v}_2) = \mathbf{0}_W = \mathbf{0}_W + \mathbf{0}_W = 0(\mathbf{v}_1) + 0(\mathbf{v}_2)$.
- **Homogeneity:** $0(\lambda\mathbf{v}_1) = \mathbf{0}_W = \lambda\mathbf{0}_W = \lambda \cdot 0(\mathbf{v}_1)$.

The zero map is linear.

2.1.2.5 Field-Dependent Examples

Example. (Complex Conjugation)

Determine whether complex conjugation $T : \mathbb{C} \rightarrow \mathbb{C}$ defined by $T(z) = \bar{z}$, where $\overline{a + bi} = a - bi$, is a linear transformation.

Solution. **As a map of $\mathbb{C}$ -vector spaces:** T is **not** linear over $\mathbb{C}$.

Consider $z = 1$ and $\alpha = i$. Then:

$$T(\alpha z) = T(i) = -i, \quad \text{but} \quad \alpha T(z) = i \cdot \bar{1} = i.$$

Since $T(i \cdot 1) = -i \neq i = i \cdot T(1)$, the homogeneity condition fails.

As a map of $\mathbb{R}$ -vector spaces: T is linear over $\mathbb{R}$.

For $z = a + bi$ and $w = c + di$ with $a, b, c, d \in \mathbb{R}$, and $\lambda \in \mathbb{R}$:

- **Additivity:** $T(z + w) = \overline{(a + c) + (b + d)i} = (a + c) - (b + d)i = (a - bi) + (c - di) = T(z) + T(w)$.
- **Homogeneity:** $T(\lambda z) = \overline{\lambda a + \lambda bi} = \lambda a - \lambda bi = \lambda(a - bi) = \lambda T(z)$.

This example illustrates that **linearity depends on the choice of scalar field**.

2.1.3 Non-Examples of Linear Transformations

It is equally important to recognize functions that are **not** linear transformations. We verify non-linearity by finding a counterexample to either (LT1) or (LT2).

Example. (Squaring)

Determine whether $T : F^2 \rightarrow F^2$ defined by $T(a_1, a_2) = (a_1^2, 0)$ is a linear transformation.

Solution. We show that T fails additivity. Consider $\mathbf{v} = (1, 0)$:

$$T(\mathbf{v}) + T(\mathbf{v}) = (1, 0) + (1, 0) = (2, 0),$$

but

$$T(\mathbf{v} + \mathbf{v}) = T(2, 0) = (4, 0).$$

Since $(2, 0) \neq (4, 0)$, additivity fails and T is not linear.

Alternatively, homogeneity fails: $T(2\mathbf{v}) = T(2, 0) = (4, 0) \neq 2 \cdot (1, 0) = (2, 0) = 2T(\mathbf{v})$.

Example. (Norm)

Determine whether $T : \mathbb{R}^n \rightarrow \mathbb{R}$ defined by $T(\mathbf{v}) = \|\mathbf{v}\|$ (the Euclidean norm) is a linear transformation.

Solution. Homogeneity fails for negative scalars. Consider $\mathbf{v} = (1, 0, \dots, 0)$ and $\lambda = -1$:

$$T(\lambda\mathbf{v}) = T(-1, 0, \dots, 0) = \|(-1, 0, \dots, 0)\| = 1,$$

but

$$\lambda T(\mathbf{v}) = (-1) \cdot \|(1, 0, \dots, 0)\| = -1.$$

Since $1 \neq -1$, homogeneity fails and T is not a linear transformation.

(The norm does satisfy $\|c\mathbf{v}\| = |c|\|\mathbf{v}\|$, but this is not the same as $c\|\mathbf{v}\|$ when $c < 0$.)

2.1.4 Basic Theorems

Theorem 2.1.1 — Zero Maps to Zero

Let $T : V \rightarrow W$ be a linear transformation. Then $T(\mathbf{0}_V) = \mathbf{0}_W$.

Proof. Using additivity with $\mathbf{v} = \mathbf{0}_V$:

$$T(\mathbf{0}_V) = T(\mathbf{0}_V + \mathbf{0}_V) = T(\mathbf{0}_V) + T(\mathbf{0}_V).$$

Adding $-T(\mathbf{0}_V)$ to both sides gives $\mathbf{0}_W = T(\mathbf{0}_V)$. ■

Remark. This theorem provides a quick test for non-linearity: if $T(\mathbf{0}) \neq \mathbf{0}$, then T is not linear.

Example. (Constant Function)

Determine whether $T : \mathbb{R} \rightarrow \mathbb{R}$ defined by $T(x) = 5$ for all $x \in \mathbb{R}$ is a linear transformation.

Solution. We use [Theorem 2.1.1](#): any linear transformation must map zero to zero.

Here, $T(0) = 5 \neq 0$.

Therefore, T is not a linear transformation.

Example. (Translation)

Determine whether $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by $T(\mathbf{v}) = \mathbf{v} + \mathbf{c}$ for some fixed nonzero $\mathbf{c} \in \mathbb{R}^2$ is a linear transformation.

Solution. Using [Theorem 2.1.1](#): $T(\mathbf{0}) = \mathbf{0} + \mathbf{c} = \mathbf{c} \neq \mathbf{0}$.

Therefore, translation by a nonzero vector is not linear.

Maps of the form $\mathbf{v} \mapsto \mathbf{A}\mathbf{v} + \mathbf{c}$ are called **affine transformations**. They are linear if and only if $\mathbf{c} = \mathbf{0}$.

Theorem 2.1.2 — Linear Maps Preserve Negation

Let $T : V \rightarrow W$ be a linear transformation. Then $T(-\mathbf{v}) = -T(\mathbf{v})$ for all $\mathbf{v} \in V$.

Proof. Using homogeneity with $\lambda = -1$:

$$T(-\mathbf{v}) = T((-1)\mathbf{v}) = (-1)T(\mathbf{v}) = -T(\mathbf{v}).$$

■

Theorem 2.1.3 — Equivalent Characterization of Linearity

Let V, W be vector spaces over F and $T : V \rightarrow W$ be a function. Then T is a linear transformation if and only if

$$T(\alpha\mathbf{v}_1 + \beta\mathbf{v}_2) = \alpha T(\mathbf{v}_1) + \beta T(\mathbf{v}_2)$$

for all $\alpha, \beta \in F$ and $\mathbf{v}_1, \mathbf{v}_2 \in V$.

Proof. ($\Rightarrow$) Suppose T is linear. Then for any $\alpha, \beta \in F$ and $\mathbf{v}_1, \mathbf{v}_2 \in V$:

$$\begin{aligned} T(\alpha\mathbf{v}_1 + \beta\mathbf{v}_2) &= T(\alpha\mathbf{v}_1) + T(\beta\mathbf{v}_2) && \text{(by LT1)} \\ &= \alpha T(\mathbf{v}_1) + \beta T(\mathbf{v}_2) && \text{(by LT2)} \end{aligned}$$

($\Leftarrow$) Suppose $T(\alpha\mathbf{v}_1 + \beta\mathbf{v}_2) = \alpha T(\mathbf{v}_1) + \beta T(\mathbf{v}_2)$ for all $\alpha, \beta \in F$ and $\mathbf{v}_1, \mathbf{v}_2 \in V$.

- **Additivity (LT1):** Setting $\alpha = \beta = 1$, we get $T(\mathbf{v}_1 + \mathbf{v}_2) = T(\mathbf{v}_1) + T(\mathbf{v}_2)$.
- **Homogeneity (LT2):** Setting $\beta = 0$, we get $T(\alpha\mathbf{v}_1 + \mathbf{0}) = \alpha T(\mathbf{v}_1) + 0 \cdot T(\mathbf{v}_2)$, i.e., $T(\alpha\mathbf{v}_1) = \alpha T(\mathbf{v}_1)$.

Thus T satisfies both (LT1) and (LT2), so T is linear. ■

Theorem 2.1.4 — Linear Maps Preserve Linear Combinations

Let $T : V \rightarrow W$ be a linear transformation. Then for any vectors $\mathbf{v}_1, \dots, \mathbf{v}_n \in V$ and scalars $a_1, \dots, a_n \in F$:

$$T\left(\sum_{i=1}^n a_i \mathbf{v}_i\right) = \sum_{i=1}^n a_i T(\mathbf{v}_i).$$

Proof. By induction on n . The base case $n = 1$ is homogeneity. For the inductive step:

$$\begin{aligned} T\left(\sum_{i=1}^{n+1} a_i \mathbf{v}_i\right) &= T\left(\sum_{i=1}^n a_i \mathbf{v}_i + a_{n+1} \mathbf{v}_{n+1}\right) \\ &= T\left(\sum_{i=1}^n a_i \mathbf{v}_i\right) + T(a_{n+1} \mathbf{v}_{n+1}) && \text{(additivity)} \\ &= \sum_{i=1}^n a_i T(\mathbf{v}_i) + a_{n+1} T(\mathbf{v}_{n+1}) && \text{(induction + homogeneity)} \\ &= \sum_{i=1}^{n+1} a_i T(\mathbf{v}_i). \end{aligned}$$
■

2.1.5 Linear Transformation Determined by Basis

Another useful fact of a linear transformation is that once we know how it acts on a basis, then we know how it acts on every vector in the domain.

Theorem 2.1.5 — Linear Transformations are Determined by Their Action on a Basis

Let $T : V \rightarrow W$ be a linear transformation and V is finite dimensional, and $B = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ be a basis for V . Then we have:

1. For every $\mathbf{v} \in V$, $f(\mathbf{v})$ is a linear combination of $f(\mathbf{v}_1), f(\mathbf{v}_2), \dots, f(\mathbf{v}_n)$.
2. Suppose the linear transformation $g : V \rightarrow W$ satisfies $g(\mathbf{v}_i) = f(\mathbf{v}_i)$ for all i . Then $g = f$.

We say ‘ f is determined by $f(\mathbf{v}_1), \dots, f(\mathbf{v}_n)$ ’ in this case.

Proof. Let $\mathbf{v} \in V$. Since B is a basis of V , we can write $\mathbf{v} = \alpha_1 \mathbf{v}_1 + \dots + \alpha_n \mathbf{v}_n$. Then

$$\begin{aligned} f(\mathbf{v}) &= f(\alpha_1 \mathbf{v}_1 + \dots + \alpha_n \mathbf{v}_n) \\ &= \alpha_1 f(\mathbf{v}_1) + \dots + \alpha_n f(\mathbf{v}_n). \end{aligned}$$

This proves the first condition.

Moreover, if $g : V \rightarrow W$ satisfies $g(\mathbf{v}_i) = f(\mathbf{v}_i)$, then:

$$\begin{aligned} g(\mathbf{v}) &= g(\alpha_1 \mathbf{v}_1 + \dots + \alpha_n \mathbf{v}_n) \\ &= \alpha_1 g(\mathbf{v}_1) + \dots + \alpha_n g(\mathbf{v}_n) \\ &= \alpha_1 f(\mathbf{v}_1) + \dots + \alpha_n f(\mathbf{v}_n) \\ &= f(\alpha_1 \mathbf{v}_1 + \dots + \alpha_n \mathbf{v}_n) \\ &= f(\mathbf{v}). \end{aligned}$$

Since $g(\mathbf{v}) = f(\mathbf{v})$ for all $\mathbf{v} \in V$, we have $f = g$. This finishes the second condition. ■

Example.

Is it possible to have a linear map $f : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ such that

$$f(1, 0) = (2, 3, 4), \quad f(1, 1) = (3, 4, 5), \quad f(2, 3) = (4, 56)?$$

Solution. Idea: Since $\{(1, 0), (1, 1)\}$ forms a basis of $\mathbb{R}^2$, these two function value would determine f completely.

Suppose such a linear transformation f exists. Observe that

$$(2, 3) = -1(1, 0) + 3(1, 1).$$

Apply f then we have:

$$\begin{aligned} f(2, 3) &= f(-1(1, 0) + 3(1, 1)) \\ &= -1f(1, 0) + 3f(1, 1) \\ &= -1(2, 3, 4) + 3(3, 4, 5) \\ &= (7, 8, 11) \\ &\neq (4, 56). \end{aligned}$$

Which is inconsistent with the third equation, therefore no such transformation exists.

Example.

Suppose we are given a linear transformation $f : \mathbb{R}[x]_{\leq 3} \rightarrow \mathbb{R}[x]_{\leq 2}$. And it is also known that

$$f(x^3) = 3x^2, \quad f(x^2) = 2x, \quad f(x) = 1, \quad f(1) = 0.$$

Solution. As $\{1, x, x^2, x^3\}$ forms a basis of $\mathbb{R}[x]_{\leq 3}$. And $D(x^k) = f(x^k)$, for $k = 0, 1, 2, 3$, as D is differentiation. We have that $D \equiv f$ is the same linear transformation.

2.2 Kernel and Image

2.2.1 Definition

Definition 2.2.1 — Kernel

Let $f : V \rightarrow W$ be a linear transformation. Define the **kernel** of f by

$$\ker(f) := \{\mathbf{v} \in V : f(\mathbf{v}) = \mathbf{0}_W\}.$$

i.e. The kernel contains everything in the domain that becomes the zero vector under f .

Definition 2.2.2 — Image

Let $f : V \rightarrow W$ be a linear transformation. Define the **image** of f by

$$\text{im}(f) := \{\mathbf{w} \in W : \mathbf{w} = f(\mathbf{v}) \text{ for some } \mathbf{v} \in V\}.$$

i.e. The image contains all the possible outputs of the function f .

Remark. The kernel is also known as the nullspace, which you might be familiar from Gilbert Strang's book. And the image is also called the range. Also, we sometimes denote the image as $f(V)$.

2.2.2 Examples

Example. (Differentiation)

Let $D : \mathbb{R}[x]_{\leq 3} \rightarrow \mathbb{R}[x]_{\leq 2}$ be defined by $D(f(x)) = f'(x)$. Find the kernel and image of D .

Solution. Kernel: We seek all polynomials $f(x) \in \mathbb{R}[x]_{\leq 3}$ such that $D(f(x)) = f'(x) = 0$.

A polynomial has derivative zero if and only if it is a constant. Thus:

$$\ker(D) = \{c : c \in \mathbb{R}\} = \text{span}\{1\}.$$

So $\dim(\ker(D)) = 1$.

Image: We need to determine which polynomials in $\mathbb{R}[x]_{\leq 2}$ can be obtained as derivatives.

Let $f(x) = a_0 + a_1x + a_2x^2 + a_3x^3 \in \mathbb{R}[x]_{\leq 3}$. Then:

$$D(f(x)) = f'(x) = a_1 + 2a_2x + 3a_3x^2.$$

As a_1, a_2, a_3 range over all real numbers, we can obtain any polynomial of degree at most 2. Therefore:

$$\text{im}(D) = \mathbb{R}[x]_{\leq 2}.$$

So $\dim(\text{im}(D)) = 3$.

Note that $\dim(\ker(D)) + \dim(\text{im}(D)) = 1 + 3 = 4 = \dim(\mathbb{R}[x]_{\leq 3})$.

Example. (Trace)

Let $\text{tr} : M_2(\mathbb{R}) \rightarrow \mathbb{R}$ be defined by $\text{tr}(\mathbf{A}) = a_{11} + a_{22}$. Find the kernel and image of tr .

Solution. Kernel: We seek all matrices $\mathbf{A} \in M_2(\mathbb{R})$ such that $\text{tr}(\mathbf{A}) = 0$.

Let $\mathbf{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. Then $\text{tr}(\mathbf{A}) = a + d = 0$, which means $d = -a$.

So the kernel consists of all matrices of the form:

$$\mathbf{A} = \begin{bmatrix} a & b \\ c & -a \end{bmatrix} = a \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} + b \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} + c \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}.$$

Therefore:

$$\ker(\text{tr}) = \text{span} \left\{ \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} \right\}.$$

So $\dim(\ker(\text{tr})) = 3$.

Image: For any $r \in \mathbb{R}$, we can find a matrix with trace r , for example $\begin{bmatrix} r & 0 \\ 0 & 0 \end{bmatrix}$.

Therefore:

$$\text{im}(\text{tr}) = \mathbb{R}.$$

So $\dim(\text{im}(\text{tr})) = 1$.

Note that $\dim(\ker(\text{tr})) + \dim(\text{im}(\text{tr})) = 3 + 1 = 4 = \dim(M_2(\mathbb{R}))$.

2.2.3 Basic Theorems

Theorem 2.2.1 — Properties of Kernel

Let $f : V \rightarrow W$ be a linear transformation. Then:

1. $\ker(f)$ is a subspace of V ;
2. f is injective iff $\ker(f) = \{\mathbf{0}_V\}$.

Proof.

1.
 - Non-emptiness: As f is linear, $f(\mathbf{0}_V) = \mathbf{0}_W$, therefore $\mathbf{0}_V \in \ker(f)$.
 - Closure under addition: Let $\mathbf{v}_1, \mathbf{v}_2 \in \ker(f)$, it implies that $f(\mathbf{v}_1) = \mathbf{0}_W$ and $f(\mathbf{v}_2) = \mathbf{0}_W$. We have

$$f(\mathbf{v}_1 + \mathbf{v}_2) = f(\mathbf{v}_1) + f(\mathbf{v}_2) = \mathbf{0}_W + \mathbf{0}_W = \mathbf{0}_W.$$

Therefore $\mathbf{v}_1 + \mathbf{v}_2 \in \ker(f)$.

- Closure under scalar multiplication: Let $\mathbf{v} \in \ker(f)$, it implies that $f(\mathbf{v}) = \mathbf{0}_W$. We have

$$f(\lambda \mathbf{v}) = \lambda f(\mathbf{v}) = \lambda \mathbf{0}_W = \mathbf{0}_W.$$

Therefore $\lambda \mathbf{v} \in \ker(f)$.

2.
 - ($\Rightarrow$) Suppose f is injective. Let $\mathbf{v} \in \ker(f)$, we have

$$\begin{aligned} f(\mathbf{v}) &= \mathbf{0}_W \\ &= f(\mathbf{0}_V) \end{aligned} \quad (\text{linearity})$$

By having $f(\mathbf{v}) = f(\mathbf{0}_V)$ and f is injective, it implies that $\mathbf{v} = \mathbf{0}_V$, so $\ker(f) = \{\mathbf{0}_V\}$.

- ($\Leftarrow$) Suppose $\ker(f) = \{\mathbf{0}_V\}$, and we consider $f(\mathbf{v}) = f(\mathbf{w})$. Reordering the terms we get $f(\mathbf{v}) - f(\mathbf{w}) = \mathbf{0}_W$, and since f is linear we have $f(\mathbf{v} - \mathbf{w}) = \mathbf{0}_W$. Therefore $\mathbf{v} - \mathbf{w} \in \ker(f) = \{\mathbf{0}_V\}$, implying that $\mathbf{v} - \mathbf{w} = \mathbf{0}_V$, hence $\mathbf{v} = \mathbf{w}$. So f is injective.

■

Theorem 2.2.2 — Properties of Image

Let $f : V \rightarrow W$ be a linear transformation. Then:

1. $\text{im}(f)$ is a subspace of W ;
2. f is surjective iff $\text{im}(f) = W$.

Proof.

1.
 - Non-emptiness: Since f is linear, $\mathbf{0}_W = f(\mathbf{0}_V)$, therefore $\mathbf{0}_W \in \text{im}(f)$.
 - Closure under addition: Let $\mathbf{w}_1, \mathbf{w}_2 \in \text{im}(f)$. So $\mathbf{w}_1 = f(\mathbf{v}_1)$, and $\mathbf{w}_2 = f(\mathbf{v}_2)$

for some $\mathbf{v}_1, \mathbf{v}_2 \in V$. Notice that

$$\mathbf{w}_1 + \mathbf{w}_2 = f(\mathbf{v}_1) + f(\mathbf{v}_2) = f(\mathbf{v}_1 + \mathbf{v}_2).$$

Therefore $\mathbf{w}_1 + \mathbf{w}_2 \in \text{im}(f)$.

- Closure under scalar multiplication: Let $\mathbf{w} \in \text{im}(f)$. So $\mathbf{w} = f(\mathbf{v})$ for some $\mathbf{v} \in V$. Notice that

$$\lambda \mathbf{w} = \lambda f(\mathbf{v}) = f(\lambda \mathbf{v}).$$

Therefore $\lambda \mathbf{w} \in \text{im}(f)$.

2. f is surjective $\Leftrightarrow \forall \mathbf{w} \in W$, we have $\mathbf{w} \in f(\mathbf{v})$ for some $\mathbf{v} \in V$, that is equivalent to say $W \subseteq \text{im}(f)$, and by the first condition we know that $\text{im}(f) \subseteq W$ is always true. Hence we have f is surjective $\Leftrightarrow W = \text{im}(f)$.

■

2.3 Rank-Nullity Theorem

2.3.1 Nullity and Rank

Definition 2.3.1 — Nullity

Let $f : V \rightarrow W$ be a linear transformation. The **nullity** of f is defined by

$$\text{nullity}(f) := \dim(\ker(f)).$$

Definition 2.3.2 — Rank

Let $f : V \rightarrow W$ be a linear transformation. The **rank** of f is defined by

$$\text{rank}(f) := \dim(\text{im}(f)).$$

Example.

Recall that for $D : \mathbb{R}[x]_{\leq 3} \rightarrow \mathbb{R}[x]_{\leq 2}$ defined by

$$D(f(x)) = f'(x),$$

we have $\ker(D) = \mathbb{R}$ and $\text{im}(D) = \mathbb{R}[x]_{\leq 2}$. Compute the nullity and rank of D .

Solution. $\text{nullity}(D) := \dim_{\mathbb{R}}(\ker(D)) = \dim_{\mathbb{R}}(\mathbb{R}) = 1.$

$\text{rank}(D) := \dim_{\mathbb{R}}(\text{im}(D)) = \dim_{\mathbb{R}}(\mathbb{R}[x]_{\leq 2}) = 3.$

Remark. For other examples of nullity and rank, review the examples in the last section, which we have it covered.

2.3.2 The Theorem

Theorem 2.3.1 — Rank-Nullity Theorem

Let V be a finite dimensional vector space and $f : V \rightarrow W$ be a linear transformation. Then we have

$$\text{nullity}(f) + \text{rank}(f) = \dim(V).$$

Proof. Let $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ be a basis of $\ker(f)$, having $\text{nullity}(f) = \dim(\ker f) = n$. Since $\ker(f) \subseteq V$, by the Basis Extension Theorem (Theorem 1.6.5), we can extend S to $S' = \{\mathbf{v}_1, \dots, \mathbf{v}_n, \mathbf{s}_1, \dots, \mathbf{s}_k\}$, a basis of V , having $\dim(V) = n + k$.

Claim. $S'' = \{f(\mathbf{s}_1), f(\mathbf{s}_2), \dots, f(\mathbf{s}_k)\}$ is a basis of $\text{im}(f)$.

Proof. We verify two properties.

Spanning. Let $\mathbf{w} \in \text{im}(f)$, we have $\mathbf{w} = f(\mathbf{v})$ for some $\mathbf{v} \in V$ by definition. As S' is a basis of V , we can write $\mathbf{v} = \sum_{i=1}^n \alpha_i \mathbf{v}_i + \sum_{i=1}^k \beta_i \mathbf{s}_i$. Then we apply f to both sides:

$$\mathbf{w} = f(\mathbf{v}) = f\left(\sum \alpha_i \mathbf{v}_i + \sum \beta_i \mathbf{s}_i\right) = \sum \alpha_i f(\mathbf{v}_i) + \sum \beta_i f(\mathbf{s}_i) = \sum \beta_i f(\mathbf{s}_i),$$

where the last equality holds since $\mathbf{v}_i \in \ker(f)$. Since $\mathbf{w}$ can be expressed by $f(\mathbf{s}_i)$'s, therefore S'' spans $\text{im}(f)$.

Linear independence. Suppose that $\alpha_1 f(\mathbf{s}_1) + \dots + \alpha_k f(\mathbf{s}_k) = \mathbf{0}$. By linearity of f , it becomes $f(\alpha_1 \mathbf{s}_1 + \dots + \alpha_k \mathbf{s}_k) = \mathbf{0}$. Therefore we have $\alpha_1 \mathbf{s}_1 + \dots + \alpha_k \mathbf{s}_k \in \ker(f)$. Moreover, as S is a basis of $\ker(f)$, we know that $\sum \alpha_i \mathbf{s}_i = \sum \beta_i \mathbf{v}_i$ for some $\beta_i \in F$. Reordering gives $\sum \alpha_i \mathbf{s}_i - \sum \beta_i \mathbf{v}_i = \mathbf{0}$. As S' is linearly independent, this forces all α_i 's and β_i 's to be 0. This completes the proof of the claim. ■

Hence, we have

$$\text{rank}(f) = \dim(\text{im}(f)) = k = (n + k) - n = \dim(V) - \text{nullity}(f). \quad \blacksquare$$

For the map $f_{\mathbf{A}} : \mathbb{R}^4 \rightarrow \mathbb{R}^3$, $f_{\mathbf{A}}(\mathbf{v}) = \mathbf{A}\mathbf{v}$, the rank and nullity can be computed exactly with SymPy. In the online version the code can be edited and run; for any matrix, the sum is the number of columns, $\dim(\mathbb{R}^4) = 4$.

```
from sympy import Matrix

A = Matrix([
    [1, 2, 3, 4],
    [2, 4, 6, 8],
    [1, 0, 1, 0],
])
rank = A.rank() # dim(im f_A)
nullity = len(A.nullspace()) # dim(ker f_A): size of a basis of the kernel
```

```
print("rank =", rank, " nullity =", nullity, " columns =", A.cols)
rank + nullity == A.cols
```